SM-X906B_M_REV0.4A

SM-X906B/C
Common

2021.11.04

- sheet01 : RF (1/3)
- sheet02 : RF (2/3)
- sheet03 : RF (3/3)
- sheet04 : BT / WIFI
- sheet05 : AP_SM8450(Secondary PCB), JTAG, UFS
- sheet06 : AP_SM8450(Secondary PCB)
- sheet07 : PMIC(Secondary PCB), IF PMIC(CC IC), D/C
- sheet08 : AUDIO
- sheet09 : CAMERA / DISPLAY
- sheet10 : SENSOR / SD_SIM / KEY / IF
- sheet11 : CP PMIC
- sheet12 : WACOM / ANT SUB PCB
- sheet13 : USB PCB
- sheet14 : Secondary PCB PAD

TITLE TabS7P
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name SHEET 0 of 14 Date Drawing Date

MAIN RF_1

MAIN1 ANT
(TRX_LB/MB/HB / TRX_N41)
WIFI : NC
VCC_1P8_AP_RF
R1000
C1004
U1000
SW1
SW2
SW3
SW4
SW12
SW34
VIO
C1I
C1O
DATA
CLK
USID
GND
SDM_RFFE0_DATA
SDM_RFFE0_CLK
RF_MAIN_1
C1010
L1022
L1023
L1027
L1028
C1013
C1014
ANT1000
C1029
L1036
C1030
C1031
ANT1002
RFS1000
ANT
COMMON
C1020
C1019
L1030
L1031
C1027
U1002
HB
LB
ground
C1168
L1106
C1169
L1105
C1012
L1025
L1026
C1037
L1041
L1042
MAIN1_LMB_TRX
U1001
RF1
RF3
SDATA
SCLK
RF2
RF4
n41_SRS1_SUB1
n41_SRS1_SUB2
n41_SRS1_SUB4
SDR1_RFFE1_DATA
SDR1_RFFE1_CLK
R1001
R1002
C1170
L1018
L1019
L1020
C1007
MAIN1_HB_TRX
@VDD_5G_SW_1P8
VREG_S10B_1P8
C1016
F1004
HIGH
LOW
GND|NC
C1047
L1050
L1051
MAIN1_MB_TRX
C1062
L1060
L1061
MAIN1_LB_TRX
L1048
VCC_1P8_AP_RF
ANT1003
C1063
L1064
L1062
L1063
MAIN1_GRIP
R1004
C1065
U1006
RFC
RF4
RF1
USID
VIO
RF3
RF2
GND
SCLK
TP1002
TP1003
SDM_RFFE1_CLK
SDM_RFFE1_DATA
C1068
C1066
C1067
L1065
L1066
MAIN1 ANT XGND1
VPH_PWR
VDD_ENDC_GRIP_LDO_3P0
R1005
U1007
VDD
VOUT
CE
GRIP3_LDO_EN
R1006
C1076
C1081
C1077
ENDC GRIP LDO
MAIN2 ANT
(TRX_N77/N78)
ANT1004
V1000
C1103
L1101
L1077
ANT1005
C1112
C1113
L1102
L1072
L1073
MAIN2_WIFI1_GRIP
C1104
CN1001
IN
C1171
L1170
L1173
VDD_WIFI_1P8
R1016
R1017
CABLE_DET_MAIN3
C1172
ZD1001
C1173
RFS1001
L1078
C1106
L1076
C1107
L1079
L1080
MAIN2_n77_78_TRX
close to PA
VCC_1P8_AP_RF
R1013
U1010
TP1004
TP1005
SDM_RFFE0_CLK
SDM_RFFE0_DATA
L1081
L1082
C1114
C1115
C1174
VREG_L4E_1P776
VRF_PA_3
C1117
C1118
C1119
C1124
C1125
C1120
C1121
U1011
PIN11->VCC FOR PIN2PIN
ANT1
ANT2
SDATA_RX
SCLK_RX
SDATA_TX
SCLK_TX
VBATT
VIO_TX
VIO_RX
VCC
NC
LNA_OUT1
LNA_OUT2
TX_IN
CPL
ANT3_SRS
C1136
MAIN2_n77_78_TRX
SUB4_n77_78_PRX_MIMO
L1095
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
C1134
C1137
C1141
C1142
L1096
C1147
L1168
PRX1_TR_IN_UN
PRX1_TR_IN_M_UN
TX_TR_OUT0_UHB
CPL1_n77_78_OUT
n78_SRS_TX0
UHB PAMID
Global N77/78
NA B48/N77
SUB3 ANT
(DRX MIMO_n77/78)
CN1000
C1157
C1158
C1159
L1099
L1098
L1097
L1169
SUB3_n77_78_DRX_MIMO
L1172
VDD_WIFI_1P8
R1014
R1015
CABLE_DET_SUB3
C1160
ZD1000
C1161
REAR CONTACT
ANT1014
ANT1007
ANT1008
ANT1009
ANT1010
ANT1011
ANT1012
ANT1013
LB LPAMID
Global B5/8/12/13/17/20/26/28/29, G850/900
NA B5/8/12/13/14/20/26/28/29/71, G850/900
VREG_S10B_1P8
VRF_MIPI_IO_1P8
L1009
VREG_L4E_1P776
VREG_LNA_1P8
L1029
VRF_MIPI_IO_1P8
C1028
TX_TR_OUT0_LB_SW_OUT
C1032
L1037
C1025
C1026
P17
P33
TX_TR_OUT1_LB
C1035
C1038
L1038
TX_TR_OUT0_GSM_LB
C1039
C1040
L1043
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
TRX_B13
TRX_B28
TRX_B20_N28A
RF_IN1
RF_IN2
2G_IN
VIO1
SCLK1
SDATA1
VIO2
SCLK2
SDATA2
TRX1
TRX2
TRX3
TRX4
VPH_PWR
VRF_PA_0
C1011
L1021
C1008
C1009
C1015
C1017
C1018
Need to check
MAIN1_LB_TRX
ANT
VBATT
VCC1
VCC2
2G_VCC1
2G_VCC2
U1003
RX_OUT1
RX_OUT2
C1033
C1034
PRX1_TR_IN_L
PRX2_TR_IN_L
LNA_AUX1
LNA_AUX2
LNA_AUX3
LNA_AUX4
LNA_IN_AUX_B13
LNA_IN_AUX_B28
LNA_IN_AUX_B20_N28A
PA_OUT1
PA_OUT2
PA_OUT3
PA_OUT4
TX_OUT_B13
TX_OUT_B28
TX_OUT_B20
CPL_OUT
CPL_LB
CPL_IN1
CPL_IN2
CPL_IN3
CPL_OMH
CPL1_n77_78_OUT
U1005
SDR1_GRFC1
VCTL
RF2
RF1
LPWR
L1057
C1054
C1055
C1056
L1056
C1052
C1053
N28_TX2
TX_TR_OUT0_LB
TX_TR_OUT0_LB_SW_OUT
B13/B14
Global : B8682(B13 only)
NA : SAYRH757MBA0C3K
L1000
L1001
L1002
C1000
L1003
L1006
L1007
L1008
TRX_B13
F1001
TX
RX
G1 G2G3G5G6
C1005
C1006
TX_OUT_B13
LNA_IN_AUX_B13
L1014
L1015
L1016
L1017
C1036
TRX_B28
L1039
L1040
F1003
C1041
C1042
TX_OUT_B28
LNA_IN_AUX_B28
L1044
L1045
L1046
L1047
B28
Global only
C1048
C1049
TRX_B20_N28A
L1052
L1053
F1005
C1057
C1058
C1059
TX_OUT_B20
LNA_IN_AUX_B20_N28A
C1060
L1058
C1061
L1059
OMH LPAMID
Global 1/2/3/4/7/25/34/38/39/40/41/66, G1800/1900, N41
NA 1/2/3/4/7/25/30/38/39/40/41/66, G1800/1900, N41
Global only
F1006
C1074
C1075
PRX_B32_n75
PRX_LNA_IN_B32_n75
C1082
C1166
C1083
C1167
TH1000
PA_THM0
THERMISTERS
VPH_PWR
C1164
C1163
C1162
VREG_LNA_1P8
VRF_PA_1
VRF_MIPI_IO_1P8
C1069
C1070
C1071
C1072
C1073
C1078
C1079
C1080
C1084
C1085
C1086
U1008
MHB_ANT1
MHB_ANT2
CPLOUT
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
SCLK
SCLK2
SDATA
SDATA2
MAIN1_MB_TRX
MAIN1_HB_TRX
C1094
CPL_OMH
SDR1_RFFE1_CLK
SDR1_RFFE0_CLK
SDR1_RFFE1_DATA
SDR1_RFFE0_DATA
MHB_VCC2
MHB_VCC1
LNA_VDD
2G_VCC2
2G_VCC1
VIO2
VIO
VBATT
MB_IN
HB_IN
HB_TX_OUT
MB_2G_IN
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
L1067
L1068
L1069
C1087
C1088
C1089
C1090
C1095
C1096
TX_TR_OUT1_MB1
TX_TR_OUT1_HB1
TX_TR_OUT0_GSM_HB
L1070
L1071
L1074
L1075
L1100
PRX1_TR_IN_MH
PRX1_TR_IN_JMH
PRX2_TR_IN_MH
PRX2_TR_IN_JMH
PRX1_TR_IN_LMH
APT Modulator
LB APT Modulator
(QET6105#0)
MHB APT Modulator
(QET6105#1)
ENDC APT Modulator
(QET6105#2)
UHB APT Modulator
(QET6105#3)
VPH_PWR
VPH_PWR_APT0
L1085
VPH_PWR_APT2
L1086
VPH_PWR_APT3
L1084
U1015
U1013
U1014
U1012
L1090
L1088
L1089
L1087
IND pin#1 to switching pad
VSW_VREG_1
VSW_VREG_2
VSW_VREG_3
GND_VREG_1
GND_VREG_2
VPH_SW
GND_BCK_1
GND_BCK_2
VPH_CLEAN
VDD_1P8
GND
GND_SUB
VREG_S
VREG_B
VSW_BCK_1
VSW_BCK_2
VSW_BCK_3
VPA_FB
TPP1
TPP2
SCLK
SDATA
USID
GND_RESERVED
C1129
C1133
L1094
C1127
C1131
L1092
C1128
C1132
L1093
C1126
C1130
L1091
VRF_PA_0
VRF_PA_1
VRF_PA_2
VRF_PA_3
C1146
C1144
C1145
C1143
C1140
C1138
C1139
C1135
C1151
C1149
C1150
C1148
C1155
C1153
C1154
C1152
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
SDR1_RFFE1_CLK
SDR1_RFFE1_DATA
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
WIFI : NC

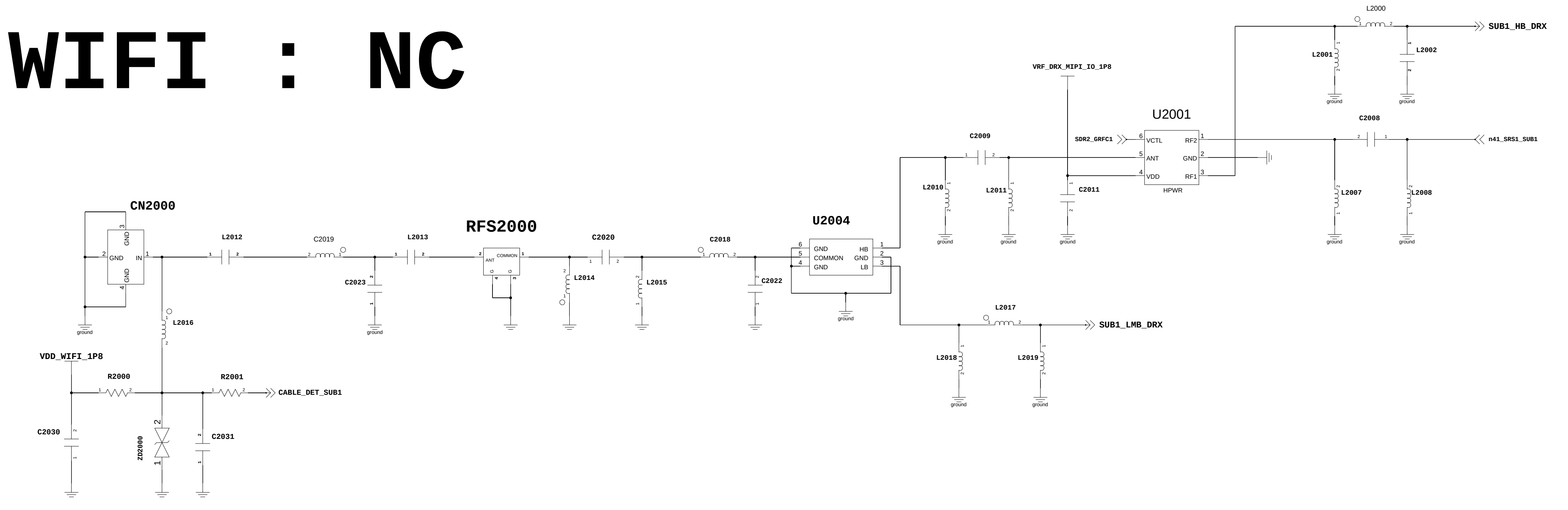
REF	NA	EUR/CHN
C1021 C1024	100pF	OPEN

REF	NA	EUR/CHN
C1049 C1059	OPEN	100pF

TITLE TabS7P
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 1 of 12
Date Drawing Date

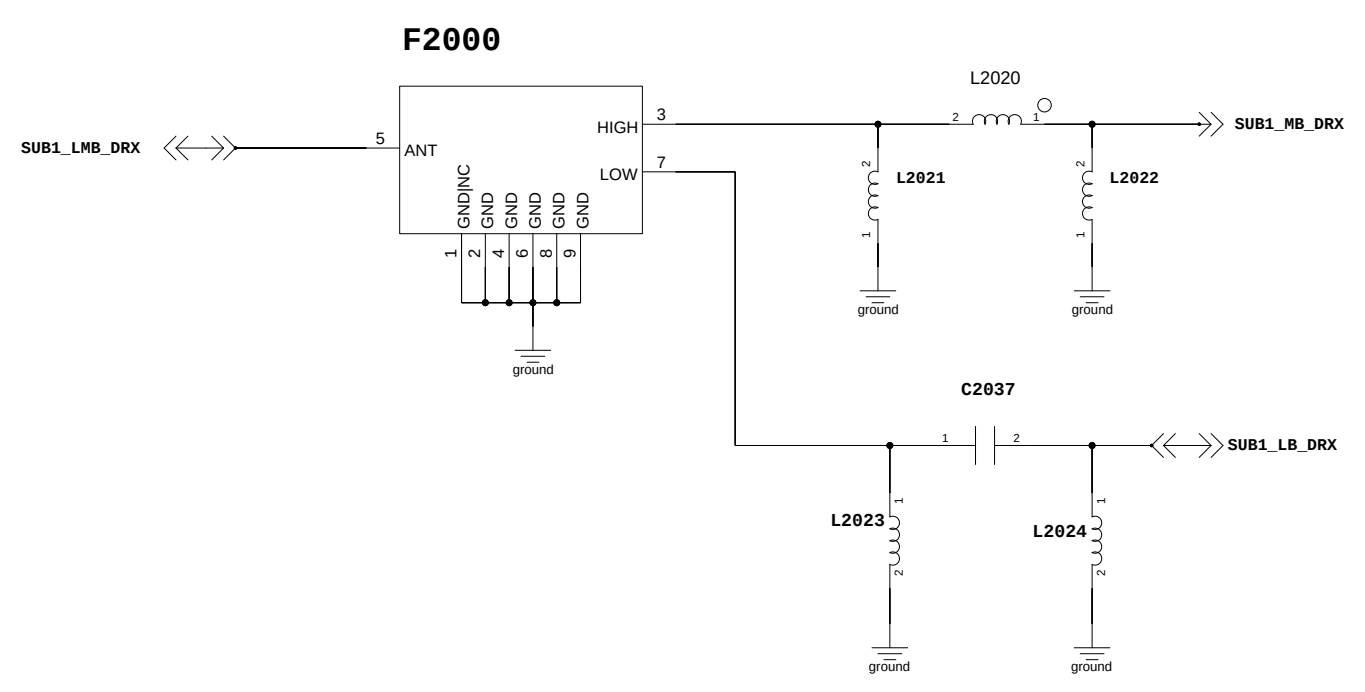
MAIN RF_2

WIFI : NC

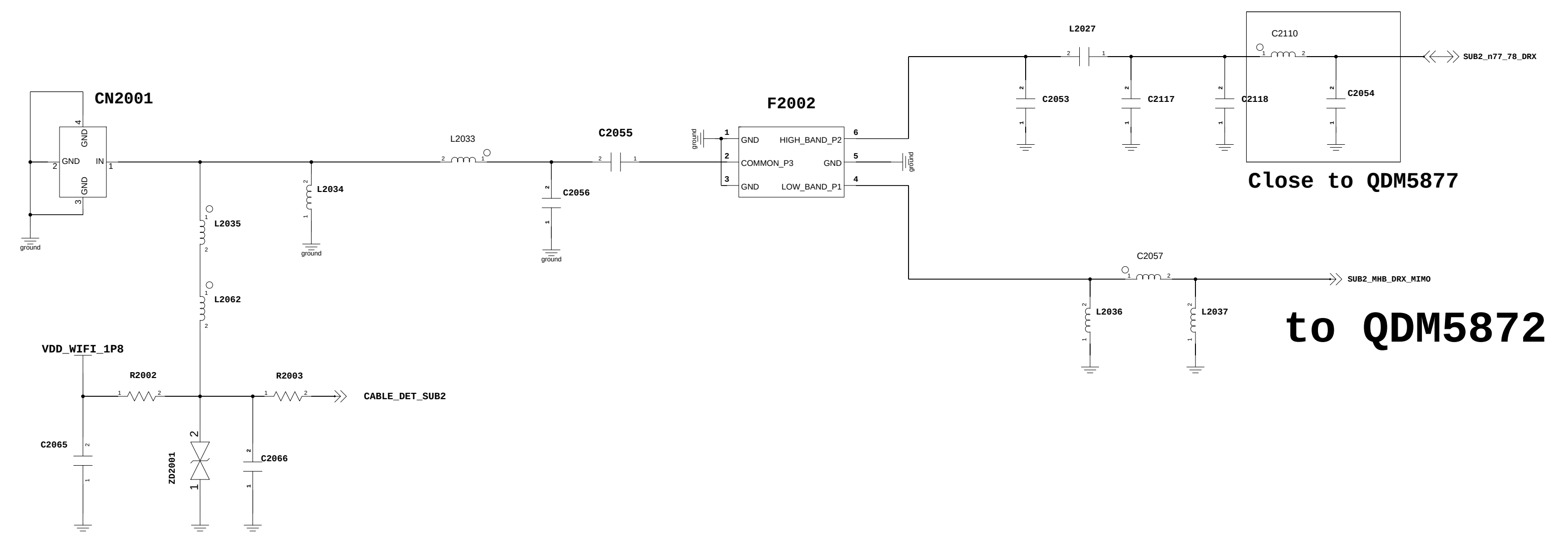


CN2000
GND
IN
L2012
C2019
L2013
RFS2000
COMMON
ANT
C2020
C2018
U2004
GND
COMMON
HB
GND
LB
C2023
L2014
L2015
C2022
L2016
VDD_WIFI_1P8
R2000
R2001
CABLE_DET_SUB1
C2030
ZD2000
C2031
C2009
SDR2_GRFC1
U2001
VCTL
ANT
VDD
RF2
GND
RF1
HPWR
VRF_DRX_MIPI_IO_1P8
L2010
L2011
C2011
L2017
SUB1_LMB_DRX
L2018
L2019
L2000
SUB1_HB_DRX
L2001
L2002
C2008
n41_SRS1_SUB1
L2007
L2008
ground

SUB1 ANT
(DRX_LB/MB/HB / DRX_n41 / ENDC_LB/MB TX)

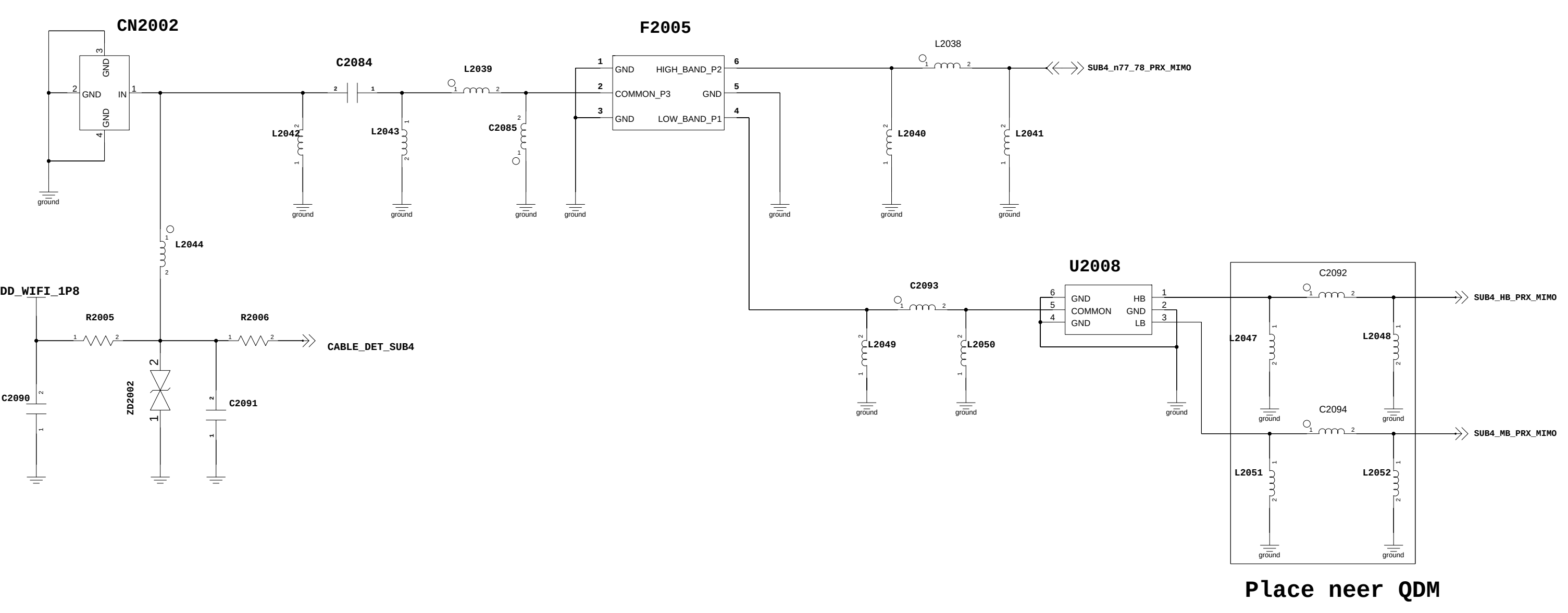


F2000
SUB1_LMB_DRX
ANT
HIGH
LOW
GND(NC)
GND
L2020
SUB1_MB_DRX
L2021
L2022
C2037
SUB1_LB_DRX
L2023
L2024
ground



CN2001
GND
IN
L2033
C2055
F2002
GND
HIGH_BAND_P2
COMMON_P3
GND
LOW_BAND_P1
L2034
C2056
L2035
L2062
VDD_WIFI_1P8
R2002
R2003
CABLE_DET_SUB2
C2065
ZD2001
C2066
L2027
C2053
C2117
C2118
C2110
C2054
SUB2_n77_78_DRX
Close to QDM5877
C2057
SUB2_MHB_DRX_MIMO
L2036
L2037
to QDM5872
ground

SUB2 ANT
(DRX_MIMO_MB/HB, DRX_n77/n78 / DRX_MIMO_n41)



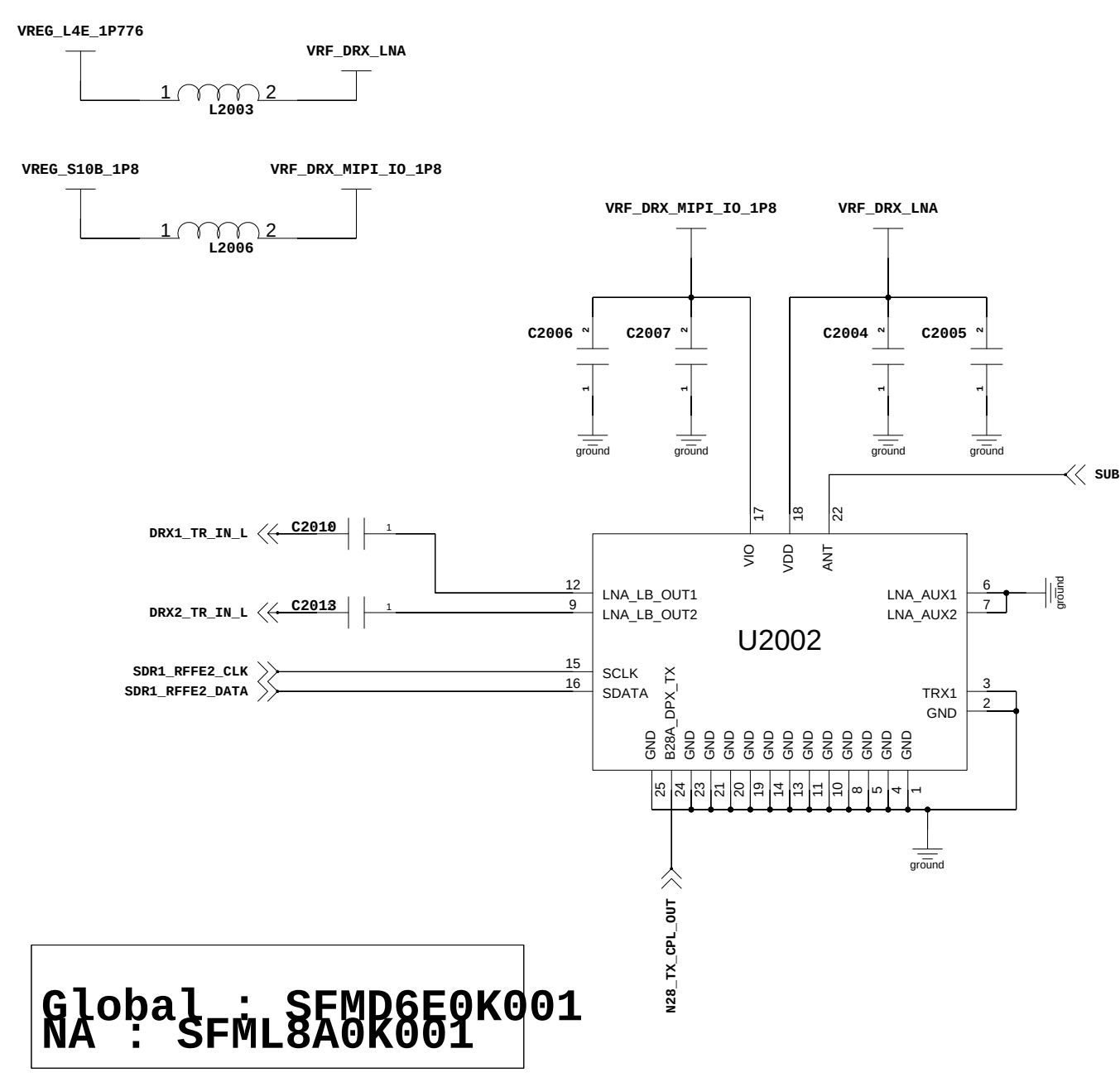
CN2002
GND
IN
C2084
L2039
F2005
GND
HIGH_BAND_P2
COMMON_P3
GND
LOW_BAND_P1
L2042
L2043
C2085
L2038
SUB4_n77_78_PRX_MIMO
L2040
L2041
L2044
VDD_WIFI_1P8
R2005
R2006
CABLE_DET_SUB4
C2090
ZD2002
C2091
C2093
U2008
GND
COMMON
HB
GND
LB
L2049
L2050
C2092
SUB4_HB_PRX_MIMO
L2047
L2048
C2094
SUB4_MB_PRX_MIMO
L2051
L2052
Place neer QDM
ground

SUB4 ANT

(PRX_MIMO_MB/HB / PRX_MIMO_n77/78 / PRX_MIMO_n41)

DRX_LB

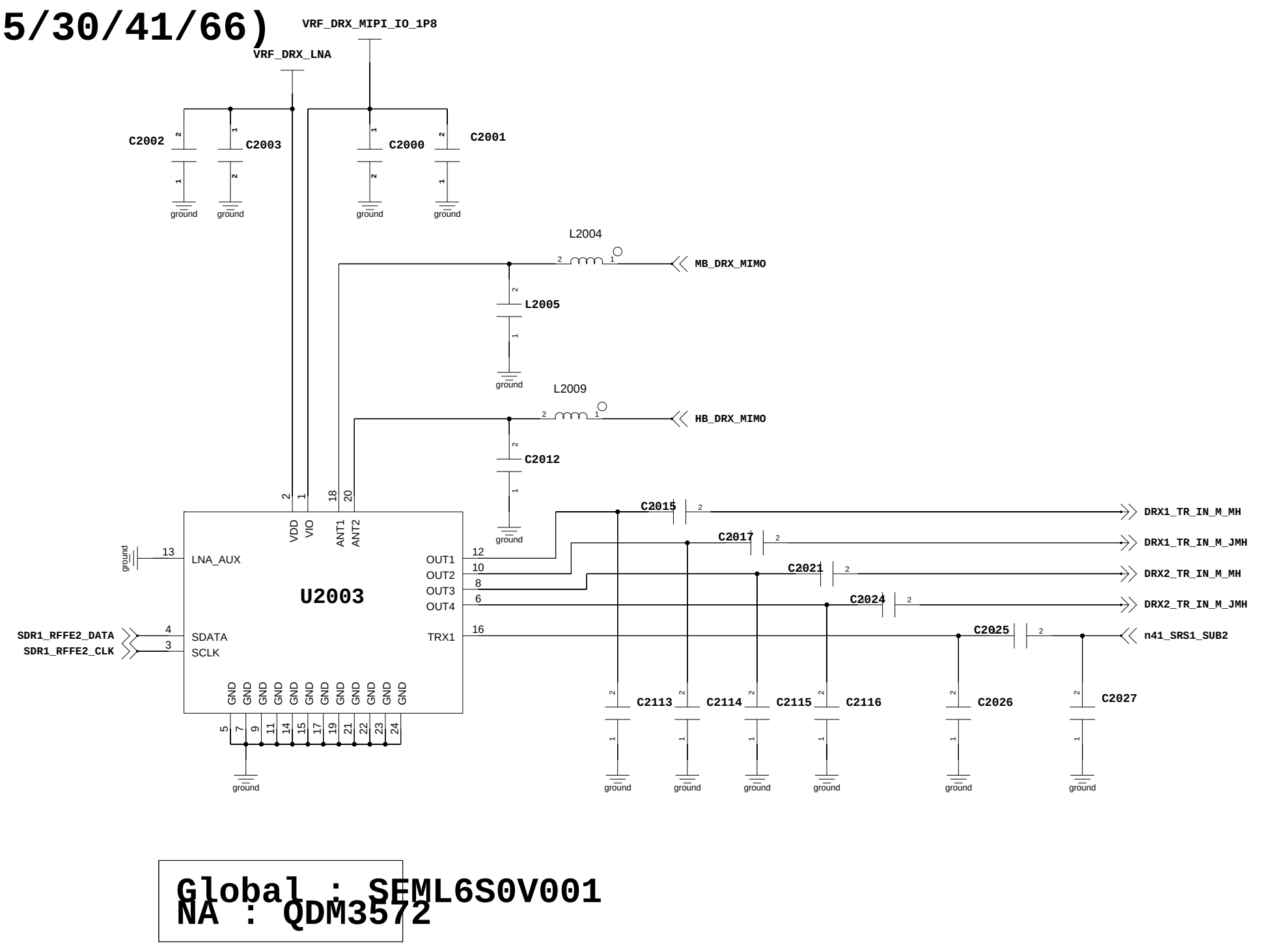
(B5/8/12/13/14/17/18/19/20/26/28/29/71_BC0/10_G850/900)



VREG_L4E_1P776
VRF_DRX_LNA
L2003
VREG_S10B_1P8
VRF_DRX_MIPI_IO_1P8
L2006
VRF_DRX_MIPI_IO_1P8
VRF_DRX_LNA
C2006
C2007
C2004
C2005
SUB1_LB_DRX
DRX1_TR_IN_L
C2010
DRX2_TR_IN_L
C2013
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
U2002
VIO
VDD
ANT
LNA_LB_OUT1
LNA_LB_OUT2
LNA_AUX1
LNA_AUX2
SCLK
SDATA
TRX1
GND
B28A_DPX_TX
N28_TX_CPL_OUT
Global : SFMD6E0K001
NA : SFML8A0K001
ground

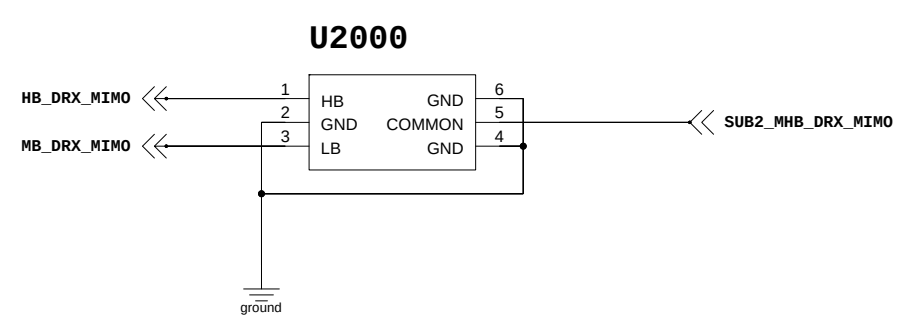
DRX_MHB_2

(B2/4/7/25/30/41/66)



VRF_DRX_MIPI_IO_1P8
VRF_DRX_LNA
C2002
C2003
C2000
C2001
L2004
MB_DRX_MIMO
L2005
L2009
HB_DRX_MIMO
C2012
U2003
VDD
VIO
ANT1
ANT2
LNA_AUX
SDATA
SCLK
OUT1
OUT2
OUT3
OUT4
TRX1
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
C2015
DRX1_TR_IN_M_MH
C2017
DRX1_TR_IN_M_JMH
C2021
DRX2_TR_IN_M_MH
C2024
DRX2_TR_IN_M_JMH
C2025
n41_SRS1_SUB2
C2113
C2114
C2115
C2116
C2026
C2027
Global : SFML6S0V001
NA : QDM3572
ground

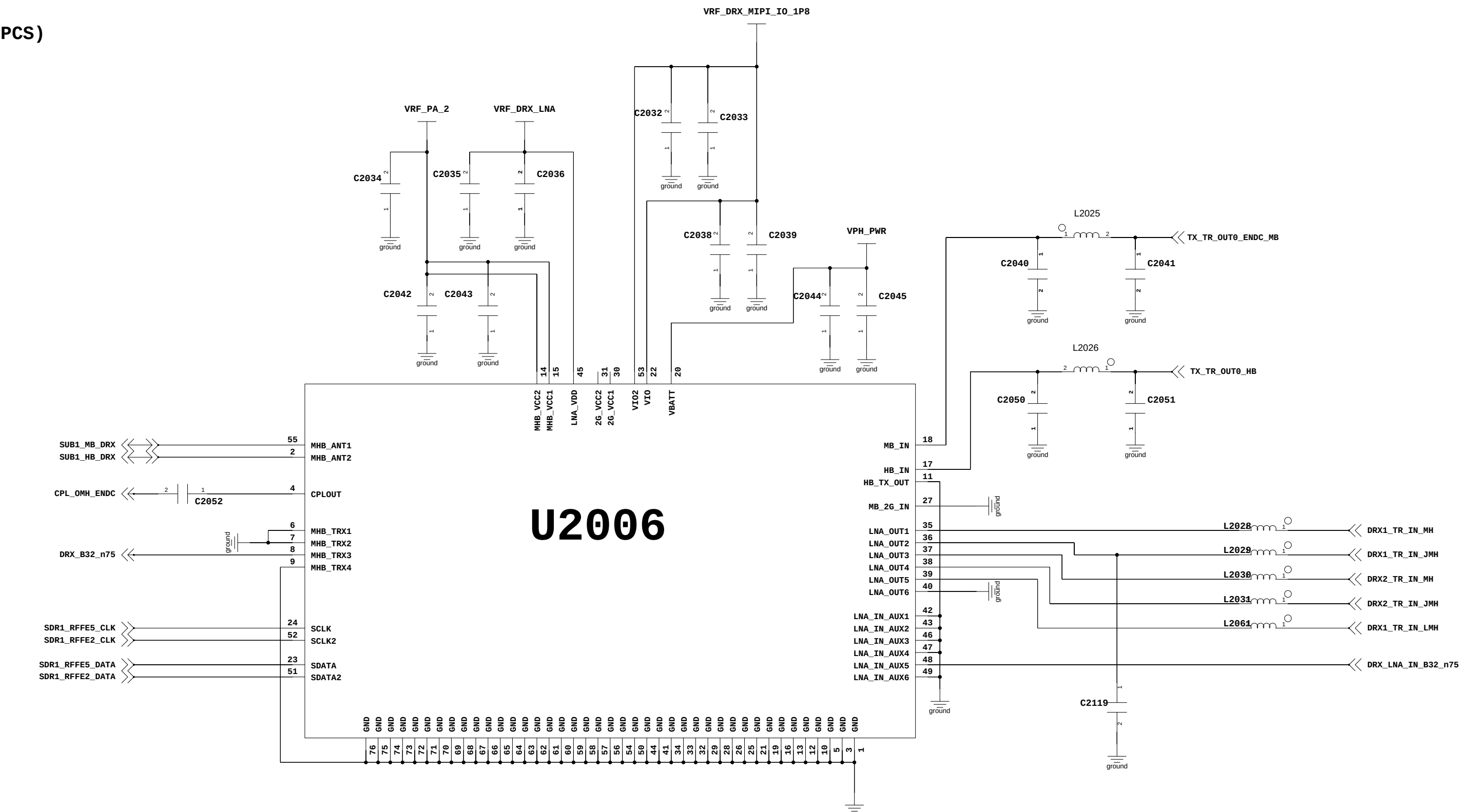
to SUB2 ANT



U2000
HB_DRX_MIMO
MB_DRX_MIMO
HB
GND
LB
GND
COMMON
GND
SUB2_MHB_DRX_MIMO
ground

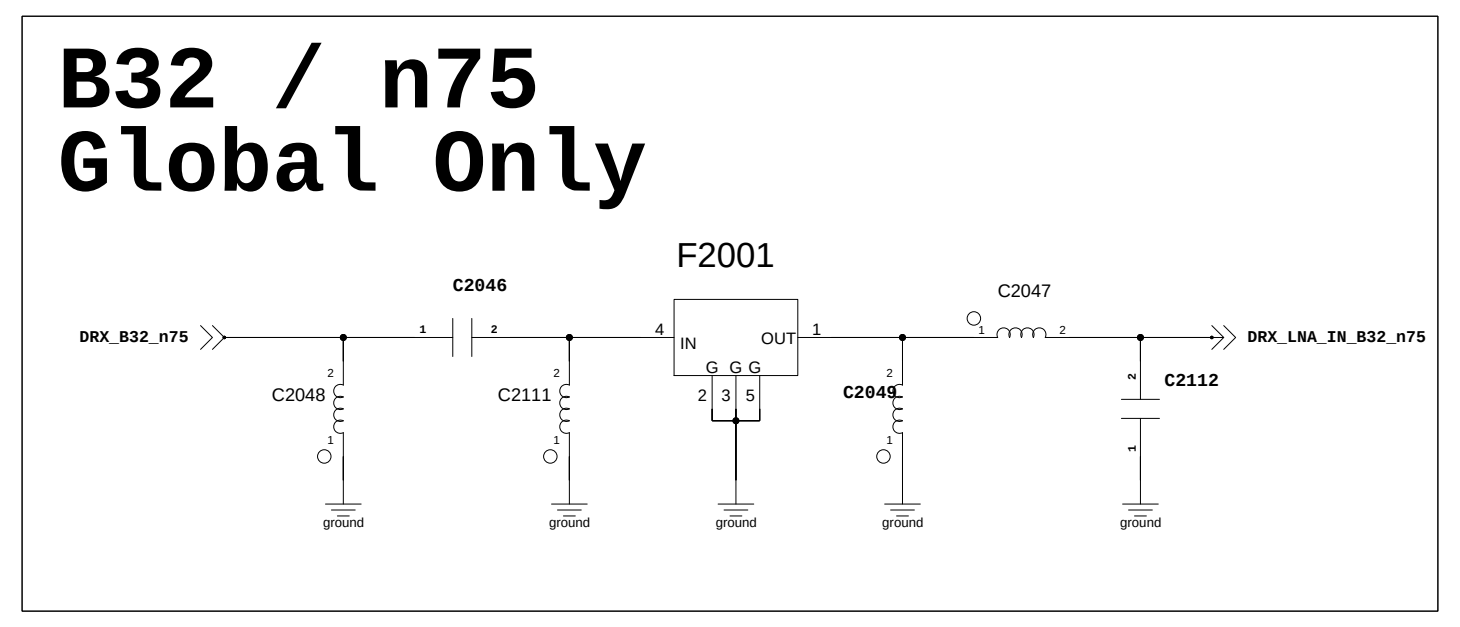
ENDC module

(B1/2/3/4/7/25/30/39/41/66_BC1_DCS/PCS)



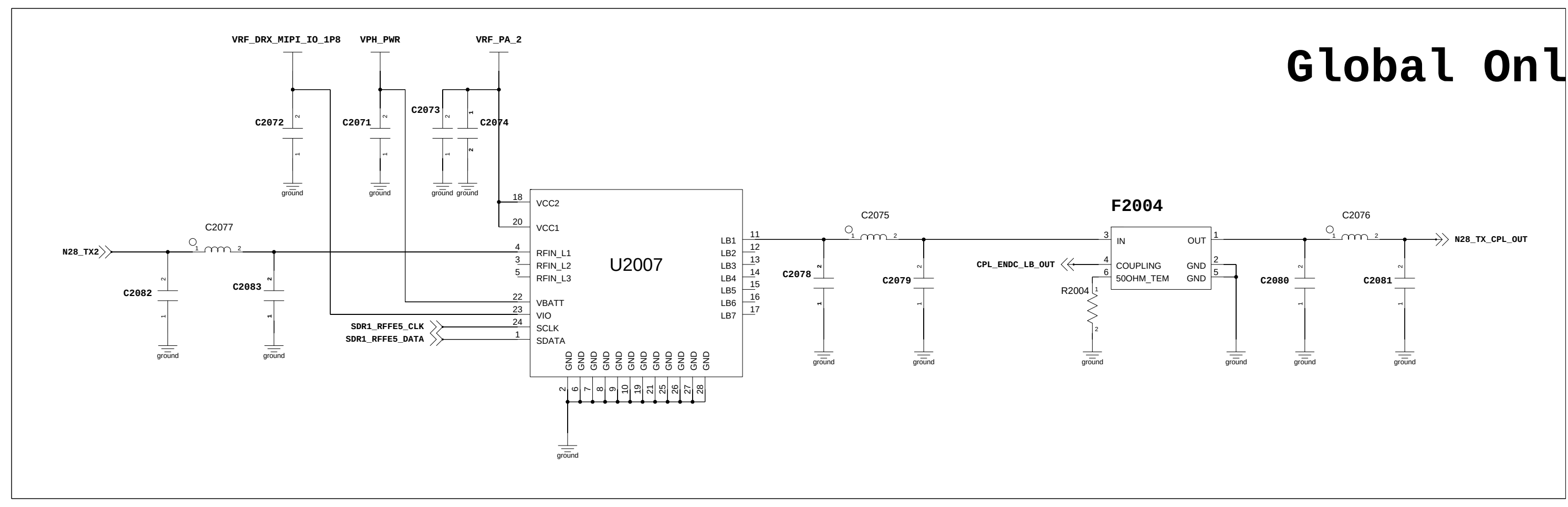
VRF_DRX_MIPI_IO_1P8
VRF_PA_2
VRF_DRX_LNA
C2032
C2033
C2034
C2035
C2036
C2038
C2039
VPH_PWR
C2044
C2045
C2042
C2043
U2006
MHB_VCC2
MHB_VCC1
LNA_VDD
2G_VCC2
2G_VCC1
VIO2
VIO
VBATT
SUB1_MB_DRX
SUB1_HB_DRX
MHB_ANT1
MHB_ANT2
CPL_OMH_ENDC
C2052
CPLOUT
DRX_B32_n75
MHB_TRX1
MHB_TRX2
MHB_TRX3
MHB_TRX4
SDR1_RFFE5_CLK
SDR1_RFFE2_CLK
SDR1_RFFE5_DATA
SDR1_RFFE2_DATA
SCLK
SCLK2
SDATA
SDATA2
MB_IN
HB_IN
HB_TX_OUT
MB_2G_IN
LNA_OUT1
LNA_OUT2
LNA_OUT3
LNA_OUT4
LNA_OUT5
LNA_OUT6
LNA_IN_AUX1
LNA_IN_AUX2
LNA_IN_AUX3
LNA_IN_AUX4
LNA_IN_AUX5
LNA_IN_AUX6
L2025
TX_TR_OUT0_ENDC_MB
C2040
C2041
L2026
TX_TR_OUT0_HB
C2050
C2051
L2028
DRX1_TR_IN_MH
L2029
DRX1_TR_IN_JMH
L2030
DRX2_TR_IN_MH
L2031
DRX2_TR_IN_JMH
L2061
DRX1_TR_IN_LMH
DRX_LNA_IN_B32_n75
C2119
GND
ground

B32 / n75
Global Only

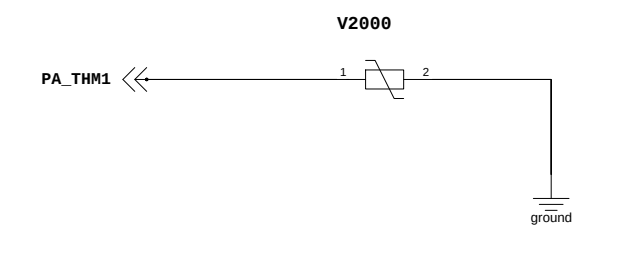


F2001
DRX_B32_n75
C2046
IN
OUT
G G G
C2047
DRX_LNA_IN_B32_n75
C2048
C2111
C2049
C2112
ground

Global Only



VRF_DRX_MIPI_IO_1P8
VPH_PWR
VRF_PA_2
C2072
C2071
C2073
C2074
N28_TX2
C2077
C2082
C2083
U2007
VCC2
VCC1
RFIN_L1
RFIN_L2
RFIN_L3
VBATT
VIO
SCLK
SDATA
LB1
LB2
LB3
LB4
LB5
LB6
LB7
GND
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
C2075
C2078
C2079
F2004
IN
OUT
COUPLING
50OHM_TEM
GND
CPL_ENDC_LB_OUT
R2004
C2076
N28_TX_CPL_OUT
C2080
C2081
ground

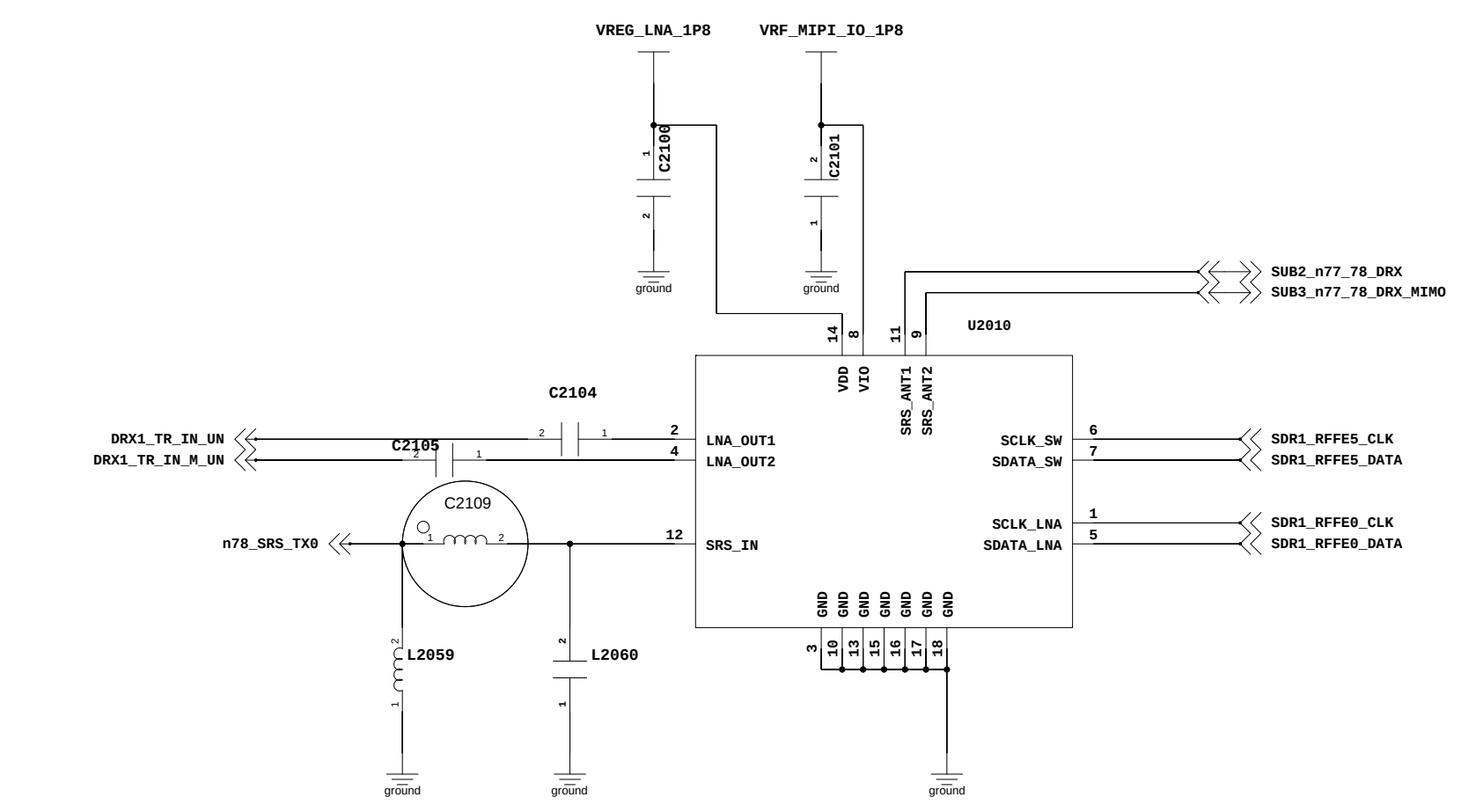


V2000
PA_THM1
ground

THERMISTERS

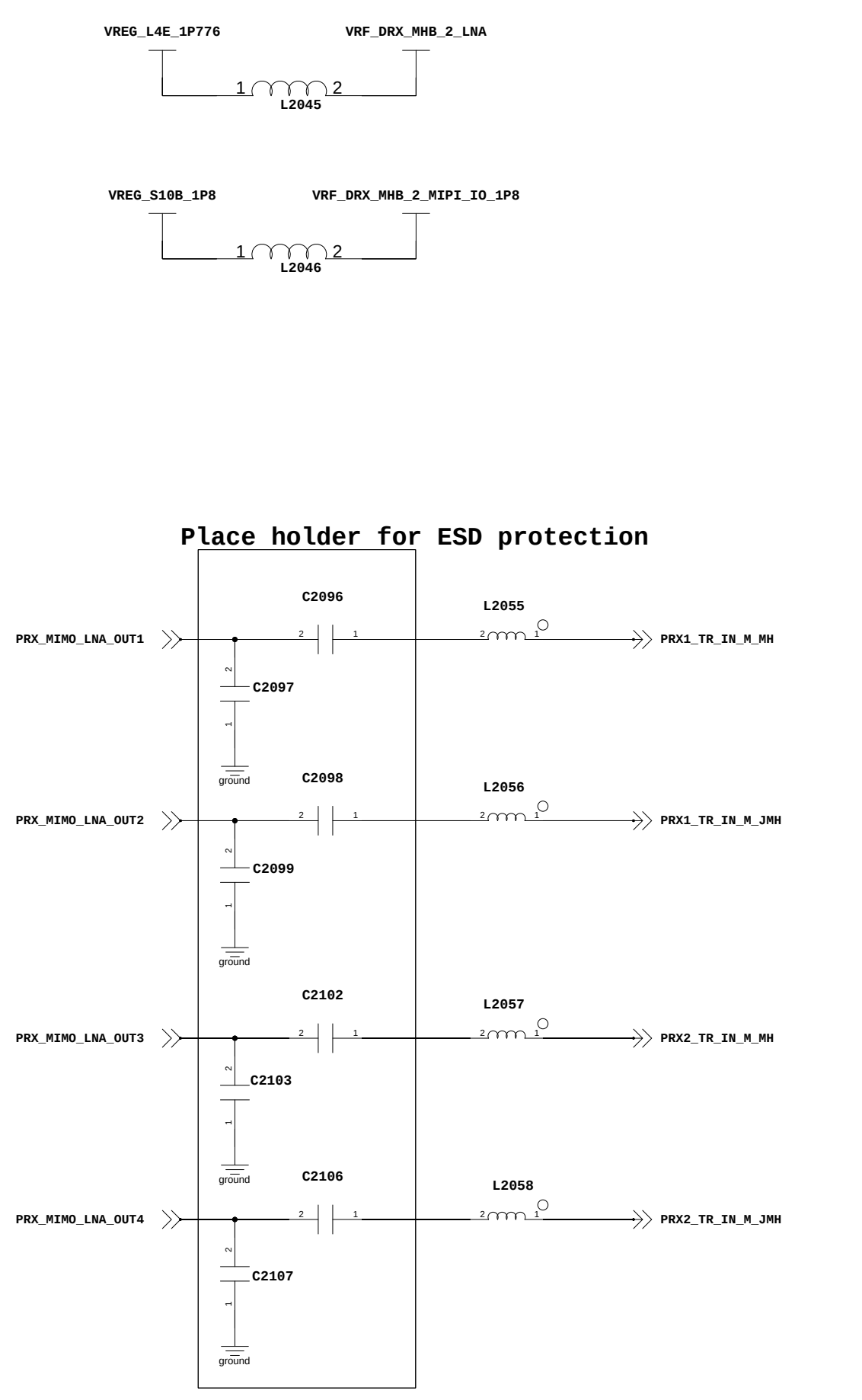
UHB

(n77/78)

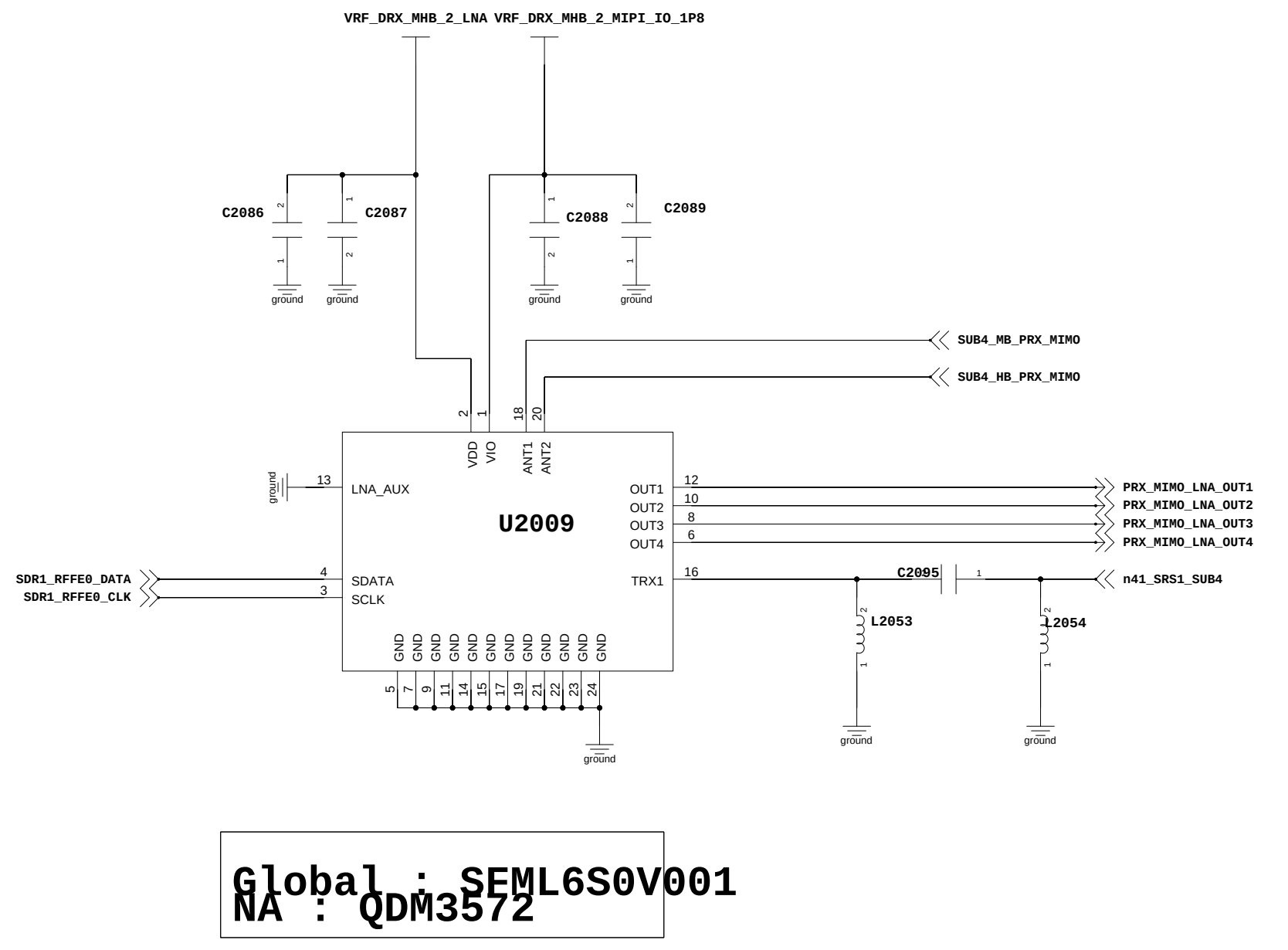


VREG_LNA_1P8
VRF_MIPI_IO_1P8
C2100
C2101
SUB2_n77_78_DRX
SUB3_n77_78_DRX_MIMO
U2010
VDD
VIO
SRS_ANT1
SRS_ANT2
C2104
DRX1_TR_IN_UN
DRX1_TR_IN_M_UN
C2105
LNA_OUT1
LNA_OUT2
C2109
n78_SRS_TX0
SRS_IN
SCLK_SW
SDATA_SW
SCLK_LNA
SDATA_LNA
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
L2059
L2060
GND
ground

to SUB4 ANT



VREG_L4E_1P776
VRF_DRX_MHB_2_LNA
L2045
VREG_S10B_1P8
VRF_DRX_MHB_2_MIPI_IO_1P8
L2046
Place holder for ESD protection
C2096
L2055
PRX_MIMO_LNA_OUT1
PRX1_TR_IN_M_MH
C2097
C2098
L2056
PRX_MIMO_LNA_OUT2
PRX1_TR_IN_M_JMH
C2099
C2102
L2057
PRX_MIMO_LNA_OUT3
PRX2_TR_IN_M_MH
C2103
C2106
L2058
PRX_MIMO_LNA_OUT4
PRX2_TR_IN_M_JMH
C2107
ground



VRF_DRX_MHB_2_LNA
VRF_DRX_MHB_2_MIPI_IO_1P8
C2086
C2087
C2088
C2089
SUB4_MB_PRX_MIMO
SUB4_HB_PRX_MIMO
U2009
VDD
VIO
ANT1
ANT2
LNA_AUX
SDATA
SCLK
OUT1
OUT2
OUT3
OUT4
TRX1
SDR1_RFFE0_DATA
SDR1_RFFE0_CLK
PRX_MIMO_LNA_OUT1
PRX_MIMO_LNA_OUT2
PRX_MIMO_LNA_OUT3
PRX_MIMO_LNA_OUT4
C2095
n41_SRS1_SUB4
L2053
L2054
Global : SFML6S0V001
NA : QDM3572
ground

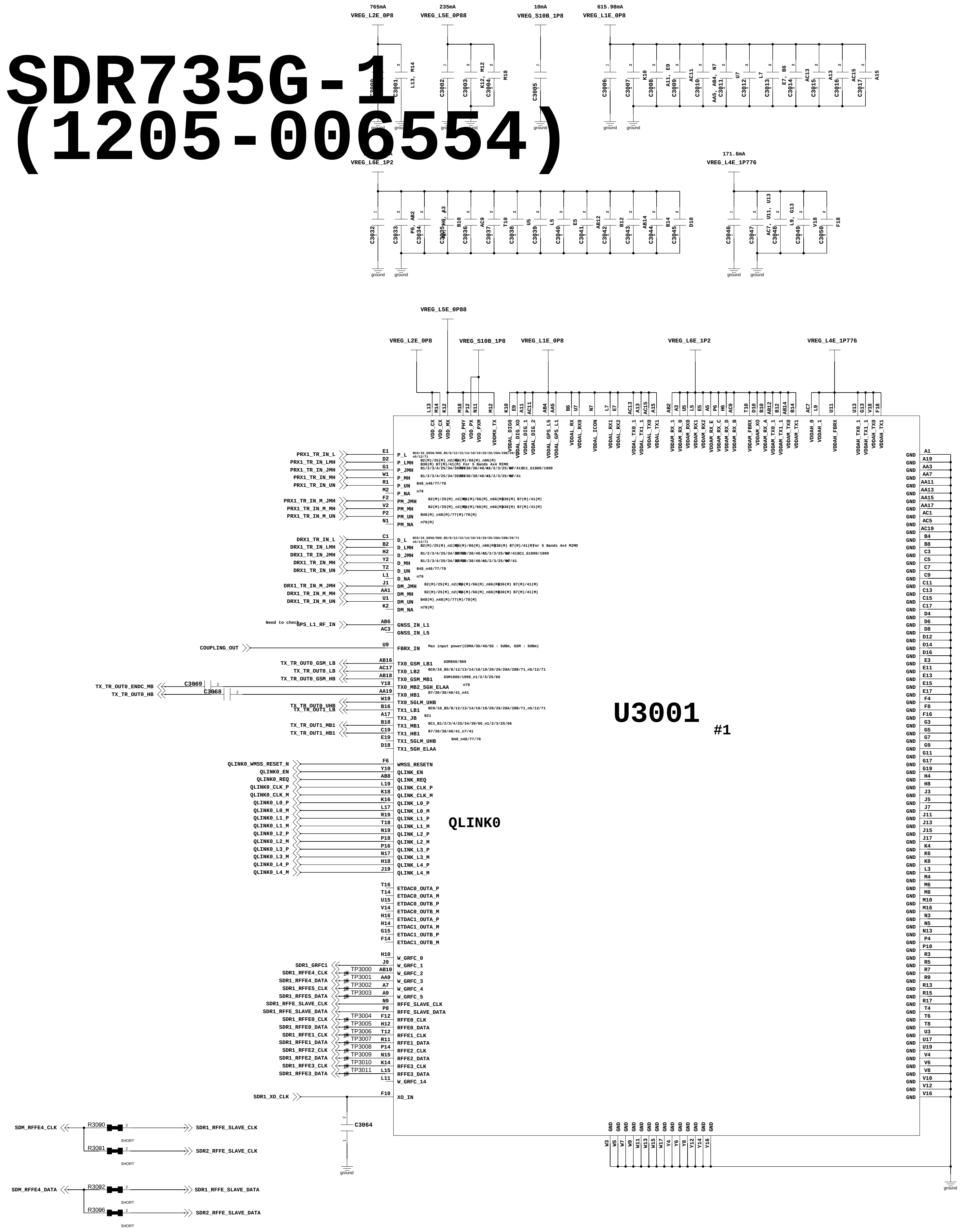
MHB MIMO_LNA OUT

TITLE TabS7P
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name SHEET 2 of 3 Date Drawing Date

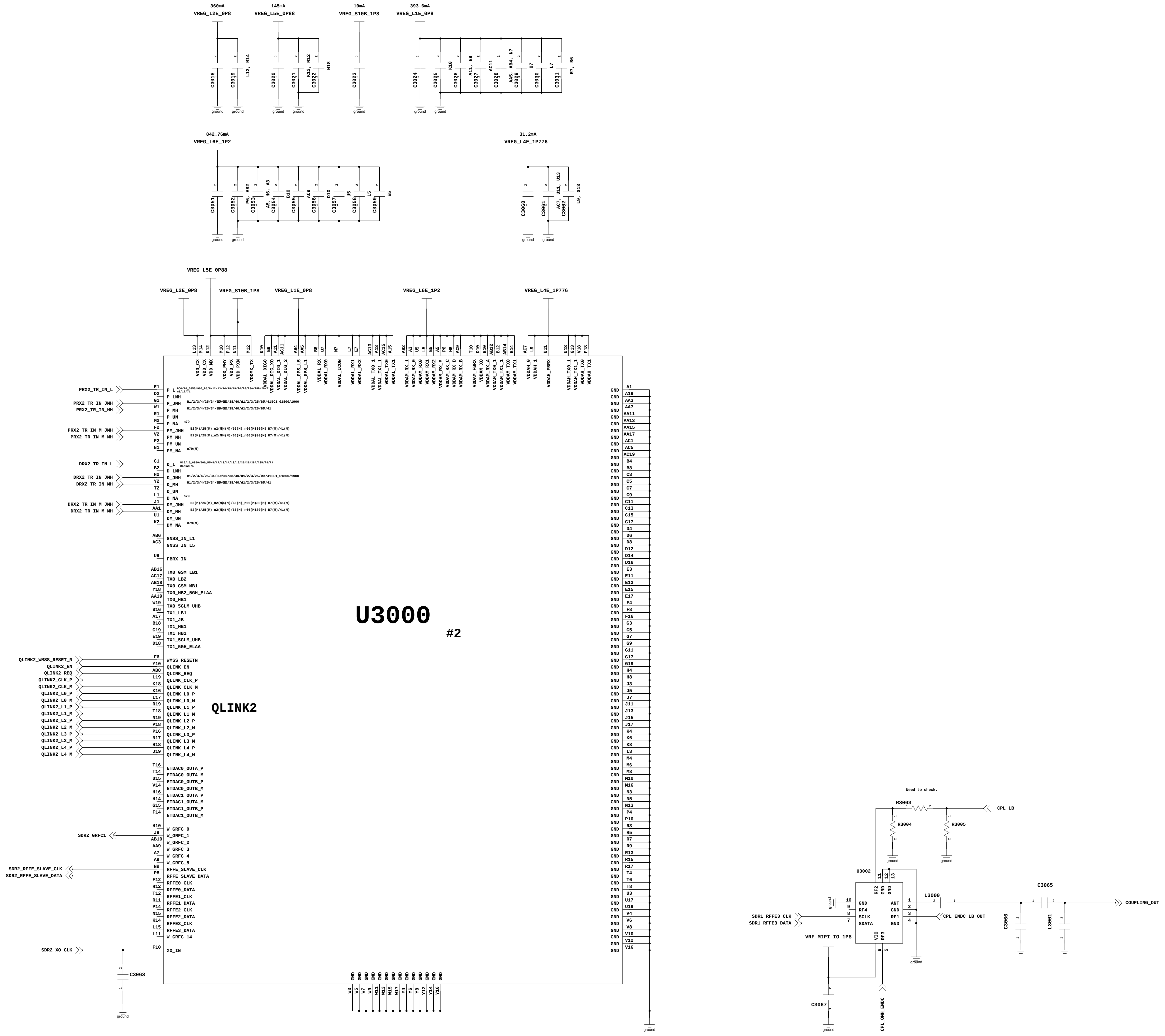
MAIN RF_3

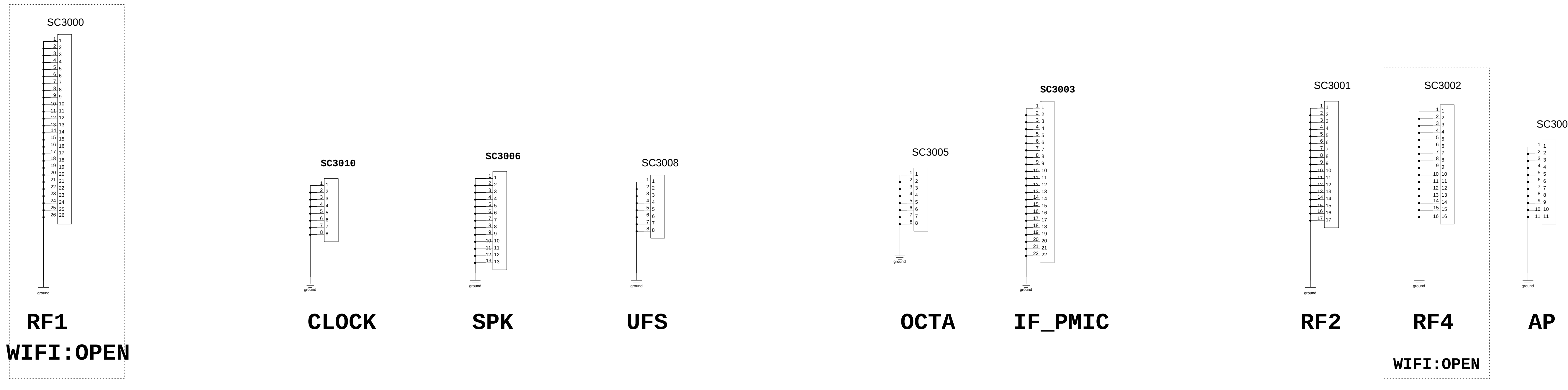
TRANSCEIVER(Dual SDR735)

WIFI : SDR735G-1
(1205-006554)


765mA
VREG_L2E_0P8
235mA
VREG_L5E_0P88
10mA
VREG_S10B_1P8
615.98mA
VREG_L1E_0P8
VREG_L6E_1P2
171.6mA
VREG_L4E_1P776
C3000
C3001
C3002
C3003
C3004
C3005
C3006
C3007
C3008
C3009
C3010
C3011
C3012
C3013
C3014
C3015
C3016
C3017
C3032
C3033
C3034
C3035
C3036
C3037
C3038
C3039
C3040
C3041
C3042
C3043
C3044
C3045
C3046
C3047
C3048
C3049
C3050
ground
U3001
#1
QLINK0
PRX1_TR_IN_L
PRX1_TR_IN_LMH
PRX1_TR_IN_JMH
PRX1_TR_IN_MH
PRX1_TR_IN_UN
PRX1_TR_IN_M_JMH
PRX1_TR_IN_M_MH
PRX1_TR_IN_M_UN
DRX1_TR_IN_L
DRX1_TR_IN_LMH
DRX1_TR_IN_JMH
DRX1_TR_IN_MH
DRX1_TR_IN_UN
DRX1_TR_IN_M_JMH
DRX1_TR_IN_M_MH
DRX1_TR_IN_M_UN
Need to check
GPS_L1_RF_IN
COUPLING_OUT
TX_TR_OUT0_GSM_LB
TX_TR_OUT0_LB
TX_TR_OUT0_GSM_HB
TX_TR_OUT0_ENDC_MB
TX_TR_OUT0_HB
C3069
C3068
TX_TR_OUT0_UHB
TX_TR_OUT1_LB
TX_TR_OUT1_MB1
TX_TR_OUT1_HB1
QLINK0_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_CLK_P
QLINK0_CLK_M
QLINK0_L0_P
QLINK0_L0_M
QLINK0_L1_P
QLINK0_L1_M
QLINK0_L2_P
QLINK0_L2_M
QLINK0_L3_P
QLINK0_L3_M
QLINK0_L4_P
QLINK0_L4_M
SDR1_GRFC1
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
SDR1_RFFE_SLAVE_CLK
SDR1_RFFE_SLAVE_DATA
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
SDR1_RFFE1_CLK
SDR1_RFFE1_DATA
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
SDR1_RFFE3_CLK
SDR1_RFFE3_DATA
TP3000
TP3001
TP3002
TP3003
TP3004
TP3005
TP3006
TP3007
TP3008
TP3009
TP3010
TP3011
SDR1_XO_CLK
C3064
SDM_RFFE4_CLK
R3000
R3001
SHORT
SDR1_RFFE_SLAVE_CLK
SDR2_RFFE_SLAVE_CLK
SDM_RFFE4_DATA
R3002
R3006
SDR1_RFFE_SLAVE_DATA
SDR2_RFFE_SLAVE_DATA
P_L
P_LMH
P_JMH
P_MH
P_UN
P_NA
PM_JMH
PM_MH
PM_UN
PM_NA
D_L
D_LMH
D_JMH
D_MH
D_UN
D_NA
DM_JMH
DM_MH
DM_UN
DM_NA
GNSS_IN_L1
GNSS_IN_L5
FBRX_IN
TX0_GSM_LB1
TX0_LB2
TX0_GSM_MB1
TX0_MB2_5GH_ELAA
TX0_HB1
TX0_5GLM_UHB
TX1_LB1
TX1_JB
TX1_MB1
TX1_HB1
TX1_5GLM_UHB
TX1_5GH_ELAA
WMSS_RESETN
QLINK_EN
QLINK_REQ
QLINK_CLK_P
QLINK_CLK_M
ETDAC0_OUTA_P
ETDAC0_OUTA_M
ETDAC0_OUTB_P
ETDAC0_OUTB_M
ETDAC1_OUTA_P
ETDAC1_OUTA_M
ETDAC1_OUTB_P
ETDAC1_OUTB_M
W_GRFC_0
W_GRFC_1
W_GRFC_2
W_GRFC_3
W_GRFC_4
W_GRFC_5
RFFE_SLAVE_CLK
RFFE_SLAVE_DATA
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
W_GRFC_14
XO_IN
GND
Max input power(CDMA/3G/4G/5G : 5dBm, GSM : 9dBm)

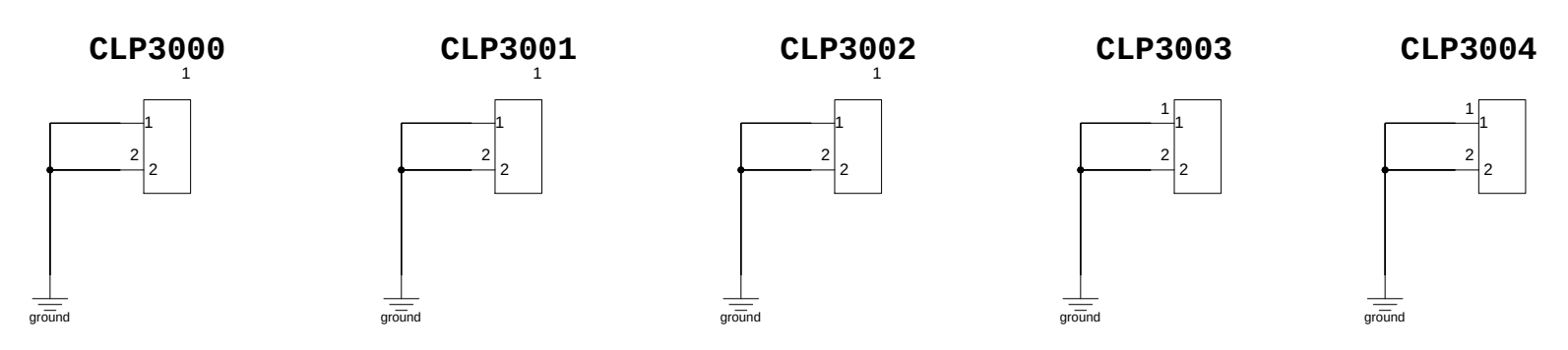
WIFI : NC


360mA
VREG_L2E_0P8
145mA
VREG_L5E_0P88
10mA
VREG_S10B_1P8
393.6mA
VREG_L1E_0P8
842.76mA
VREG_L6E_1P2
31.2mA
VREG_L4E_1P776
C3018
C3019
C3020
C3021
C3022
C3023
C3024
C3025
C3026
C3027
C3028
C3029
C3030
C3031
C3051
C3052
C3053
C3054
C3055
C3056
C3057
C3058
C3059
C3060
C3061
C3062
ground
U3000
#2
QLINK2
PRX2_TR_IN_L
PRX2_TR_IN_JMH
PRX2_TR_IN_MH
PRX2_TR_IN_M_JMH
PRX2_TR_IN_M_MH
DRX2_TR_IN_L
DRX2_TR_IN_JMH
DRX2_TR_IN_MH
DRX2_TR_IN_M_JMH
DRX2_TR_IN_M_MH
QLINK2_WMSS_RESET_N
QLINK2_EN
QLINK2_REQ
QLINK2_CLK_P
QLINK2_CLK_M
QLINK2_L0_P
QLINK2_L0_M
QLINK2_L1_P
QLINK2_L1_M
QLINK2_L2_P
QLINK2_L2_M
QLINK2_L3_P
QLINK2_L3_M
QLINK2_L4_P
QLINK2_L4_M
SDR2_GRFC1
SDR2_RFFE_SLAVE_CLK
SDR2_RFFE_SLAVE_DATA
SDR2_XO_CLK
C3063
Need to check.
R3003
R3004
R3005
CPL_LB
U3002
RF2
GND
SCLK
SDATA
RF4
ANT
RF1
VIO
RF3
SDR1_RFFE3_CLK
SDR1_RFFE3_DATA
VRF_MIPI_IO_1P8
C3067
CPL_OMH_ENDC
CPL_ENDC_LB_OUT
L3000
C3065
C3066
L3001
COUPLING_OUT

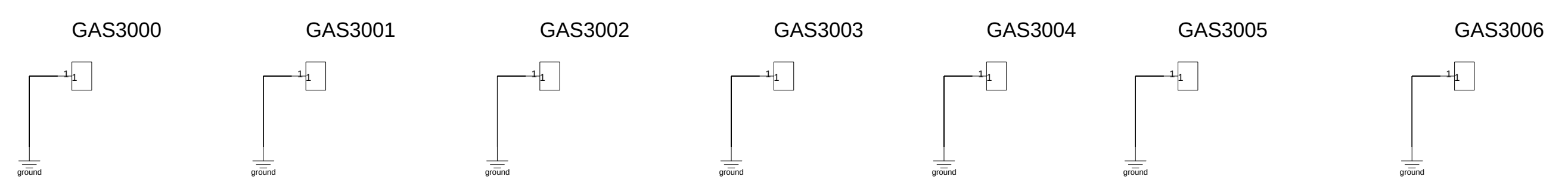

SC3000
SC3010
SC3006
SC3008
SC3005
SC3003
SC3001
SC3002
SC3009
ground
RF1
WIFI:OPEN
CLOCK
SPK
UFS
OCTA
IF_PMIC
RF2
RF4
WIFI:OPEN
AP

TOP S/C

BOTTOM S/C


CLP3000
CLP3001
CLP3002
CLP3003
CLP3004
ground

Cable CLIP

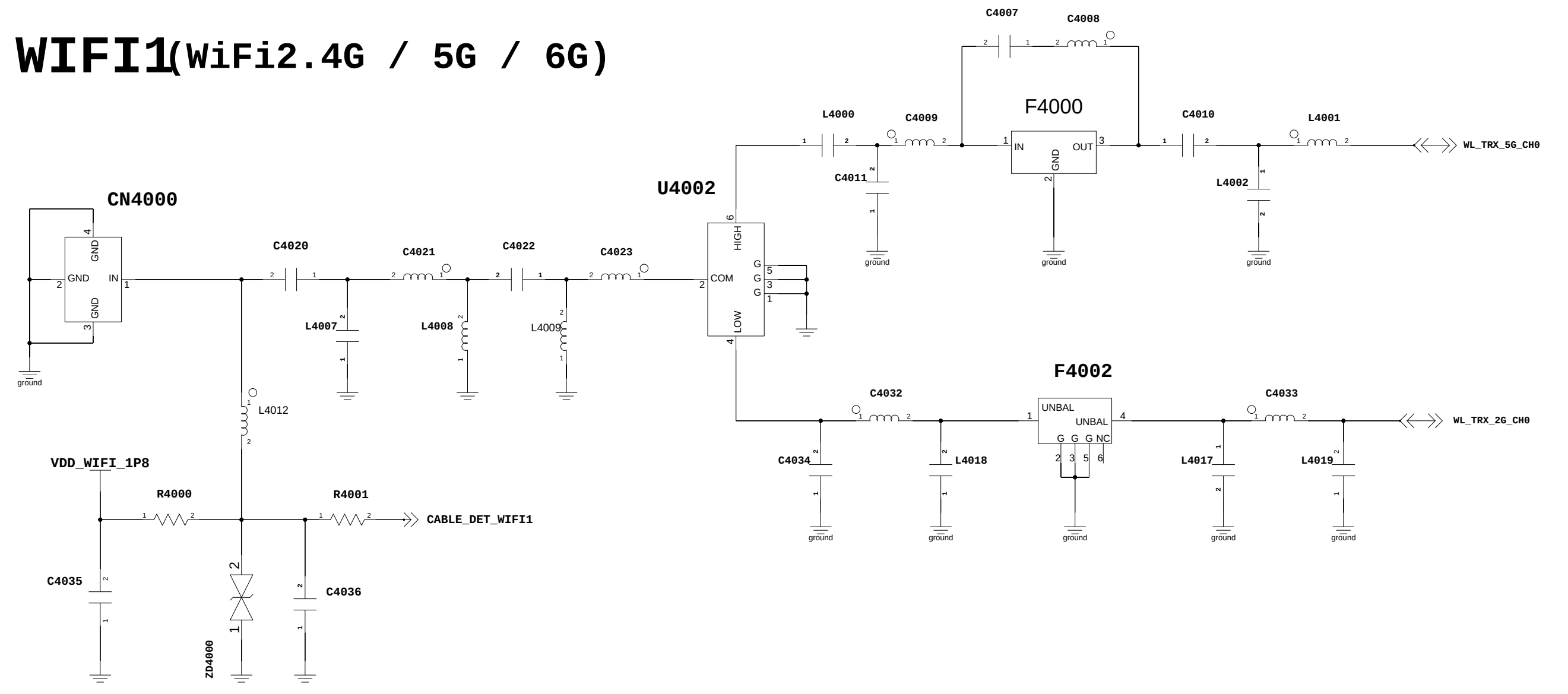

GAS3000
GAS3001
GAS3002
GAS3003
GAS3004
GAS3005
GAS3006
ground

Front CONTACT

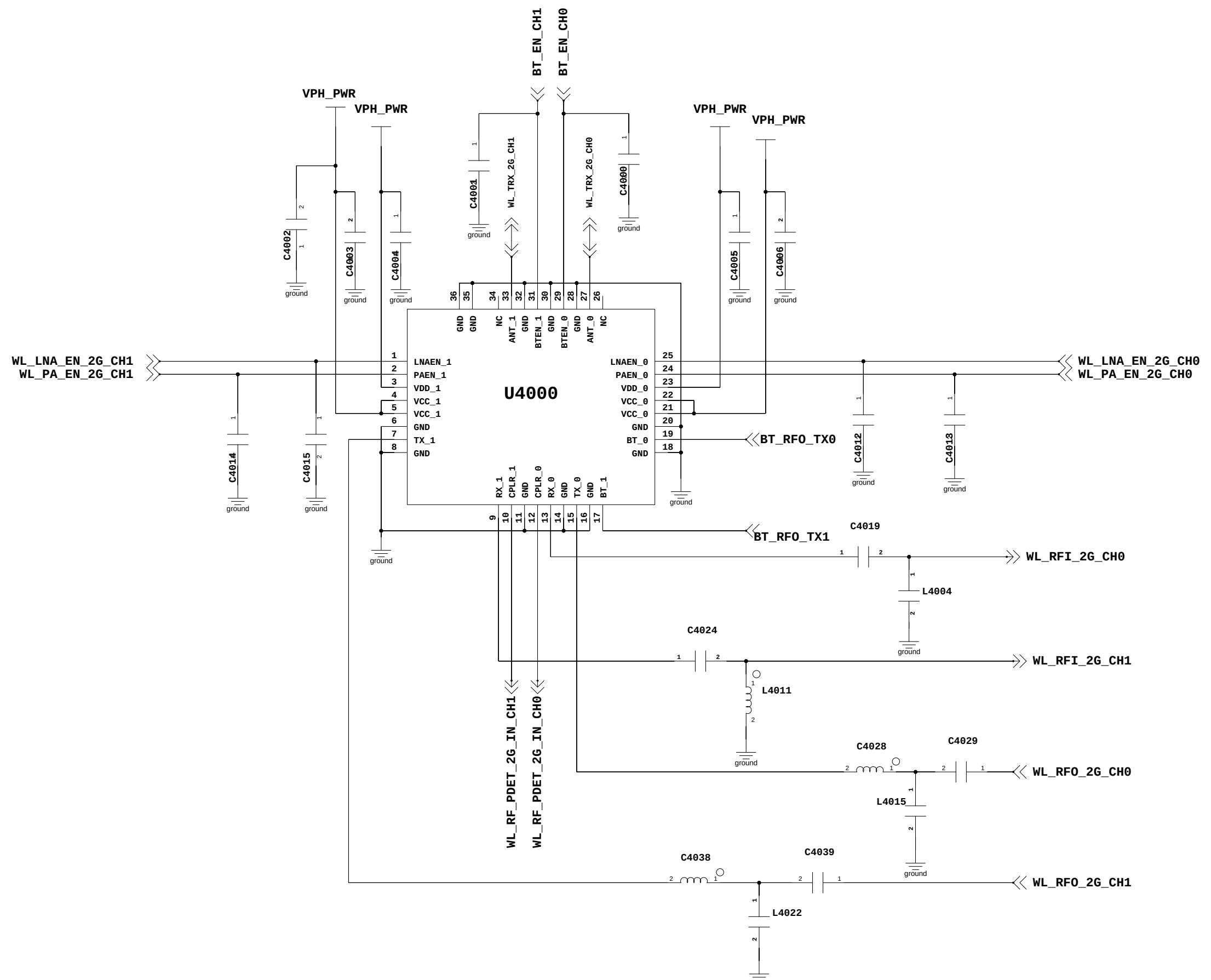
TITLE	TabS7P
SIZE A1	DWG NO PDM NUMBER
REV	Rev
DRAWN BY	User Name
SHEET	3 of 16
Date	Drawing Date

Connectivity

WIFI1(WiFi2.4G / 5G / 6G)

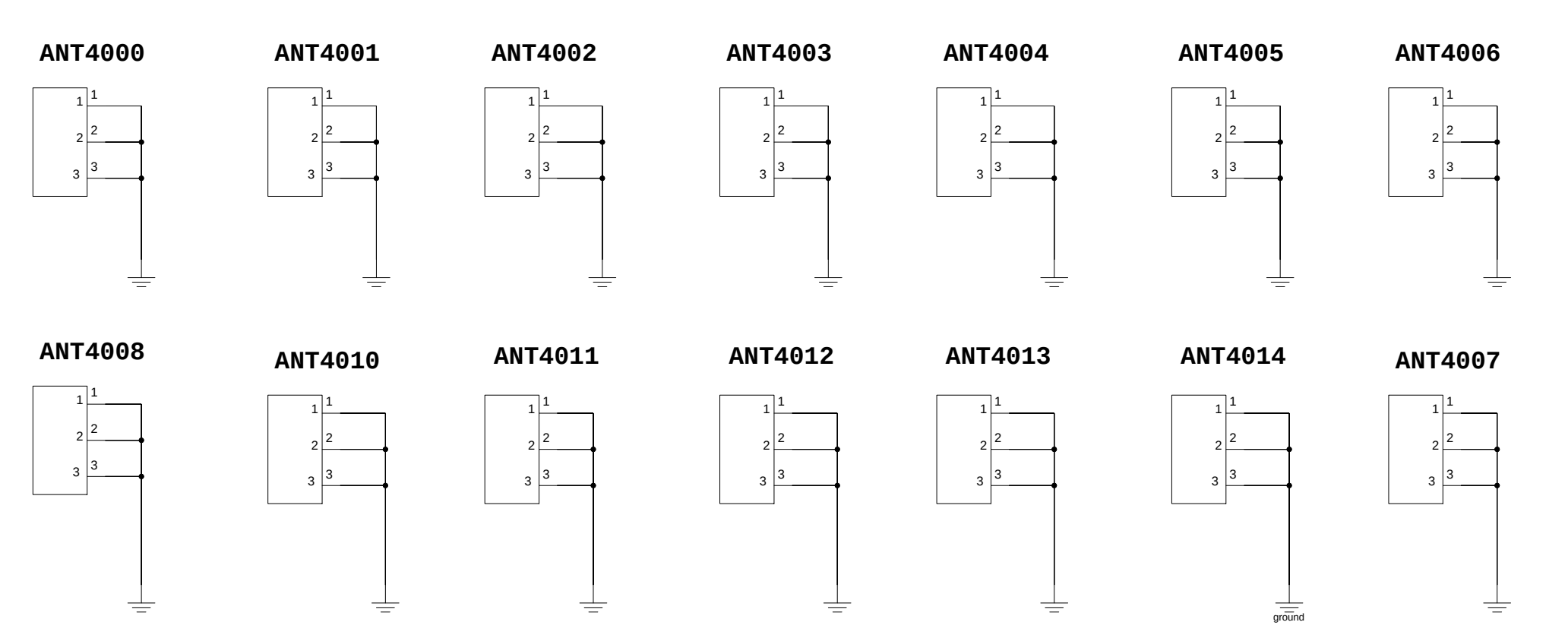


CN4000
GND
IN
C4020
C4021
C4022
C4023
L4007
L4008
L4009
U4002
HIGH
COM
LOW
G G G
L4000
C4009
C4011
F4000
IN
OUT
GND
C4007
C4008
C4010
L4001
L4002
WL_TRX_5G_CH0
F4002
UNBAL
UNBAL
G G G NC
C4032
C4033
C4034
L4018
L4017
L4019
WL_TRX_2G_CH0
L4012
VDD_WIFI_1P8
R4000
R4001
CABLE_DET_WIFI1
C4035
C4036
ZD4000
ground



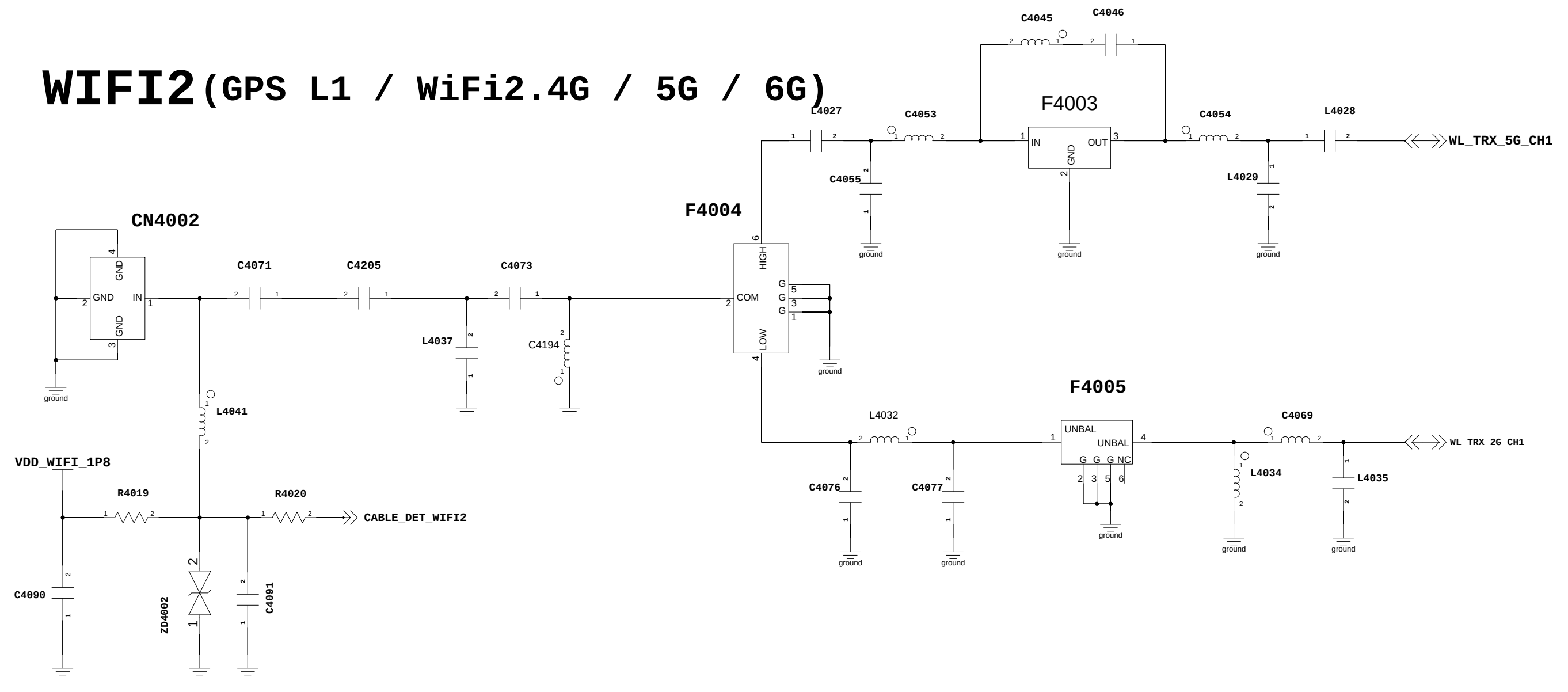
BT_EN_CH1
BT_EN_CH0
VPH_PWR
VPH_PWR
VPH_PWR
VPH_PWR
C4001
WL_TRX_2G_CH1
WL_TRX_2G_CH0
C4000
C4002
C4003
C4004
C4005
C4006
WL_LNA_EN_2G_CH1
WL_PA_EN_2G_CH1
U4000
LNAEN_1
PAEN_1
VDD_1
VCC_1
VCC_1
GND
TX_1
GND
LNAEN_0
PAEN_0
VDD_0
VCC_0
VCC_0
GND
BT_0
GND
RX_1
CPLR_1
CPLR_0
RX_0
TX_0
BT_1
ANT_1
BTEN_1
BTEN_0
ANT_0
NC
WL_LNA_EN_2G_CH0
WL_PA_EN_2G_CH0
C4012
C4013
C4014
C4015
BT_RFO_TX0
BT_RFO_TX1
C4019
WL_RFI_2G_CH0
L4004
C4024
L4011
WL_RFI_2G_CH1
C4028
C4029
L4015
WL_RFO_2G_CH0
C4038
C4039
L4022
WL_RFO_2G_CH1
WL_RF_PDET_2G_IN_CH1
WL_RF_PDET_2G_IN_CH0
ground

REAR CONTACT

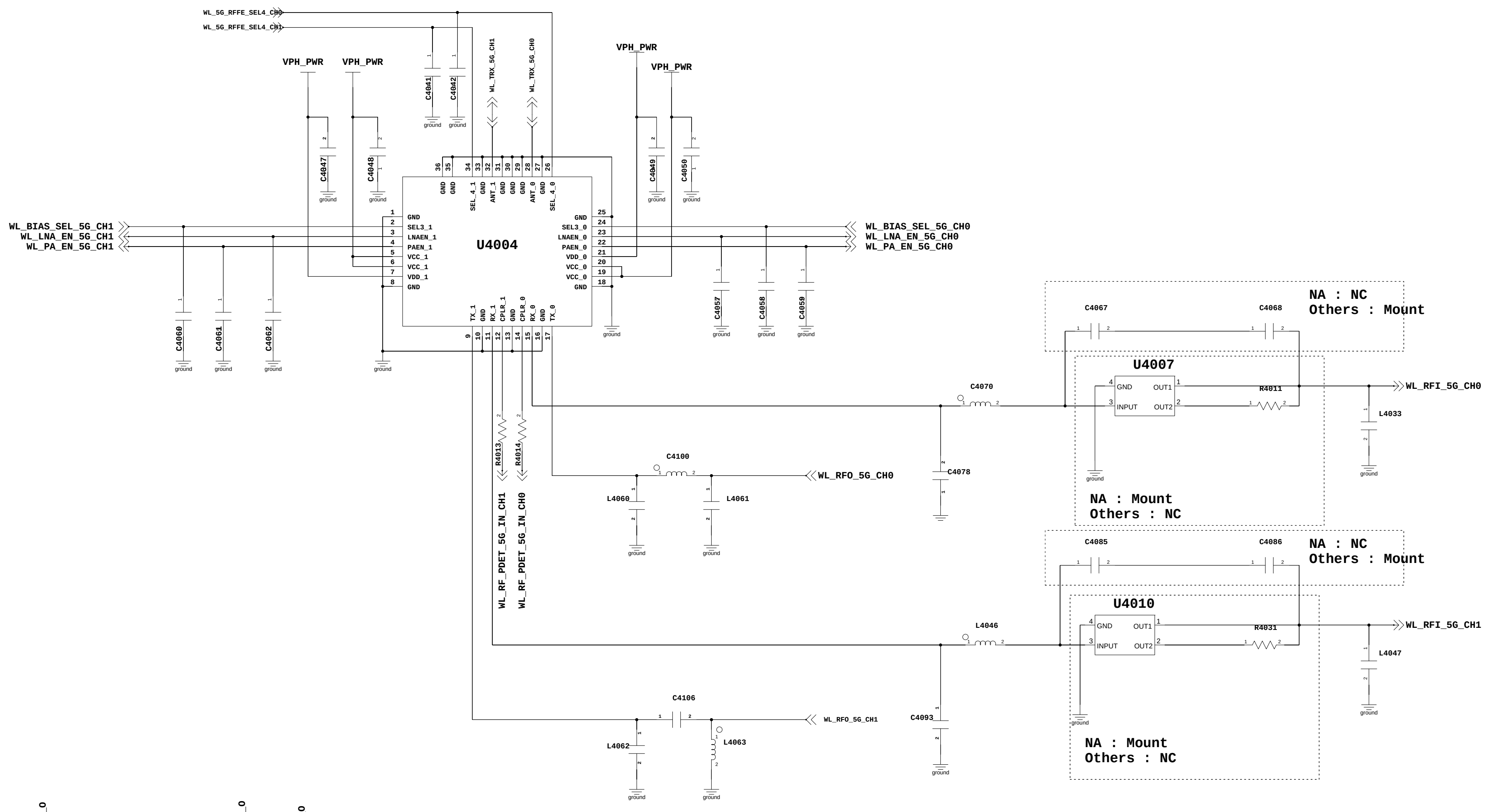


ANT4000
ANT4001
ANT4002
ANT4003
ANT4004
ANT4005
ANT4006
ANT4008
ANT4010
ANT4011
ANT4012
ANT4013
ANT4014
ANT4007
1
2
3
ground

WIFI2(GPS L1 / WiFi2.4G / 5G / 6G)

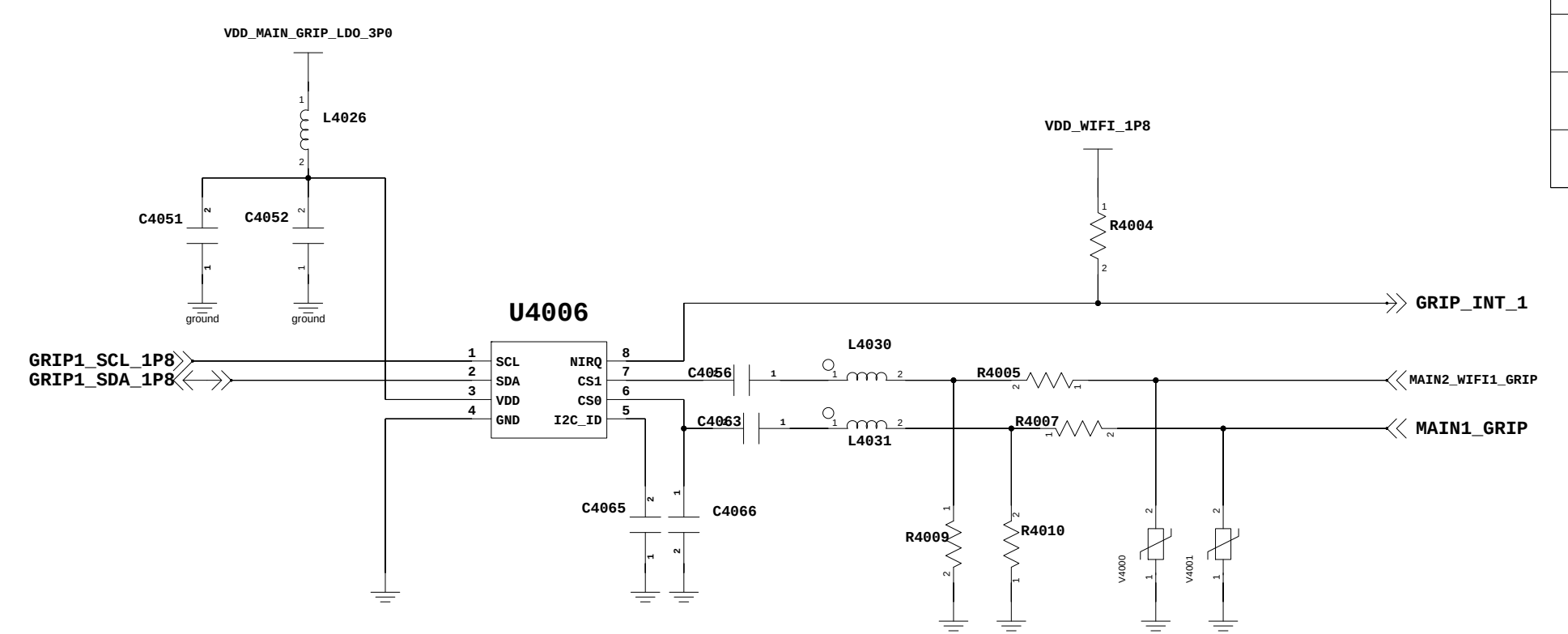


CN4002
GND
IN
C4071
C4205
C4073
L4037
C4194
F4004
HIGH
COM
LOW
L4027
C4053
C4055
F4003
IN
OUT
GND
C4045
C4046
C4054
L4028
L4029
WL_TRX_5G_CH1
F4005
UNBAL
UNBAL
L4032
C4069
C4076
C4077
L4034
L4035
WL_TRX_2G_CH1
L4041
VDD_WIFI_1P8
R4019
R4020
CABLE_DET_WIFI2
C4090
ZD4002
C4091
ground



WL_5G_RFFE_SEL4_CH0
WL_5G_RFFE_SEL4_CH1
VPH_PWR
VPH_PWR
VPH_PWR
VPH_PWR
C4041
C4042
WL_TRX_5G_CH1
WL_TRX_5G_CH0
C4047
C4048
C4049
C4050
WL_BIAS_SEL_5G_CH1
WL_LNA_EN_5G_CH1
WL_PA_EN_5G_CH1
U4004
GND
SEL3_1
LNAEN_1
PAEN_1
VCC_1
VCC_1
VDD_1
GND
GND
SEL3_0
LNAEN_0
PAEN_0
VDD_0
VCC_0
VCC_0
GND
SEL_4_1
ANT_1
ANT_0
SEL_4_0
TX_1
RX_1
CPLR_1
CPLR_0
RX_0
TX_0
WL_BIAS_SEL_5G_CH0
WL_LNA_EN_5G_CH0
WL_PA_EN_5G_CH0
C4057
C4058
C4059
C4060
C4061
C4062
R4013
R4014
WL_RF_PDET_5G_IN_CH1
WL_RF_PDET_5G_IN_CH0
C4067
C4068
NA : NC
Others : Mount
U4007
GND
INPUT
OUT1
OUT2
R4011
C4070
C4078
WL_RFI_5G_CH0
L4033
NA : Mount
Others : NC
C4100
L4060
L4061
WL_RFO_5G_CH0
C4085
C4086
NA : NC
Others : Mount
U4010
GND
INPUT
OUT1
OUT2
R4031
L4046
C4093
WL_RFI_5G_CH1
L4047
NA : Mount
Others : NC
C4106
L4062
L4063
WL_RFO_5G_CH1
ground

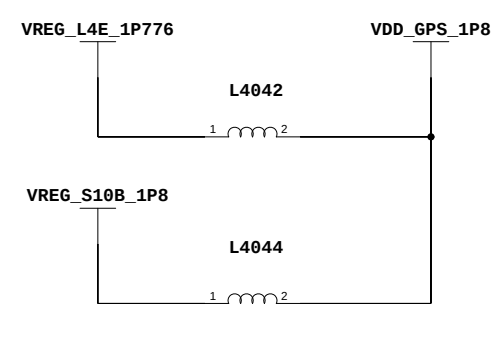
MAIN_GRIP_SENSOR(GRIP #2_#3)



VDD_MAIN_GRIP_LDO_3P0
L4026
C4051
C4052
ground
U4006
GRIP1_SCL_1P8
GRIP1_SDA_1P8
SCL
SDA
VDD
GND
NIRQ
CS1
CS0
I2C_ID
C4065
C4066
C4056
C4063
L4030
L4031
VDD_WIFI_1P8
R4004
GRIP_INT_1
R4005
R4007
MAIN2_WIFI1_GRIP
MAIN1_GRIP
R4009
R4010
V4000
V4001

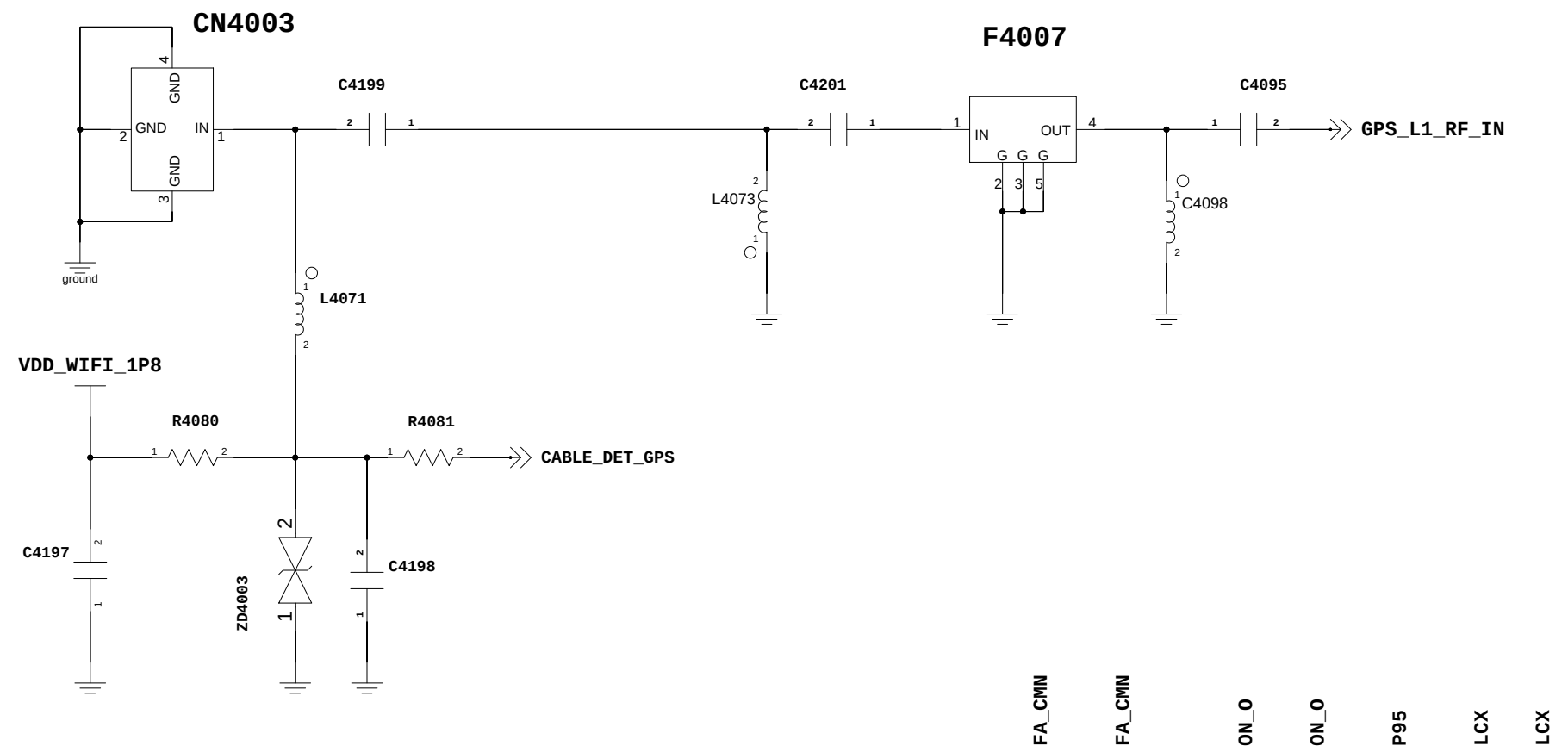
	Sub6	WIFI
C4063	O	X
L4031	O	X
R4010	O	X
R4007	O	X
C4066	X	68pF

WIFI : NC



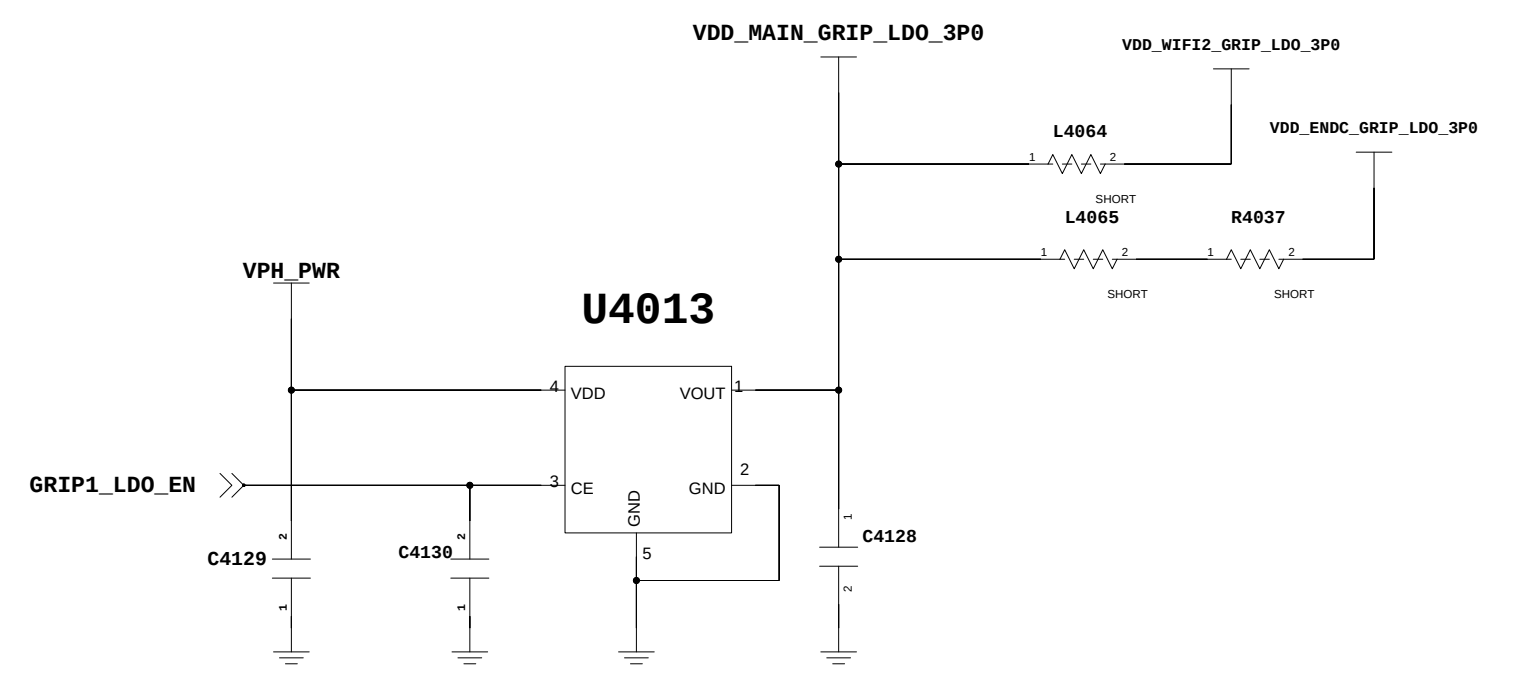
VREG_L4E_1P776
VDD_GPS_1P8
L4042
VREG_S10B_1P8
L4044

GPS



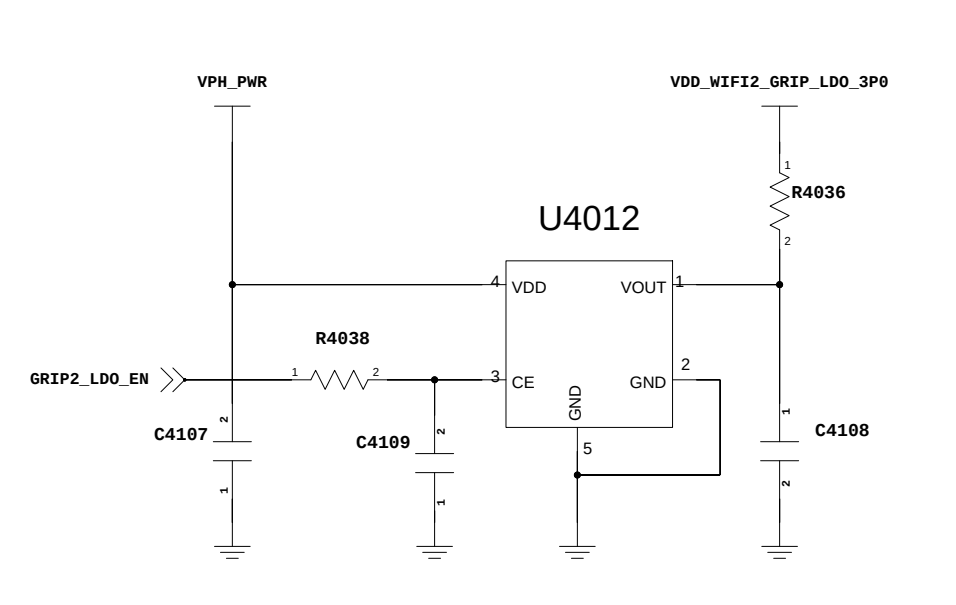
CN4003
GND
IN
C4199
C4201
L4073
F4007
IN
OUT
G G G
C4095
GPS_L1_RF_IN
C4098
L4071
VDD_WIFI_1P8
R4080
R4081
CABLE_DET_GPS
C4197
ZD4003
C4198
ground

MAIN GRIP LDO

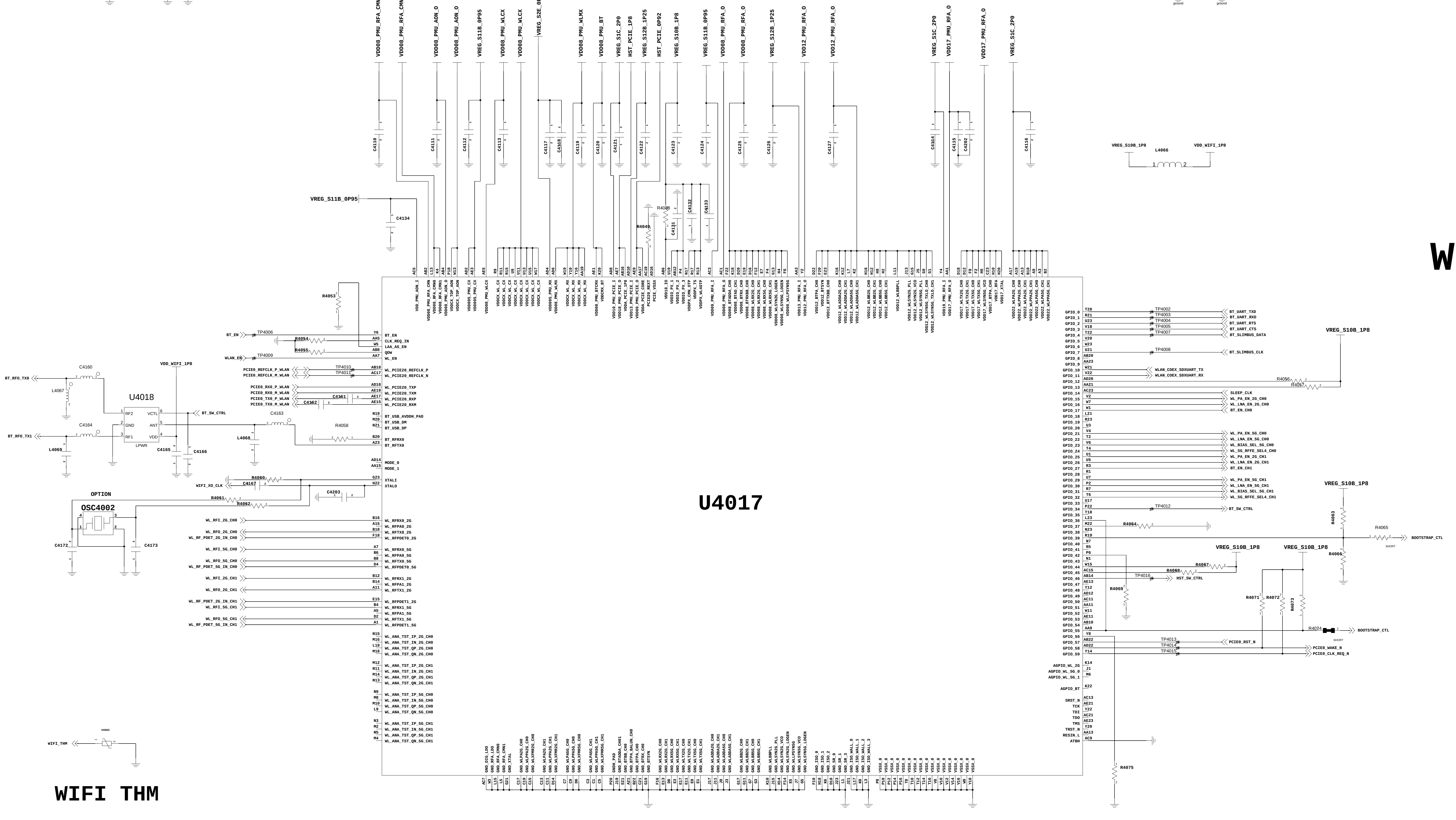


VDD_MAIN_GRIP_LDO_3P0
VDD_WIFI2_GRIP_LDO_3P0
L4064
SHORT
VDD_ENDC_GRIP_LDO_3P0
L4065
R4037
SHORT
SHORT
VPH_PWR
U4013
VDD
VOUT
CE
GND
GRIP1_LDO_EN
C4129
C4130
C4128

WIFI GRIP LDO



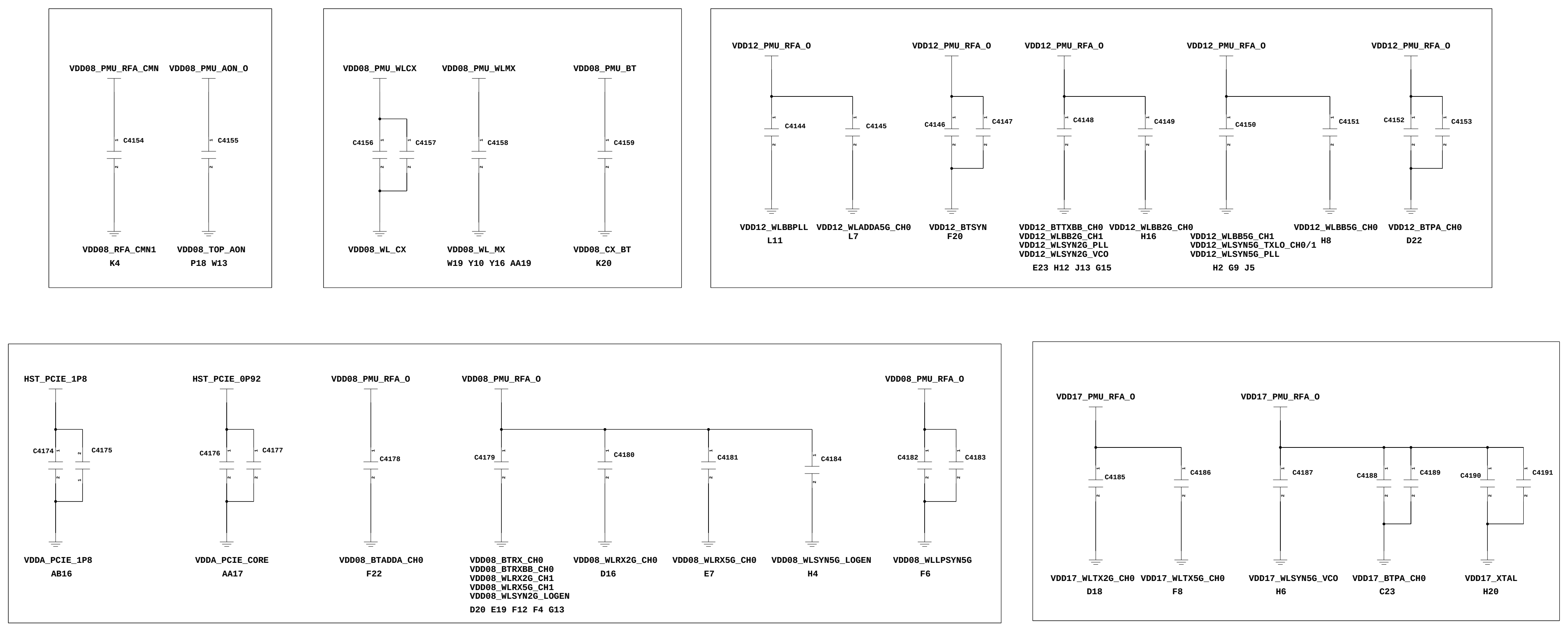
VPH_PWR
VDD_WIFI2_GRIP_LDO_3P0
R4036
U4012
VDD
VOUT
CE
GND
R4038
GRIP2_LDO_EN
C4107
C4109
C4108



VDD08_PMU_RFA_CMN
VDD08_PMU_RFA_CMN
VDD08_PMU_AON_O
VDD08_PMU_AON_O
VREG_S11B_0P95
VDD08_PMU_WLCX
VDD08_PMU_WLCX
VREG_S2E_0P85
VDD08_PMU_WLMX
VDD08_PMU_BT
VREG_S1C_2P0
HST_PCIE_1P8
VREG_S12B_1P25
HST_PCIE_0P92
VREG_S10B_1P8
VREG_S11B_0P95
VDD08_PMU_RFA_O
VDD08_PMU_RFA_O
VREG_S12B_1P25
VDD12_PMU_RFA_O
VDD12_PMU_RFA_O
VREG_S1C_2P0
VDD17_PMU_RFA_O
VDD17_PMU_RFA_O
VREG_S1C_2P0
C4110
C4111
C4112
C4113
C4117
C4118
C4119
C4120
C4121
C4122
C4123
C4124
C4125
C4126
C4127
C4114
C4115
C4202
C4116
VREG_S11B_0P95
C4134
R4048
R4049
C4131
C4132
C4133
VREG_S10B_1P8
L4066
VDD_WIFI_1P8
U4017
R4053
BT_EN
TP4006
R4054
R4055
WLAN_EN
TP4009
BT_EN
CLK_REQ_IN
LAA_AS_EN
QOW
WL_EN
PCIE0_REFCLK_P_WLAN
PCIE0_REFCLK_M_WLAN
TP4010
TP4011
WL_PCIE20_REFCLK_P
WL_PCIE20_REFCLK_N
PCIE0_RX0_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_TX0_P_WLAN
PCIE0_TX0_M_WLAN
C4161
C4162
WL_PCIE20_TXP
WL_PCIE20_TXM
WL_PCIE20_RXP
WL_PCIE20_RXM
BT_USB_AVDDH_PAD
BT_USB_DM
BT_USB_DP
BT_RFRX0
BT_RFTX0
MODE_0
MODE_1
XTALI
XTALO
VDD_WIFI_1P8
BT_RFO_TX0
C4160
L4067
U4018
RF2
GND
RF1
VCTL
ANT
VDD
LPWR
BT_SW_CTRL
C4164
BT_RFO_TX1
L4069
C4165
C4166
C4163
L4068
R4058
R4060
C4167
WIFI_XO_CLK
C4203
OPTION
OSC4002
R4061
R4062
C4172
C4173
WL_RFI_2G_CH0
WL_RFO_2G_CH0
WL_RF_PDET_2G_IN_CH0
WL_RFI_5G_CH0
WL_RFO_5G_CH0
WL_RF_PDET_5G_IN_CH0
WL_RFI_2G_CH1
WL_RFO_2G_CH1
WL_RF_PDET_2G_IN_CH1
WL_RFI_5G_CH1
WL_RFO_5G_CH1
WL_RF_PDET_5G_IN_CH1
WL_RFRX0_2G
WL_RFPA0_2G
WL_RFTX0_2G
WL_RFPDET0_2G
WL_RFRX0_5G
WL_RFPA0_5G
WL_RFTX0_5G
WL_RFPDET0_5G
WL_RFRX1_2G
WL_RFPA1_2G
WL_RFTX1_2G
WL_RFPDET1_2G
WL_RFRX1_5G
WL_RFPA1_5G
WL_RFTX1_5G
WL_RFPDET1_5G
WL_ANA_TST_IP_2G_CH0
WL_ANA_TST_IN_2G_CH0
WL_ANA_TST_QP_2G_CH0
WL_ANA_TST_QN_2G_CH0
WL_ANA_TST_IP_2G_CH1
WL_ANA_TST_IN_2G_CH1
WL_ANA_TST_QP_2G_CH1
WL_ANA_TST_QN_2G_CH1
WL_ANA_TST_IP_5G_CH0
WL_ANA_TST_IN_5G_CH0
WL_ANA_TST_QP_5G_CH0
WL_ANA_TST_QN_5G_CH0
WL_ANA_TST_IP_5G_CH1
WL_ANA_TST_IN_5G_CH1
WL_ANA_TST_QP_5G_CH1
WL_ANA_TST_QN_5G_CH1
TP4002
TP4003
TP4004
TP4005
TP4007
BT_UART_TXD
BT_UART_RXD
BT_UART_RTS
BT_UART_CTS
BT_SLIMBUS_DATA
TP4008
BT_SLIMBUS_CLK
WLAN_COEX_SDXUART_TX
WLAN_COEX_SDXUART_RX
VREG_S10B_1P8
R4056
R4057
SLEEP_CLK
WL_PA_EN_2G_CH0
WL_LNA_EN_2G_CH0
BT_EN_CH0
WL_PA_EN_5G_CH0
WL_LNA_EN_5G_CH0
WL_BIAS_SEL_5G_CH0
WL_5G_RFFE_SEL4_CH0
WL_PA_EN_2G_CH1
WL_LNA_EN_2G_CH1
BT_EN_CH1
WL_PA_EN_5G_CH1
WL_LNA_EN_5G_CH1
WL_BIAS_SEL_5G_CH1
WL_5G_RFFE_SEL4_CH1
TP4012
BT_SW_CTRL
R4064
VREG_S10B_1P8
R4063
R4065
SHORT
BOOTSTRAP_CTL
R4066
VREG_S10B_1P8
VREG_S10B_1P8
R4067
R4068
TP4016
HST_SW_CTRL
R4069
R4071
R4072
R4073
R4074
SHORT
BOOTSTRAP_CTL
TP4013
TP4014
TP4015
PCIE0_RST_N
PCIE0_WAKE_N
PCIE0_CLK_REQ_N
AGPIO_WL_2G
AGPIO_WL_5G_0
AGPIO_WL_5G_1
AGPIO_BT
SRST_N
TCK
TDI
TDO
TMS
TRST_N
RESIN_L
ATBH
R4075
WIFI_THM
V4002

WIFI THM

WCN6856 Decap



VDD08_PMU_RFA_CMN
VDD08_PMU_AON_O
C4154
C4155
VDD08_RFA_CMN1
K4
VDD08_TOP_AON
P18 W13
VDD08_PMU_WLCX
VDD08_PMU_WLMX
VDD08_PMU_BT
C4156
C4157
C4158
C4159
VDD08_WL_CX
VDD08_WL_MX
W19 Y10 Y16 AA19
VDD08_CX_BT
K20
VDD12_PMU_RFA_O
C4144
C4145
C4146
C4147
C4148
C4149
C4150
C4151
C4152
C4153
VDD12_WLBBPLL
L11
VDD12_WLADDA5G_CH0
L7
VDD12_BTSYN
F20
VDD12_BTTXBB_CH0
VDD12_WLBB2G_CH1
VDD12_WLSYN2G_PLL
VDD12_WLSYN2G_VCO
E23 H12 J13 G15
VDD12_WLBB2G_CH0
H16
VDD12_WLBB5G_CH1
VDD12_WLSYN5G_TXLO_CH0/1
VDD12_WLSYN5G_PLL
H2 G9 J5
VDD12_WLBB5G_CH0
H8
VDD12_BTPA_CH0
D22
HST_PCIE_1P8
HST_PCIE_0P92
VDD08_PMU_RFA_O
C4174
C4175
C4176
C4177
C4178
C4179
C4180
C4181
C4184
C4182
C4183
VDDA_PCIE_1P8
AB16
VDDA_PCIE_CORE
AA17
VDD08_BTADDA_CH0
F22
VDD08_BTRX_CH0
VDD08_BTRXBB_CH0
VDD08_WLRX2G_CH1
VDD08_WLRX5G_CH1
VDD08_WLSYN2G_LOGEN
D20 E19 F12 F4 G13
VDD08_WLRX2G_CH0
D16
VDD08_WLRX5G_CH0
E7
VDD08_WLSYN5G_LOGEN
H4
VDD08_WLLPSYN5G
F6
VDD17_PMU_RFA_O
C4185
C4186
C4187
C4188
C4189
C4190
C4191
VDD17_WLTX2G_CH0
D18
VDD17_WLTX5G_CH0
F8
VDD17_WLSYN5G_VCO
H6
VDD17_BTPA_CH0
C23
VDD17_XTAL
H20

TITLE	TabS7P
SIZE A1	DWG NO PDM NUMBER
REV	Rev
DRAWN BY	User Name
SHEET	4 of 14
Date	Drawing Date

SM8450
(INTERFACE/GPIO)

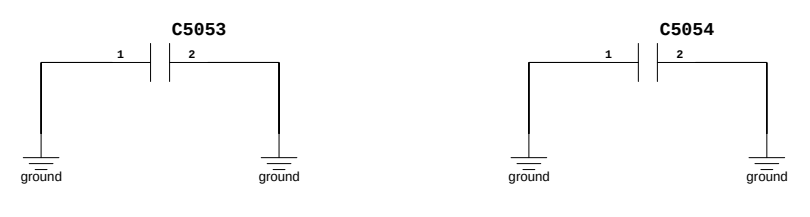
System

System

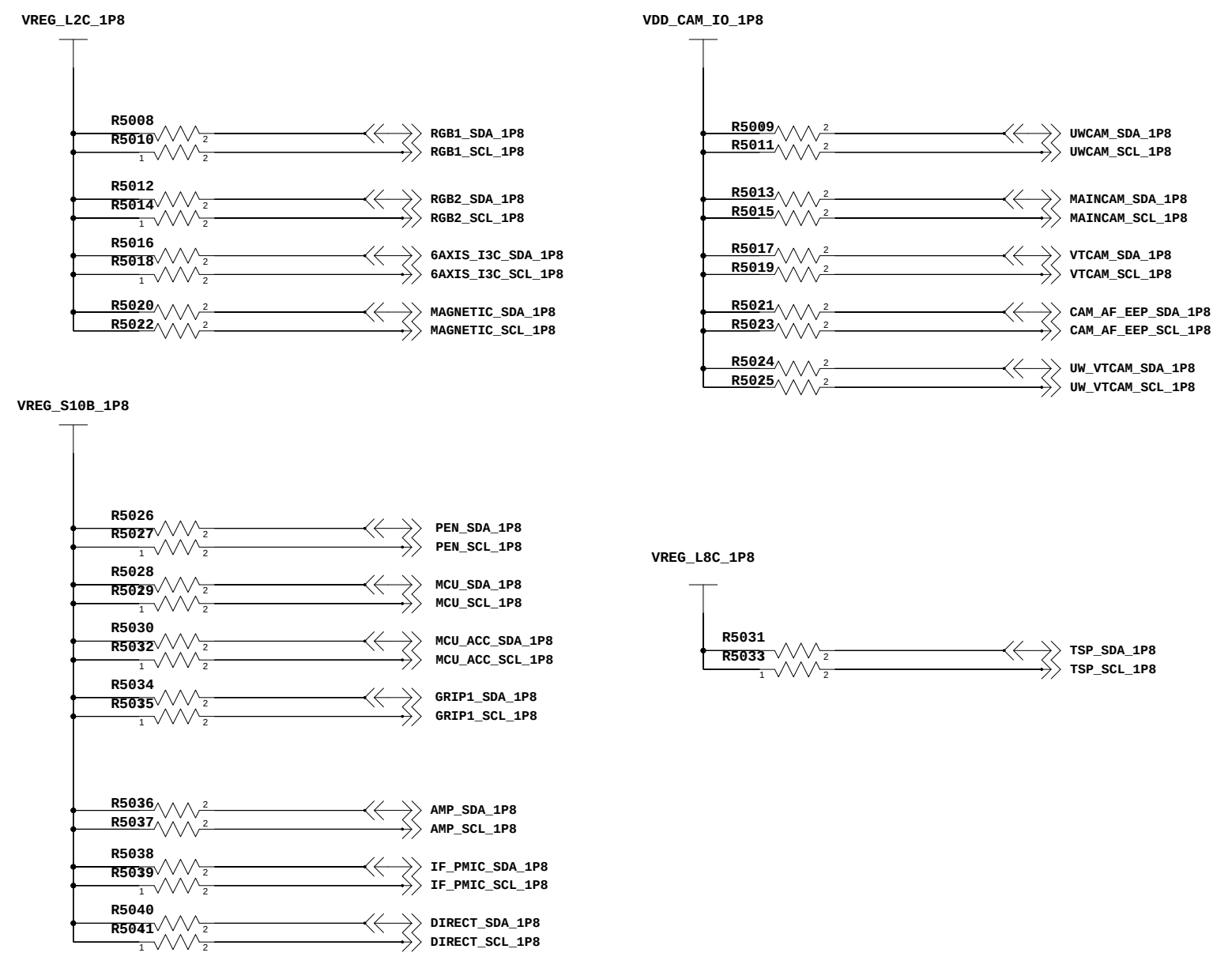
CORE

on SECONDARY PCB

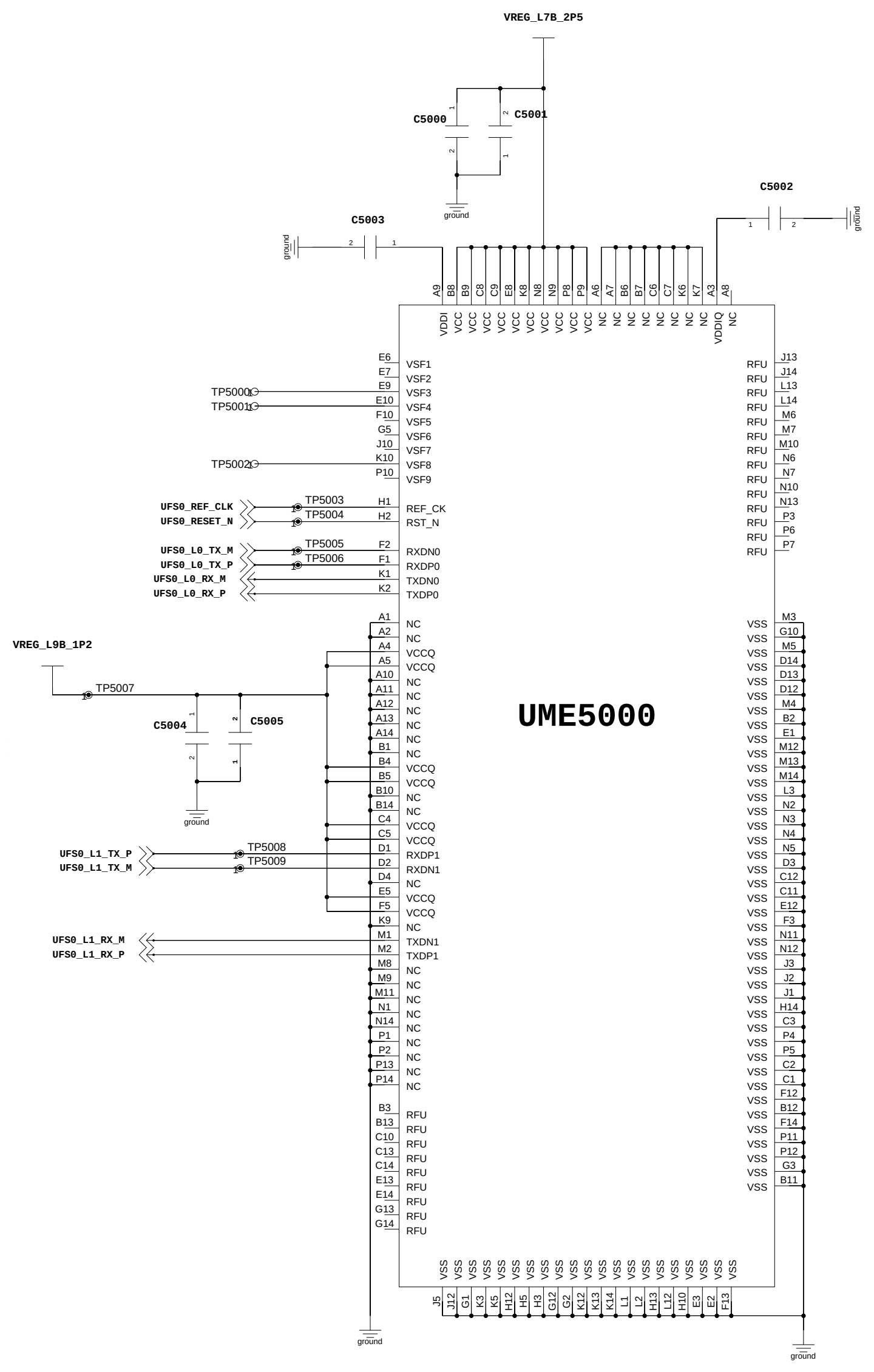
DUMMY CAP



C5053
C5054
1
2
ground



VREG_L2C_1P8
R5008
R5010
RGB1_SDA_1P8
RGB1_SCL_1P8
R5012
R5014
RGB2_SDA_1P8
RGB2_SCL_1P8
R5016
R5018
6AXIS_I3C_SDA_1P8
6AXIS_I3C_SCL_1P8
R5020
R5022
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
VREG_S10B_1P8
R5026
R5027
PEN_SDA_1P8
PEN_SCL_1P8
R5028
R5029
MCU_SDA_1P8
MCU_SCL_1P8
R5030
R5032
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
R5034
R5035
GRIP1_SDA_1P8
GRIP1_SCL_1P8
R5036
R5037
AMP_SDA_1P8
AMP_SCL_1P8
R5038
R5039
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
R5040
R5041
DIRECT_SDA_1P8
DIRECT_SCL_1P8
VDD_CAM_IO_1P8
R5009
R5011
UWCAM_SDA_1P8
UWCAM_SCL_1P8
R5013
R5015
MAINCAM_SDA_1P8
MAINCAM_SCL_1P8
R5017
R5019
VTCAM_SDA_1P8
VTCAM_SCL_1P8
R5021
R5023
CAM_AF_EEP_SDA_1P8
CAM_AF_EEP_SCL_1P8
R5024
R5025
UW_VTCAM_SDA_1P8
UW_VTCAM_SCL_1P8
VREG_L8C_1P8
R5031
R5033
TSP_SDA_1P8
TSP_SCL_1P8

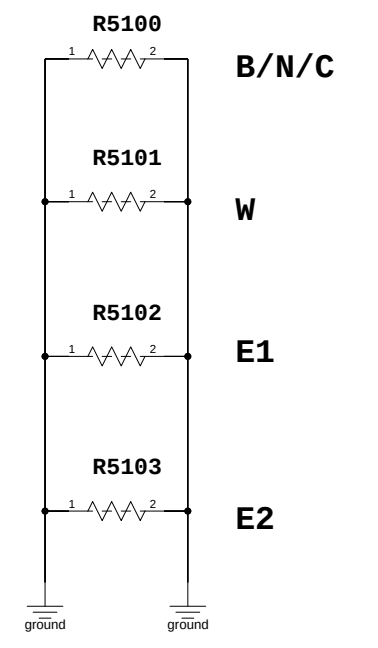


VREG_L7B_2P5
C5000
C5001
C5002
C5003
ground
UME5000
VDDI
VCC
VCCQ
VDDIQ
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
VSF7
VSF8
VSF9
TP5000
TP5001
TP5002
UFS0_REF_CLK
UFS0_RESET_N
TP5003
TP5004
REF_CK
RST_N
UFS0_L0_TX_M
UFS0_L0_TX_P
UFS0_L0_RX_M
UFS0_L0_RX_P
TP5005
TP5006
RXDN0
RXDP0
TXDN0
TXDP0
VREG_L9B_1P2
TP5007
C5004
C5005
UFS0_L1_TX_P
UFS0_L1_TX_M
TP5008
TP5009
RXDP1
RXDN1
UFS0_L1_RX_M
UFS0_L1_RX_P
TXDN1
TXDP1
NC
RFU
VSS

UFS : 128G / 256G / 512G / 1TB

REGION

ANT PCB	SHORT
X906(B/N/C)	R5100
X900(W)	R5101
ETC1 (E1)	R5102
ETC2 (E2)	R5103



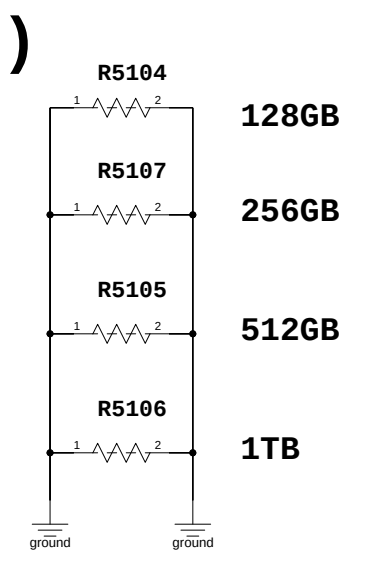
R5100
B/N/C
R5101
W
R5102
E1
R5103
E2
ground

DRAM_POP (S_PCB)

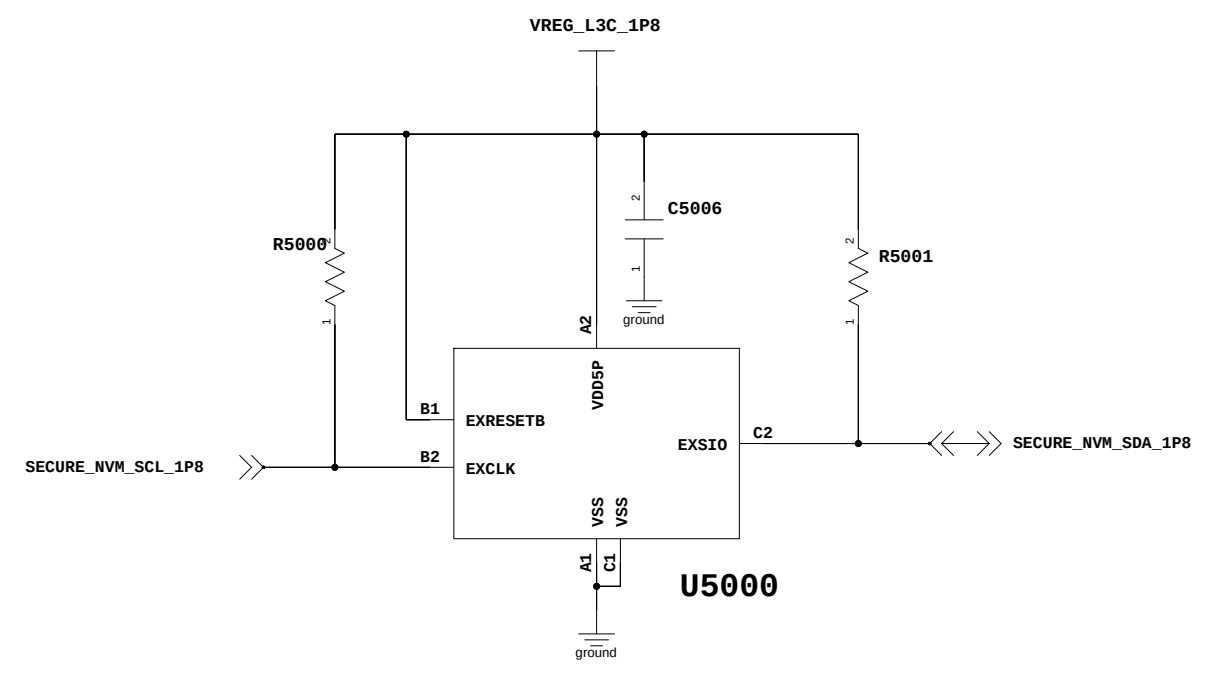
8GB	1105-003137
12GB	1105-003133
16GB	1105-003144

UFS Memory (M_PCB)

128GB	1107-002751
256GB	1107-002753
512GB	1107-002855
1TB	

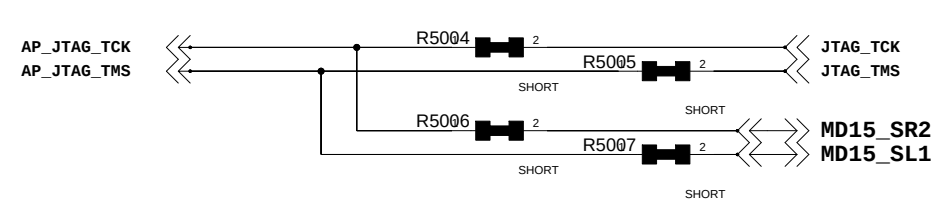


R5104
128GB
R5107
256GB
R5105
512GB
R5106
1TB
ground

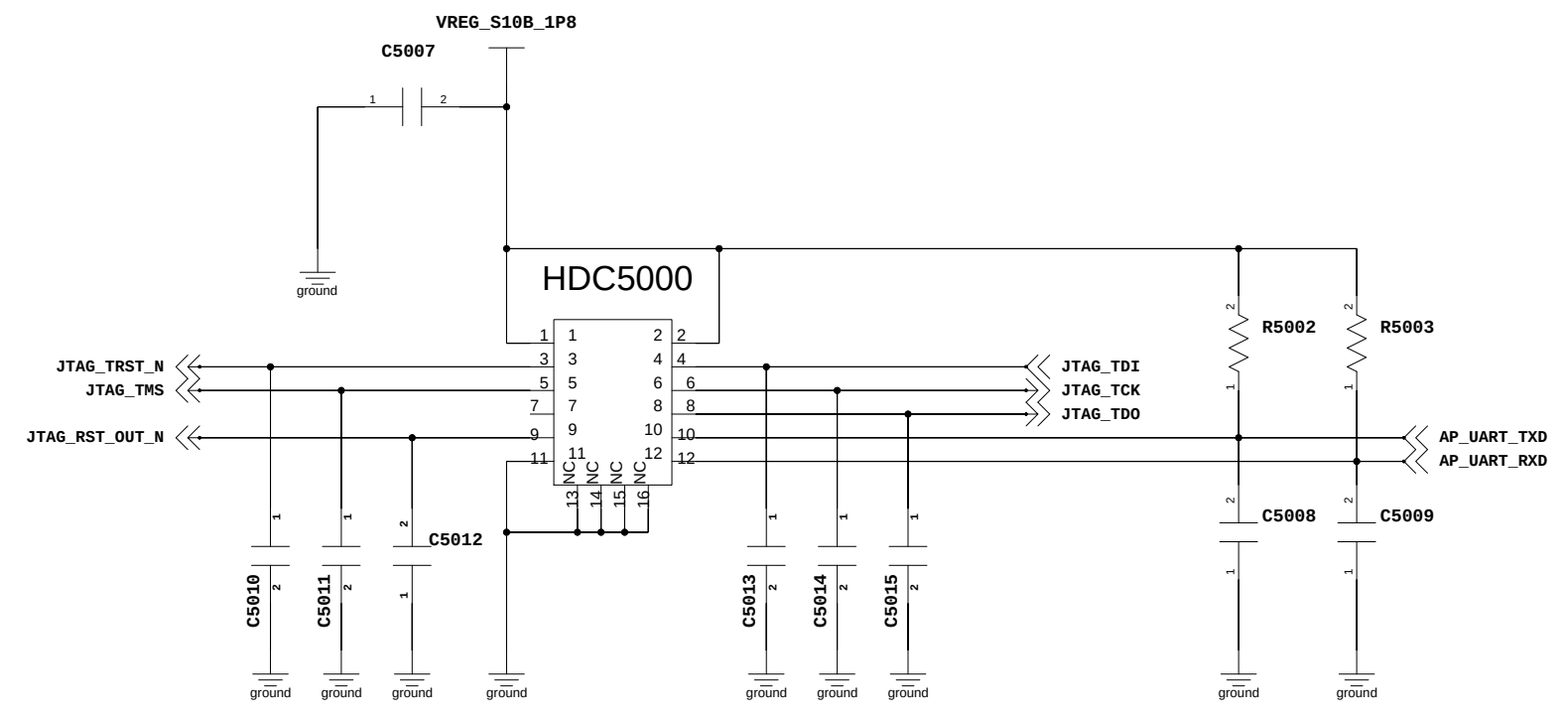


VREG_L3C_1P8
R5000
R5001
C5006
ground
A2
VDDSP
B1
EXRESETB
B2
EXCLK
EXSIO
C2
VSS
A1
C1
SECURE_NVM_SCL_1P8
SECURE_NVM_SDA_1P8
U5000

SECURE NVM

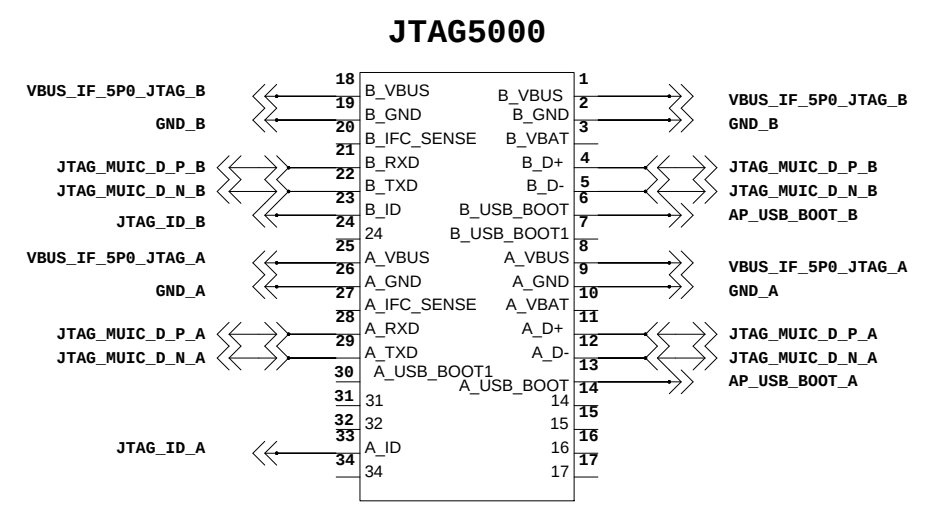


AP_JTAG_TCK
AP_JTAG_TMS
R5094
R5095
R5096
R5097
SHORT
JTAG_TCK
JTAG_TMS
MD15_SR2
MD15_SL1



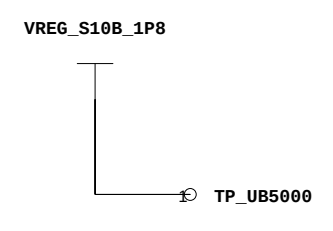
VREG_S10B_1P8
C5007
HDC5000
JTAG_TRST_N
JTAG_TMS
JTAG_RST_OUT_N
JTAG_TDI
JTAG_TCK
JTAG_TDO
AP_UART_TXD
AP_UART_RXD
R5002
R5003
C5008
C5009
C5010
C5011
C5012
C5013
C5014
C5015
ground

AP JTAG CONNECTOR
(3711-007173 / AXE612124)

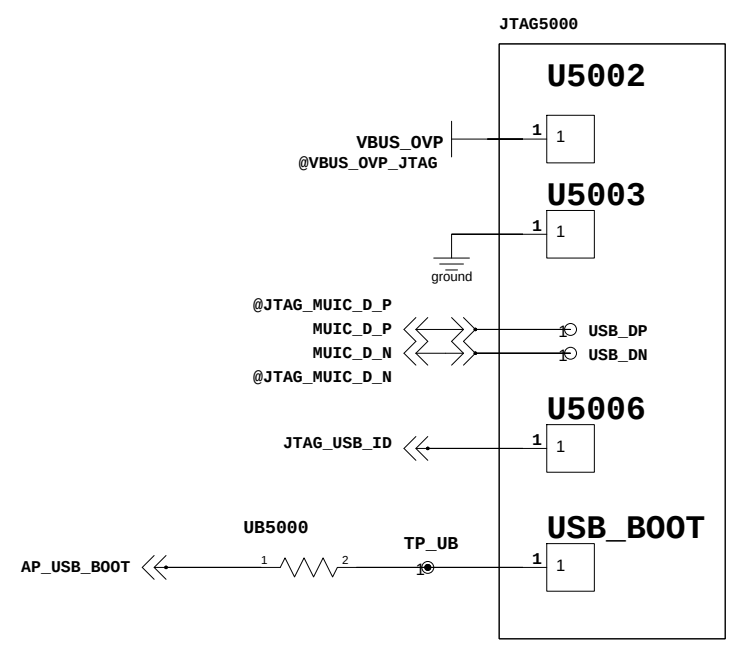


JTAG5000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
JTAG_ID_A
B_VBUS
B_GND
B_IFC_SENSE
B_RXD
B_TXD
B_ID
A_VBUS
A_GND
A_IFC_SENSE
A_RXD
A_TXD
A_USB_BOOT1
A_ID
B_VBAT
B_D+
B_D-
B_USB_BOOT
B_USB_BOOT1
A_VBAT
A_D+
A_D-
A_USB_BOOT
AP_USB_BOOT_B
AP_USB_BOOT_A

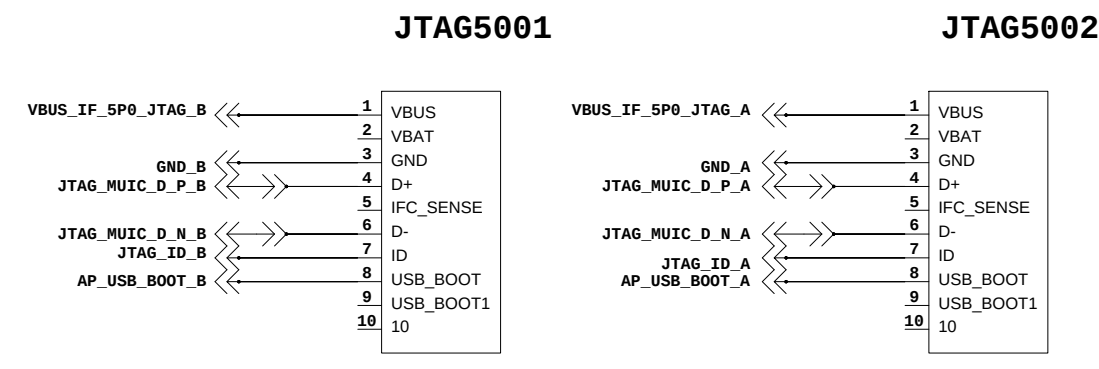
CAUTION: PLACEMENT
Near to TP AP_USB_BOOT



VREG_S10B_1P8
TP_UB5000



JTAG5000
U5002
VBUS_OVP
@VBUS_OVP_JTAG
U5003
ground
@JTAG_MUIC_D_P
MUIC_D_P
MUIC_D_N
@JTAG_MUIC_D_N
USB_DP
USB_DN
U5006
JTAG_USB_ID
UB5000
AP_USB_BOOT
TP_UB
USB_BOOT

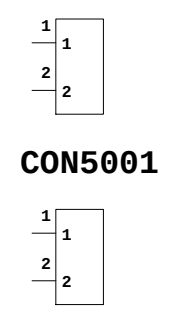


JTAG5001
JTAG5002
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
AP_USB_BOOT_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
JTAG_ID_A
AP_USB_BOOT_A
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT
USB_BOOT1
10

PCB array JTAG

CON5000

CON5001



2D BARCODE
SMD NC

TITLE TabS7P		
SIZE A1	DWG NO PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 5 of 15	Date Drawing Date

SM8450 Power

(POWER, GND)

CORE

on SECONDARY PCB

TITLE Tab S7+
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 6 of 14
Date Drawing Date

P PMIC set & CLK Generator

PMIC (PM8450/8350/8350C)

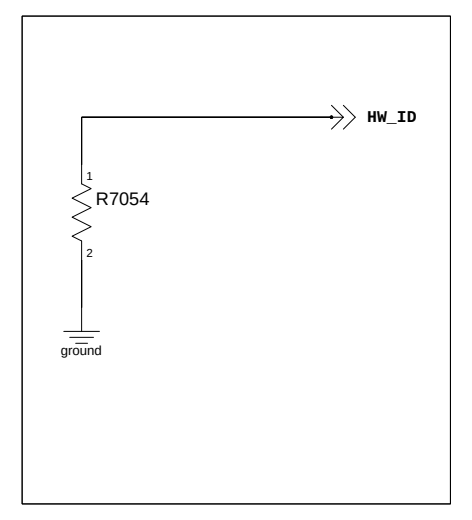
!! CAUTION !!
Place Cin 10uF Bulk Caps at corner of PM8350 and as close to PM8350 as possible to achieve this requirement.

CORE

on SECONDARY PCB

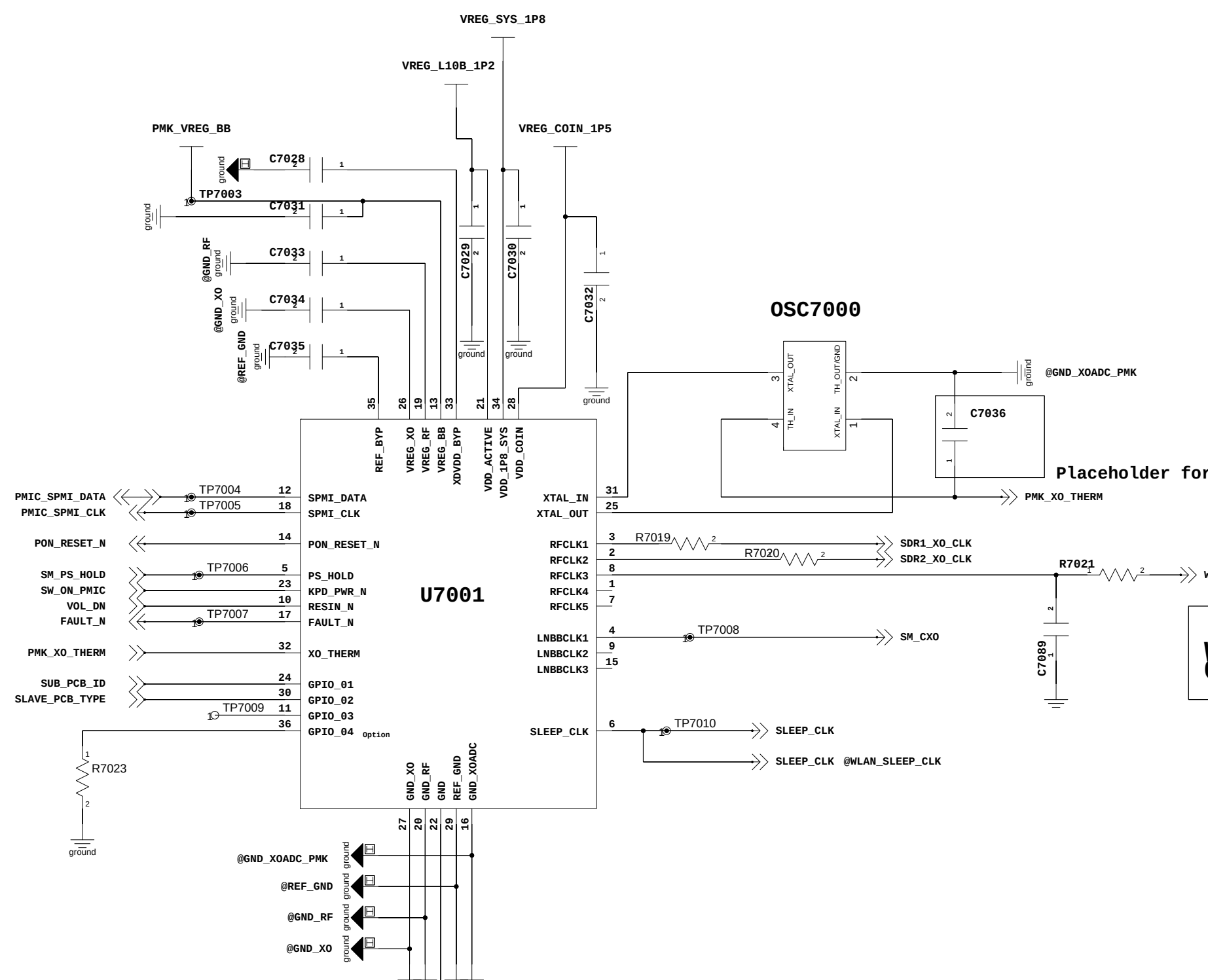
Power System

H/W ID


HW_ID
R7054
ground

H/W ID table(PM8350 AMUX_2)

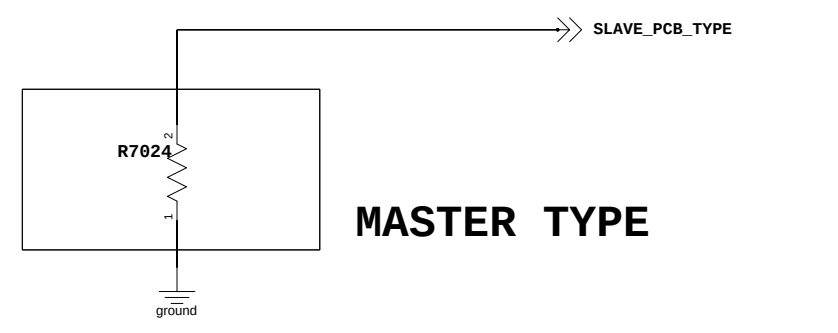
HW ID	BOARD REVISION	Value	Mech
00	TAB S8 ULTRA 14 REV0.0(SW bring-up)	6.04K	
01	TAB S8 ULTRA 14 REV0.1	9.1K	
02	TAB S8 ULTRA 14 REV0.2	12.4K	
03	TAB S8 ULTRA 14 REV0.3	18K	
04	TAB S8 ULTRA 14 REV0.3_CP	22K	
05	TAB S8 ULTRA 14 REV0.4	30K	
06		39K	
07		47K	
08		56K	
09		68K	
10		82K	
11		100K	
12		120K	
13		150K	
14		180K	
15		200K	


VREG_SYS_1P8
VREG_L10B_1P2
VREG_COIN_1P5
PMK_VREG_BB
C7028
TP7003
C7031
C7033
C7034
C7035
@GND_RF
@GND_XO
@REF_GND
C7029
C7030
C7032
OSC7000
XTAL_OUT
TH_OUT/GND
TH_IN
XTAL_IN
@GND_XOADC_PMK
C7036
Placeholder for 1nF
PMK_XO_THERM
U7001
REF_BYP
VREG_XO
VREG_RF
VREG_BB
XDVDD_BYP
VDD_ACTIVE
VDD_1P8_SYS
VDD_COIN
PMIC_SPMI_DATA
PMIC_SPMI_CLK
TP7004
TP7005
SPMI_DATA
SPMI_CLK
PON_RESET_N
SM_PS_HOLD
SW_ON_PMIC
VOL_DN
FAULT_N
TP7006
TP7007
PS_HOLD
KPD_PWR_N
RESIN_N
FAULT_N
PMK_XO_THERM
XO_THERM
SUB_PCB_ID
SLAVE_PCB_TYPE
TP7009
GPIO_01
GPIO_02
GPIO_03
GPIO_04
Option
R7023
ground
XTAL_IN
XTAL_OUT
RFCLK1
RFCLK2
RFCLK3
RFCLK4
RFCLK5
R7019
R7020
SDR1_XO_CLK
SDR2_XO_CLK
R7021
WIFI_XO_CLK
C7089
LNBBCLK1
LNBBCLK2
LNBBCLK3
TP7008
SM_CXO
SLEEP_CLK
TP7010
SLEEP_CLK
SLEEP_CLK @WLAN_SLEEP_CLK
GND_XO
GND_RF
GND
REF_GND
GND_XOADC
@GND_XOADC_PMK
@REF_GND
@GND_RF
@GND_XO
@GND_XO
@GND_RF
@REF_GND
@GND_XOADC_PMK
ground

WIFI Option
Open R7020

Option pin state (PMK8450; 76.8MHz XO)

OPTION	LPDDR5	SD card LDO	GPIO05B Active	R value
8	O	enable	HIGH	120K
9	O	enable	LOW	150K
10 (Reserved)				
11 (Reserved)				
12	O	skip	HIGH	360K
13	O	skip	LOW	560K
14 (Reserved)				
15 (Reserved)				


SLAVE_PCB_TYPE
R7024
ground

MASTER TYPE

MASTER_TYPE

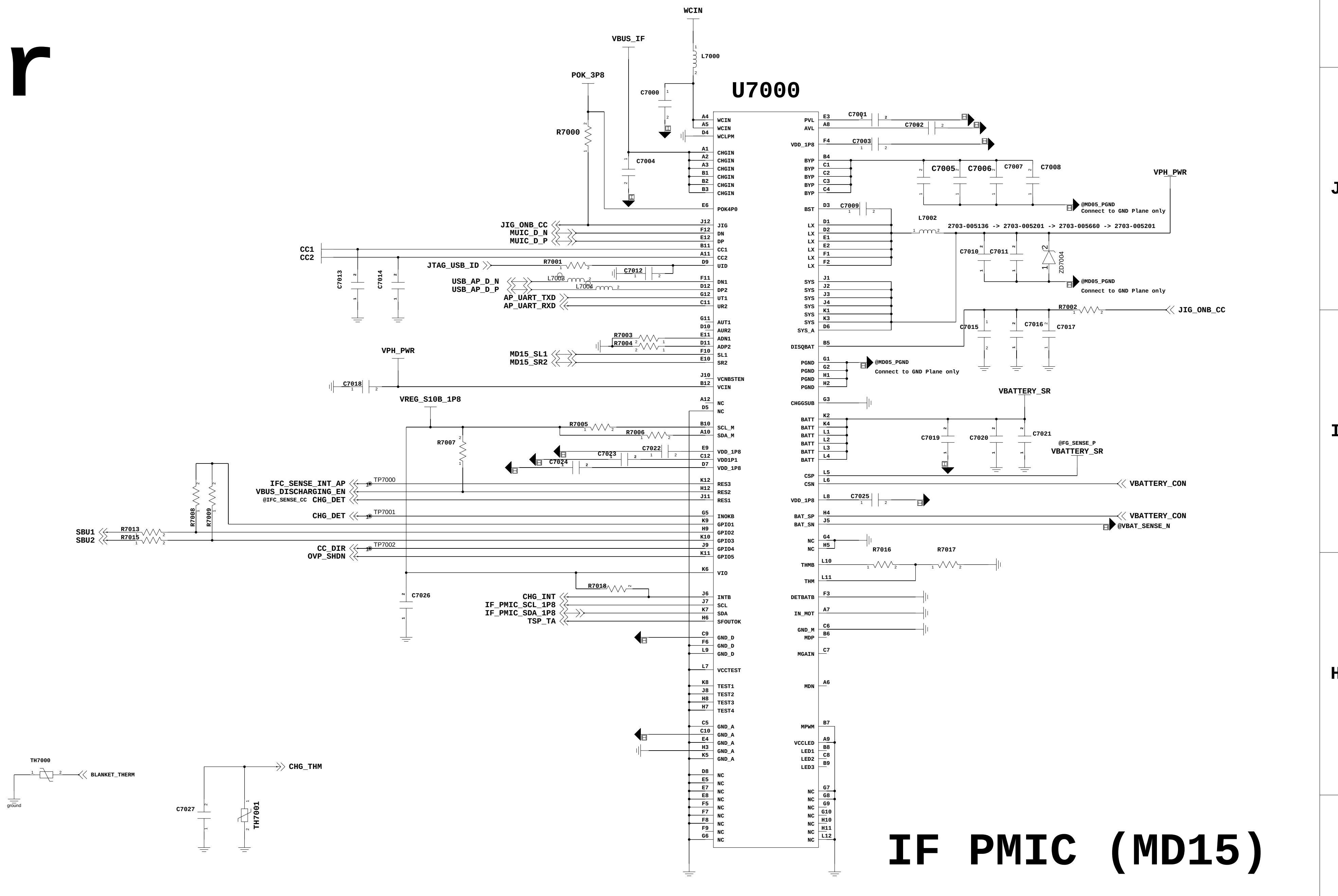
PMK8450 GPIO_02	ADC
EUR(Sub_6)	68K
USA(Sub_6)	100K
WIFI	150K
mmWave	200K
EUR with eSE	560K
For JIG BINARY	OPEN

!! CAUTION !!
Place Cin 10uF Bulk Caps at corner of PM8350C and as close to PM8350C as possible to achieve this requirement.

CORE

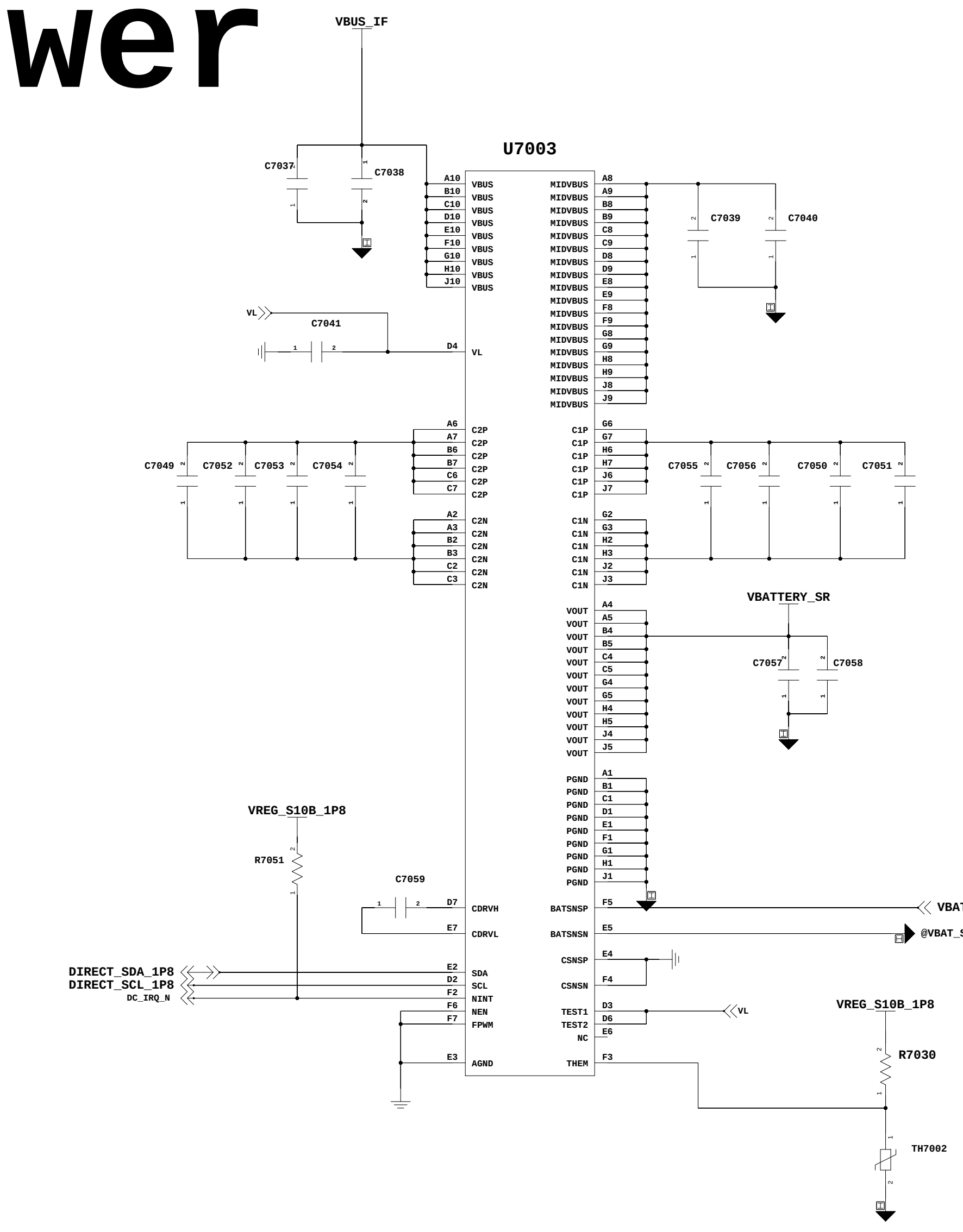
on SECONDARY PCB

Power


WCIN
VBUS_IF
L7000
POK_3P8
C7000
U7000
R7000
C7004
WCIN
WCIN
WCLPM
CHGIN
POK4P0
JIG_ONB_CC
MUIC_D_N
MUIC_D_P
JIG
DN
DP
CC1
CC2
CC1
CC2
UID
JTAG_USB_ID
R7001
C7012
C7013
C7014
USB_AP_D_N
USB_AP_D_P
L7003
L7004
DN1
DP2
AP_UART_TXD
AP_UART_RXD
UT1
UR2
AUT1
AUR2
ADN1
ADP2
R7003
R7004
MD15_SL1
MD15_SR2
SL1
SR2
VPH_PWR
C7018
VCNBSTEN
VCIN
VREG_S10B_1P8
NC
R7005
R7006
R7007
SCL_M
SDA_M
C7022
C7023
C7024
VDD_1P8
VDD1P1
IFC_SENSE_INT_AP
VBUS_DISCHARGING_EN
@IFC_SENSE_CC
CHG_DET
TP7000
TP7001
TP7002
RES3
RES2
RES1
INOKB
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
SBU1
SBU2
R7013
R7015
R7008
R7009
CC_DIR
OVP_SHDN
VIO
C7026
R7018
CHG_INT
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
TSP_TA
INTB
SCL
SDA
SFOUTOK
GND_D
VCCTEST
TEST1
TEST2
TEST3
TEST4
GND_A
TH7000
BLANKET_THERM
CHG_THM
C7027
TH7001
PVL
AVL
C7001
C7002
C7003
VDD_1P8
BYP
C7005
C7006
C7007
C7008
VPH_PWR
@MD05_PGND
Connect to GND Plane only
BST
C7009
LX
L7002
2703-005136 -> 2703-005201 -> 2703-005660 -> 2703-005201
C7010
C7011
ZD7004
SYS
SYS_A
DISQBAT
C7015
C7016
C7017
R7002
JIG_ONB_CC
PGND
CHGGSUB
VBATTERY_SR
BATT
C7019
C7020
C7021
@FG_SENSE_P
CSP
CSN
VBATTERY_CON
C7025
BAT_SP
BAT_SN
@VBAT_SENSE_N
THMB
THM
R7016
R7017
DETBATB
IN_MOT
GND_M
MDP
MGAIN
MDN
MPWM
VCCLED
LED1
LED2
LED3

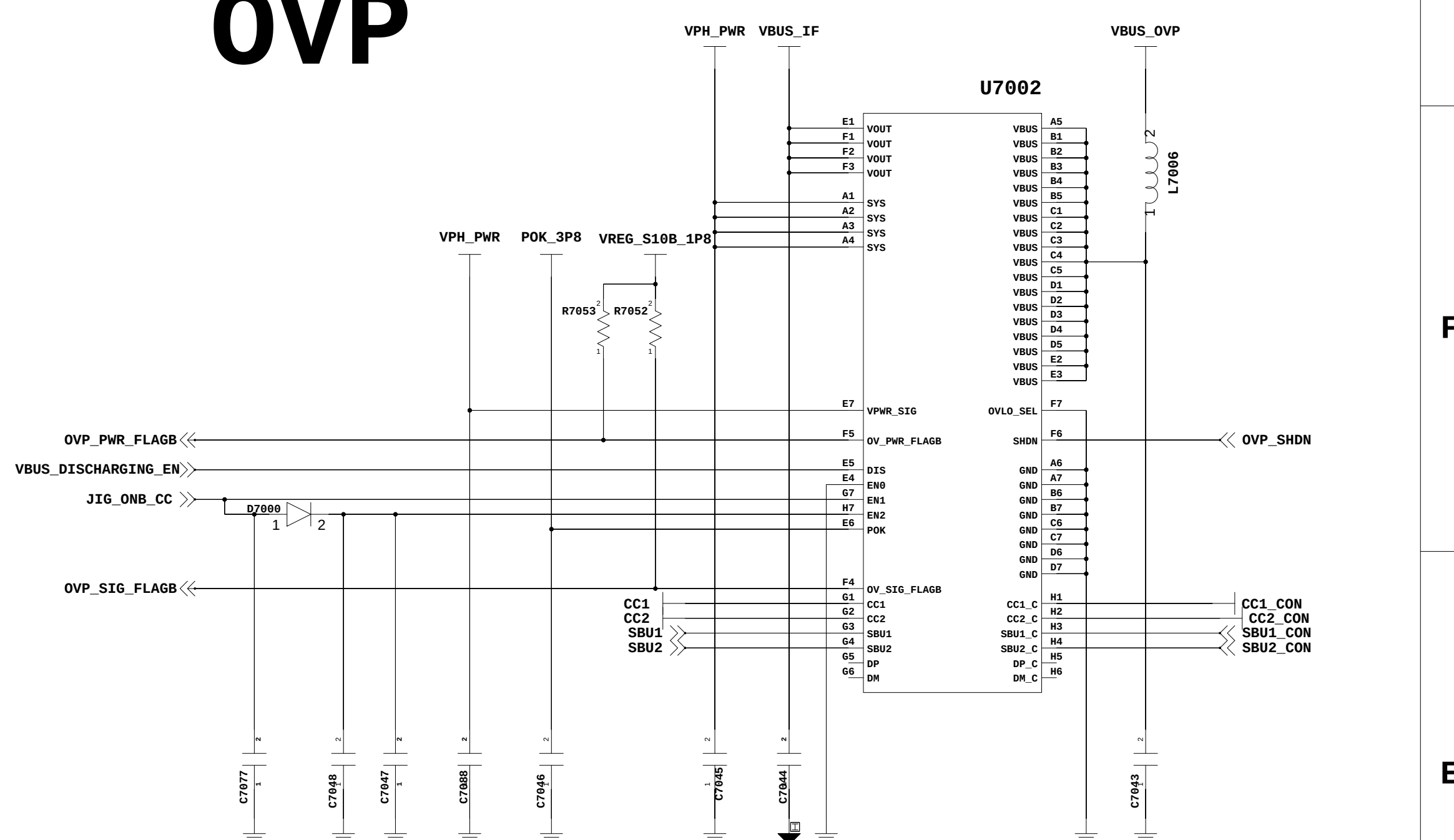
IF PMIC (MD15)

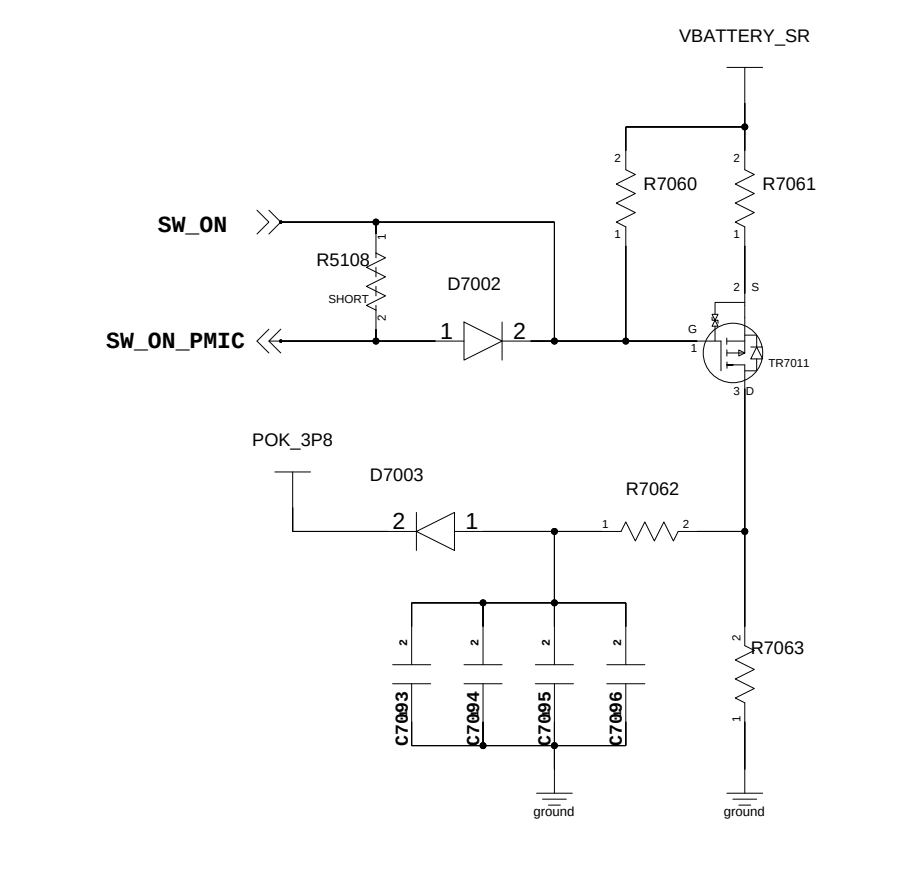
Power

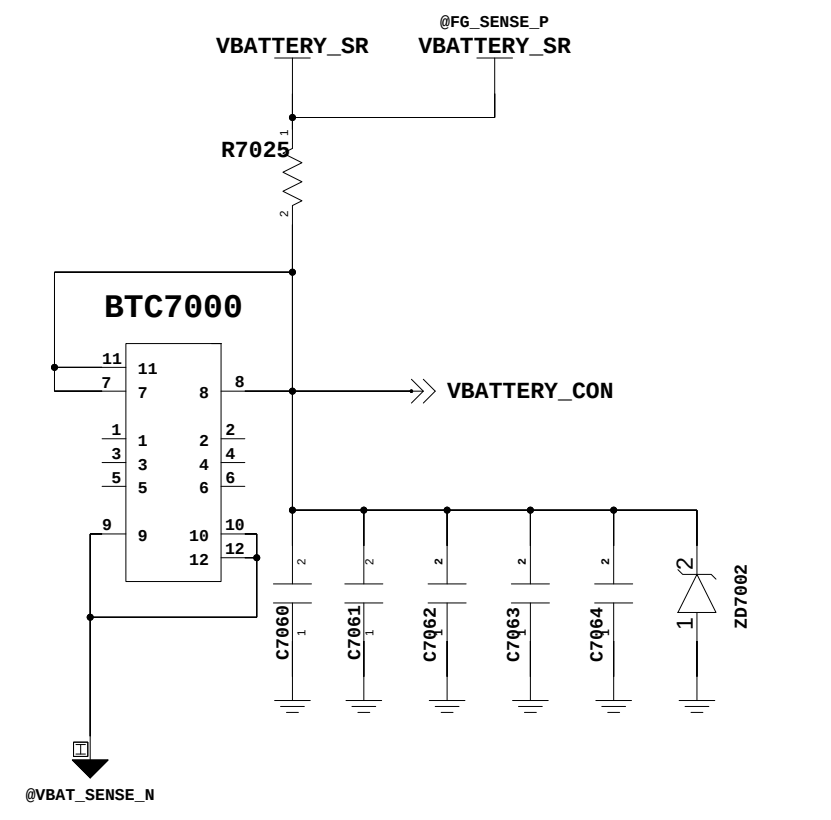

VBUS_IF
C7037
C7038
U7003
VBUS
MIDVBUS
C7039
C7040
VL
C7041
C2P
C2N
C1P
C1N
C7049
C7052
C7053
C7054
C7055
C7056
C7050
C7051
VBATTERY_SR
VOUT
C7057
C7058
PGND
VREG_S10B_1P8
R7051
C7059
CDRVH
CDRVL
BATSNSP
BATSNSN
VBATTERY_CON
@VBAT_SENSE_N
CSNSP
CSNSN
DIRECT_SDA_1P8
DIRECT_SCL_1P8
DC_IRQ_N
SDA
SCL
NINT
NEN
FPWM
TEST1
TEST2
NC
AGND
THEM
VL
VREG_S10B_1P8
R7030
TH7002

Direct Charging

OVP

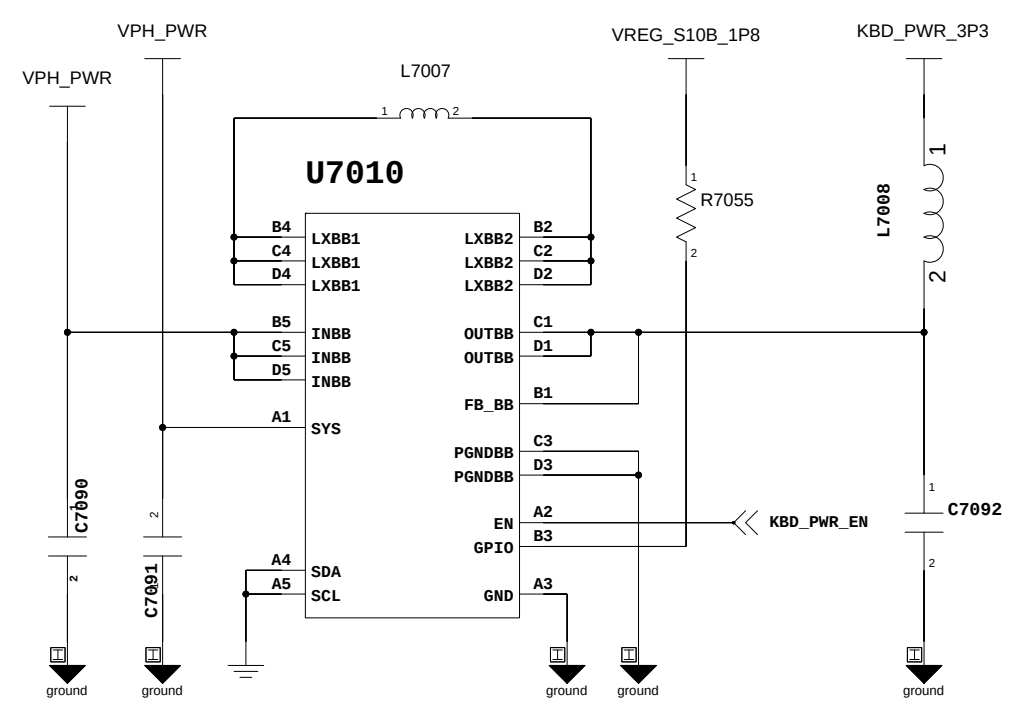

VPH_PWR
VBUS_IF
VBUS_OVP
U7002
VOUT
SYS
VBUS
L7006
VPH_PWR
POK_3P8
VREG_S10B_1P8
R7053
R7052
VPWR_SIG
OVLO_SEL
OVP_PWR_FLAGB
OV_PWR_FLAGB
SHDN
OVP_SHDN
VBUS_DISCHARGING_EN
DIS
EN0
EN1
EN2
POK
JIG_ONB_CC
D7000
GND
OVP_SIG_FLAGB
OV_SIG_FLAGB
CC1
CC2
SBU1
SBU2
DP
DM
CC1_C
CC2_C
SBU1_C
SBU2_C
DP_C
DM_C
CC1_CON
CC2_CON
SBU1_CON
SBU2_CON
C7077
C7048
C7047
C7088
C7046
C7045
C7044
C7043

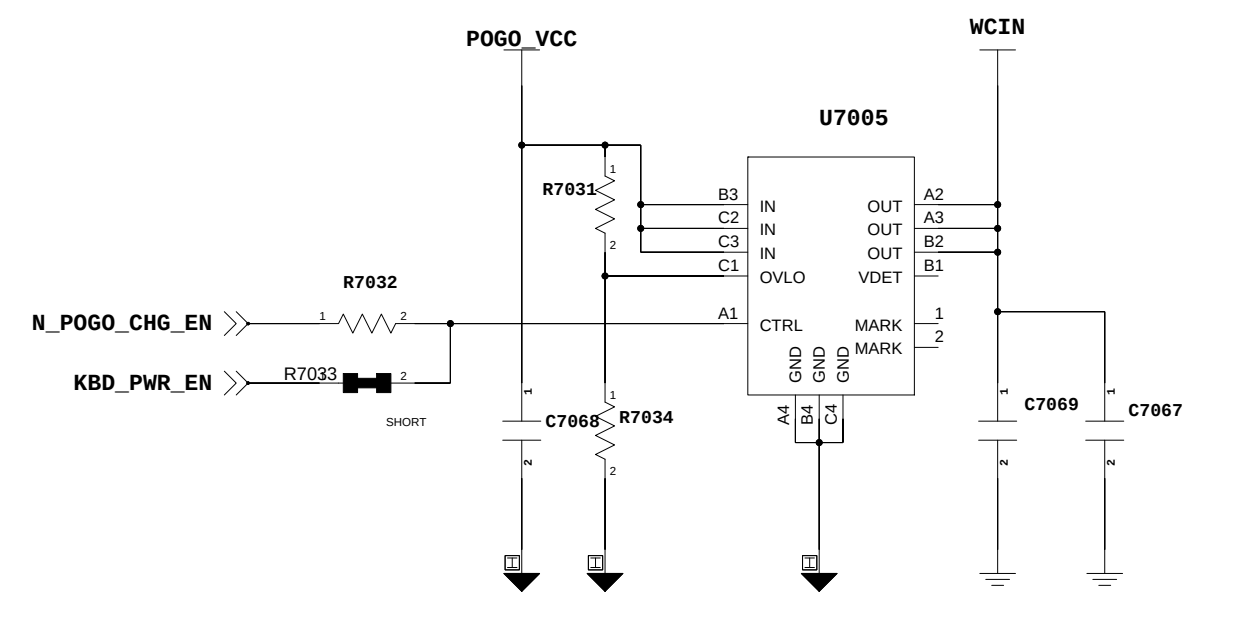

VBATTERY_SR
SW_ON
R5108
SHORT
D7002
R7060
R7061
SW_ON_PMIC
TR7011
POK_3P8
D7003
R7062
R7063
C7093
C7094
C7095
C7096
ground

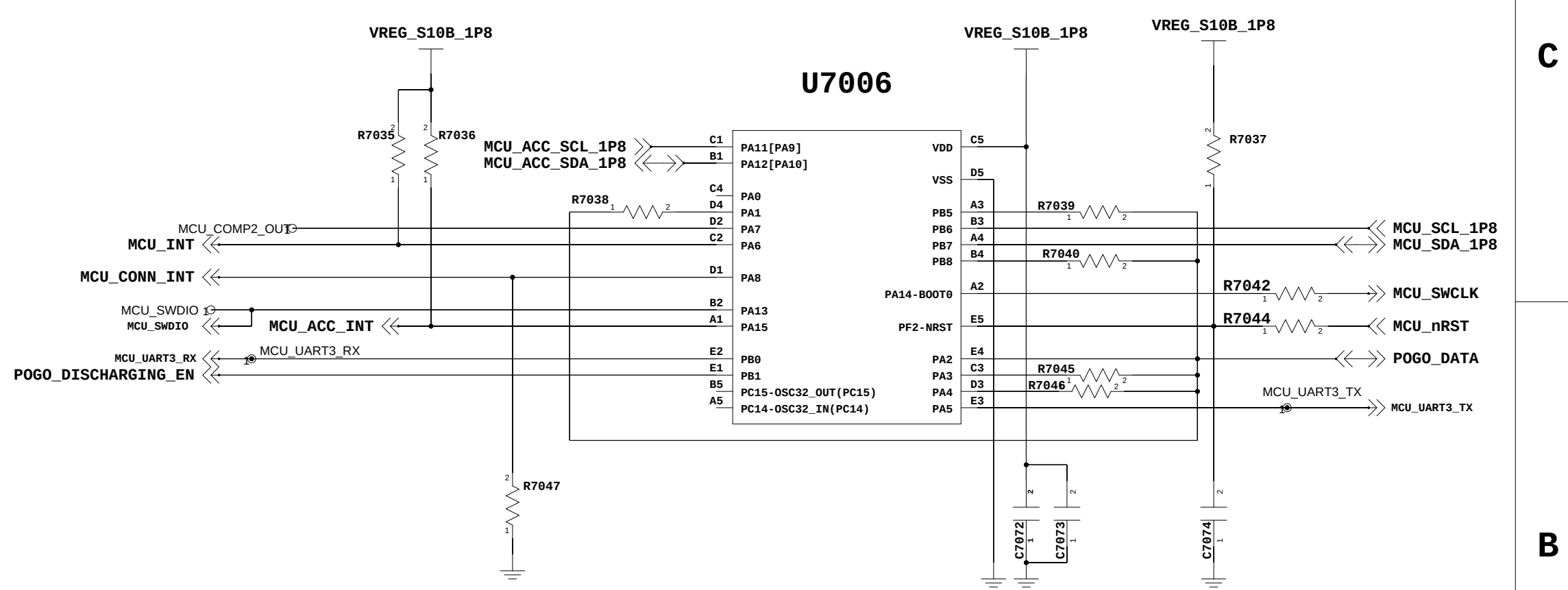

VBATTERY_SR
@FG_SENSE_P
VBATTERY_SR
R7025
BTC7000
VBATTERY_CON
C7060
C7061
C7062
C7063
C7064
ZD7002
@VBAT_SENSE_N

Ship Mode Off **Battery Con**

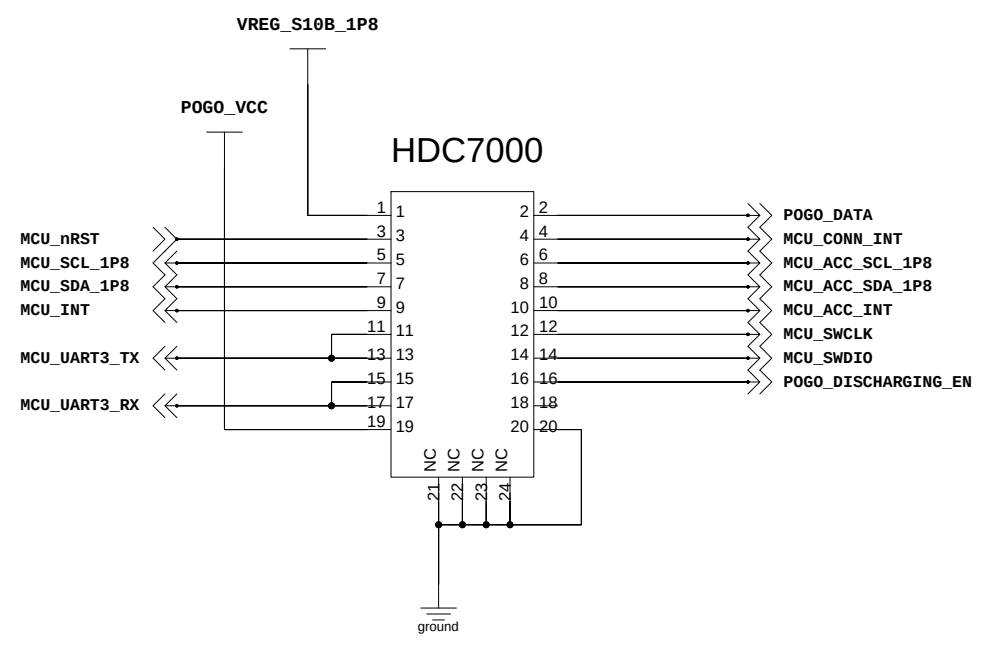
POGO


VPH_PWR
VPH_PWR
L7007
VREG_S10B_1P8
KBD_PWR_3P3
U7010
LXBB1
LXBB2
INBB
OUTBB
R7055
L7008
SYS
FB_BB
PGNDBB
EN
GPIO
KBD_PWR_EN
C7092
SDA
SCL
GND
C7090
C7091
ground

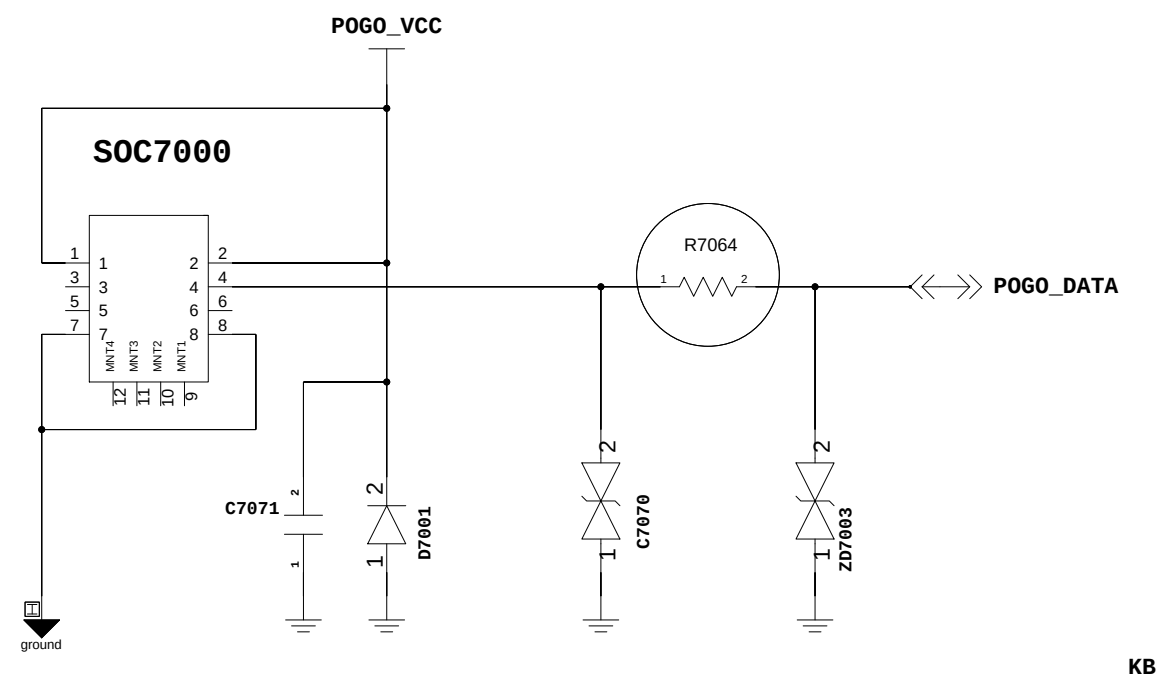

POGO_VCC
WCIN
U7005
R7031
IN
OUT
OVLO
VDET
N_POGO_CHG_EN
R7032
CTRL
MARK
KBD_PWR_EN
R7033
SHORT
C7068
R7034
GND
C7069
C7067


VREG_S10B_1P8
VREG_S10B_1P8
VREG_S10B_1P8
U7006
R7035
R7036
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
PA11[PA9]
PA12[PA10]
VDD
VSS
R7038
PA0
PA1
PA7
PA6
MCU_COMP2_OUT
MCU_INT
MCU_CONN_INT
PA8
MCU_SWDIO
MCU_SWDIO
MCU_ACC_INT
PA13
PA15
MCU_UART3_RX
MCU_UART3_RX
POGO_DISCHARGING_EN
PB0
PB1
PC15-OSC32_OUT(PC15)
PC14-OSC32_IN(PC14)
PB5
PB6
PB7
PB8
PA14-BOOT0
PF2-NRST
PA2
PA3
PA4
PA5
R7037
R7039
R7040
MCU_SCL_1P8
MCU_SDA_1P8
R7042
MCU_SWCLK
R7044
MCU_nRST
POGO_DATA
R7045
R7046
MCU_UART3_TX
MCU_UART3_TX
R7047
C7072
C7073
C7074

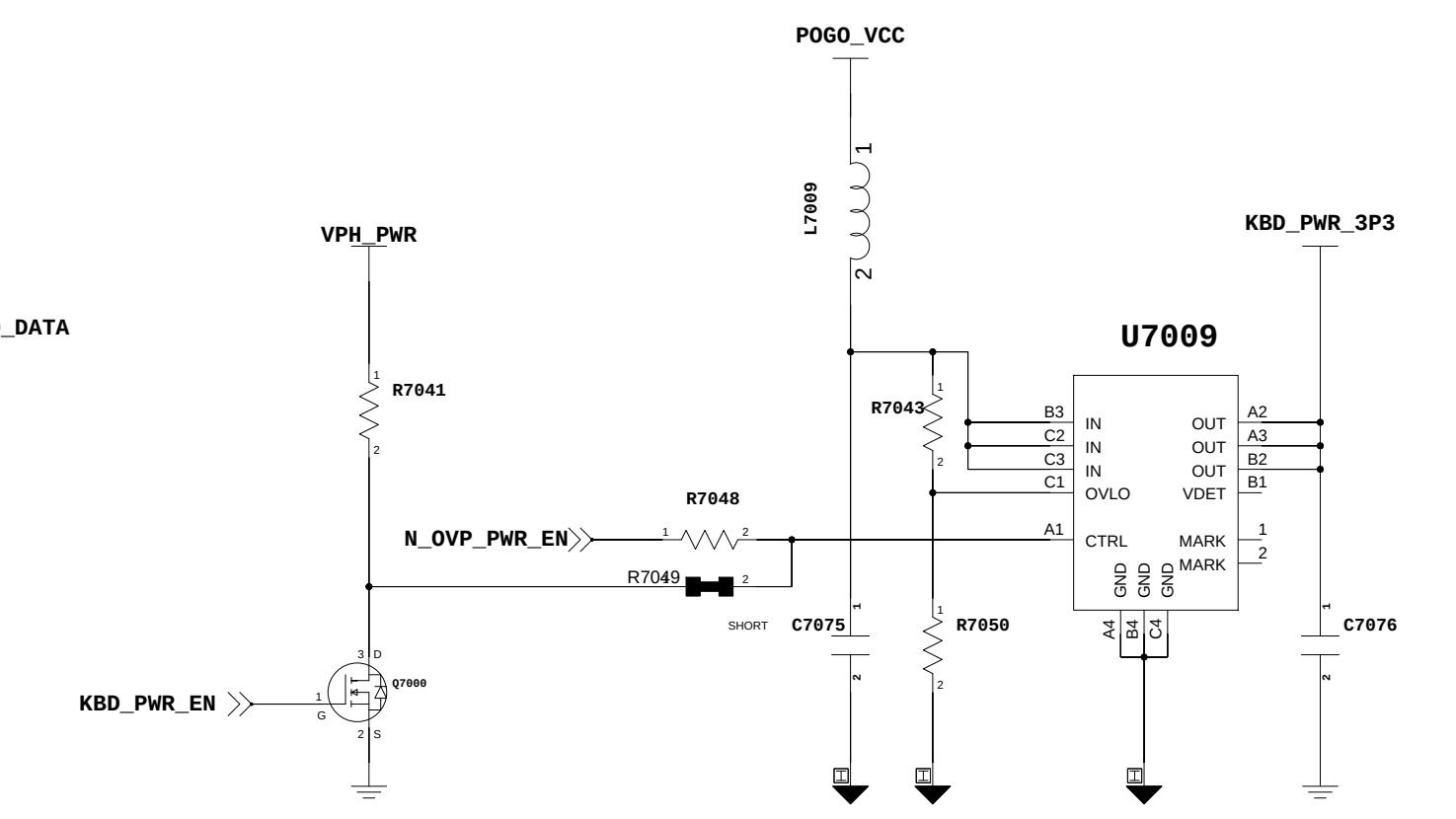
POGO MCU

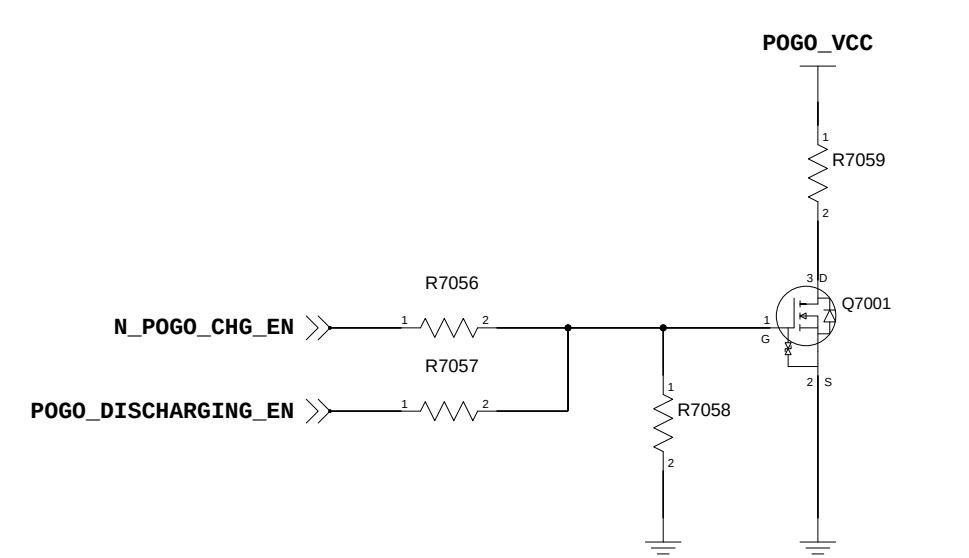

VREG_S10B_1P8
POGO_VCC
HDC7000
MCU_nRST
MCU_SCL_1P8
MCU_SDA_1P8
MCU_INT
MCU_UART3_TX
MCU_UART3_RX
POGO_DATA
MCU_CONN_INT
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_ACC_INT
MCU_SWCLK
MCU_SWDIO
POGO_DISCHARGING_EN
NC
ground

MCU DEBUG CON


POGO_VCC
SOC7000
R7064
POGO_DATA
C7071
D7001
C7070
ZD7003

POGO BtoB Con.


VPH_PWR
POGO_VCC
KBD_PWR_3P3
L7009
U7009
R7041
R7043
IN
OUT
OVLO
VDET
N_OVP_PWR_EN
R7048
R7049
SHORT
CTRL
MARK
C7075
R7050
GND
C7076
KBD_PWR_EN
Q7000


POGO_VCC
R7059
Q7001
N_POGO_CHG_EN
R7056
R7057
POGO_DISCHARGING_EN
R7058

TITLE	MODEL NAME		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 1 of 2	Date Drawing Date	

AUDIO

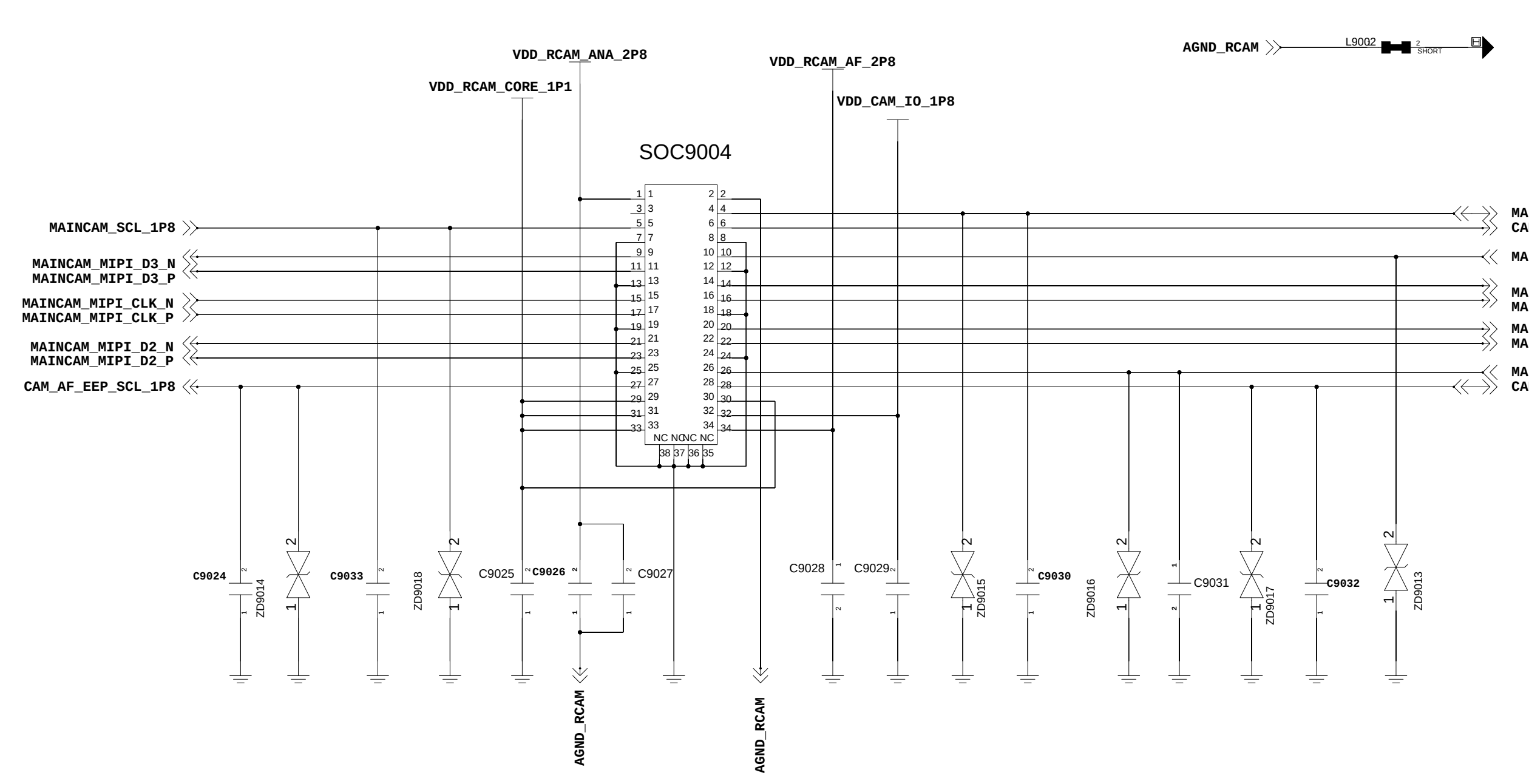
VREG_L12C_1P8
ANT8004 ANT8005
R8002
MAIN_MIC_DATA
C8001
ZD8001
C8003
ANT8006 ANT8007
MAIN_MIC_CLK
ZD8003
MAIN MIC
VREG_L12C_1P8
ANT8002 ANT8003
R8001
SUB_MIC_DATA
C8004
ZD8000
C8002
ANT8008 ANT8009
SUB_MIC_CLK
ZD8004
SUB MIC
VREG_L12C_1P8
ANT8000 ANT8001
R8000
3RD_MIC_DATA
C8000
ZD8002
C8005
ANT8010 ANT8011
3RD_MIC_CLK
ZD8005
3rd MIC
VBATTERY_SR
VPH_PWR
R8003
SHORT
Place On array dummy
(SPK AMP SMD F/T)
VREG_S10B_1P8
C8008
C8009
C8011
R8005
SHORT
C8013
R8006
AMP_RST_N
R8007
R8008
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
R8010
AMP_INT_N
R8011
AMP_SYNC
R8013
C8031
U8001
RESET
I2C_SCL/SPI_MISO
I2C_SDA/SPI_MOSI
SPI_SS
SPI_SCK
ASP_DOUT
ASP_DIN
ASP_BCLK
ASP_FSYNC
INT/GPIO2
SYNC/GPIO1
VDD_LV
VREF_FILT
VDD_BST
SWIRE_CLK
SWIRE_DATA
ADDR
VDD_A
LDO_FILT
VDD_BATT
SW
VDD_SPK
OUTP_SPK
OUTN_SPK
VSNSP_SPK
VSNSN_SPK
GPIO3
GND_A
GND_B
GND_P
C8015
L8001
VBATTERY_SR
C8018
C8019
TA8001
SPK_TL_OUT_P
SPK_TL_OUT_N
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_SENSE_N
C8032
C8033
C8034
C8035
SPK TOP LEFT AMP
VREG_S10B_1P8
C8006
C8007
C8010
R8004
SHORT
C8012
U8000
R8009
R8012
C8020
C8014
L8000
VBATTERY_SR
C8016
C8017
TA8000
SPK_TR_OUT_P
SPK_TR_OUT_N
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_SENSE_N
C8027
C8028
C8029
C8030
SPK TOP RIGHT AMP
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_P
L8002
ANT8012 ANT8013
L8003
SPK_BR_OUT_SENSE_N
SPK_BR_OUT_N
C8021
C8022
C8023
ZD8006
ZD8007
C8024
C8025
C8026
SPK BR CON
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_P
L8004
ANT8014 ANT8015
L8005
SPK_TR_OUT_SENSE_N
SPK_TR_OUT_N
C8036
C8037
C8038
ZD8008
ZD8009
C8039
C8040
C8041
SPK TR CON
VREG_S10B_1P8
C8044
C8045
C8047
R8015
C8049
U8003
TP8000
TP8001
TP8002
R8017
R8019
SHORT
C8071
C8051
L8007
VBATTERY_SR
C8054
C8055
TA8003
L8010
L8011
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
C8066
C8067
C8068
C8069
C8072
C8075
C8078
C8074
SPK BOTTOM LEFT AMP
VREG_S10B_1P8
C8042
C8043
C8046
R8014
C8048
U8002
R8016
R8018
C8070
C8050
L8006
VBATTERY_SR
C8052
C8053
TA8002
SPK_BR_OUT_P
SPK_BR_OUT_N
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_SENSE_N
C8056
C8057
C8058
C8059
SPK BOTTOM RIGHT AMP
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_P
L8008
ANT8016 ANT8017
L8009
SPK_TL_OUT_SENSE_N
SPK_TL_OUT_N
C8060
C8061
C8062
ZD8010
ZD8011
C8063
C8064
C8065
SPK TL CON

TITLE TabS7P

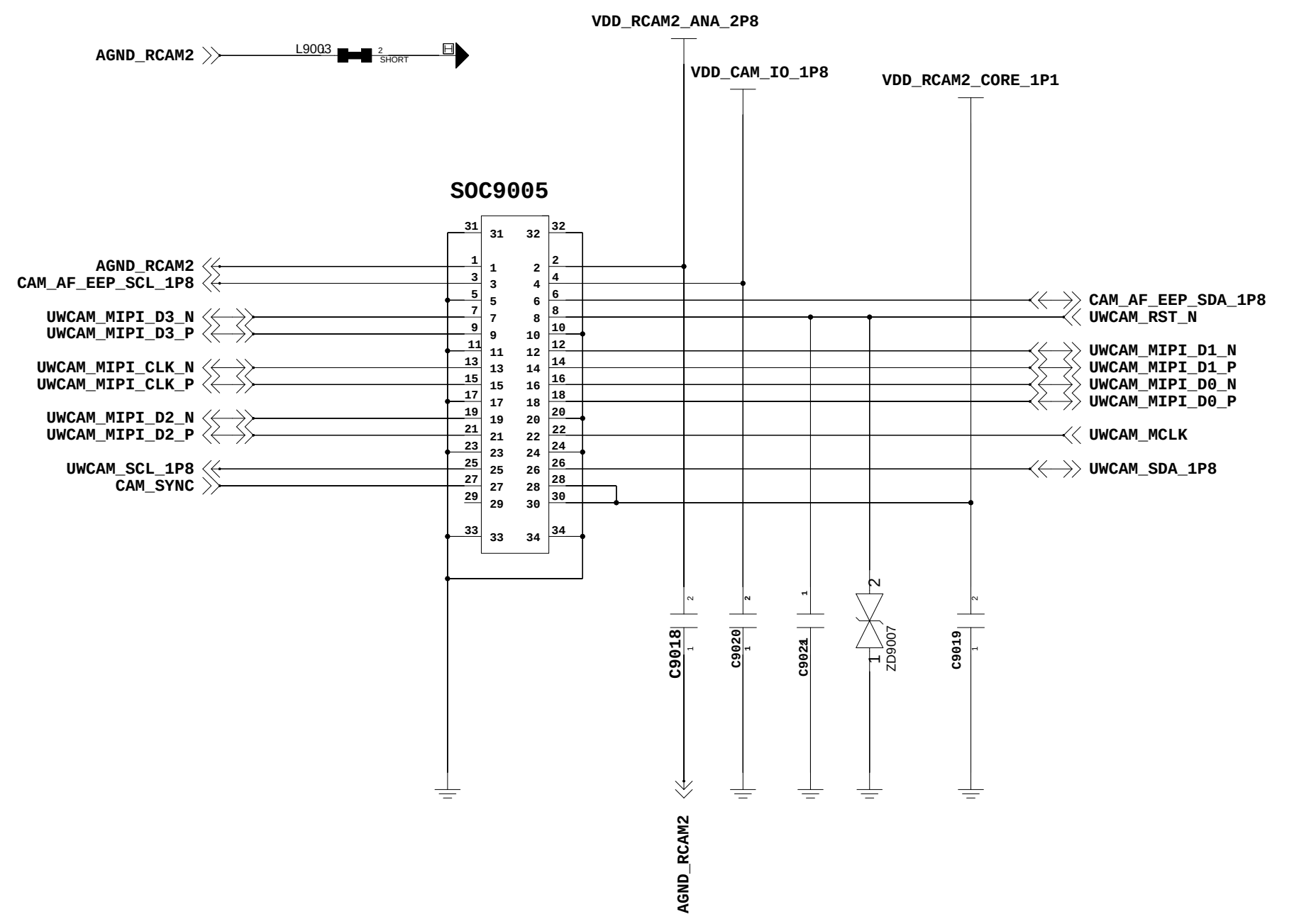
SIZE A1 | DWG NO PDM NUMBER | REV Rev

DRAWN BY User Name | SHEET 8 of 16 | Date Drawing Date

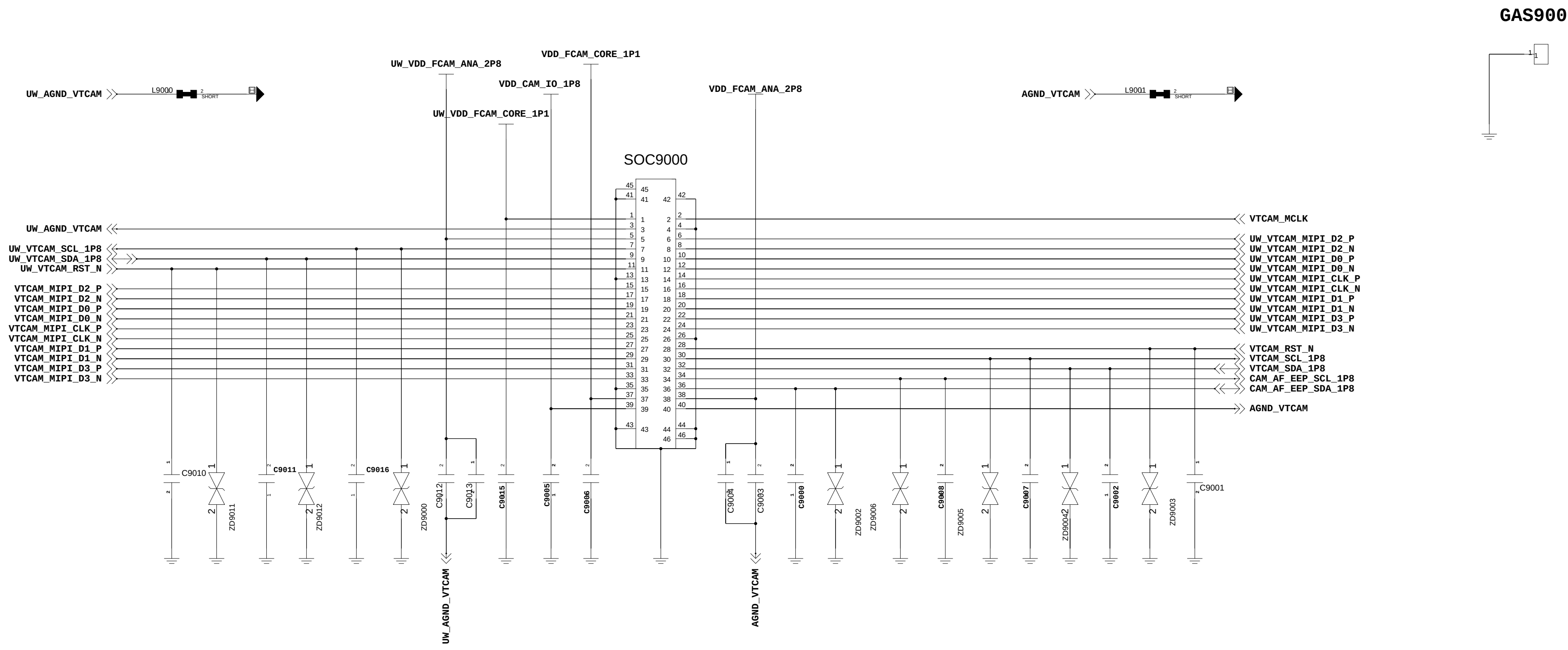
CAMERA


SOC9004
VDD_RCAM_ANA_2P8
VDD_RCAM_CORE_1P1
VDD_RCAM_AF_2P8
VDD_CAM_IO_1P8
AGND_RCAM
L9002
SHORT
MAINCAM_SCL_1P8
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
CAM_AF_EEP_SCL_1P8
MAINCAM_SDA_1P8
CAM_SYNC
MAINCAM_MCLK
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_RST_N
CAM_AF_EEP_SDA_1P8
C9024
ZD9014
C9033
ZD9018
C9025
C9026
C9027
C9028
C9029
ZD9015
C9030
ZD9016
C9031
ZD9017
C9032
ZD9013
AGND_RCAM
AGND_RCAM

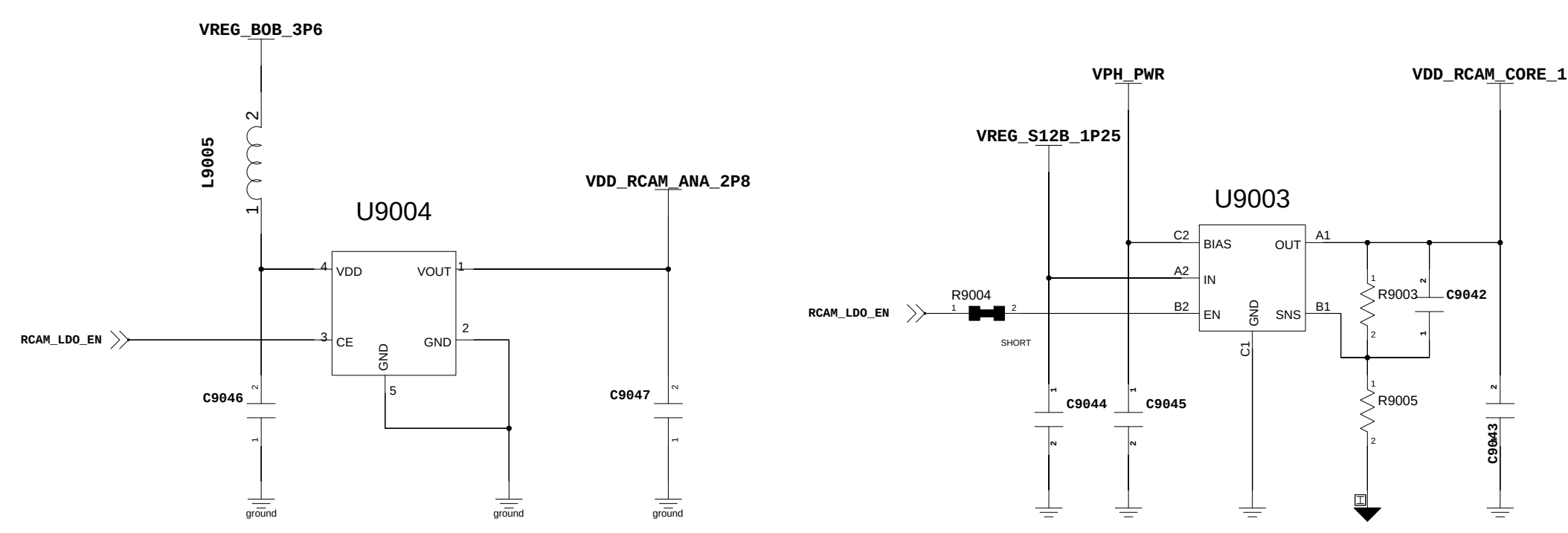
13M_CAM (RCAM)


SOC9005
AGND_RCAM2
L9003
SHORT
VDD_RCAM2_ANA_2P8
VDD_CAM_IO_1P8
VDD_RCAM2_CORE_1P1
AGND_RCAM2
CAM_AF_EEP_SCL_1P8
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_CLK_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_SCL_1P8
CAM_SYNC
CAM_AF_EEP_SDA_1P8
UWCAM_RST_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_D1_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
UWCAM_MCLK
UWCAM_SDA_1P8
C9018
C9020
C9024
ZD9007
C9019
AGND_RCAM2

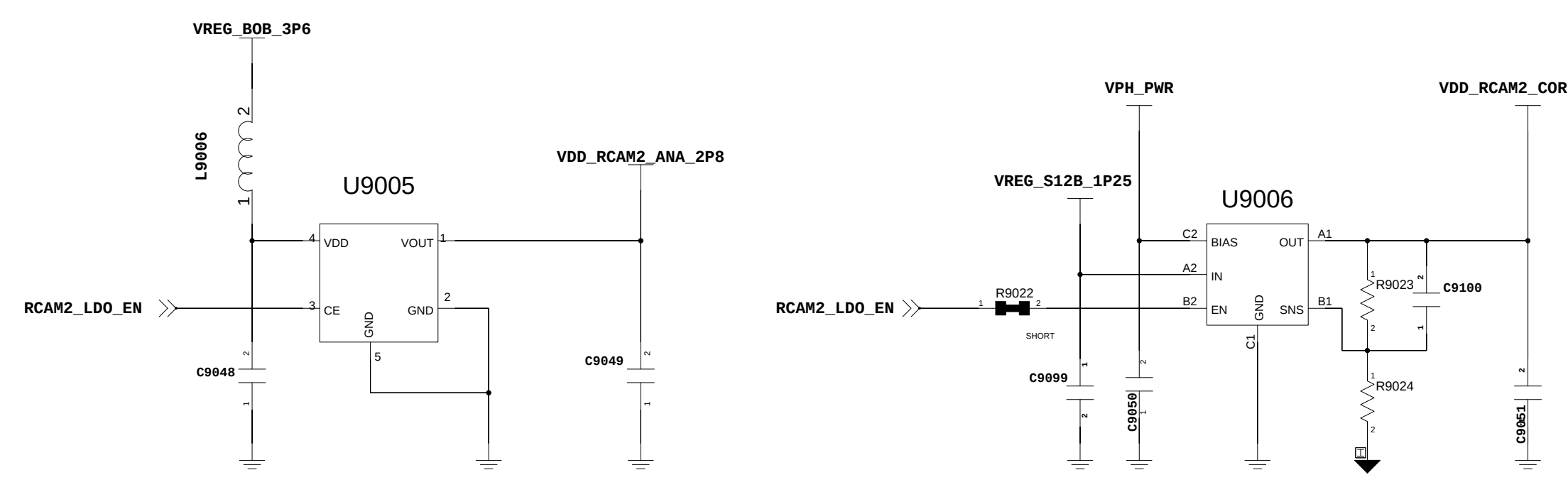
6M_UW CAM (RCAM2)


GAS9000
UW_AGND_VTCAM
L9000
SHORT
UW_VDD_FCAM_ANA_2P8
VDD_FCAM_CORE_1P1
VDD_CAM_IO_1P8
UW_VDD_FCAM_CORE_1P1
VDD_FCAM_ANA_2P8
AGND_VTCAM
L9001
SOC9000
UW_AGND_VTCAM
UW_VTCAM_SCL_1P8
UW_VTCAM_SDA_1P8
UW_VTCAM_RST_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_D0_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D3_P
VTCAM_MIPI_D3_N
VTCAM_MCLK
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D1_P
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_RST_N
VTCAM_SCL_1P8
VTCAM_SDA_1P8
CAM_AF_EEP_SCL_1P8
CAM_AF_EEP_SDA_1P8
AGND_VTCAM
C9010
ZD9011
C9011
ZD9012
C9016
ZD9000
C9012
C9013
C9015
C9005
C9006
C9004
C9003
C9000
ZD9002
ZD9006
C9008
ZD9005
C9007
ZD9004
C9002
ZD9003
C9001
UW_AGND_VTCAM
AGND_VTCAM

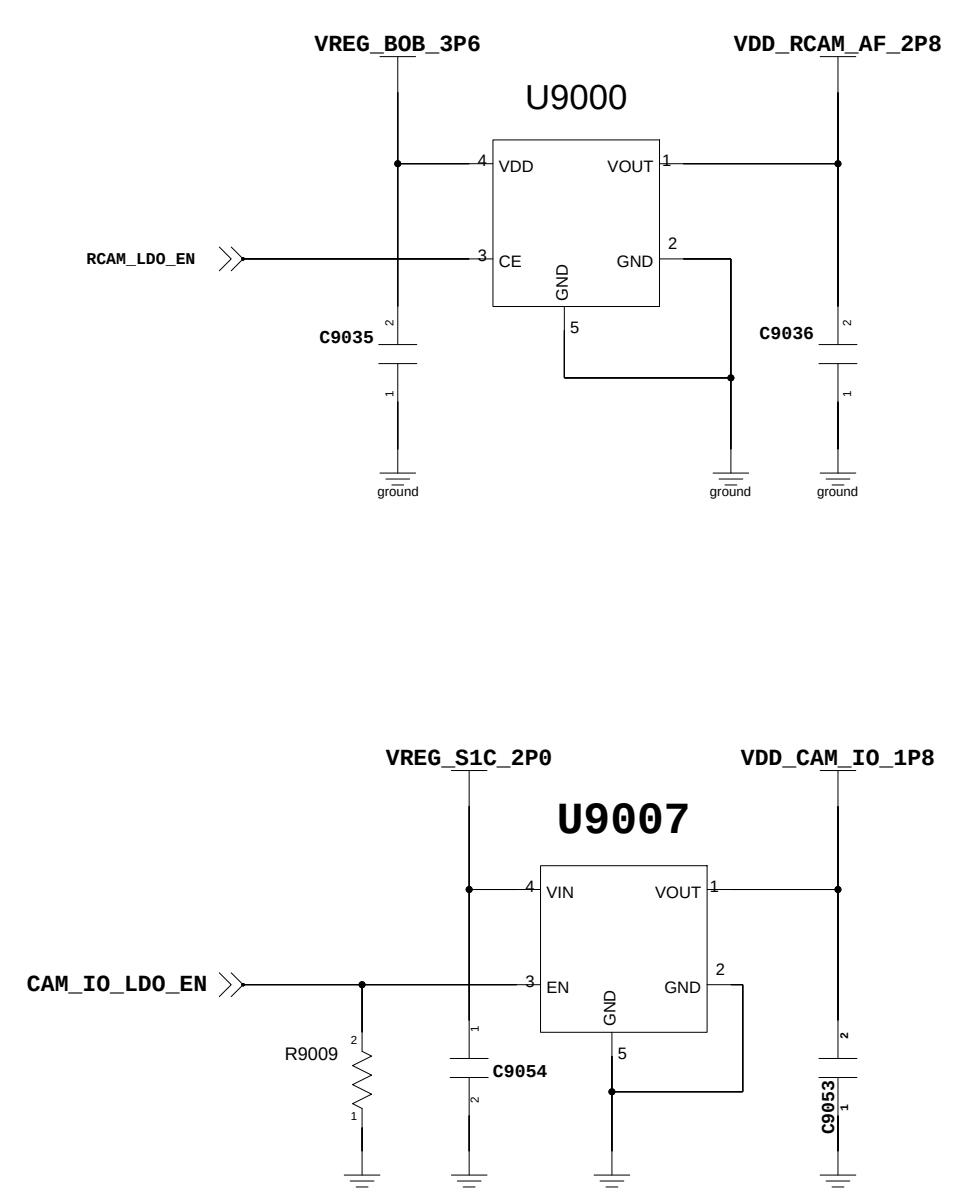
12M Dual FCAM

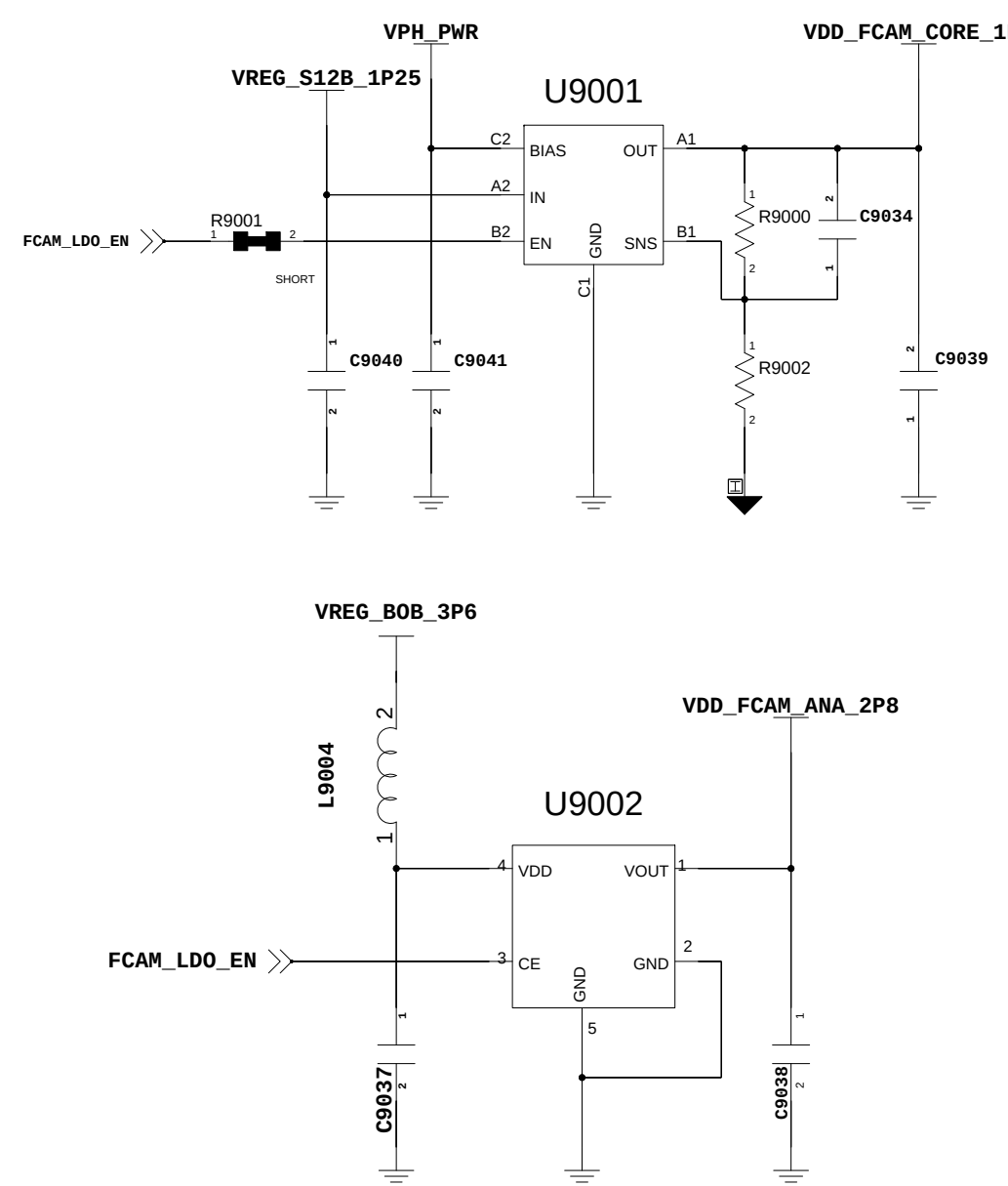

VREG_BOB_3P6
L9005
U9004
VDD
VOUT
CE
GND
VDD_RCAM_ANA_2P8
RCAM_LDO_EN
C9046
C9047
ground
VPH_PWR
VREG_S12B_1P25
U9003
BIAS
IN
EN
OUT
SNS
VDD_RCAM_CORE_1P1
R9004
SHORT
R9003
C9042
R9005
C9044
C9045
C9043

13M_CAM (RCAM)_LDO

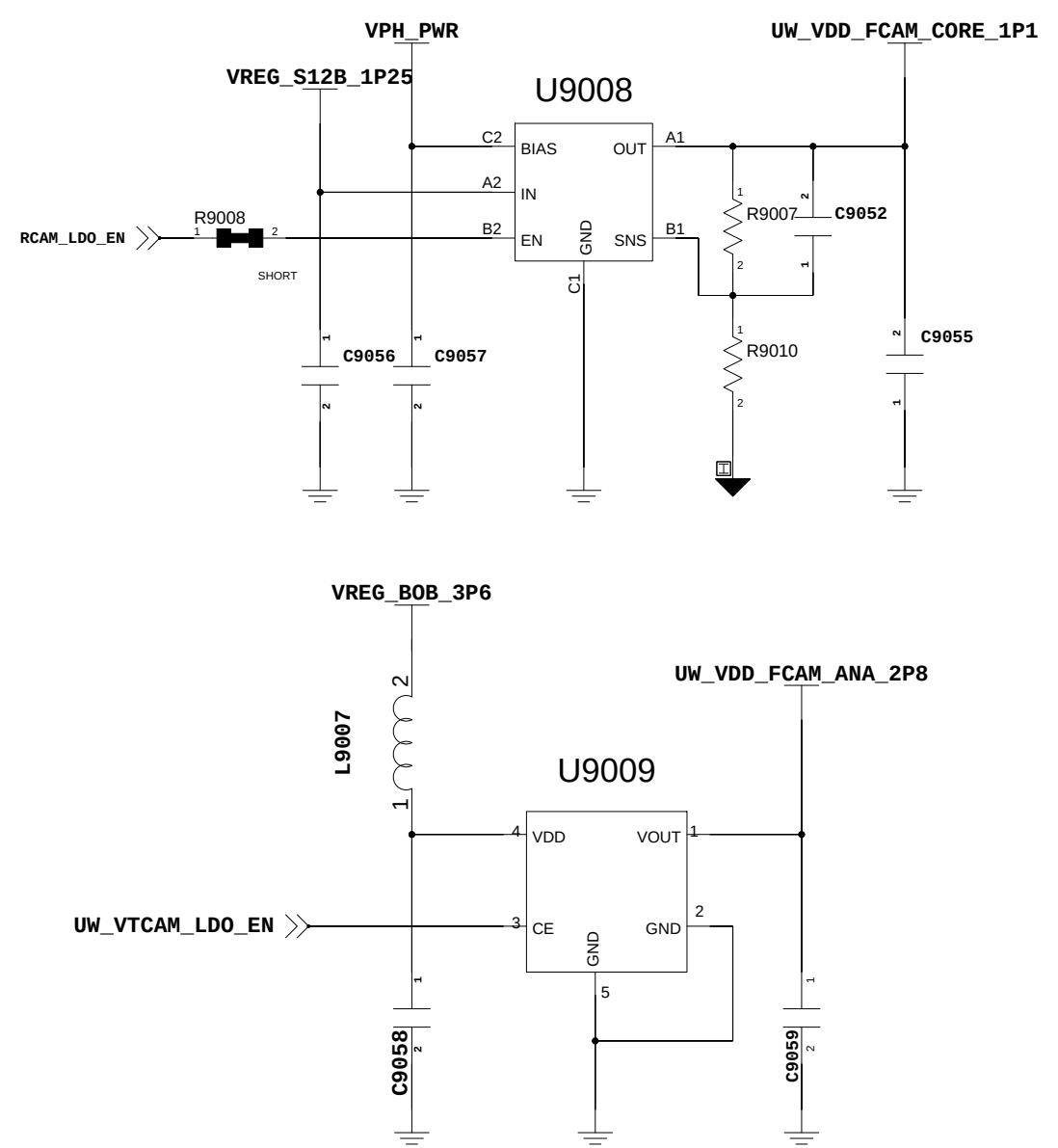

VREG_BOB_3P6
L9006
U9005
VDD
VOUT
CE
GND
VDD_RCAM2_ANA_2P8
RCAM2_LDO_EN
C9048
C9049
VPH_PWR
VREG_S12B_1P25
U9006
BIAS
IN
EN
OUT
SNS
VDD_RCAM2_CORE_1P1
R9022
SHORT
R9023
C9100
R9024
C9099
C9050
C9051

6M_UW CAM (RCAM2)_LDO

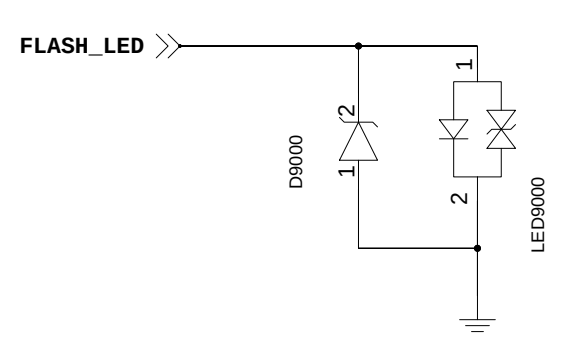

VREG_BOB_3P6
U9000
VDD_RCAM_AF_2P8
VDD
VOUT
CE
GND
RCAM_LDO_EN
C9035
C9036
ground
VREG_S1C_2P0
U9007
VDD_CAM_IO_1P8
VIN
VOUT
EN
GND
CAM_IO_LDO_EN
R9009
C9054
C9053


VPH_PWR
VREG_S12B_1P25
U9001
VDD_FCAM_CORE_1P1
BIAS
IN
EN
OUT
SNS
FCAM_LDO_EN
R9001
SHORT
R9000
C9034
R9002
C9040
C9041
C9039
VREG_BOB_3P6
L9004
U9002
VDD_FCAM_ANA_2P8
VDD
VOUT
CE
GND
FCAM_LDO_EN
C9037
C9038

VTCAM_LDO

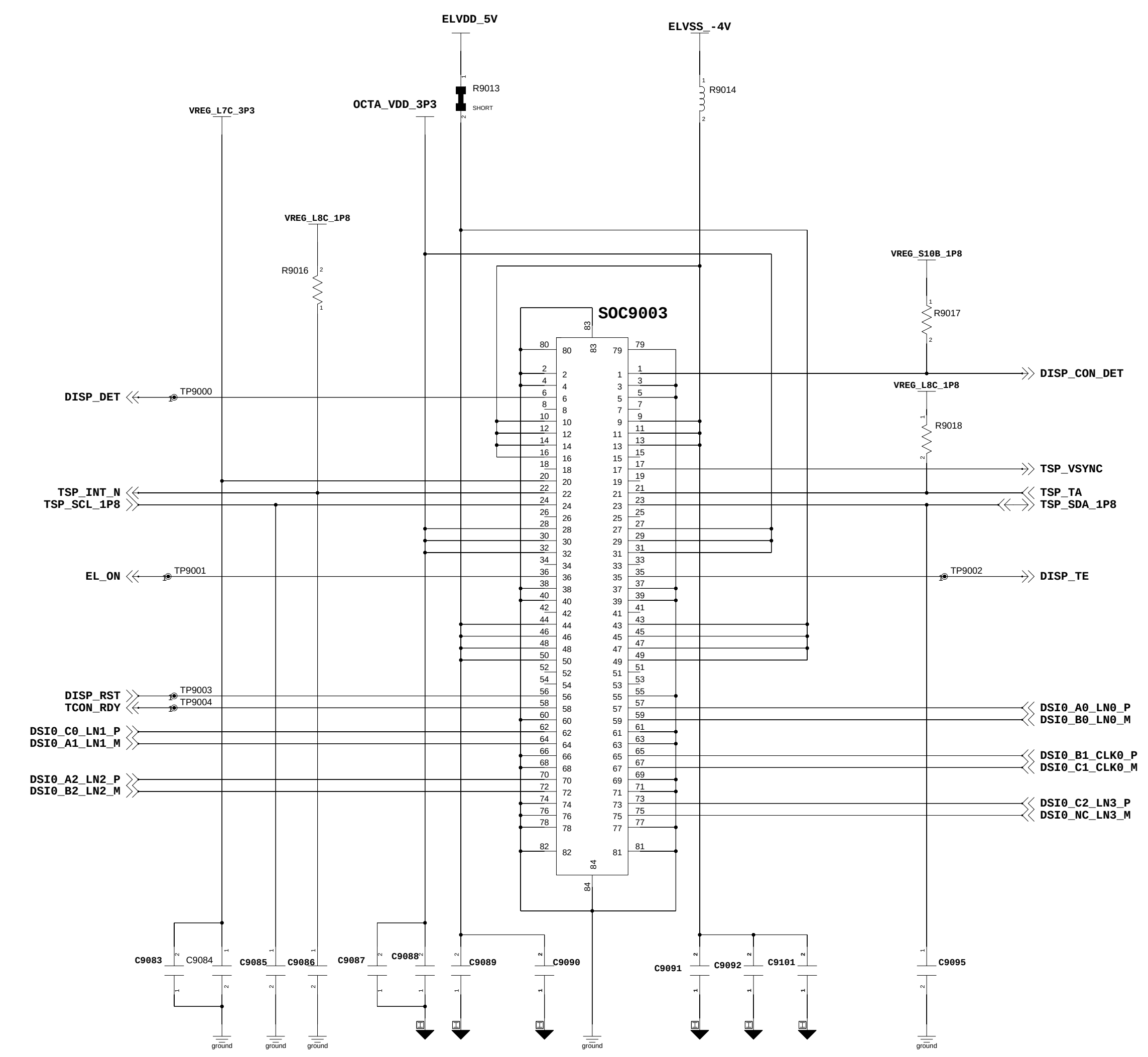

VPH_PWR
VREG_S12B_1P25
U9008
UW_VDD_FCAM_CORE_1P1
BIAS
IN
EN
OUT
SNS
RCAM_LDO_EN
R9008
SHORT
R9007
C9052
R9010
C9056
C9057
C9055
VREG_BOB_3P6
L9007
U9009
UW_VDD_FCAM_ANA_2P8
VDD
VOUT
CE
GND
UW_VTCAM_LDO_EN
C9058
C9059

UW VT CAM_LDO

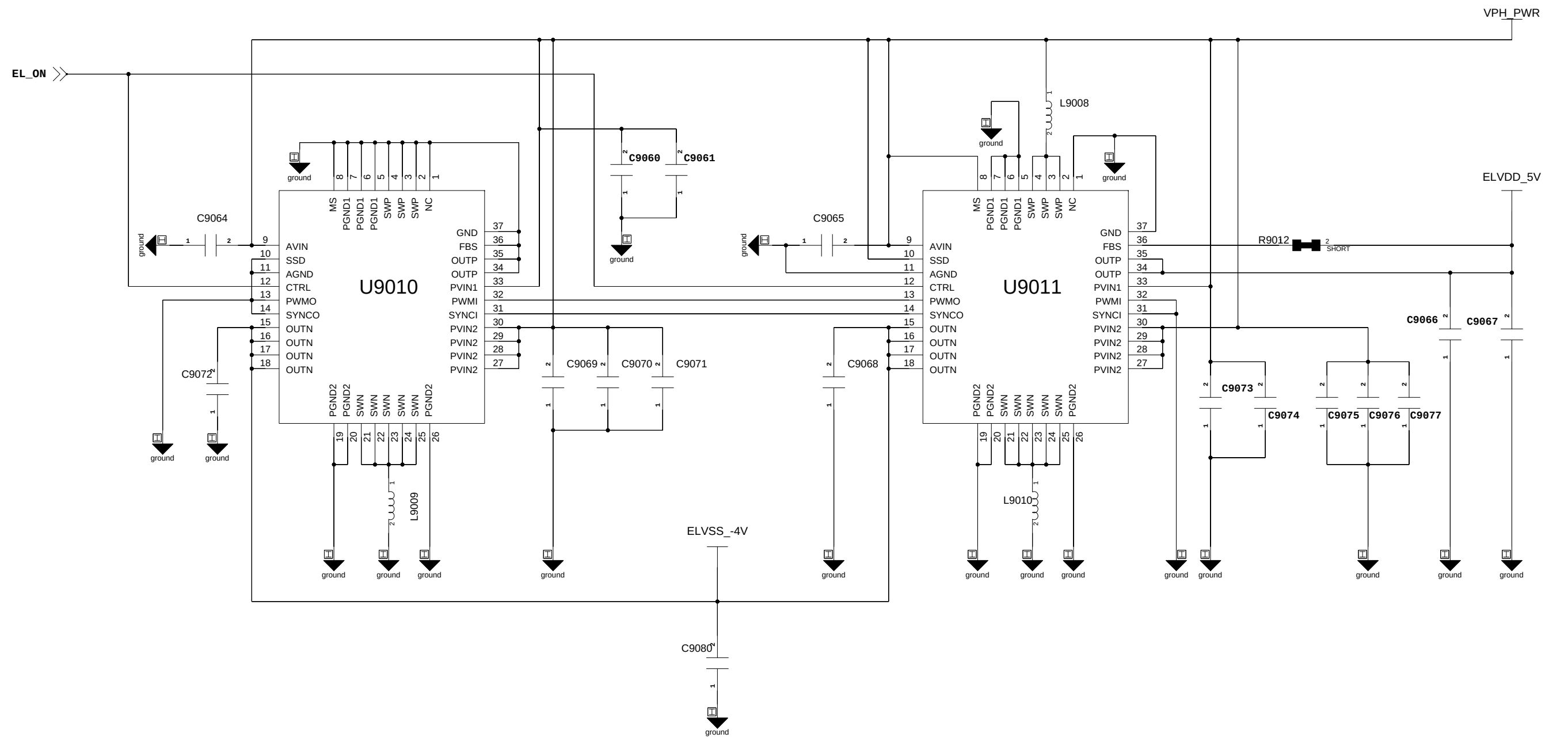

FLASH_LED
D9000
LED9000

FLASH LED

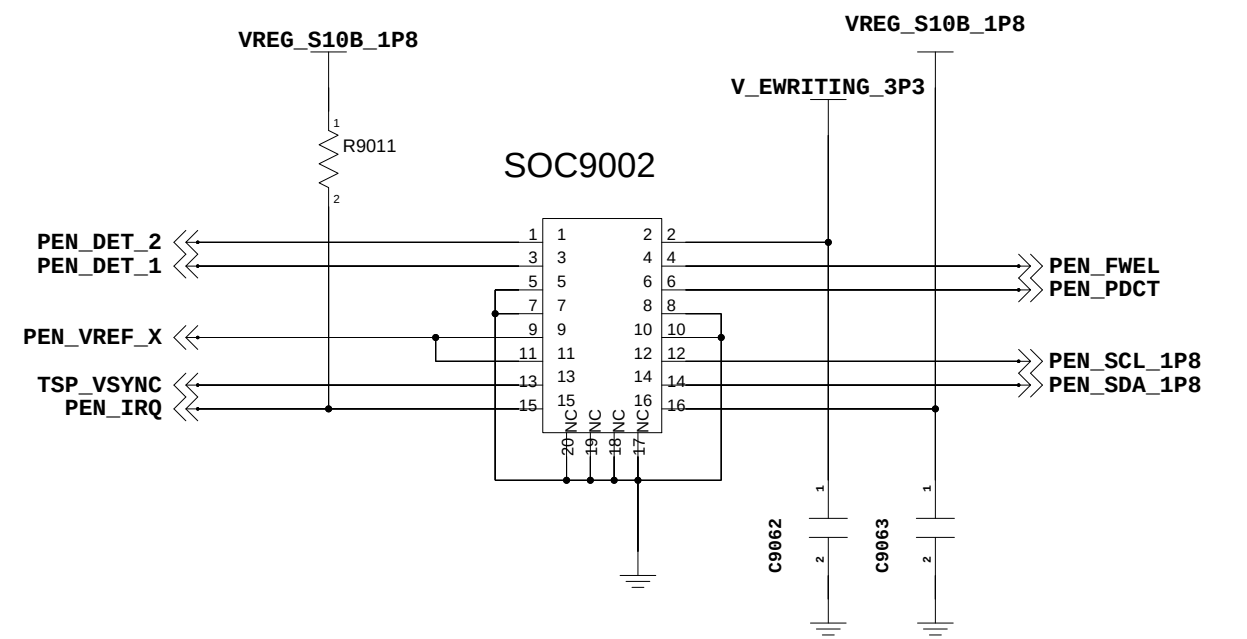
DISPLAY


ELVDD_5V
ELVSS_-4V
R9013
SHORT
R9014
VREG_L7C_3P3
OCTA_VDD_3P3
VREG_L8C_1P8
R9016
VREG_S10B_1P8
R9017
VREG_L8C_1P8
R9018
SOC9003
DISP_DET
TP9000
TSP_INT_N
TSP_SCL_1P8
EL_ON
TP9001
DISP_RST
TP9003
TCON_RDY
TP9004
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_A2_LN2_P
DSI0_B2_LN2_M
DISP_CON_DET
TSP_VSYNC
TSP_TA
TSP_SDA_1P8
DISP_TE
TP9002
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_B1_CLK0_P
DSI0_C1_CLK0_M
DSI0_C2_LN3_P
DSI0_NC_LN3_M
C9083
C9084
C9085
C9086
C9087
C9088
C9089
C9090
C9091
C9092
C9101
C9095
ground

OCTA Con.

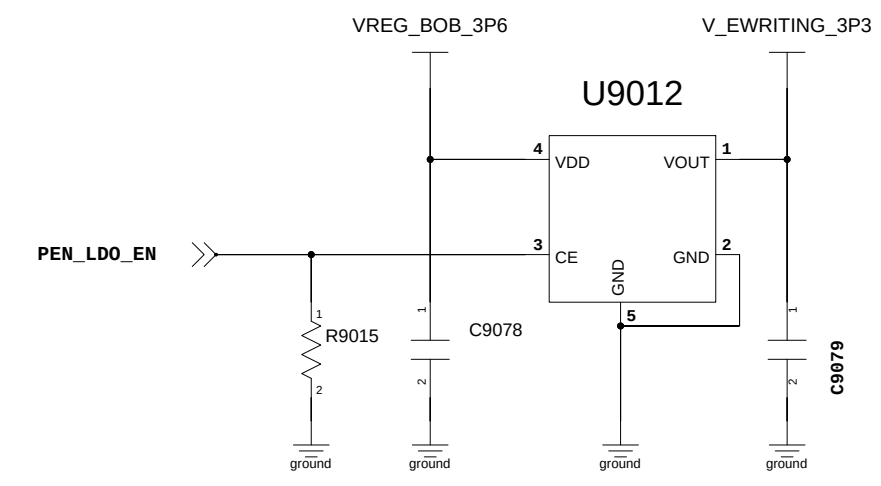

EL_ON
VPH_PWR
U9010
U9011
AVIN
SSD
AGND
CTRL
PWMO
SYNCO
OUTN
GND
FBS
OUTP
PVIN1
PWMI
SYNCI
PVIN2
MS
PGND1
SWP
NC
PGND2
SWN
C9064
C9065
C9060
C9061
C9069
C9070
C9071
C9068
C9072
L9008
L9009
L9010
R9012
SHORT
ELVDD_5V
C9066
C9067
C9073
C9074
C9075
C9076
C9077
ELVSS_-4V
C9080
ground

OCTA POWER

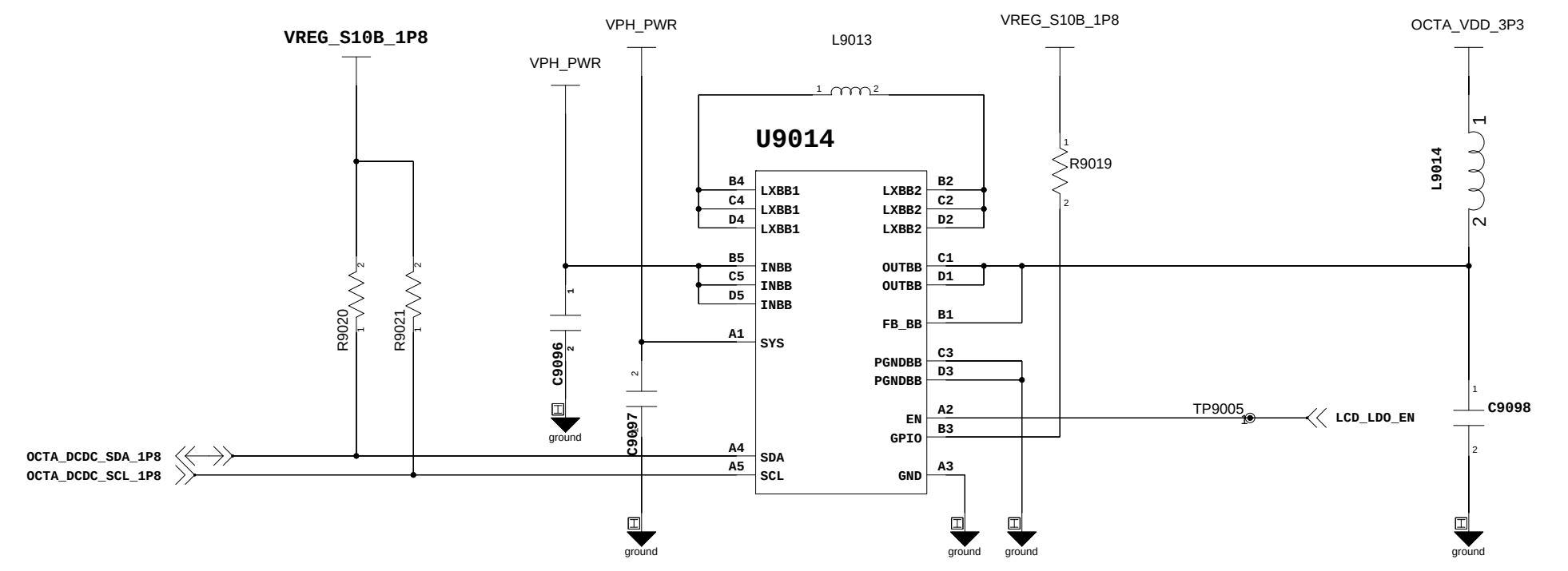

VREG_S10B_1P8
R9011
V_EWRITING_3P3
VREG_S10B_1P8
SOC9002
PEN_DET_2
PEN_DET_1
PEN_VREF_X
TSP_VSYNC
PEN_IRQ
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
C9062
C9063

WACOM CONN

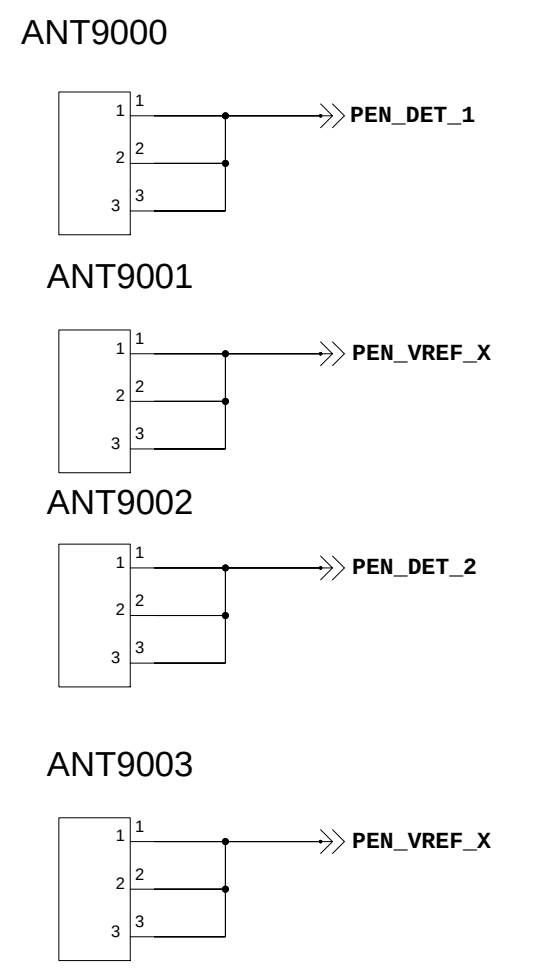
(MAIN <-> OCTA / WACOM)


VREG_BOB_3P6
V_EWRITING_3P3
U9012
VDD
VOUT
CE
GND
PEN_LDO_EN
R9015
C9078
C9079
ground

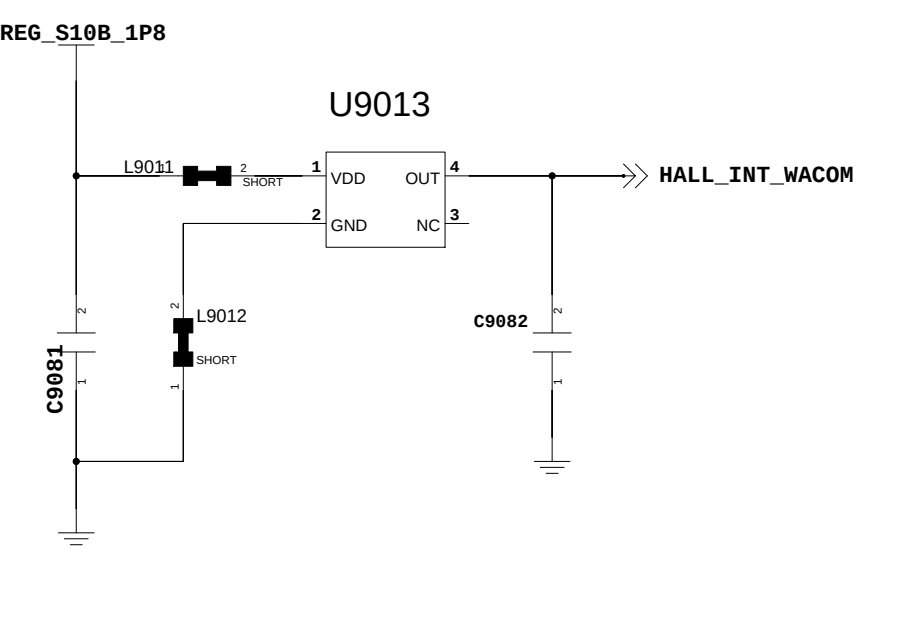
PEN LDO


VREG_S10B_1P8
VPH_PWR
VPH_PWR
L9013
VREG_S10B_1P8
OCTA_VDD_3P3
U9014
LXBB1
INBB
SYS
SDA
SCL
LXBB2
OUTBB
FB_BB
PGNDBB
EN
GPIO
GND
R9020
R9021
R9019
C9096
C9097
L9014
C9098
TP9005
LCD_LDO_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
ground

TCON BUCK BOOSTER


ANT9000
PEN_DET_1
ANT9001
PEN_VREF_X
ANT9002
PEN_DET_2
ANT9003
PEN_VREF_X

PEN CHG C/T

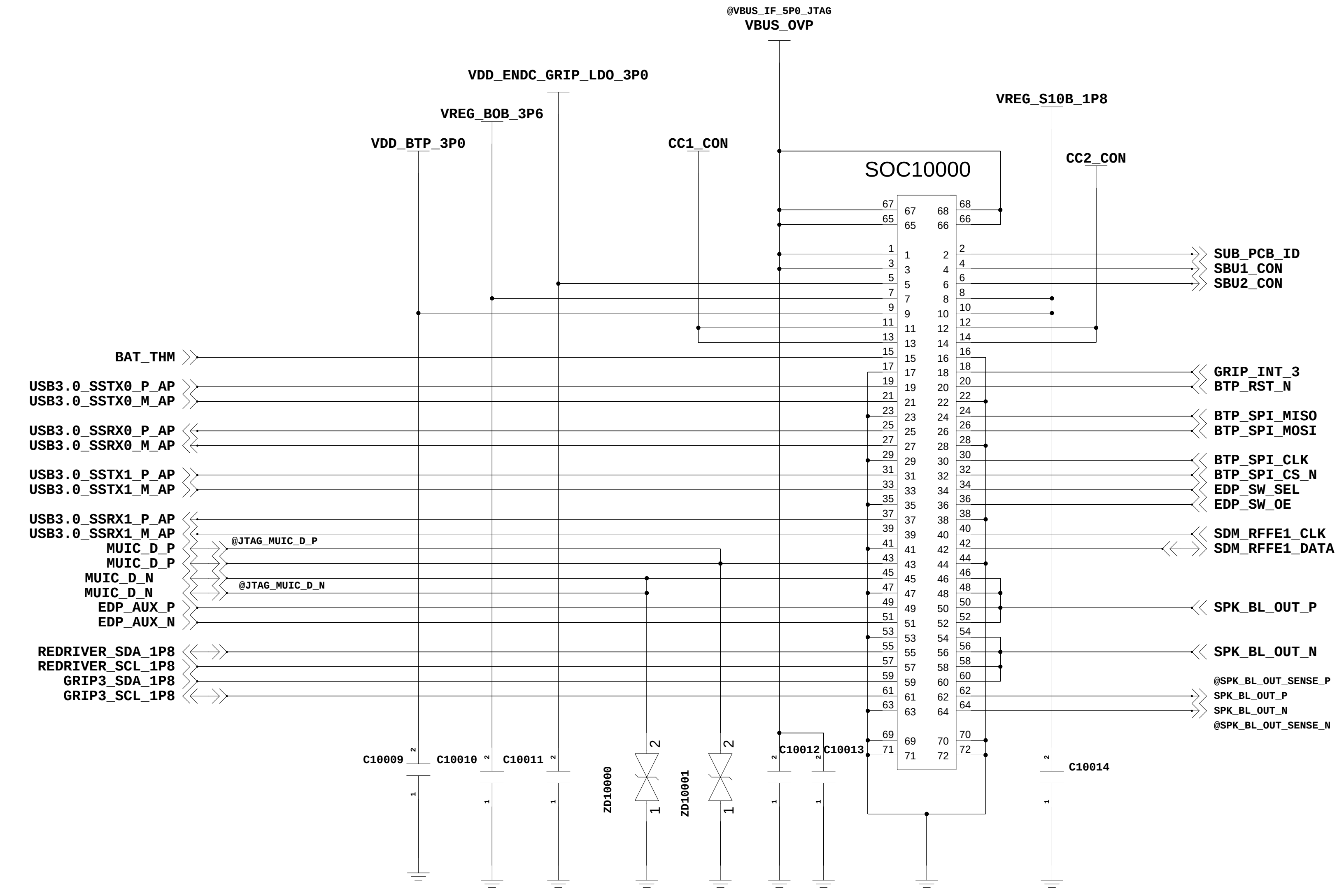

VREG_S10B_1P8
U9013
L9011
SHORT
VDD
GND
OUT
NC
HALL_INT_WACOM
L9012
C9081
C9082

COVER_DET HALL_IC

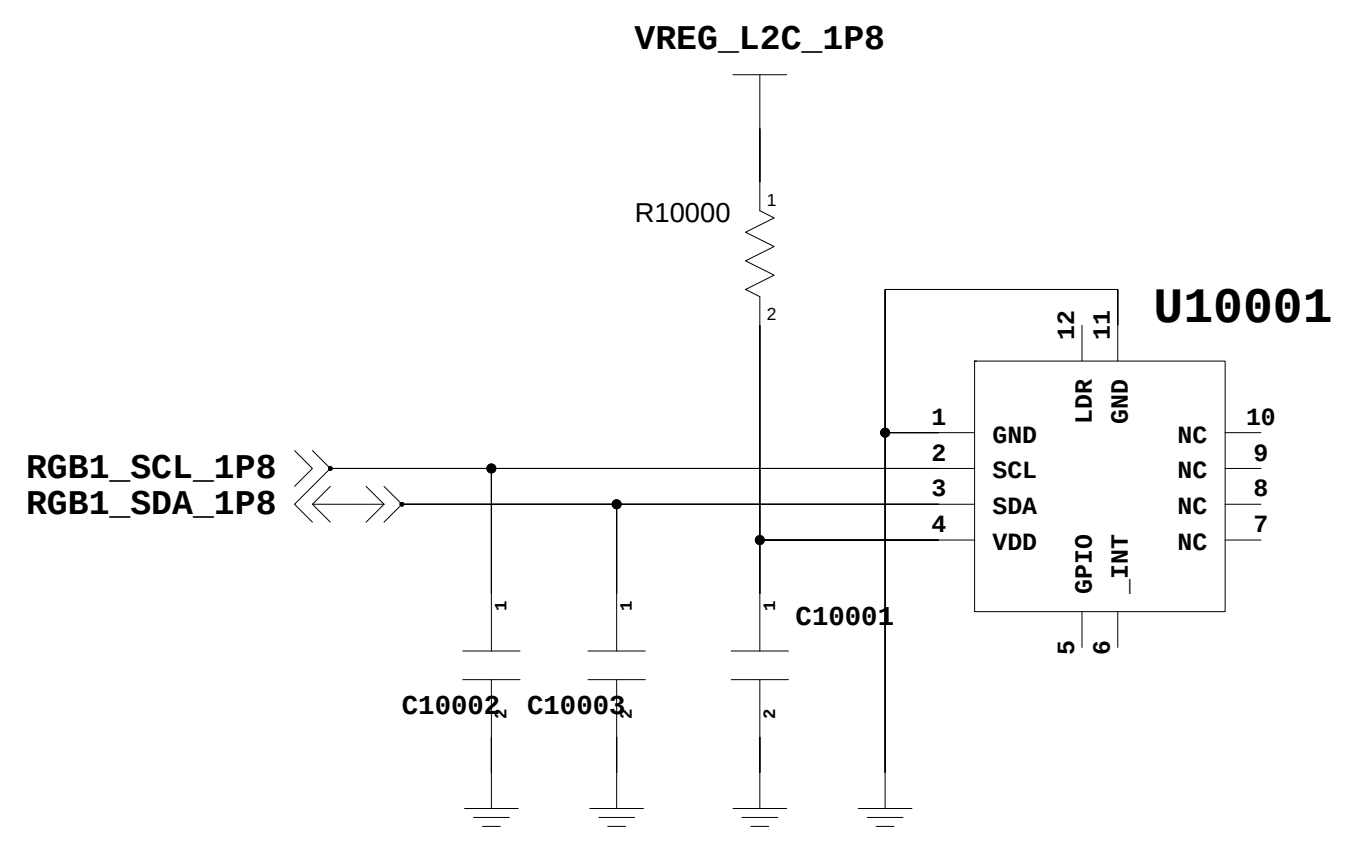
TITLE TabS7P
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name SHEET 9 of 14 Date Drawing Date

SENSOR / SD/SIM / KEY / I/F

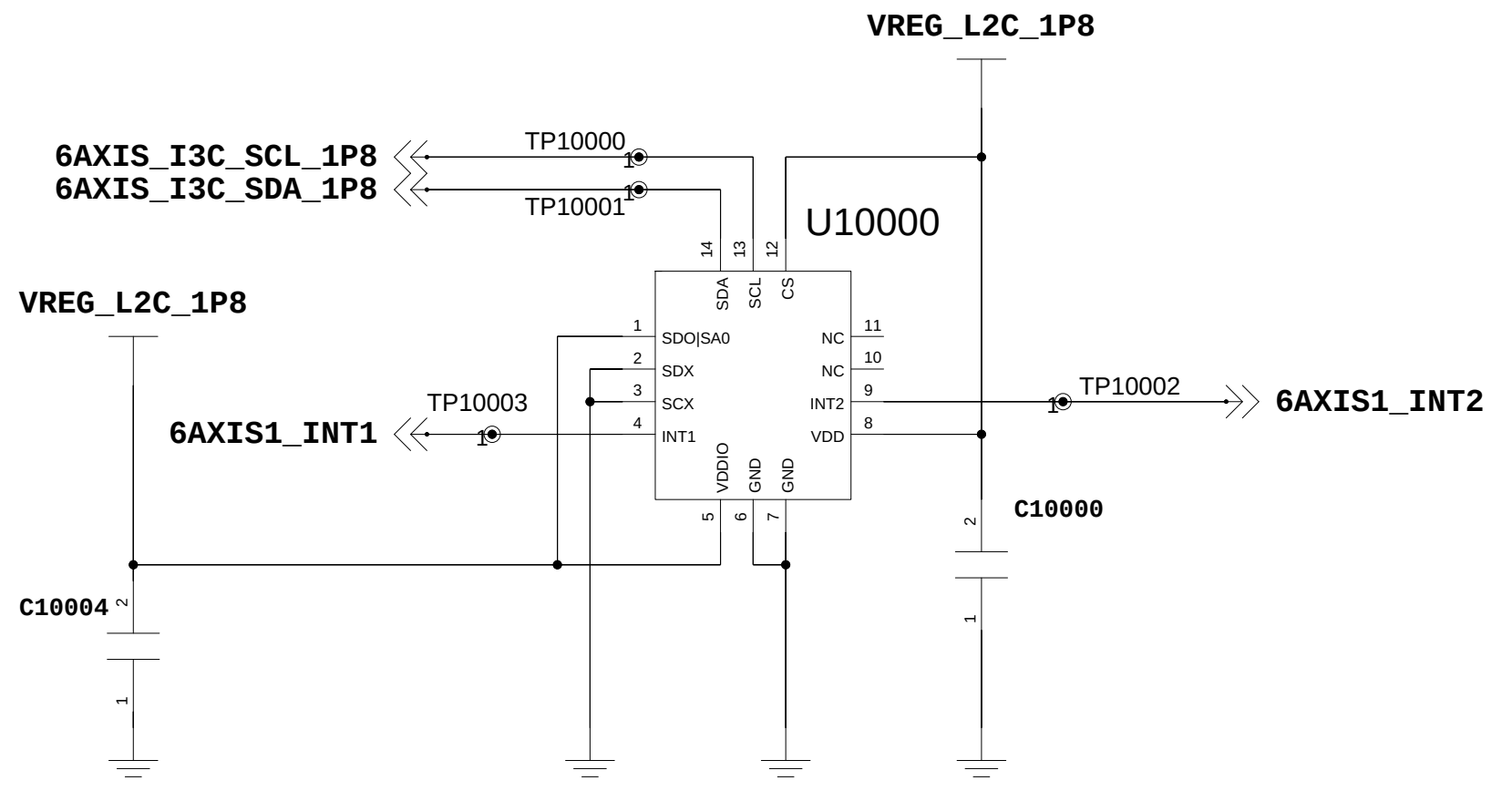
I/F CONN (MAIN <-> USB)


@VBUS_IF_5P0_JTAG
VBUS_OVP
VDD_ENDC_GRIP_LDO_3P0
VREG_BOB_3P6
VDD_BTP_3P0
CC1_CON
VREG_S10B_1P8
CC2_CON
SOC10000
SUB_PCB_ID
SBU1_CON
SBU2_CON
BAT_THM
USB3.0_SSTX0_P_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX0_M_AP
USB3.0_SSTX1_P_AP
USB3.0_SSTX1_M_AP
USB3.0_SSRX1_P_AP
USB3.0_SSRX1_M_AP
MUIC_D_P
MUIC_D_P
MUIC_D_N
MUIC_D_N
@JTAG_MUIC_D_P
@JTAG_MUIC_D_N
EDP_AUX_P
EDP_AUX_N
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GRIP_INT_3
BTP_RST_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_SPI_CLK
BTP_SPI_CS_N
EDP_SW_SEL
EDP_SW_OE
SDM_RFFE1_CLK
SDM_RFFE1_DATA
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
C10009
C10010
C10011
ZD10000
ZD10001
C10012
C10013
C10014

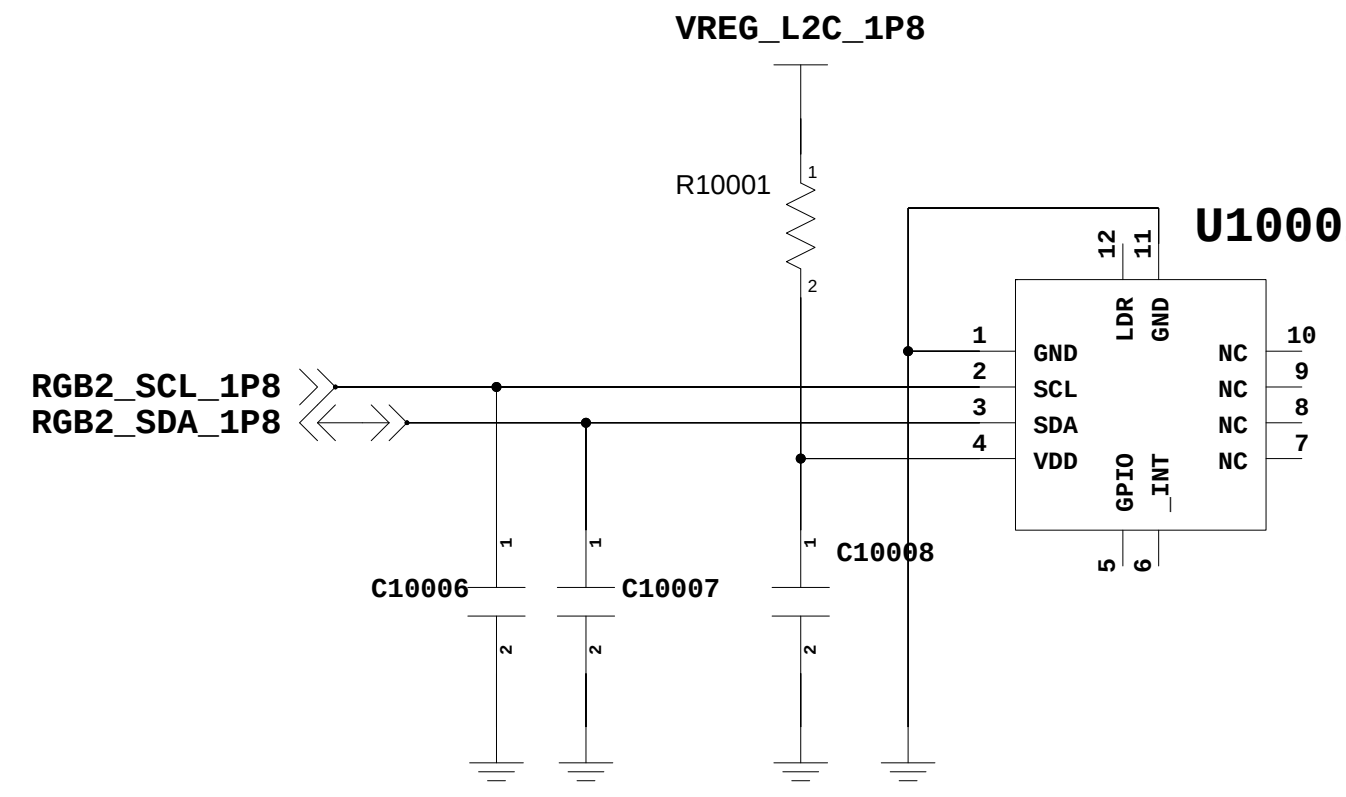
RGB Sensor 1


VREG_L2C_1P8
R10000
U10001
RGB1_SCL_1P8
RGB1_SDA_1P8
GND
SCL
SDA
VDD
LDR
GND
GPIO
_INT
NC
NC
NC
NC
C10001
C10002
C10003

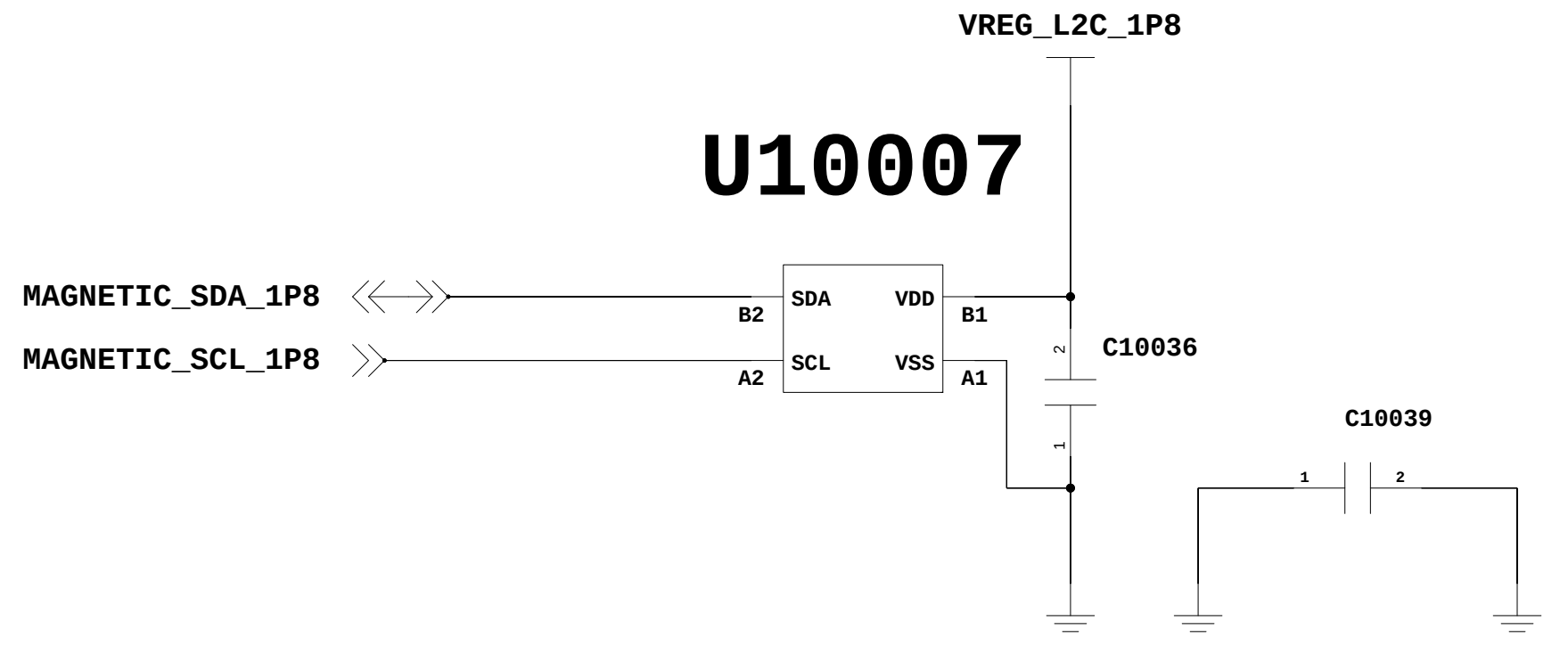
6 Axis (GYRO,ACCEL)


VREG_L2C_1P8
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
TP10000
TP10001
U10000
VREG_L2C_1P8
6AXIS1_INT1
TP10003
TP10002
6AXIS1_INT2
SDO|SA0
SDX
SCX
INT1
SDA
SCL
CS
NC
NC
INT2
VDD
VDDIO
GND
GND
C10000
C10004

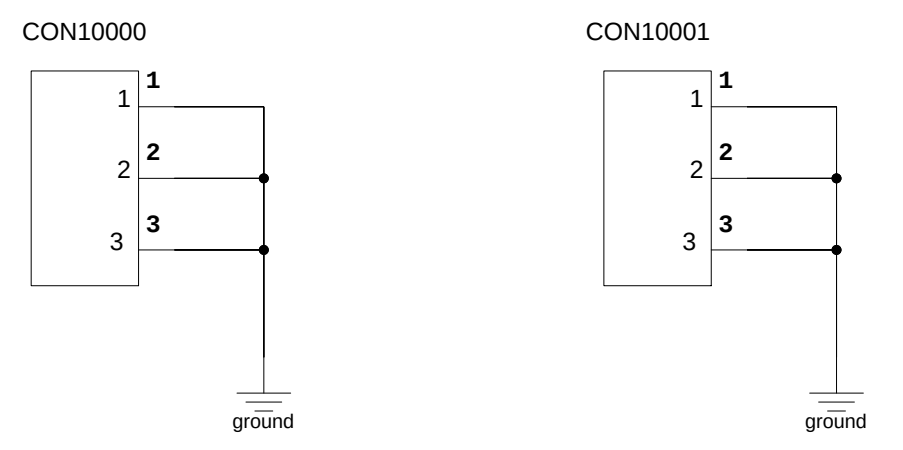
RGB Sensor 2


VREG_L2C_1P8
R10001
U10002
RGB2_SCL_1P8
RGB2_SDA_1P8
GND
SCL
SDA
VDD
LDR
GND
GPIO
_INT
NC
NC
NC
NC
C10006
C10007
C10008

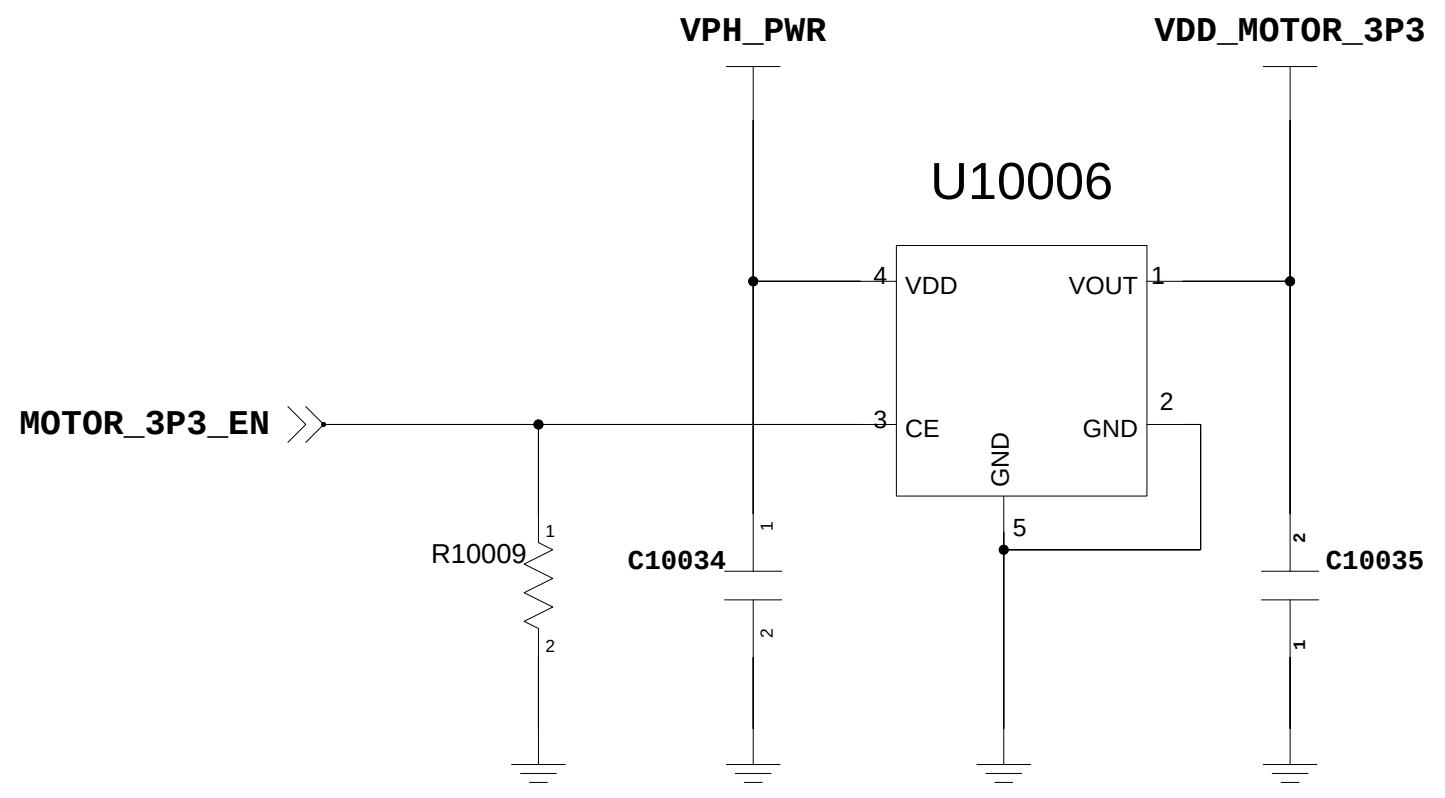
Magnetic SENSOR


VREG_L2C_1P8
U10007
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
SDA
SCL
VDD
VSS
B2
A2
B1
A1
C10036
C10039

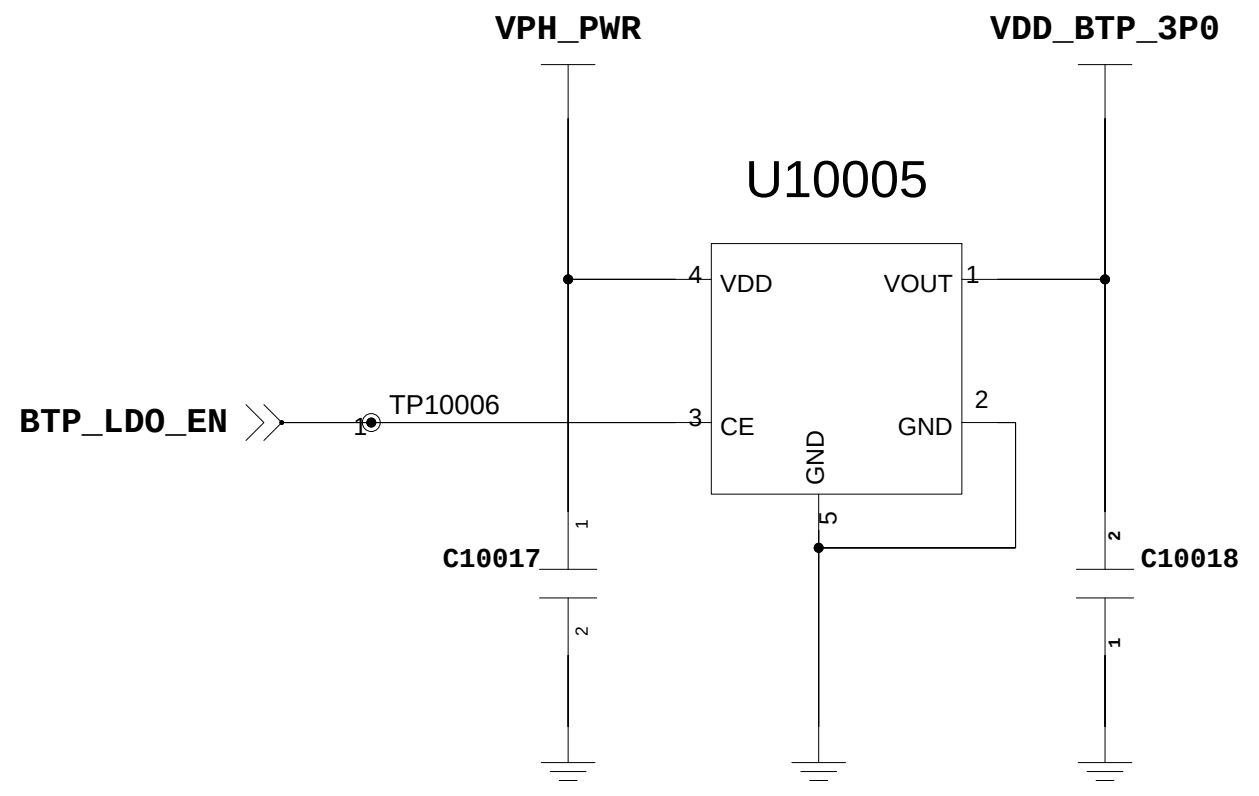
SD/SIM FPCB CONTACT


CON10000
CON10001
ground
ground

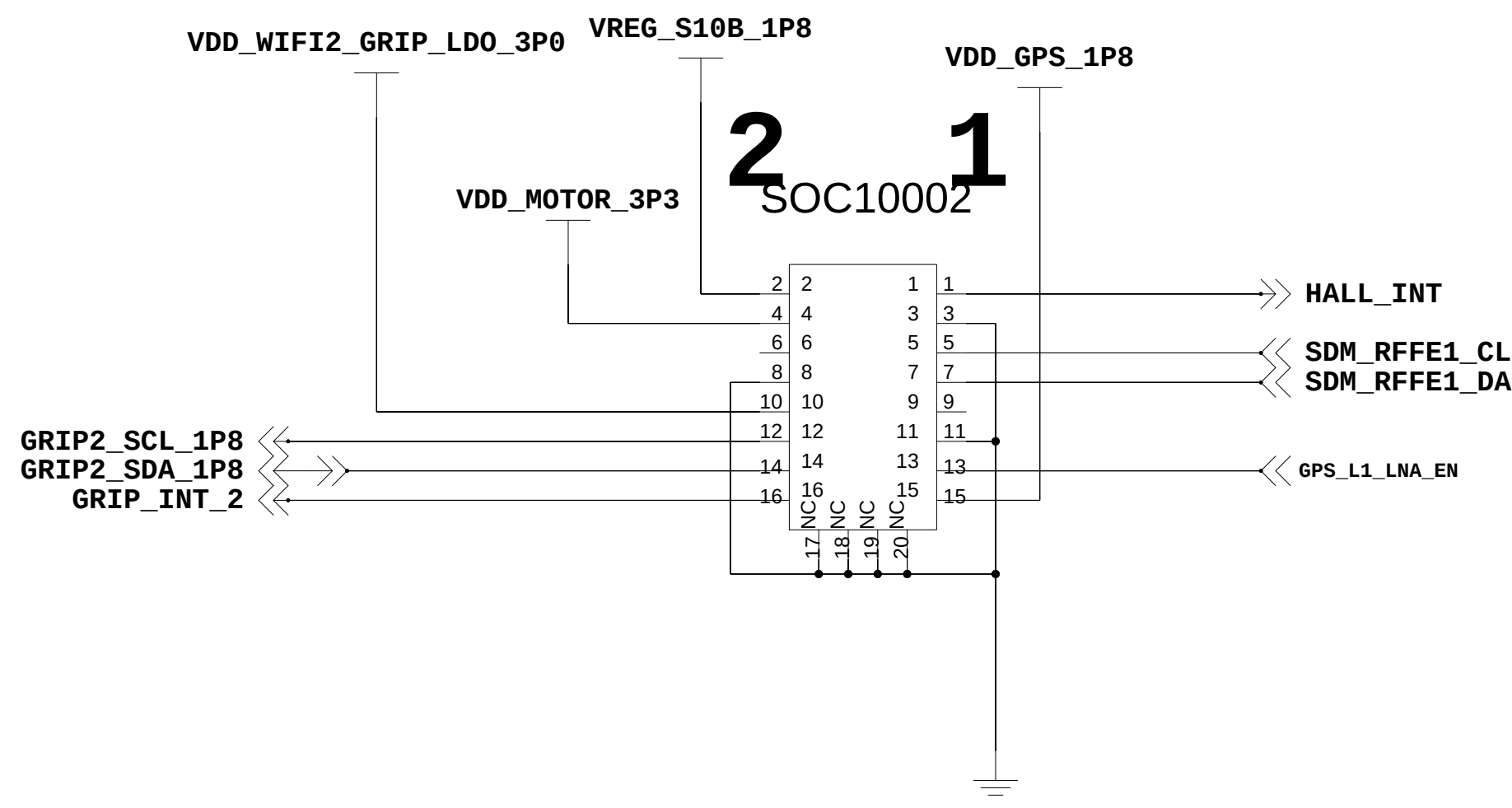
MOTOR LDO


VPH_PWR
VDD_MOTOR_3P3
U10006
MOTOR_3P3_EN
VDD
VOUT
CE
GND
GND
R10009
C10034
C10035

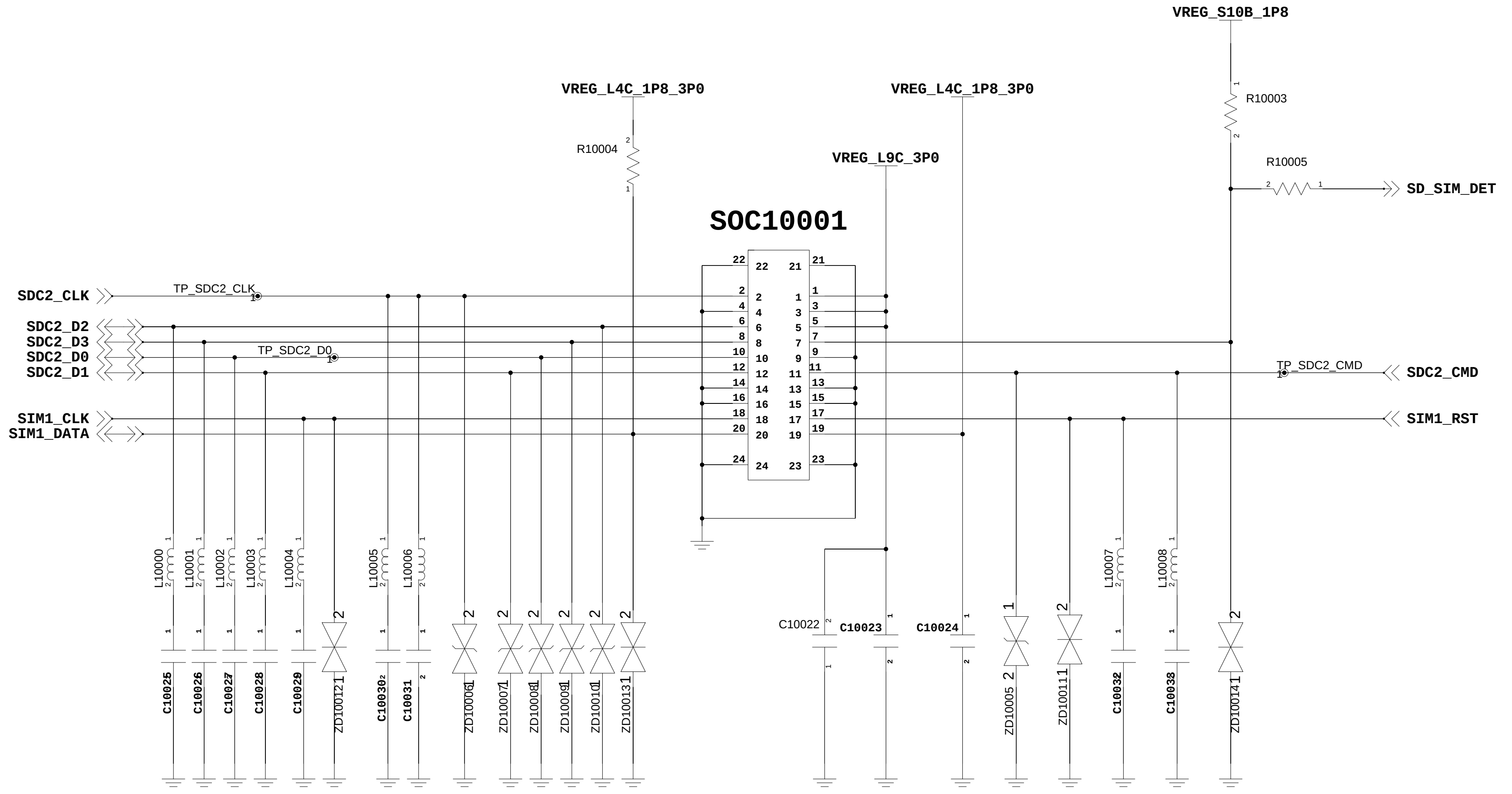
FOD LDO


VPH_PWR
VDD_BTP_3P0
U10005
BTP_LDO_EN
TP10006
VDD
VOUT
CE
GND
GND
C10017
C10018

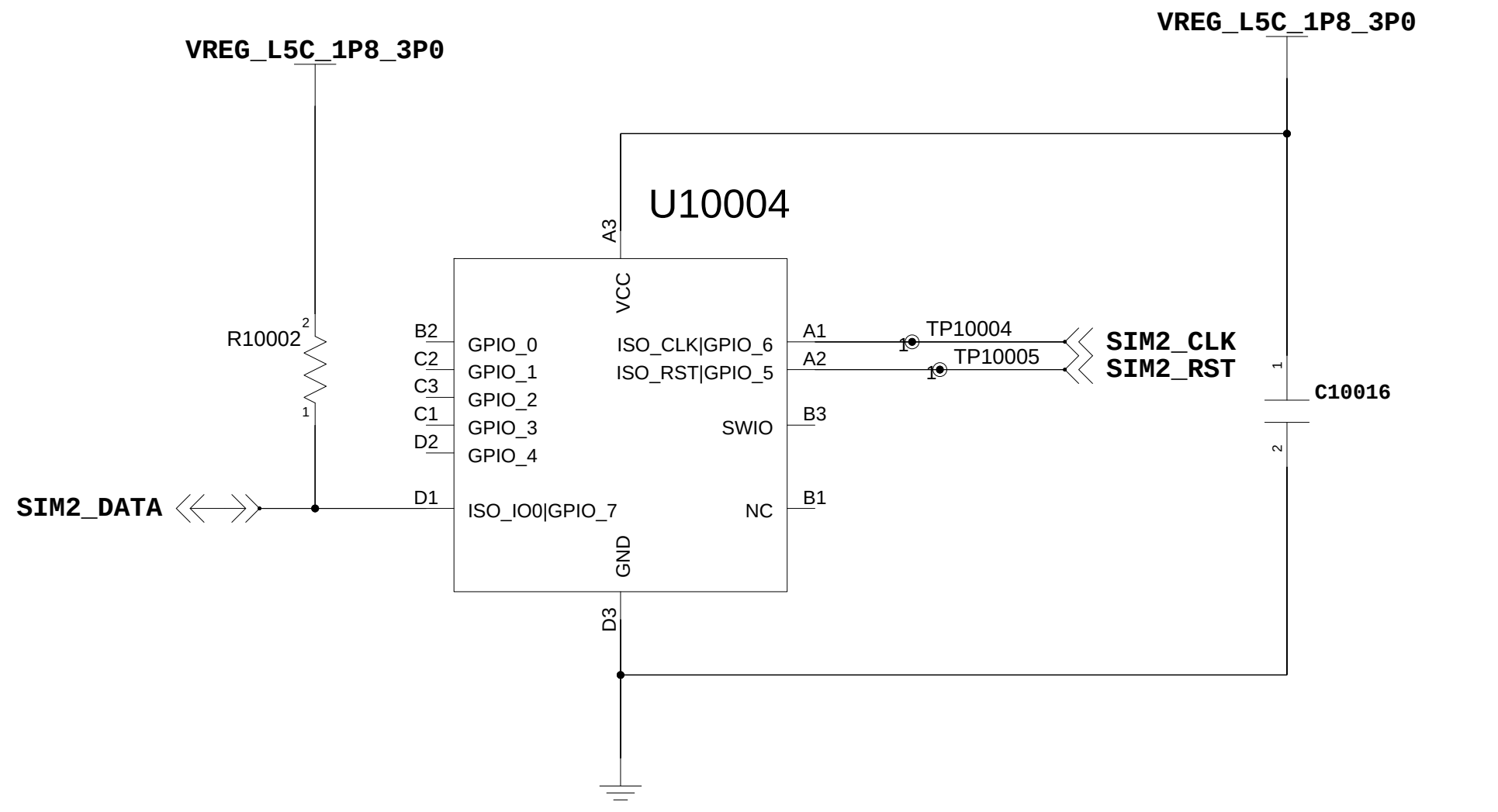
SUB CONN (MAIN <-> ANT SUB)


VDD_WIFI2_GRIP_LDO_3P0
VREG_S10B_1P8
VDD_GPS_1P8
2
1
VDD_MOTOR_3P3
SOC10002
HALL_INT
SDM_RFFE1_CLK
SDM_RFFE1_DATA
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
GPS_L1_LNA_EN

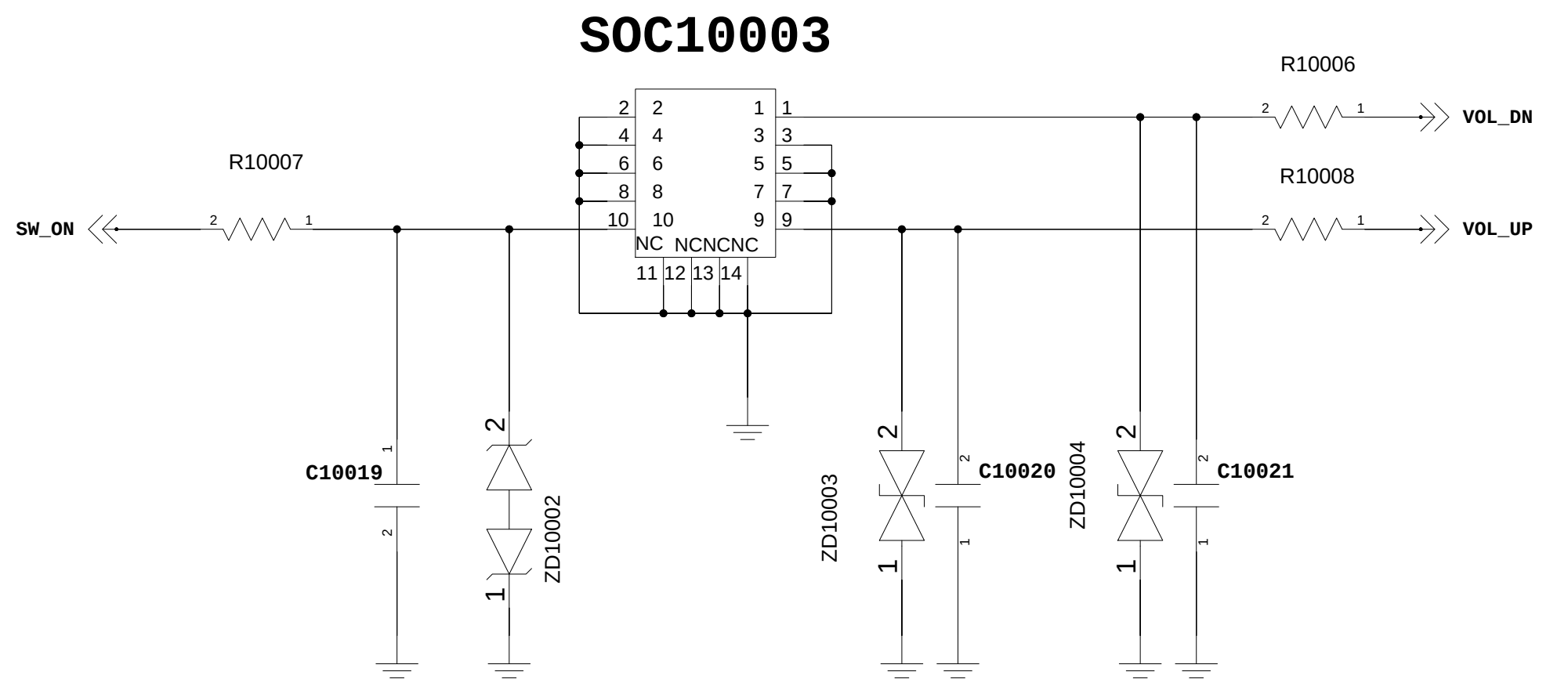
SIM/SD SOCKET Conn.


VREG_S10B_1P8
VREG_L4C_1P8_3P0
VREG_L4C_1P8_3P0
VREG_L9C_3P0
R10004
R10003
R10005
SD_SIM_DET
SOC10001
SDC2_CLK
TP_SDC2_CLK
SDC2_D2
SDC2_D3
SDC2_D0
TP_SDC2_D0
SDC2_D1
SIM1_CLK
SIM1_DATA
TP_SDC2_CMD
SDC2_CMD
SIM1_RST
L10000
L10001
L10002
L10003
L10004
L10005
L10006
L10007
L10008
C10025
C10026
C10027
C10028
C10029
C10030
C10031
C10022
C10023
C10024
C10032
C10033
ZD10012
ZD10006
ZD10007
ZD10008
ZD10009
ZD10010
ZD10013
ZD10005
ZD10011
ZD10014

eSIM - NC

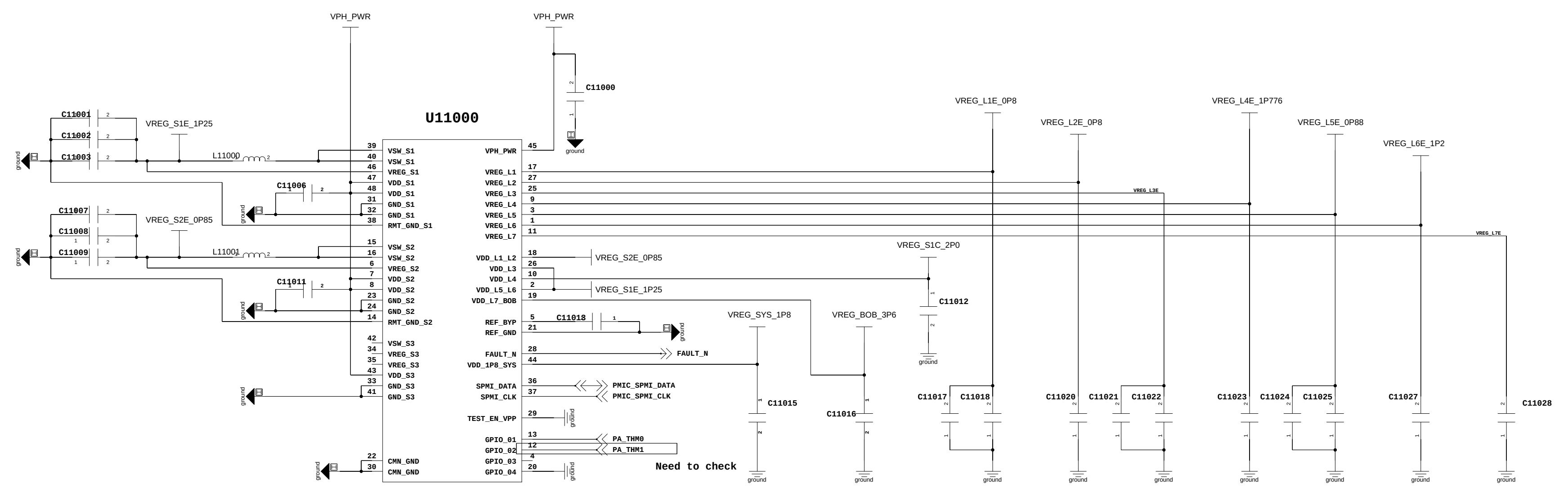

VREG_L5C_1P8_3P0
VREG_L5C_1P8_3P0
U10004
R10002
GPIO_0
GPIO_1
GPIO_2
GPIO_3
GPIO_4
ISO_IO0|GPIO_7
ISO_CLK|GPIO_6
ISO_RST|GPIO_5
SWIO
NC
VCC
GND
TP10004
TP10005
SIM2_CLK
SIM2_RST
SIM2_DATA
C10016

Key CONN


SOC10003
R10006
R10007
R10008
VOL_DN
VOL_UP
SW_ON
C10019
ZD10002
ZD10003
C10020
ZD10004
C10021

TITLE	TabS7P		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 10 of 15	Date Drawing Date	

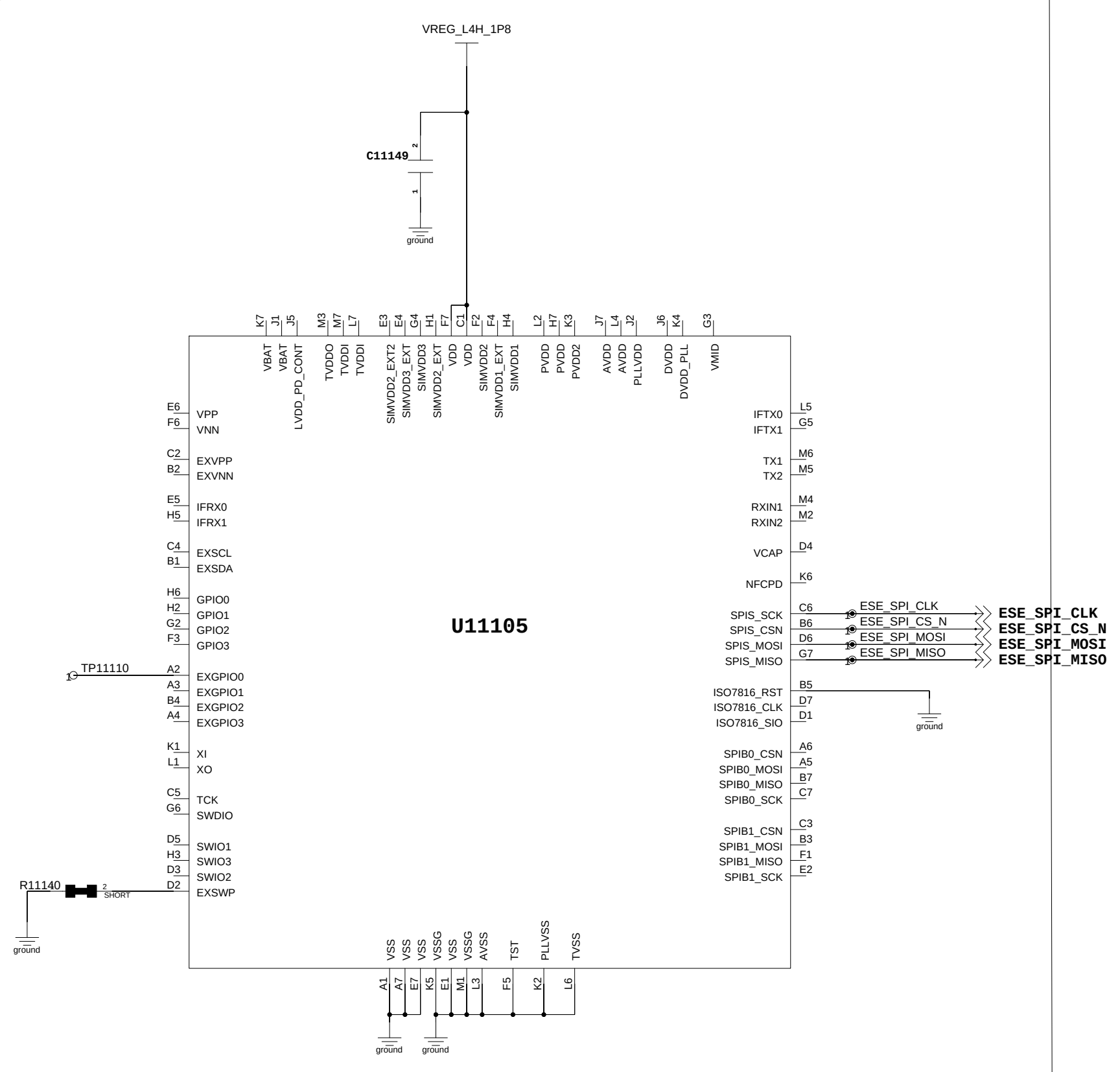
RF PMIC



U11000
VPH_PWR
VPH_PWR
C11000
ground
C11001
C11002
C11003
VREG_S1E_1P25
L11000
C11006
C11007
C11008
C11009
VREG_S2E_0P85
L11001
C11011
39 VSW_S1
40 VSW_S1
46 VREG_S1
47 VDD_S1
48 VDD_S1
31 GND_S1
32 GND_S1
38 RMT_GND_S1
15 VSW_S2
16 VSW_S2
6 VREG_S2
7 VDD_S2
8 VDD_S2
23 GND_S2
24 GND_S2
14 RMT_GND_S2
42 VSW_S3
34 VREG_S3
35 VREG_S3
43 VDD_S3
33 GND_S3
41 GND_S3
22 CMN_GND
30 CMN_GND
VPH_PWR 45
VREG_L1 17
VREG_L2 27
VREG_L3 25
VREG_L4 9
VREG_L5 3
VREG_L6 1
VREG_L7 11
VDD_L1_L2 18
VDD_L3 26
VDD_L4 10
VDD_L5_L6 2
VDD_L7_BOB 19
REF_BYP 5
REF_GND 21
FAULT_N 28
VDD_1P8_SYS 44
SPMI_DATA 36
SPMI_CLK 37
TEST_EN_VPP 29
GPIO_01 13
GPIO_02 12
GPIO_03 4
GPIO_04 20
VREG_S2E_0P85
VREG_S1E_1P25
C11018
FAULT_N
PMIC_SPMI_DATA
PMIC_SPMI_CLK
PA_THM0
PA_THM1
Need to check
VREG_SYS_1P8
VREG_BOB_3P6
VREG_S1C_2P0
C11012
C11015
C11016
VREG_L1E_0P8
VREG_L2E_0P8
VREG_L3E
VREG_L4E_1P776
VREG_L5E_0P88
VREG_L6E_1P2
VREG_L7E
C11017
C11018
C11020
C11021
C11022
C11023
C11024
C11025
C11027
C11028
ground

PMR735A

ESE



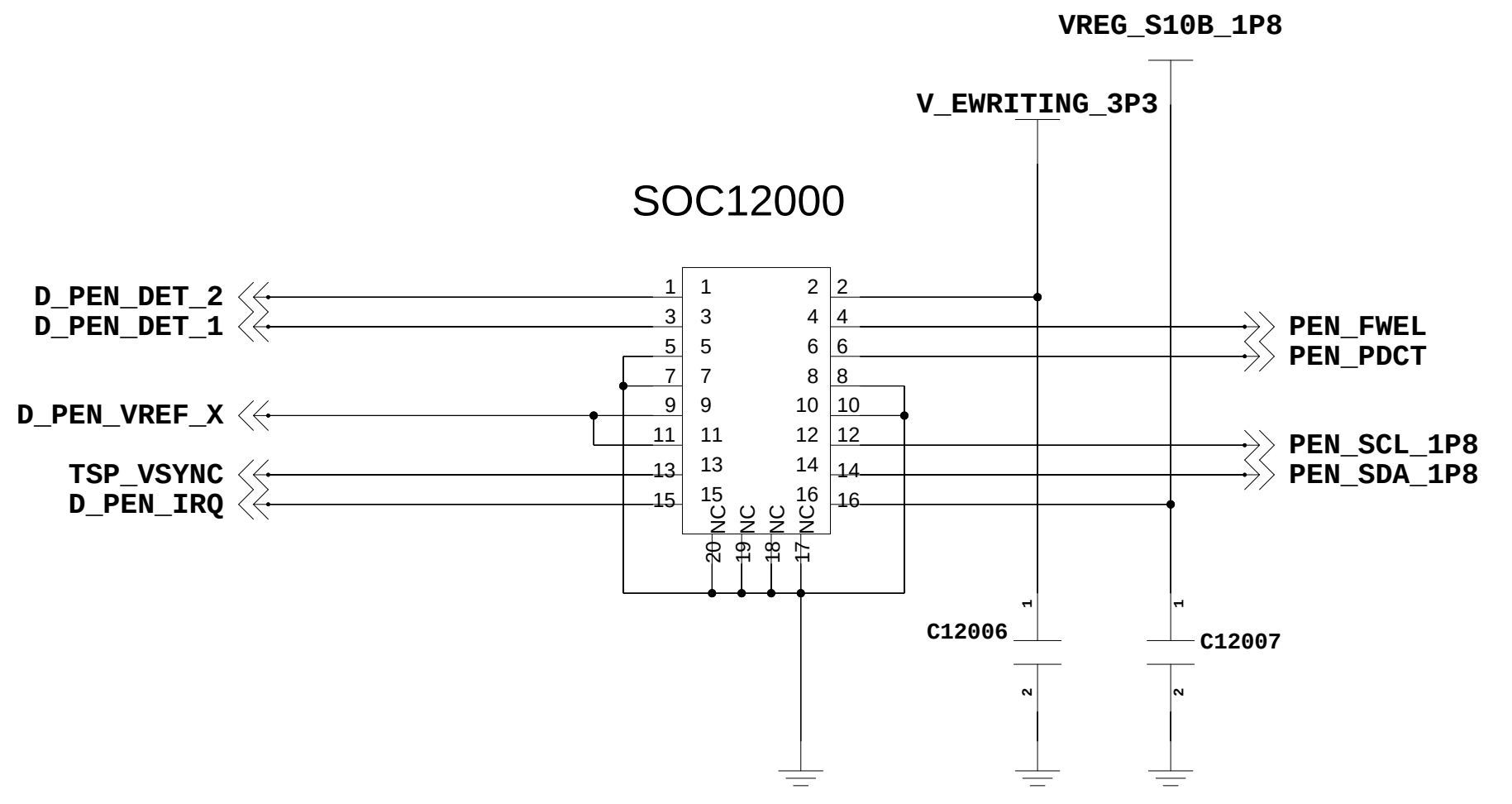
VREG_L4H_1P8
C11149
ground
U11105
K7 VBAT
J1 VBAT
J5 LVDD_PD_CONT
M3 TVDDO
M7 TVDDI
L7 TVDDI
E3 SIMVDD2_EXT2
E4 SIMVDD3_EXT
G4 SIMVDD3
H1 SIMVDD2_EXT
F7 VDD
C1 VDD
F2 SIMVDD2
F4 SIMVDD1_EXT
H4 SIMVDD1
L2 PVDD
H7 PVDD
K3 PVDD2
J7 AVDD
L4 AVDD
J2 PLLVDD
J6 DVDD
K4 DVDD_PLL
G3 VMID
E6 VPP
F6 VNN
C2 EXVPP
B2 EXVNN
E5 IFRX0
H5 IFRX1
C4 EXSCL
B1 EXSDA
H6 GPIO0
H2 GPIO1
G2 GPIO2
F3 GPIO3
A2 EXGPIO0
A3 EXGPIO1
B4 EXGPIO2
A4 EXGPIO3
K1 XI
L1 XO
C5 TCK
G6 SWDIO
D5 SWIO1
H3 SWIO3
D3 SWIO2
D2 EXSWP
TP11110
R11140
SHORT
ground
IFTX0 L5
IFTX1 G5
TX1 M6
TX2 M5
RXIN1 M4
RXIN2 M2
VCAP D4
NFCPD K6
SPIS_SCK C6
SPIS_CSN B6
SPIS_MOSI D6
SPIS_MISO G7
ISO7816_RST B5
ISO7816_CLK D7
ISO7816_SIO D1
SPIB0_CSN A6
SPIB0_MOSI A5
SPIB0_MISO B7
SPIB0_SCK C7
SPIB1_CSN C3
SPIB1_MOSI B3
SPIB1_MISO F1
SPIB1_SCK E2
ESE_SPI_CLK
ESE_SPI_CS_N
ESE_SPI_MOSI
ESE_SPI_MISO
VSS A1
VSS A7
VSS E7
VSSG K5
VSS E1
VSSG M1
AVSS L3
TST F5
PLLVSS K2
TVSS L6
ground

eSE (SUB6_B2B)

TITLE	TabS7P		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 11 of 12	Date Drawing Date	

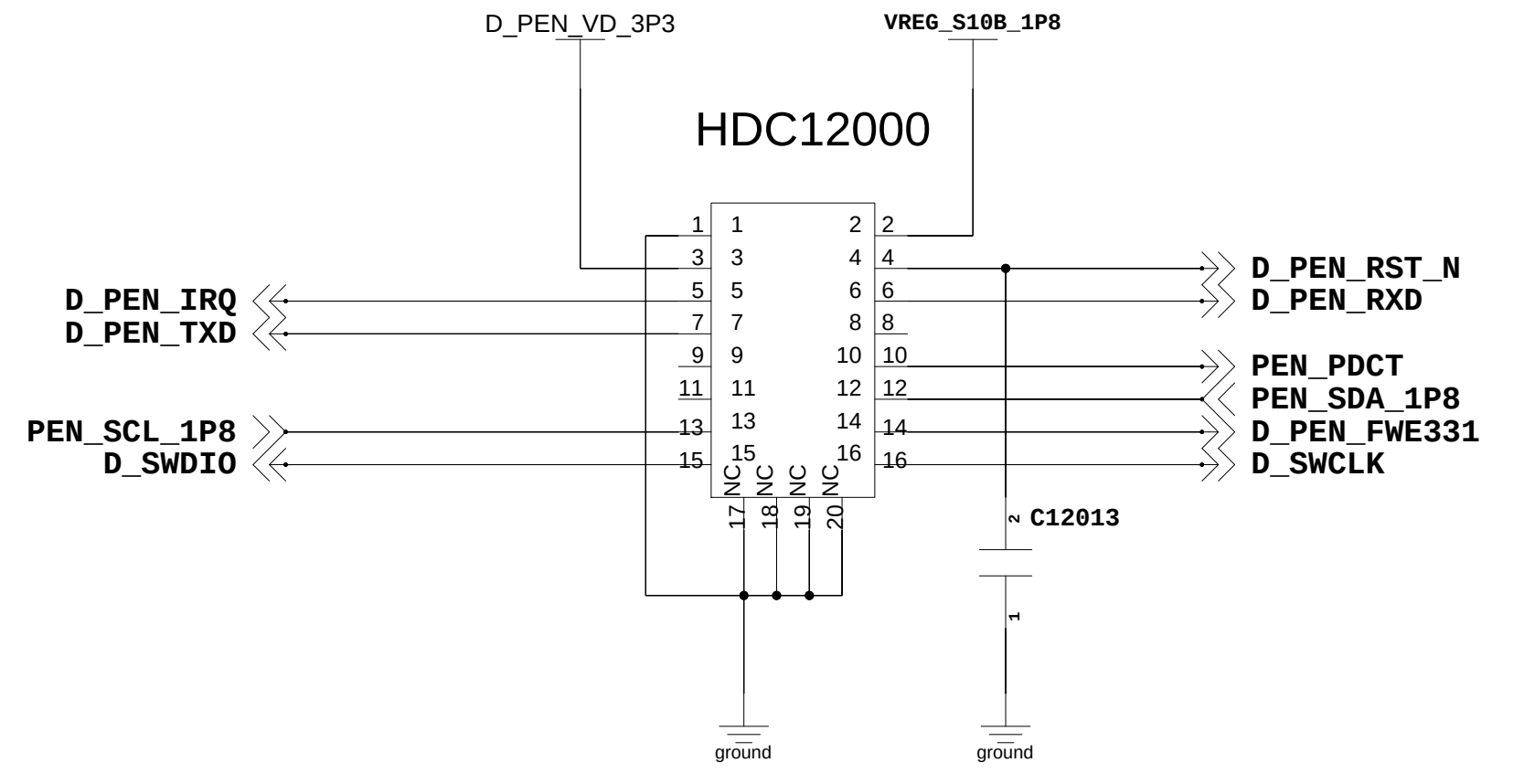
WACOM PCB

WACOM I/F CON.

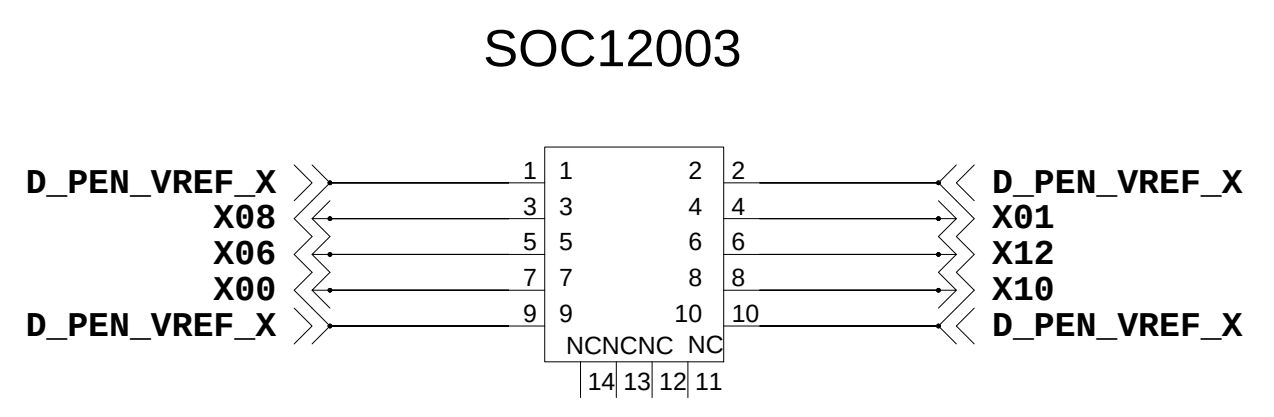

VREG_S10B_1P8
V_EWRITING_3P3
SOC12000
D_PEN_DET_2
D_PEN_DET_1
D_PEN_VREF_X
TSP_VSYNC
D_PEN_IRQ
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
NC
NC
NC
NC
C12006
C12007

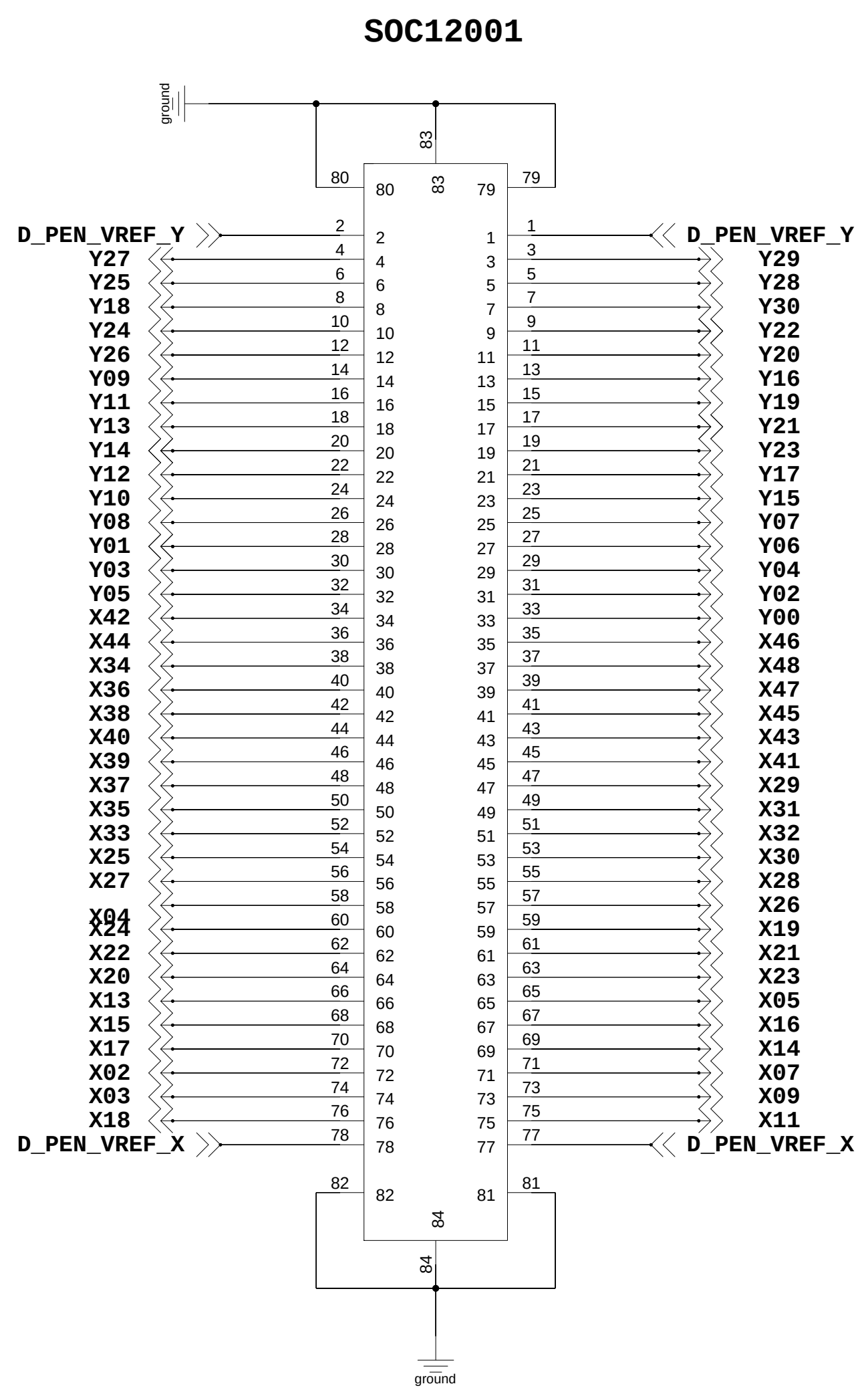
MBOM : NC

DEBUGER

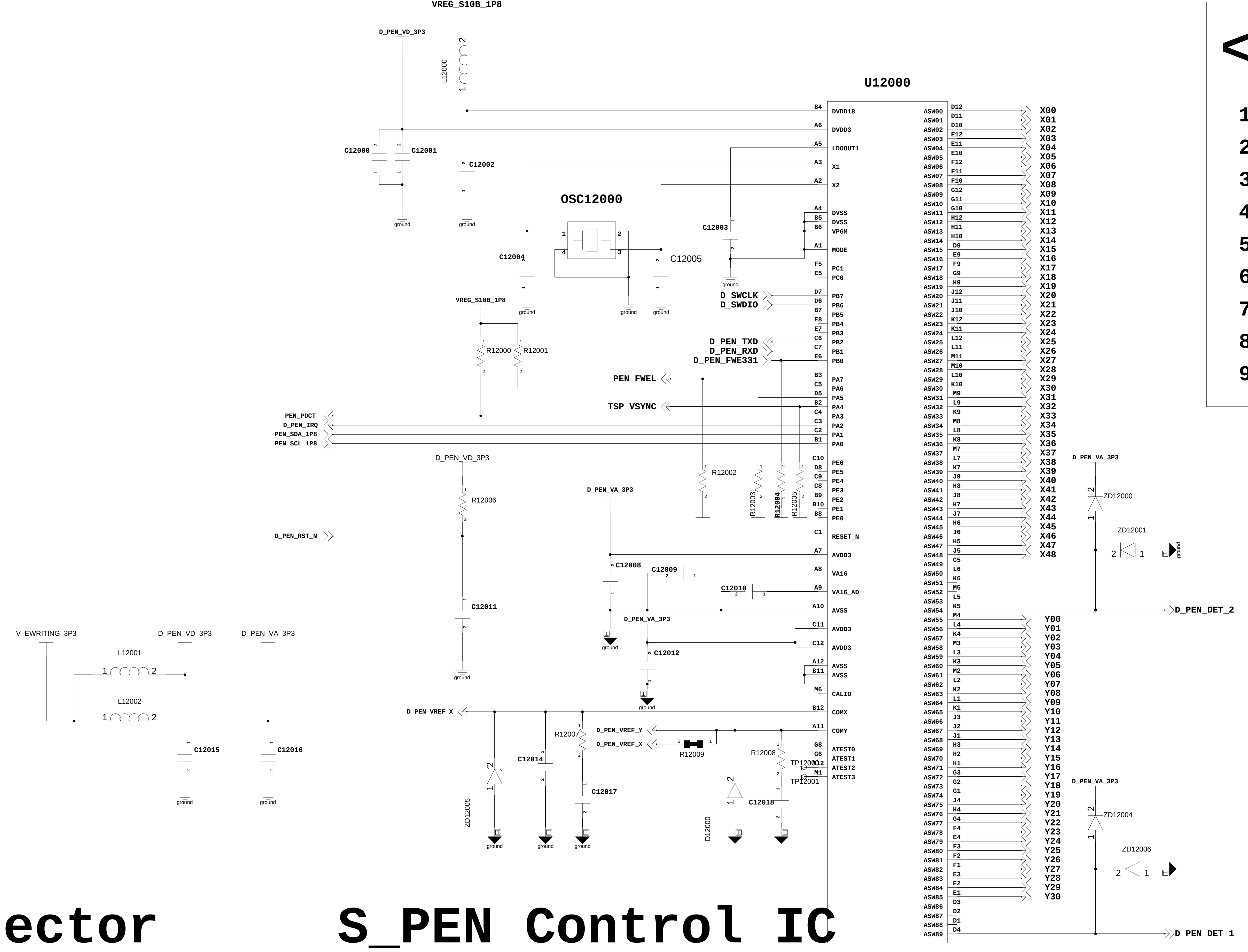

D_PEN_VD_3P3
VREG_S10B_1P8
HDC12000
D_PEN_IRQ
D_PEN_TXD
PEN_SCL_1P8
D_SWDIO
D_PEN_RST_N
D_PEN_RXD
PEN_PDCT
PEN_SDA_1P8
D_PEN_FWE331
D_SWCLK
NC
NC
NC
NC
C12013
ground
ground

78+16 pin S-PEN Connector


SOC12003
D_PEN_VREF_X
X08
X06
X00
D_PEN_VREF_X
D_PEN_VREF_X
X01
X12
X10
D_PEN_VREF_X
NCNCNC NC


SOC12001
ground
D_PEN_VREF_Y
Y27
Y25
Y18
Y24
Y26
Y09
Y11
Y13
Y14
Y12
Y10
Y08
Y01
Y03
Y05
X42
X44
X34
X36
X38
X40
X39
X37
X35
X33
X25
X27
X04
X24
X22
X20
X13
X15
X17
X02
X03
X18
D_PEN_VREF_X
D_PEN_VREF_Y
Y29
Y28
Y30
Y22
Y20
Y16
Y19
Y21
Y23
Y17
Y15
Y07
Y06
Y04
Y02
Y00
X46
X48
X47
X45
X43
X41
X29
X31
X32
X30
X28
X26
X19
X21
X23
X05
X16
X14
X07
X09
X11
D_PEN_VREF_X
ground

S_PEN Control IC


VREG_S10B_1P8
D_PEN_VD_3P3
L12000
C12000
C12001
C12002
ground
OSC12000
C12004
C12005
C12003
U12000
DVDD18
DVDD3
LDOOUT1
X1
X2
DVSS
DVSS
VPGM
MODE
PC1
PC0
PB7
PB6
PB5
PB4
PB3
PB2
PB1
PB0
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
PE6
PE5
PE4
PE3
PE2
PE1
PE0
RESET_N
AVDD3
VA16
VA16_AD
AVSS
AVDD3
AVDD3
AVSS
AVSS
CALIO
COMX
COMY
ATEST0
ATEST1
ATEST2
ATEST3
D_SWCLK
D_SWDIO
D_PEN_TXD
D_PEN_RXD
D_PEN_FWE331
PEN_FWEL
TSP_VSYNC
PEN_PDCT
D_PEN_IRQ
PEN_SDA_1P8
PEN_SCL_1P8
R12000
R12001
R12002
R12003
R12004
R12005
R12006
D_PEN_RST_N
D_PEN_VA_3P3
C12008
C12009
C12010
C12011
C12012
D_PEN_VREF_X
D_PEN_VREF_Y
D_PEN_VREF_X
R12007
R12009
R12008
C12014
C12017
C12018
ZD12005
D12000
TP12000
TP12001
ZD12000
ZD12001
ZD12004
ZD12006
D_PEN_DET_2
D_PEN_DET_1
V_EWRITING_3P3
L12001
L12002
C12015
C12016
X00
X01
X02
X03
X04
X05
X06
X07
X08
X09
X10
X11
X12
X13
X14
X15
X16
X17
X18
X19
X20
X21
X22
X23
X24
X25
X26
X27
X28
X29
X30
X31
X32
X33
X34
X35
X36
X37
X38
X39
X40
X41
X42
X43
X44
X45
X46
X47
X48
Y00
Y01
Y02
Y03
Y04
Y05
Y06
Y07
Y08
Y09
Y10
Y11
Y12
Y13
Y14
Y15
Y16
Y17
Y18
Y19
Y20
Y21
Y22
Y23
Y24
Y25
Y26
Y27
Y28
Y29
Y30

(DC mode)

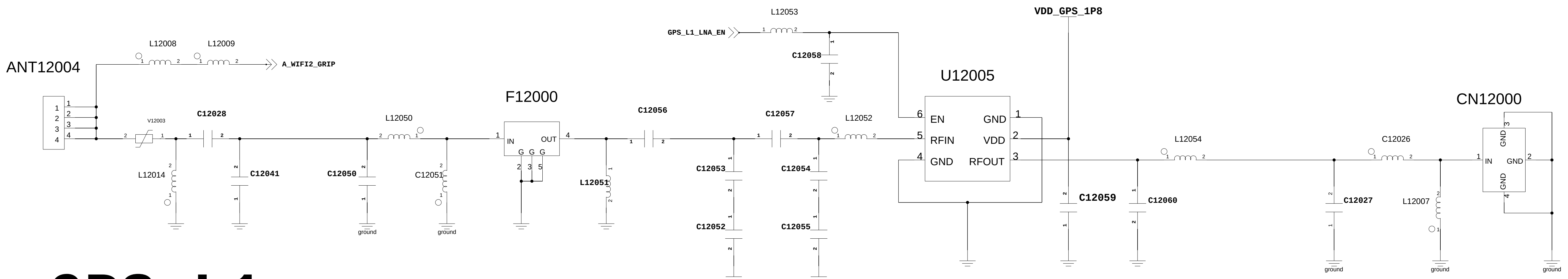
< SMD F/T >

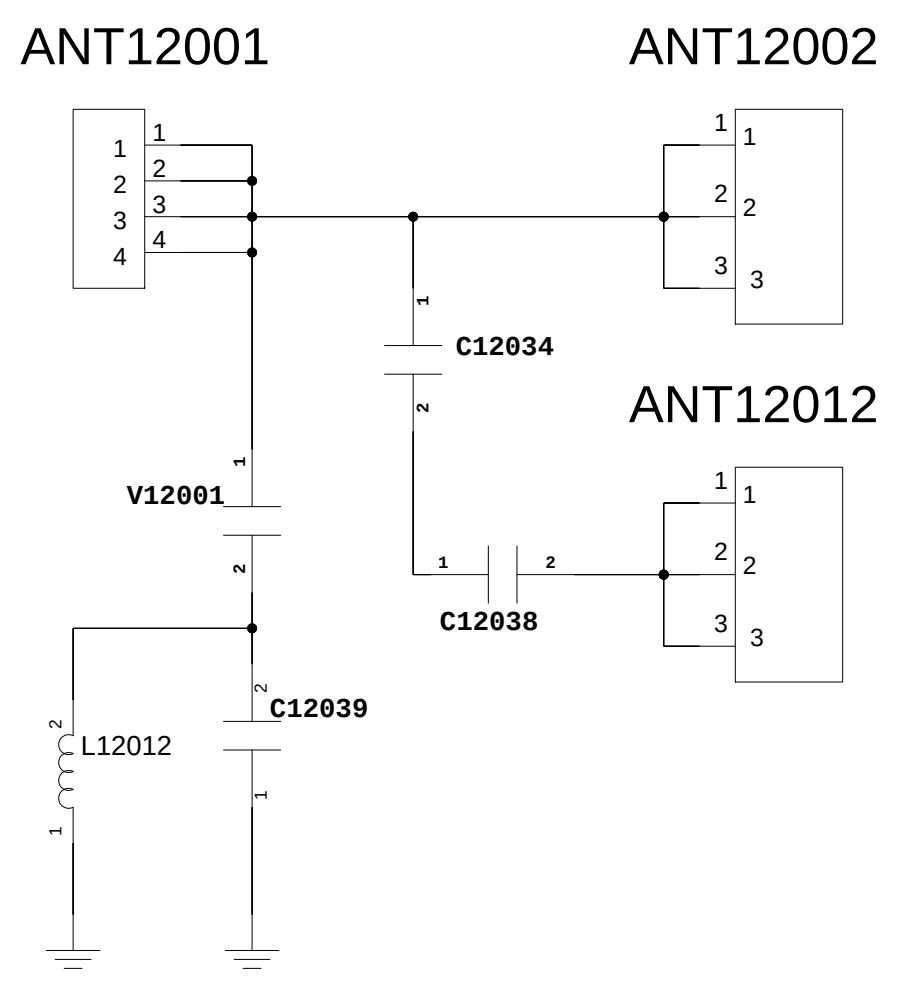
1. V_EWRITING_3P3
2. VREG_S10B_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GPS_L1_LNA_EN
8. VDD_GPS_1P8
9. GND

ANT SUB PCB

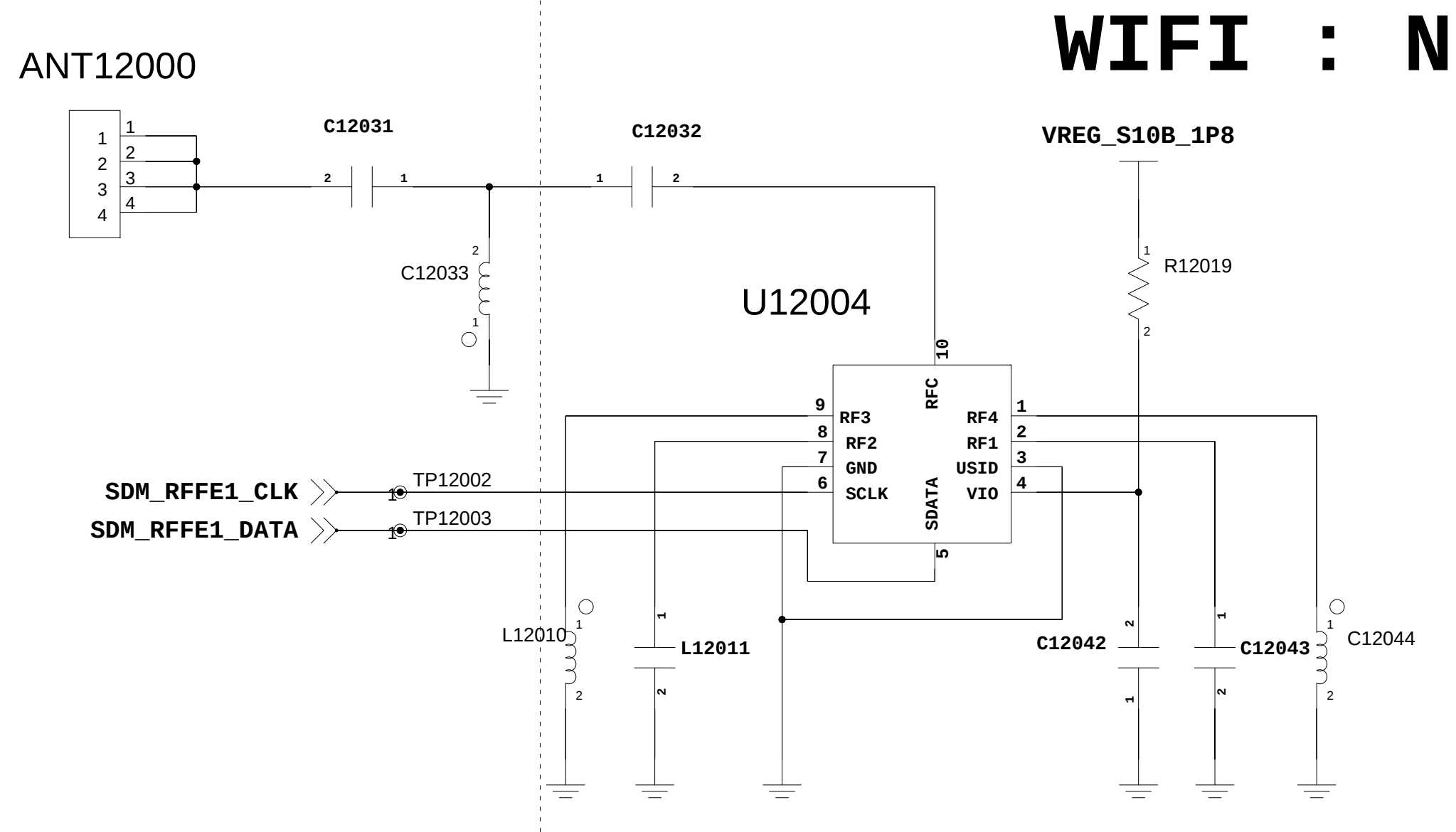
(MAIN1 ANT XGND / WIFI2 ANT GRIP)

GPS_L1

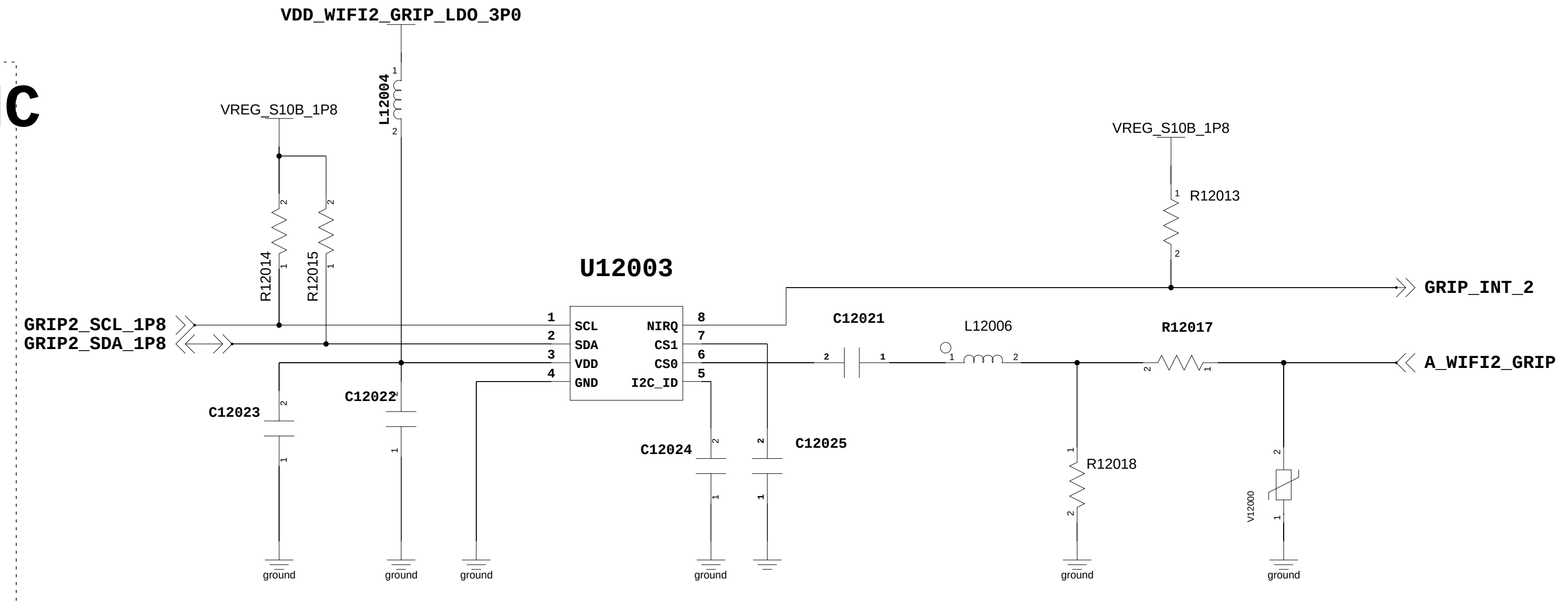

ANT12004
L12008
L12009
A_WIFI2_GRIP
V12003
C12028
L12014
C12041
C12050
L12050
C12051
F12000
IN
OUT
G G G
L12051
GPS_L1_LNA_EN
L12053
C12058
C12056
C12057
L12052
C12053
C12054
C12052
C12055
U12005
EN
RFIN
GND
GND
VDD
RFOUT
VDD_GPS_1P8
C12059
C12060
L12054
C12026
C12027
L12007
CN12000
IN
GND
GND
GND
ground


ANT12001
ANT12002
ANT12012
C12034
V12001
C12038
C12039
L12012

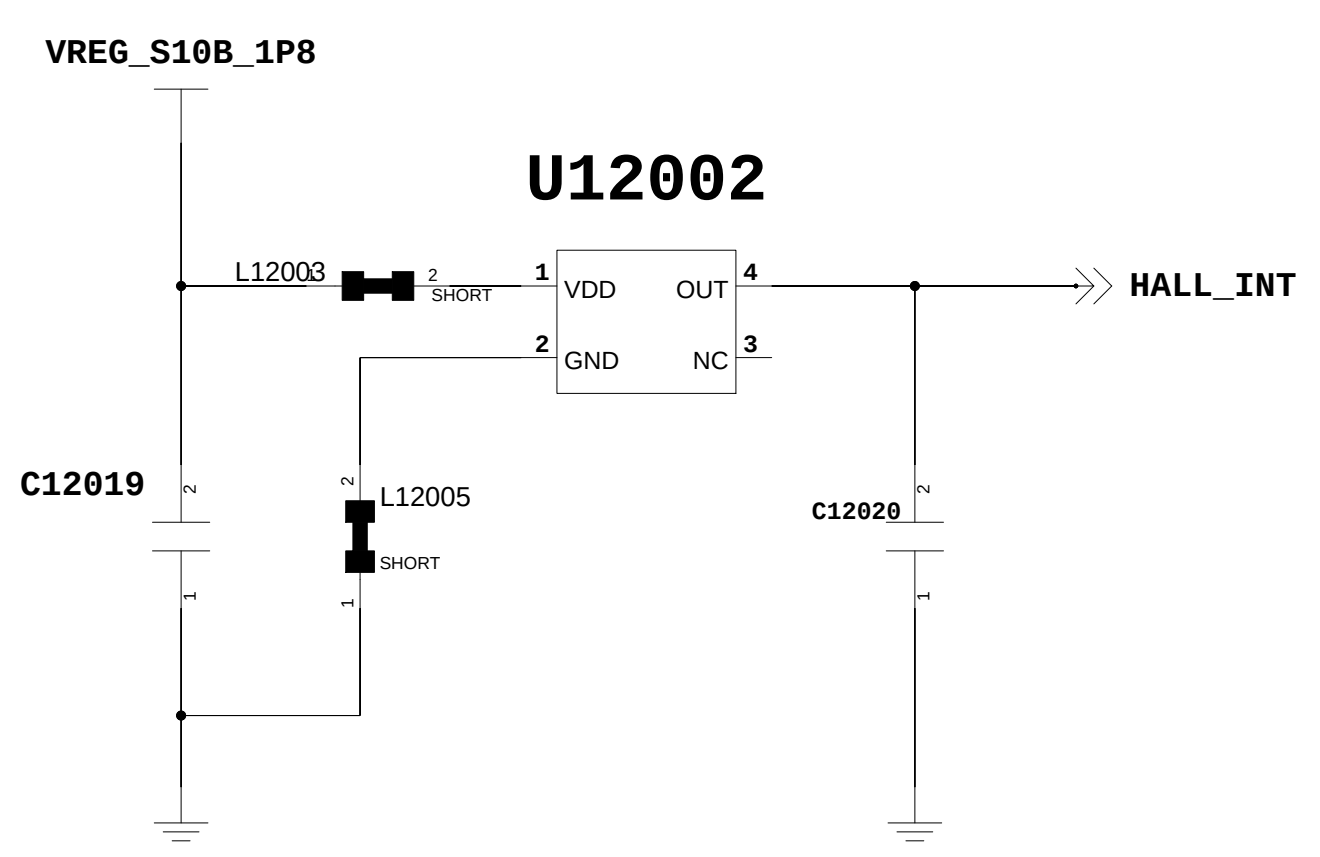
WIFI : NC


ANT12000
C12031
C12032
C12033
VREG_S10B_1P8
R12019
U12004
RF3
RF2
GND
SCLK
RFC
SDATA
RF4
RF1
USID
VIO
SDM_RFFE1_CLK
SDM_RFFE1_DATA
TP12002
TP12003
L12010
L12011
C12042
C12043
C12044

WIFI2_GRIP_SENSOR (GRIP#1)

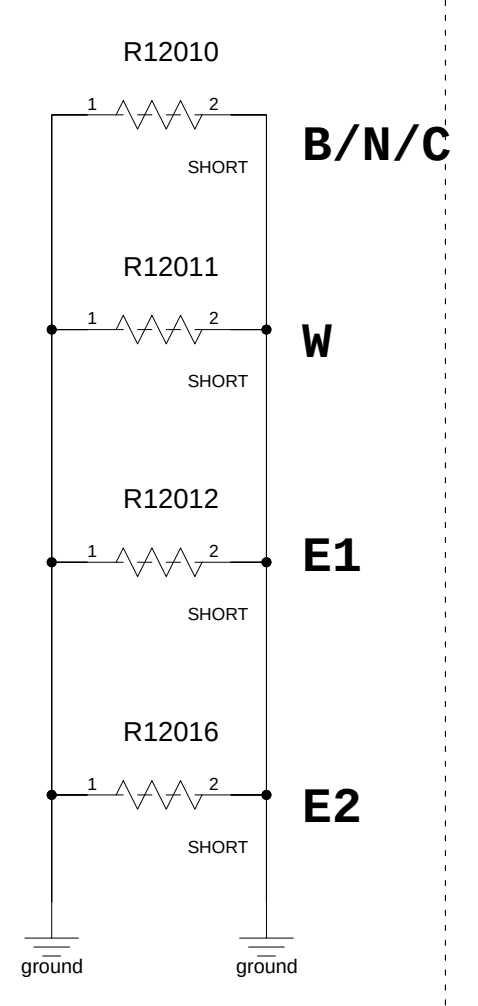

VDD_WIFI2_GRIP_LDO_3P0
L12004
VREG_S10B_1P8
R12014
R12015
GRIP2_SCL_1P8
GRIP2_SDA_1P8
U12003
SCL
SDA
VDD
GND
NIRQ
CS1
CS0
I2C_ID
C12023
C12022
C12024
C12025
C12021
L12006
R12017
R12018
V12000
VREG_S10B_1P8
R12013
GRIP_INT_2
A_WIFI2_GRIP
ground

COVER_DET HALL_IC


VREG_S10B_1P8
U12002
L12003
SHORT
VDD
OUT
GND
NC
HALL_INT
C12019
L12005
SHORT
C12020

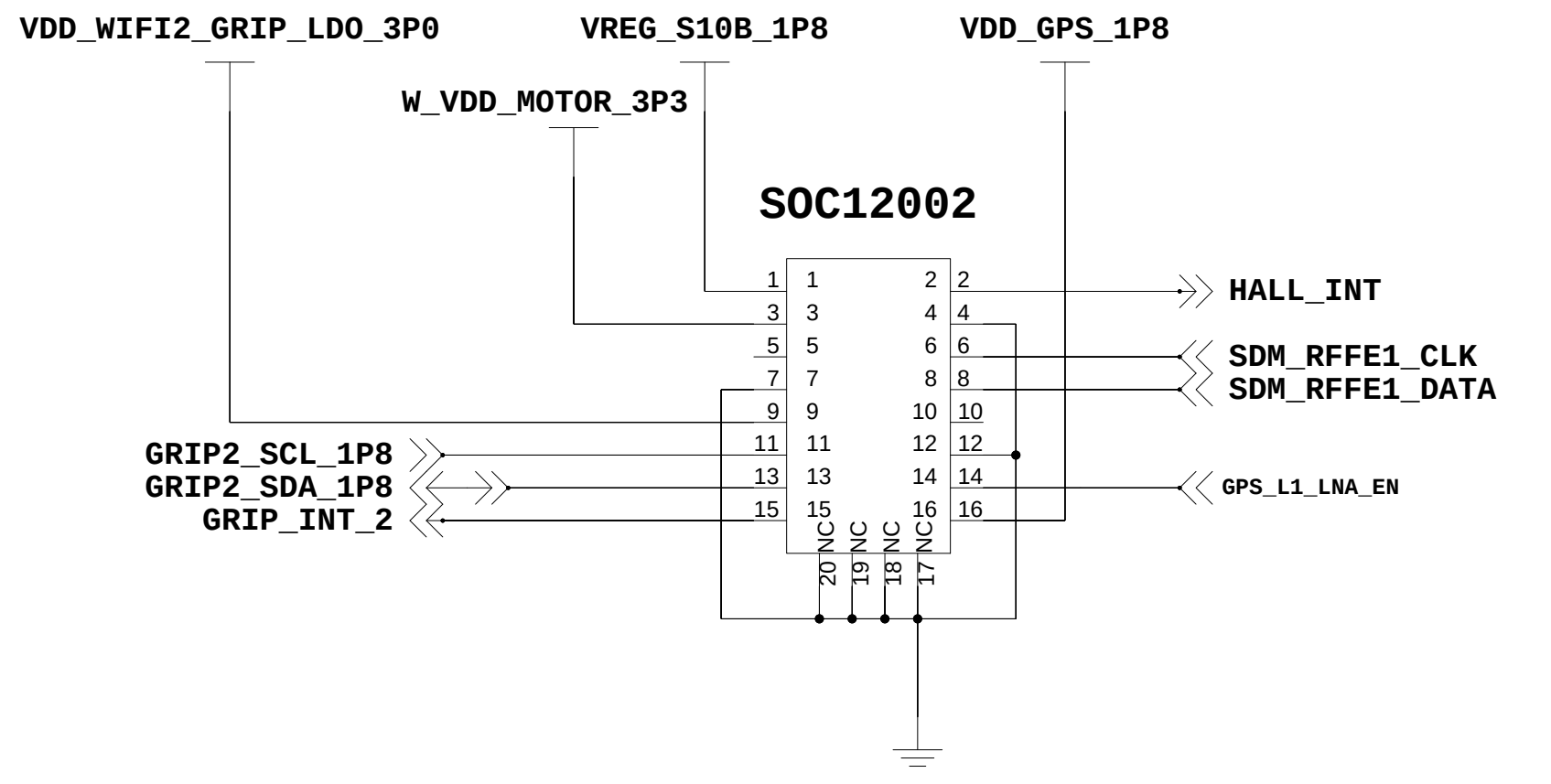
ANT PCB REGION

ANT PCB	SHORT
X906B(B/N/C)	R12010
X900(W)	R12011
ETC_1(E1)	R12012
ETC_2(E2)	R12016

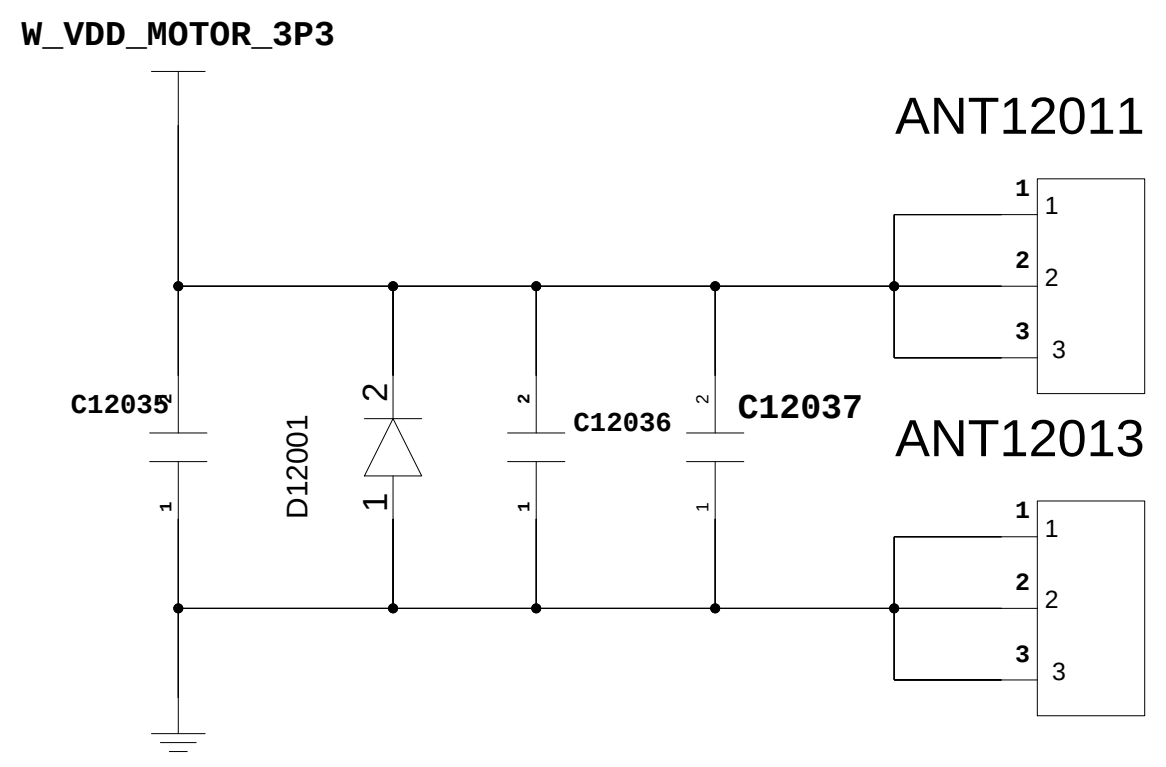

R12010
SHORT
B/N/C
R12011
SHORT
W
R12012
SHORT
E1
R12016
SHORT
E2
ground
ground

< SMD F/T >

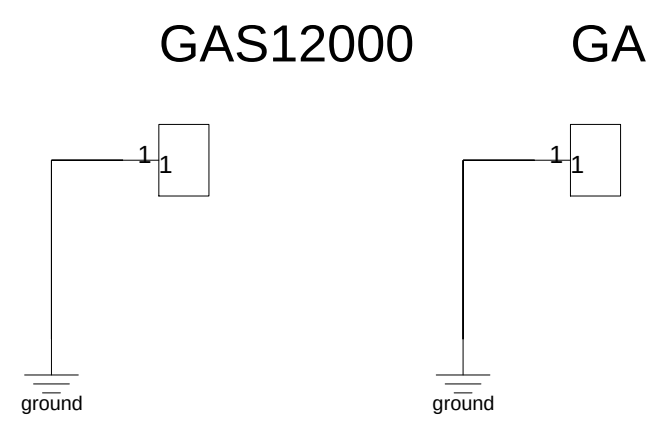
1. VDD_WIFI2_GRIP_LDO_3P0
2. GRIP2_SCL_1P8
3. GRIP2_SDA_1P8
4. GRIP_INT_2
5. SDM_RFFE1_CLK
6. SDM_RFFE1_DATA
7. VREG_S10B_1P8
8. HALL_INT
9. GPS_L1_LNA_EN
10. VDD_GPS_1P8
11. GND


VDD_WIFI2_GRIP_LDO_3P0
VREG_S10B_1P8
VDD_GPS_1P8
W_VDD_MOTOR_3P3
SOC12002
HALL_INT
SDM_RFFE1_CLK
SDM_RFFE1_DATA
GPS_L1_LNA_EN
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
NC
NC
NC
NC

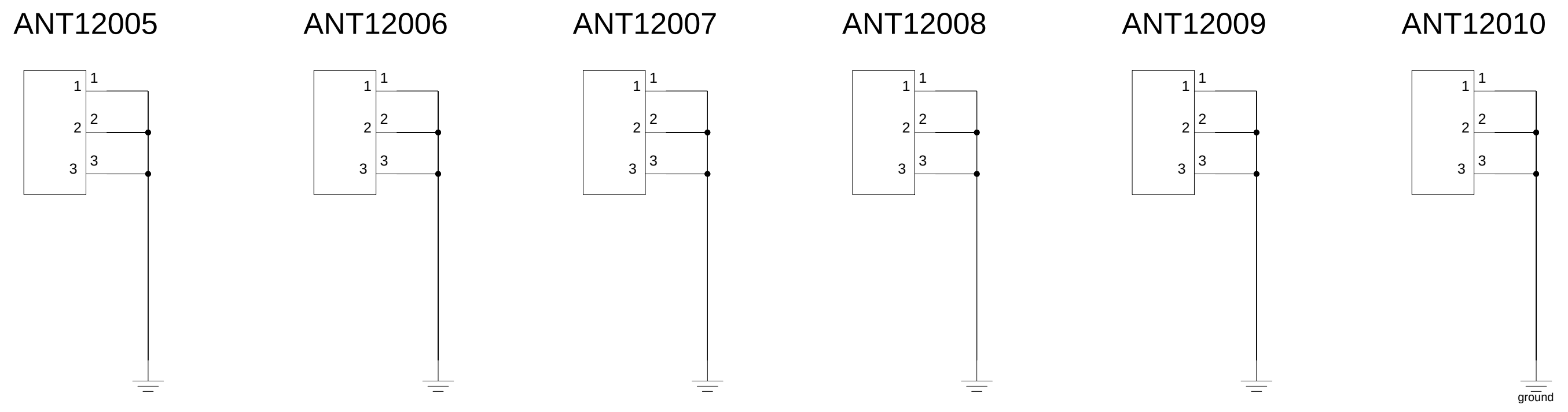
MOTOR


W_VDD_MOTOR_3P3
C12035
D12001
C12036
C12037
ANT12011
ANT12013

Front CONTACT

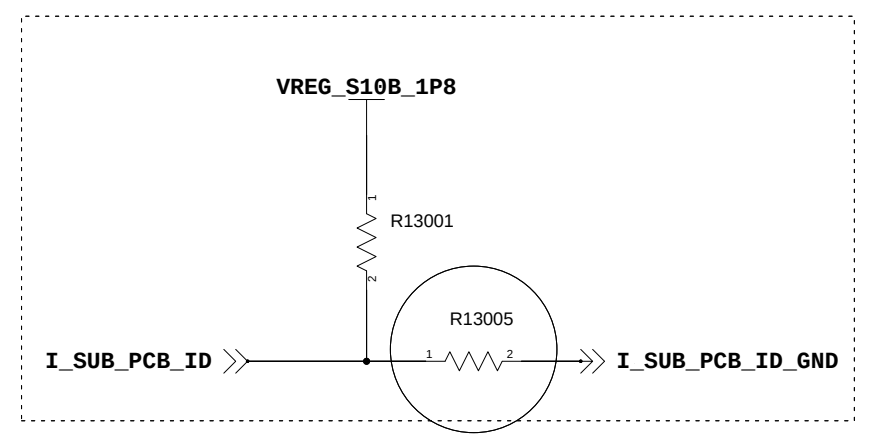

GAS12000
GAS12001
ground
ground

REAR CONTACT

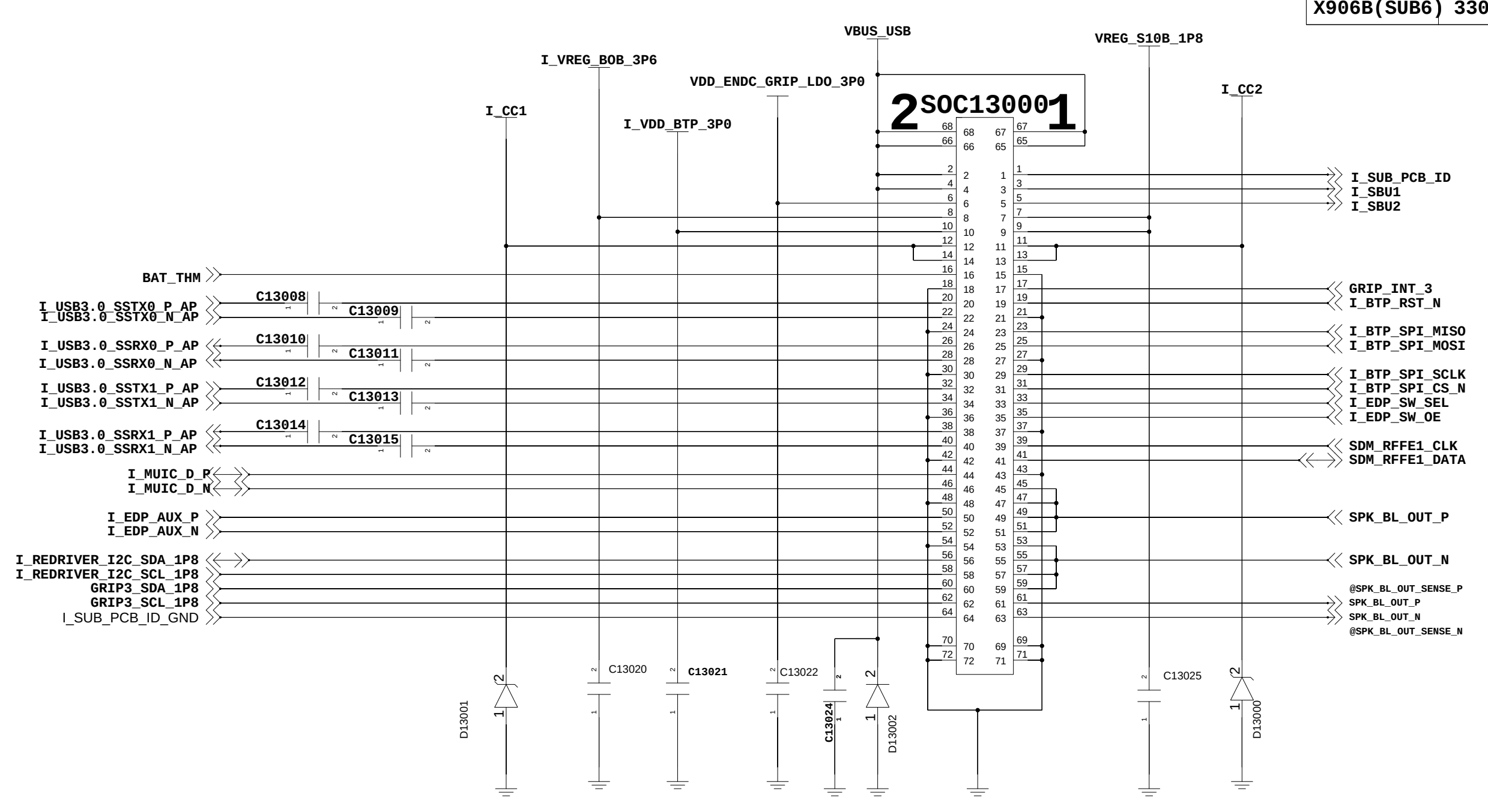

ANT12005
ANT12006
ANT12007
ANT12008
ANT12009
ANT12010
ground

TITLE TabS7P
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 12 of 14
Date Drawing Date

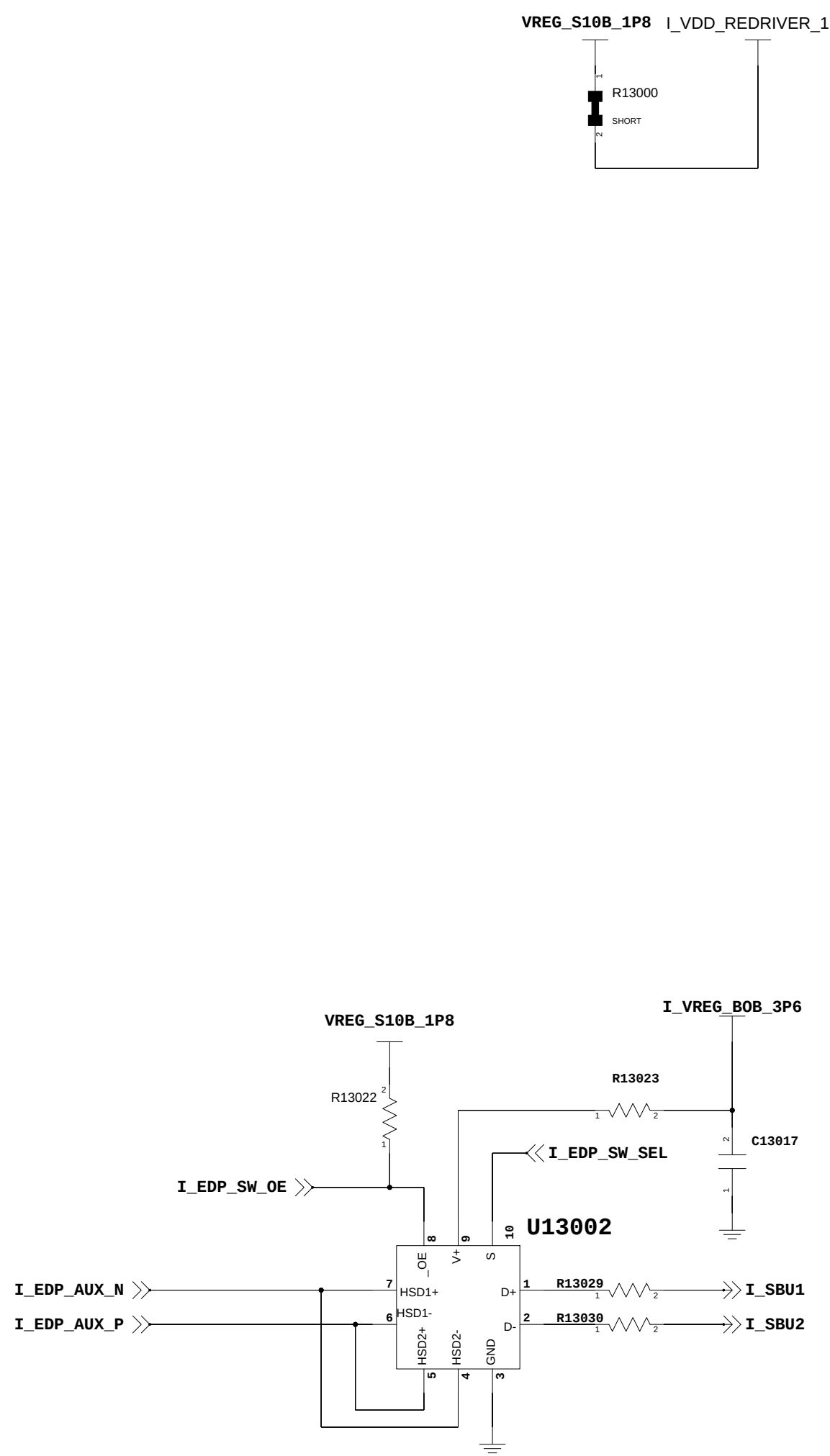
USB PCB


VREG_S10B_1P8
R13001
R13005
I_SUB_PCB_ID
I_SUB_PCB_ID_GND

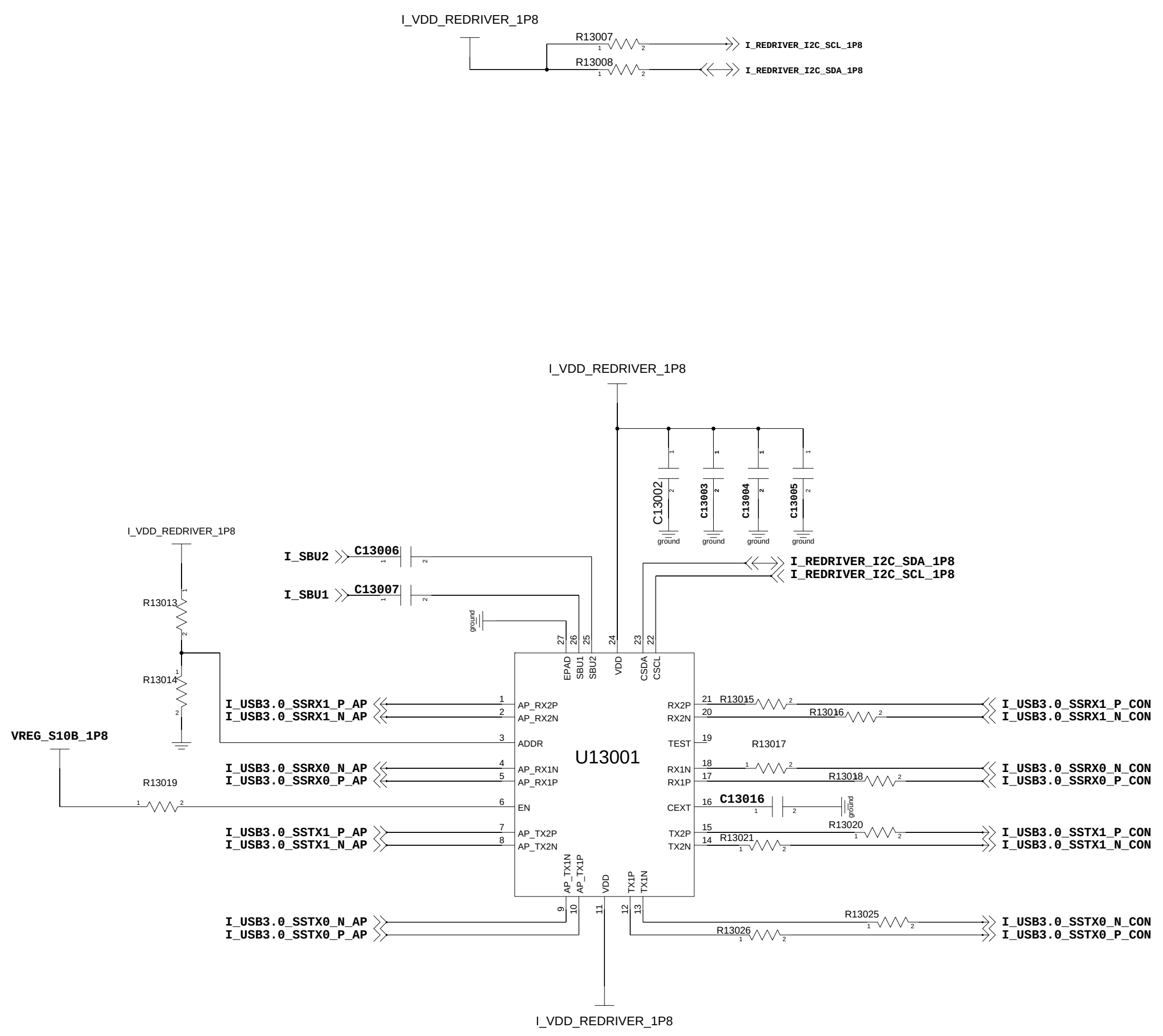
USB PCB	R13005	ADC	Range	ID
	22K	324	0 - 600	0
X900(WIFI)	68K	728	601 - 900	4
	150K	1080	901 - 1200	5
X906B(SUB6)	330K	1381	1201 - 1500	6


VBUS_USB
I_VREG_BOB_3P6
VDD_ENDC_GRIP_LDO_3P0
I_VDD_BTP_3P0
VREG_S10B_1P8
I_CC1
I_CC2
SOC13000
BAT_THM
C13008
C13009
C13010
C13011
C13012
C13013
C13014
C13015
I_USB3.0_SSTX0_P_AP
I_USB3.0_SSTX0_N_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSRX0_N_AP
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_N_AP
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_N_AP
I_MUIC_D_P
I_MUIC_D_N
I_EDP_AUX_P
I_EDP_AUX_N
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
GRIP3_SDA_1P8
GRIP3_SCL_1P8
I_SUB_PCB_ID_GND
I_SUB_PCB_ID
I_SBU1
I_SBU2
GRIP_INT_3
I_BTP_RST_N
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
I_EDP_SW_SEL
I_EDP_SW_OE
SDM_RFFE1_CLK
SDM_RFFE1_DATA
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
D13001
C13020
C13021
C13022
C13024
D13002
C13025
D13000

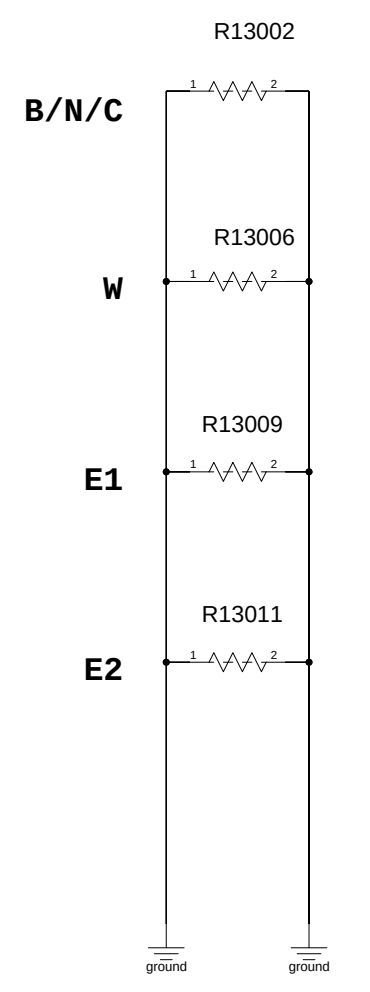
I/F CONN (USB)


VREG_S10B_1P8
I_VDD_REDRIVER_1P8
R13000
SHORT
VREG_S10B_1P8
I_VREG_BOB_3P6
R13022
R13023
C13017
I_EDP_SW_SEL
I_EDP_SW_OE
U13002
I_EDP_AUX_N
I_EDP_AUX_P
HSD1+
HSD1-
HSD2+
HSD2-
OE
V+
S
D+
D-
GND
R13029
R13030
I_SBU1
I_SBU2

DP Switch


I_VDD_REDRIVER_1P8
R13007
R13008
I_REDRIVER_I2C_SCL_1P8
I_REDRIVER_I2C_SDA_1P8
I_VDD_REDRIVER_1P8
C13002
C13003
C13004
C13005
I_SBU2
I_SBU1
C13006
C13007
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
R13013
R13014
VREG_S10B_1P8
R13019
U13001
EPAD
SBU1
SBU2
VDD
CSDA
CSCL
AP_RX2P
AP_RX2N
ADDR
AP_RX1N
AP_RX1P
EN
AP_TX2P
AP_TX2N
AP_TX1N
AP_TX1P
TX1P
TX1N
RX2P
RX2N
TEST
RX1N
RX1P
CEXT
TX2P
TX2N
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_N_AP
I_USB3.0_SSRX0_N_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_N_AP
I_USB3.0_SSTX0_N_AP
I_USB3.0_SSTX0_P_AP
R13015
R13016
R13017
R13018
C13016
R13020
R13021
R13025
R13026
I_USB3.0_SSRX1_P_CON
I_USB3.0_SSRX1_N_CON
I_USB3.0_SSRX0_N_CON
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSTX1_P_CON
I_USB3.0_SSTX1_N_CON
I_USB3.0_SSTX0_N_CON
I_USB3.0_SSTX0_P_CON
I_VDD_REDRIVER_1P8

REDRIVER & DP

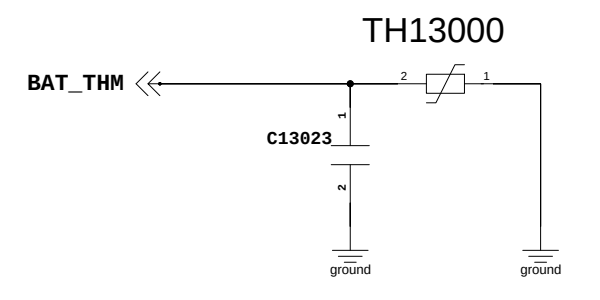

R13002
B/N/C
R13006
W
R13009
E1
R13011
E2

USB PCB REGION

USB PCB	SHORT
SUB6(B/N/C)	R13002
WIFI(W)	R13006
ETC1(E1)	R13009
ETC2(E2)	R13011

< USB PCB - SMD F/T

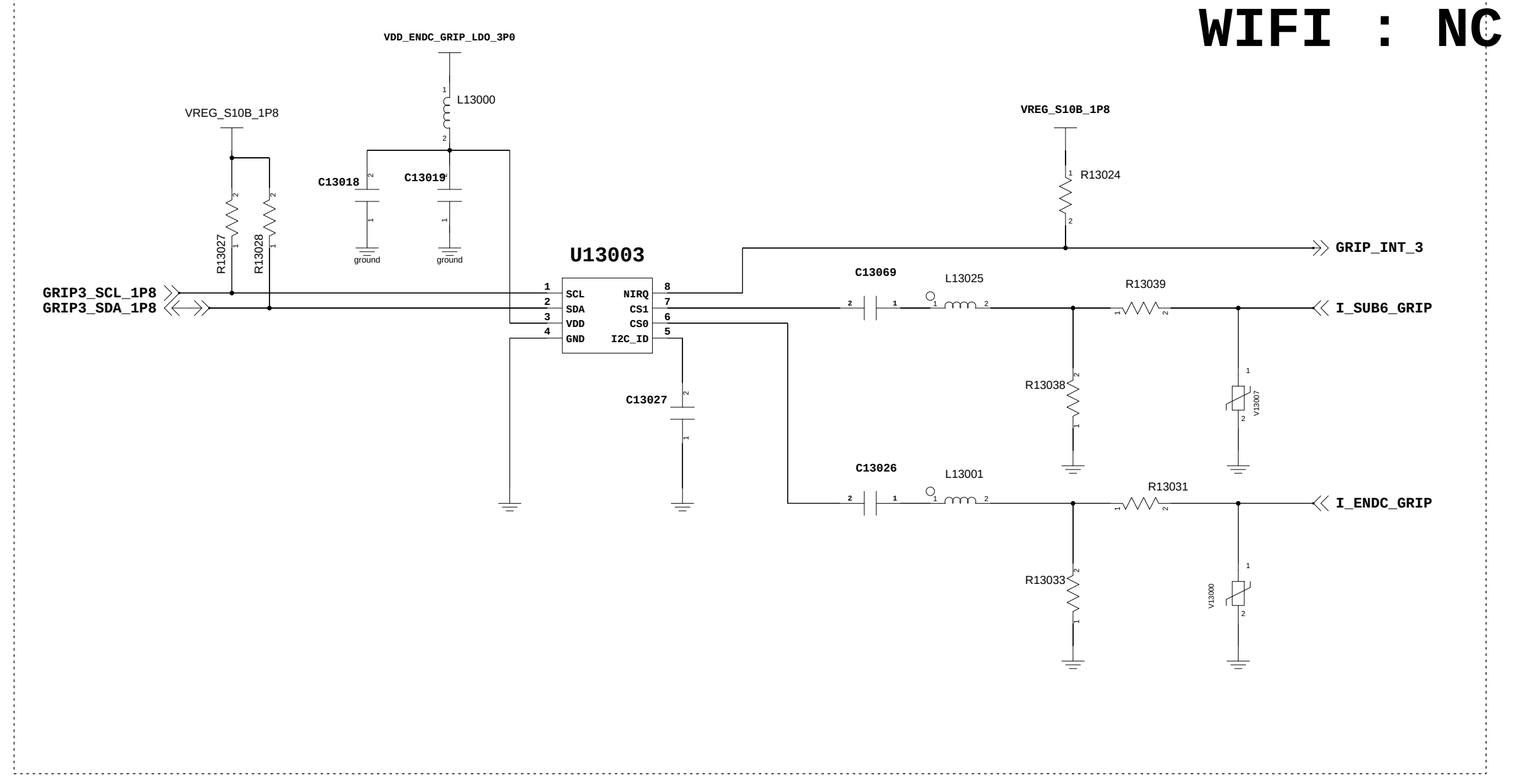
1. VREG_S10B_1P8
2. BAT_THM
3. SDM_RFFE1_CLK
4. SDM_RFFE1_DATA
5. GND
6. SPK_BL_OUT_P
7. SPK_BL_OUT_SENSE_P
8. SPK_BL_OUT_N
9. SPK_BL_OUT_SENSE_N
10. VDD_ENDC_GRIP_LDO_3P0
11. GRIP_INT_3
12. GRIP_SDA_1P8
13. GRIP_SCL_1P8

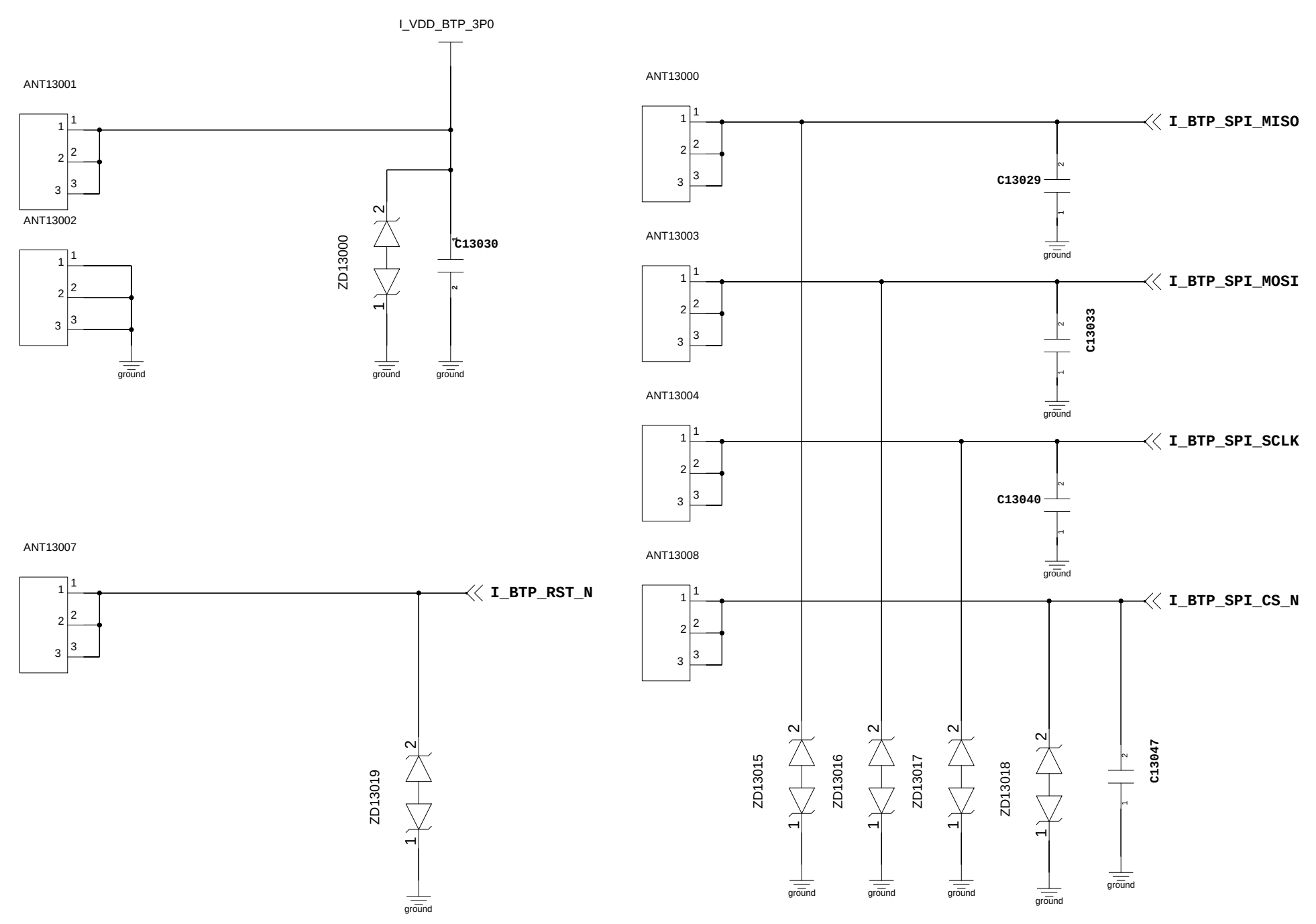

TH13000
BAT_THM
C13023

BAT THM

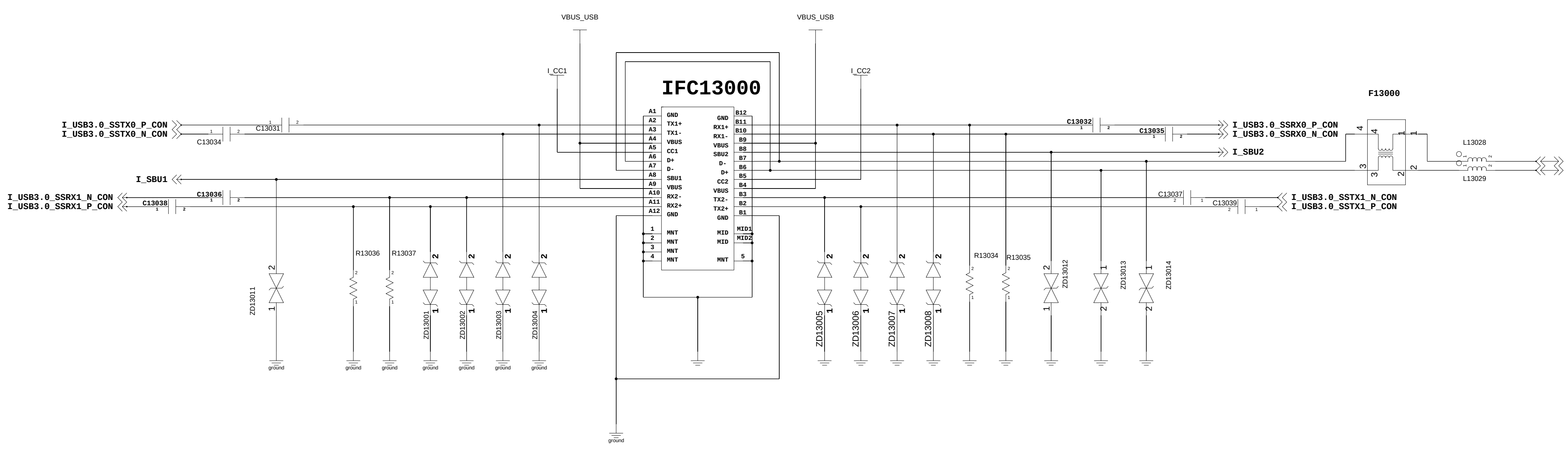
ENDC_GRIP_SENSOR(GRIP #4

WIFI : NC

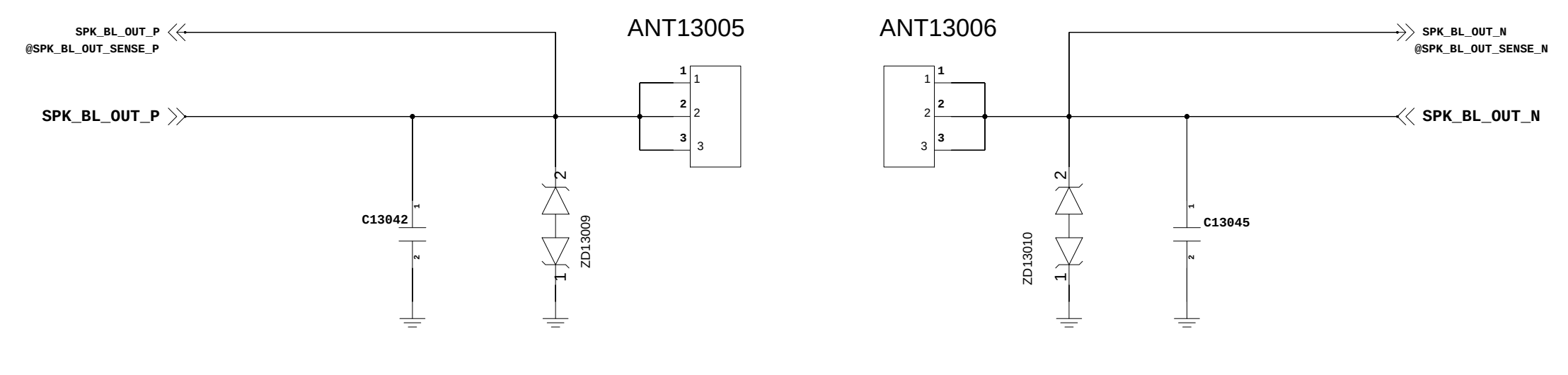

VDD_ENDC_GRIP_LDO_3P0
L13000
VREG_S10B_1P8
C13018
C13019
R13027
R13028
VREG_S10B_1P8
R13024
U13003
SCL
SDA
VDD
GND
NIRQ
CS1
CS0
I2C_ID
GRIP3_SCL_1P8
GRIP3_SDA_1P8
GRIP_INT_3
C13069
L13025
R13039
I_SUB6_GRIP
R13038
V13007
C13027
C13026
L13001
R13031
I_ENDC_GRIP
R13033
V13000


I_VDD_BTP_3P0
ANT13001
ANT13002
ZD13000
C13030
ANT13000
C13029
I_BTP_SPI_MISO
ANT13003
C13033
I_BTP_SPI_MOSI
ANT13004
C13040
I_BTP_SPI_SCLK
ANT13007
I_BTP_RST_N
ZD13019
ANT13008
I_BTP_SPI_CS_N
ZD13015
ZD13016
ZD13017
ZD13018
C13047

FOD CON.

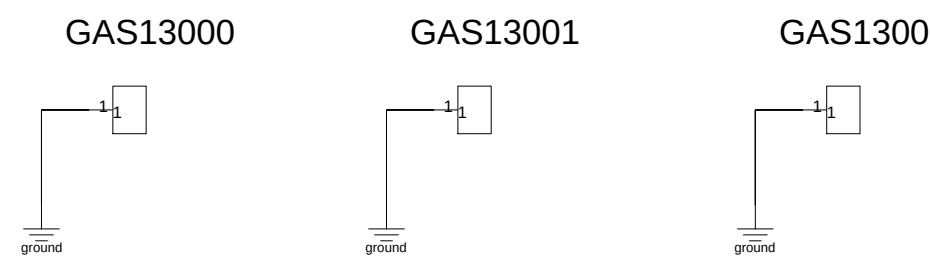

VBUS_USB
VBUS_USB
I_CC1
I_CC2
IFC13000
I_USB3.0_SSTX0_P_CON
I_USB3.0_SSTX0_N_CON
C13031
C13034
I_SBU1
I_USB3.0_SSRX1_N_CON
I_USB3.0_SSRX1_P_CON
C13036
C13038
GND
TX1+
TX1-
VBUS
CC1
D+
D-
SBU1
VBUS
RX2-
RX2+
GND
RX1+
RX1-
SBU2
CC2
TX2-
TX2+
MNT
MID
MID1
MID2
C13032
C13035
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSRX0_N_CON
I_SBU2
C13037
C13039
I_USB3.0_SSTX1_N_CON
I_USB3.0_SSTX1_P_CON
F13000
L13028
L13029
I_MUIC_D_N
I_MUIC_D_P
ZD13011
R13036
R13037
ZD13001
ZD13002
ZD13003
ZD13004
ZD13005
ZD13006
ZD13007
ZD13008
R13034
R13035
ZD13012
ZD13013
ZD13014
ground

Type-C Conn.

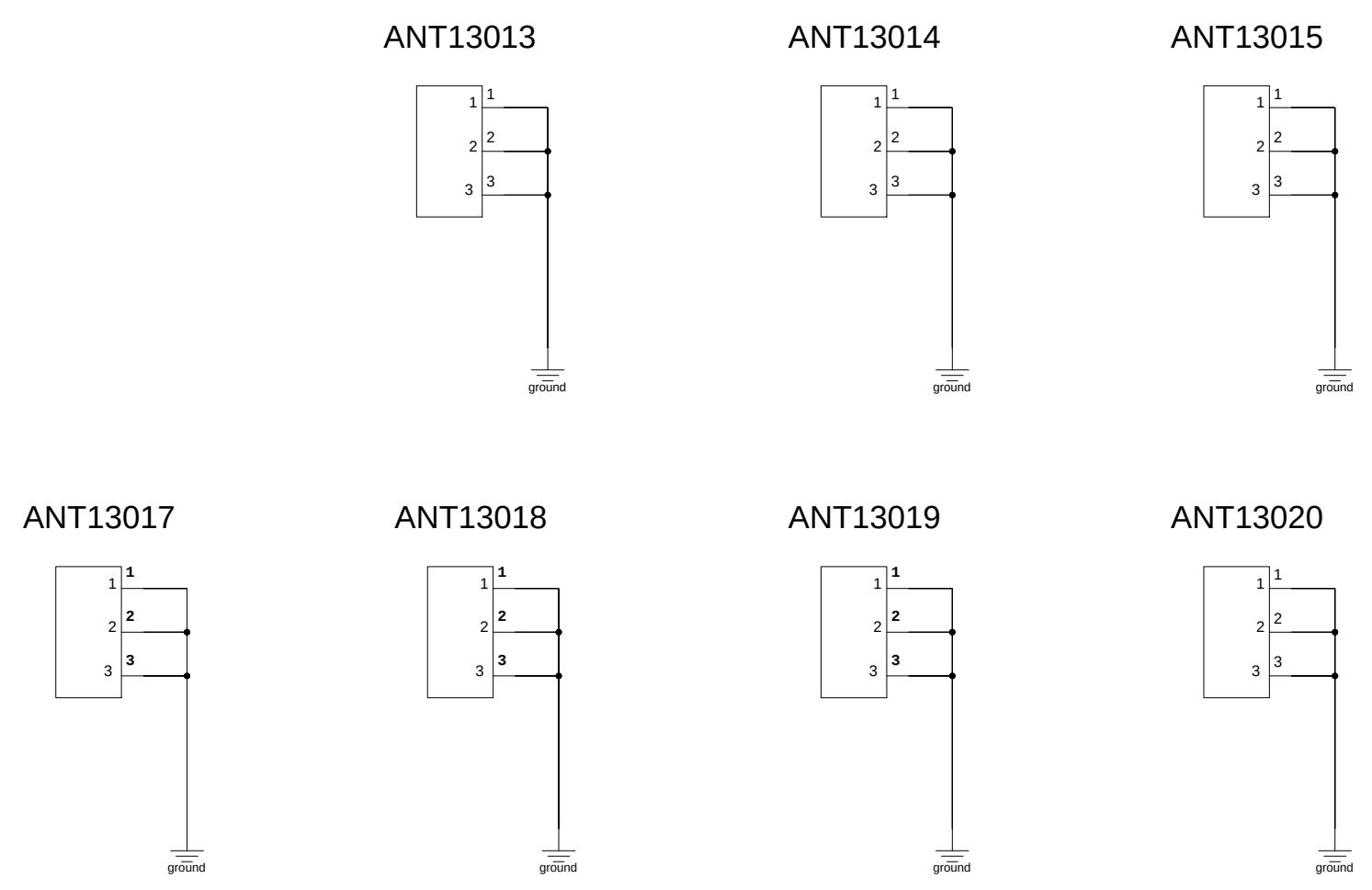

SPK_BL_OUT_P
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
ANT13005
ANT13006
C13042
ZD13009
ZD13010
C13045
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
SPK_BL_OUT_N

SPK BL CON.

BOTHHOLE/CONTACT

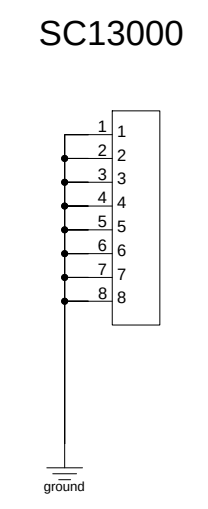

GAS13000
GAS13001
GAS13002

Front CONTACT


ANT13013
ANT13014
ANT13015
ANT13017
ANT13018
ANT13019
ANT13020

REAR CONTACT(0.9t)

SHIELD CAN

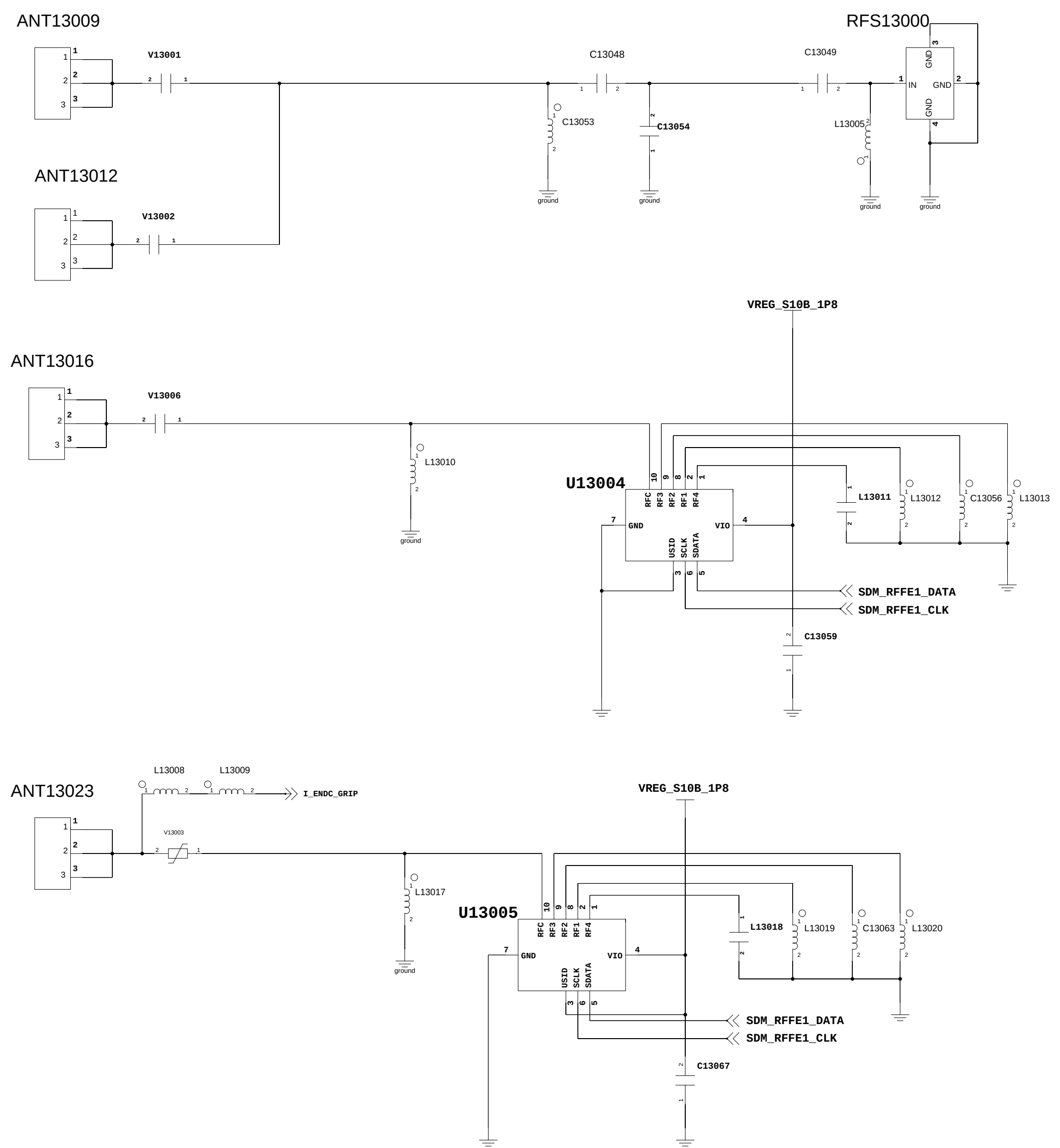


Re-Driver

< BOTTOM S/C >

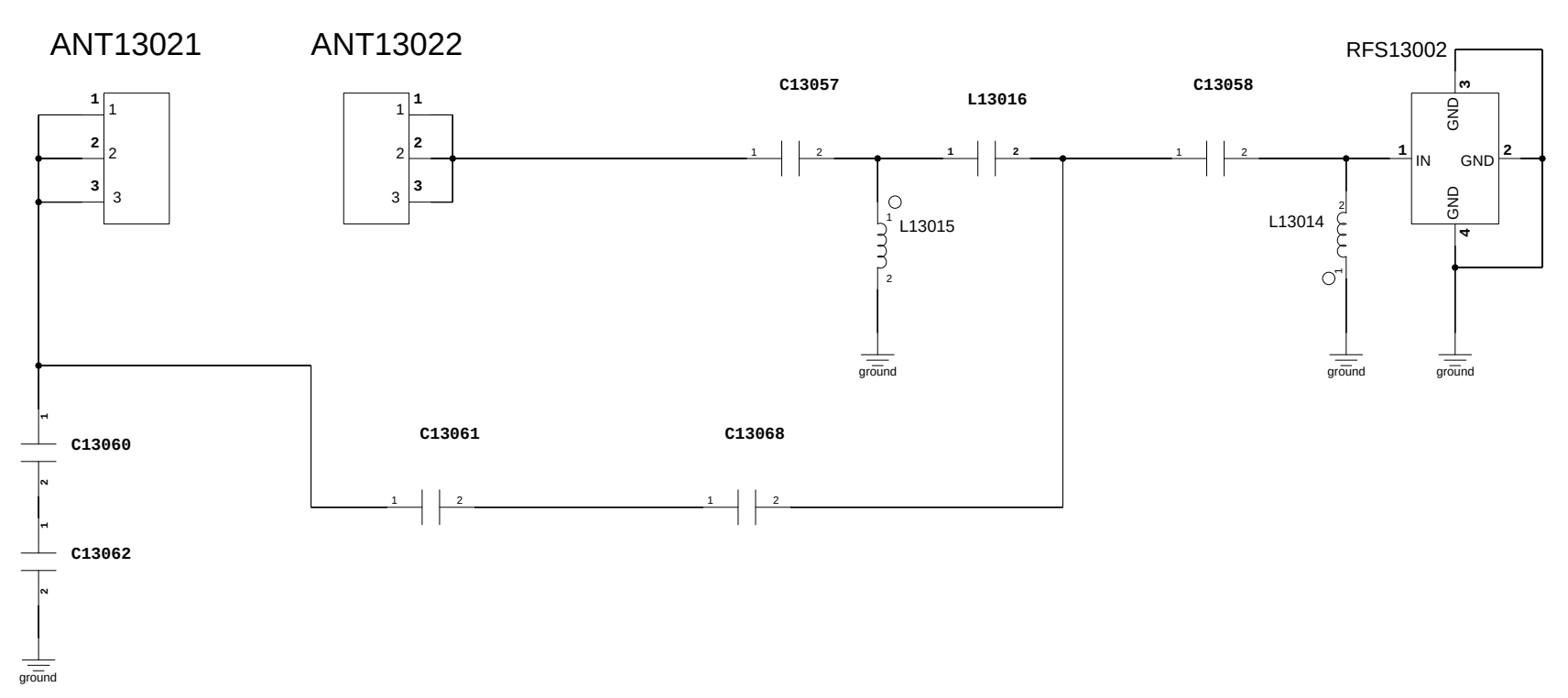
SUB1 ANT

(DRX_LB/MB/HB / DRX_n41 / ENDC_LB/MB TX)


ANT13009
V13001
C13048
C13049
RFS13000
C13053
C13054
L13005
ANT13012
V13002
VREG_S10B_1P8
ANT13016
V13006
L13010
U13004
RFC
RF3
RF2
RF1
RF4
GND
VIO
USID
SCLK
SDATA
L13011
L13012
C13056
L13013
SDM_RFFE1_DATA
SDM_RFFE1_CLK
C13059
ANT13023
L13008
L13009
I_ENDC_GRIP
V13003
VREG_S10B_1P8
L13017
U13005
L13018
L13019
C13063
L13020
SDM_RFFE1_DATA
SDM_RFFE1_CLK
C13067

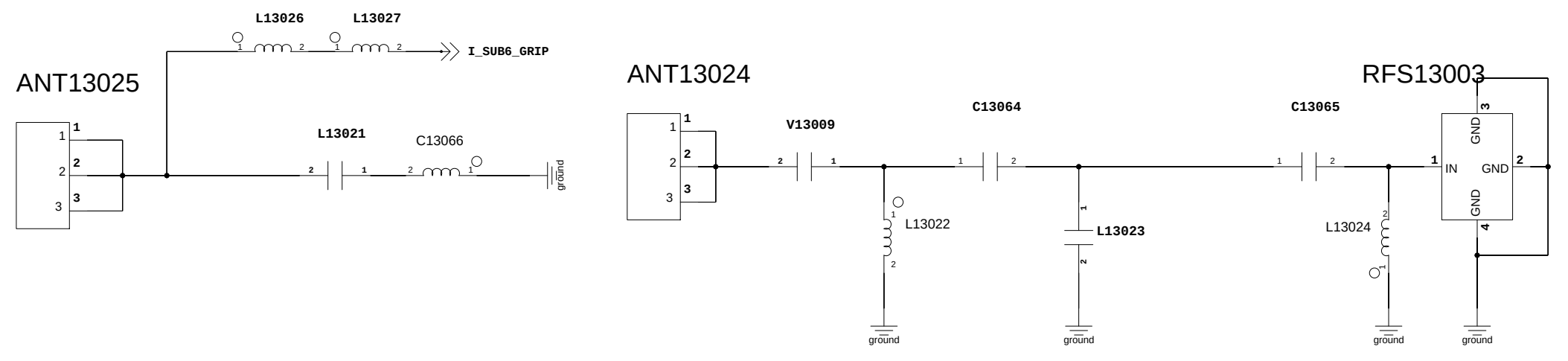
SUB2 ANT

(DRX_MIMO_MB/HB / DRX_n77/n78 / DRX_MIMO_n41)

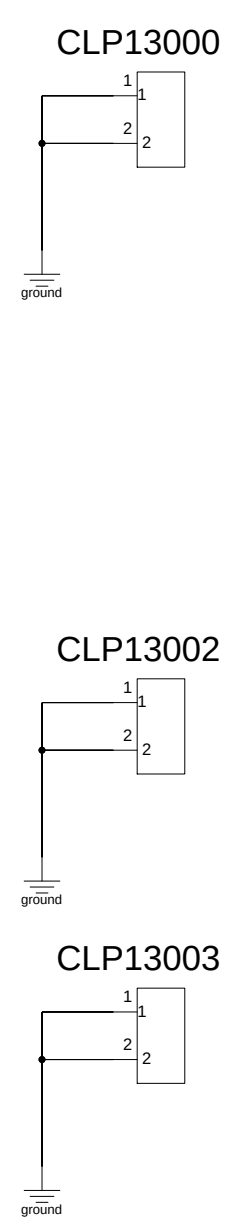

ANT13021
ANT13022
C13057
L13016
C13058
RFS13002
L13015
L13014
C13060
C13062
C13061
C13068

SUB4 ANT

(PRX_MIMO_MB/HB / PRX_MIMO_n77/n78 / PRX_MIMO_n41)


ANT13025
L13026
L13027
I_SUB6_GRIP
L13021
C13066
ANT13024
V13009
C13064
C13065
RFS13003
L13022
L13023
L13024

WIFI : NC


CLP13000
CLP13002
CLP13003

Cable CLIP

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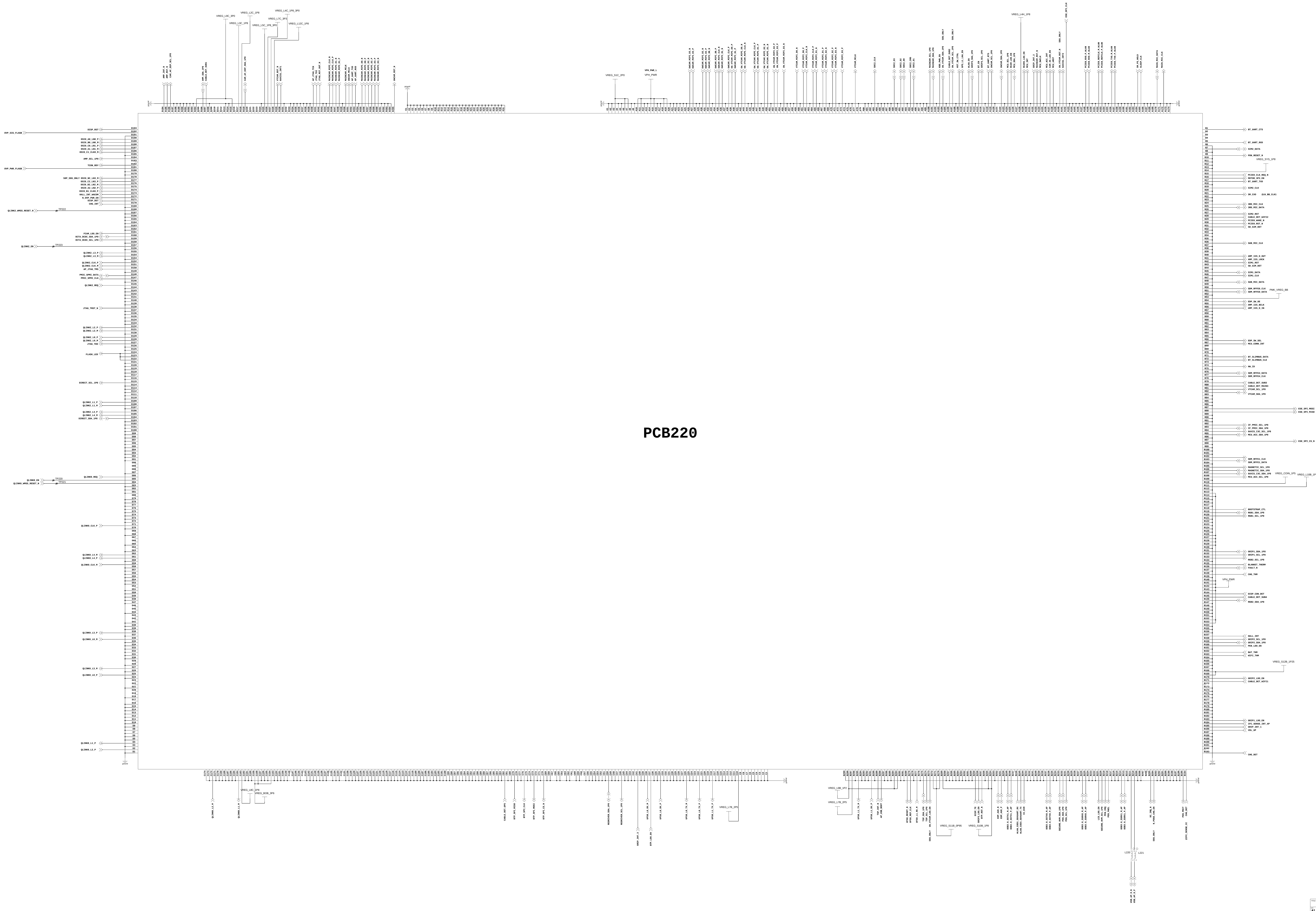
A1
PDM NUMBER
Rev
User Name
13 of 17
Drawing Date

COREPCB PAD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VREG_L2C_1P8
VREG_L3C_1P8
VREG_L4C_1P8_3P0
VREG_L7C_3P3
VREG_L5C_1P8_3P0
VREG_L12C_1P8
AMP_INT_N
AMP_RST_N
CAM_AF_EEP_SCL_1P8
AMP_SDA_1P8
CABLE_DET_SUB1
CAM_AF_EEP_SDA_1P8
VTCAM_RST_N
6AXIS1_INT1
AP_JTAG_TCK
JTAG_TDI
JTAG_RST_OUT_N
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MCLK
UWCAM_MCLK
AP_UART_TXD
AP_UART_RXD
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D3_N
UWCAM_RST_N
ground
VREG_S1C_2P0
VPH_PWR_L
VPH_PWR
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D0_P
UWCAM_MIPI_D1_P
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D3_N
VTCAM_MIPI_D3_P
VTCAM_MCLK
SDC2_CLK
SDC2_D3
SDC2_D2
SDC2_D0
SDC2_CMD
SDC2_D1
MAINCAM_SCL_1P8
MAINCAM_SDA_1P8
KBD_PWR_EN
S8U_ONLY
UW_VTCAM_SDA_1P8
CABLE_DET_SUB2
S8U_ONLY
UW_VTCAM_SCL_1P8
HST_SW_CTRL
GPS_L1_LNA_EN
WLAN_EN
GRIP3_SDA_1P8
BT_EN
GRIP3_SCL_1P8
BT_UART_RTS
UWCAM_SCL_1P8
UWCAM_SDA_1P8
MCU_SCL_1P8
RCAM_LDO_EN
MCU_SDA_1P8
VREG_L4H_1P8
RCAM2_LDO_EN
MCU_INT
GRIP_INT_2
MAINCAM_RST_N
MCU_SWCLK
MCU_ACC_INT
CAM_IO_LDO_EN
MCU_NRST
S8U_ONLY
UW_VTCAM_RST_N
6AXIS1_INT2
ESE_SPI_CLK
PCIE0_RX0_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_TX0_M_WLAN
PCIE0_TX0_P_WLAN
SM_PS_HOLD
SLEEP_CLK
MAIN_MIC_DATA
MAIN_MIC_CLK
DISP_RST
OVP_SIG_FLAGB
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_C1_CLK0_M
AMP_SCL_1P8
TCON_RDY
OVP_PWR_FLAGB
S8P_S8U_ONLY DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_B2_LN2_M
DSI0_A2_LN2_P
DSI0_B1_CLK0_P
HALL_INT_WACOM
N_OVP_PWR_EN
DISP_DET
CHG_INT
QLINK2_WMSS_RESET_N
TP222
FCAM_LDO_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
QLINK2_EN
TP223
QLINK2_L3_P
QLINK2_L3_M
QLINK2_CLK_P
QLINK2_CLK_M
AP_JTAG_TMS
PMIC_SPMI_DATA
PMIC_SPMI_CLK
QLINK2_REQ
JTAG_TRST_N
QLINK2_L2_P
QLINK2_L2_M
QLINK2_L0_P
QLINK2_L0_M
JTAG_TDO
FLASH_LED
DIRECT_SCL_1P8
QLINK2_L1_P
QLINK2_L1_M
QLINK2_L4_P
QLINK2_L4_M
DIRECT_SDA_1P8
QLINK0_REQ
QLINK0_EN
TP220
QLINK0_WMSS_RESET_N
TP221
QLINK0_CLK_P
QLINK0_L4_M
QLINK0_L4_P
QLINK0_CLK_M
QLINK0_L3_P
QLINK0_L0_M
QLINK0_L3_M
QLINK0_L0_P
QLINK0_L1_P
QLINK0_L2_P
BT_UART_CTS
BT_UART_RXD
SIM2_DATA
PON_RESET_N
VREG_SYS_1P8
PCIE0_CLK_REQ_N
MOTOR_3P3_EN
BT_UART_TXD
SIM2_CLK
SM_CXO
@LN_BB_CLK1
3RD_MIC_CLK
3RD_MIC_DATA
SIM2_RST
CABLE_DET_WIFI2
PCIE0_WAKE_N
PCIE0_RST_N
SD_SIM_DET
SUB_MIC_CLK
AMP_I2S_D_OUT
AMP_I2S_LRCK
SIM1_RST
SD_SIM_DET
SIM1_DATA
SIM1_CLK
SUB_MIC_DATA
SDM_RFFE0_CLK
SDM_RFFE0_DATA
PMK_VREG_BB
EDP_SW_OE
AMP_I2S_BCLK
AMP_I2S_D_IN
EDP_SW_SEL
MCU_CONN_INT
BT_SLIMBUS_DATA
BT_SLIMBUS_CLK
HW_ID
SDM_RFFE4_DATA
SDM_RFFE4_CLK
CABLE_DET_SUB3
CABLE_DET_MAIN3
VTCAM_SCL_1P8
VTCAM_SDA_1P8
ESE_SPI_MOSI
ESE_SPI_MISO
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
6AXIS_I3C_SCL_1P8
MCU_ACC_SDA_1P8
ESE_SPI_CS_N
SDM_RFFE1_CLK
SDM_RFFE1_DATA
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
6AXIS_I3C_SDA_1P8
MCU_ACC_SCL_1P8
VREG_COIN_1P5
VREG_L10B_1P2
BOOTSTRAP_CTL
RGB1_SDA_1P8
RGB1_SCL_1P8
GRIP1_SDA_1P8
GRIP1_SCL_1P8
RGB2_SCL_1P8
BLANKET_THERM
FAULT_N
CHG_THM
VPH_PWR
DISP_CON_DET
CABLE_DET_SUB4
RGB2_SDA_1P8
HALL_INT
GRIP2_SCL_1P8
GRIP2_SDA_1P8
PEN_LDO_EN
BAT_THM
WIFI_THM
VREG_S12B_1P25
GRIP2_LDO_EN
CABLE_DET_WIFI1
GRIP1_LDO_EN
IFC_SENSE_INT_AP
GRIP_INT_1
VOL_UP
CHG_DET
QLINK0_L2_M
QLINK0_L1_M
VREG_L8C_1P8
VREG_BOB_3P6
CABLE_DET_GPS
BTP_SPI_MISO
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_CS_N
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
GRIP_INT_3
UFS0_L0_RX_M
BTP_LDO_EN
UFS0_L0_RX_P
UFS0_L0_TX_M
UFS0_L0_TX_P
UFS0_L1_TX_P
VREG_L7B_2P5
VREG_L9B_1P2
VREG_L7B_2P5
UFS0_L1_TX_M
UFS0_L1_RX_P
TSP_INT_N
AP_USB_BOOT
UFS0_RESET_N
UFS0_REF_CLK
UFS0_L1_RX_M
TSP_SDA_1P8
TSP_SCL_1P8
UW_VTCAM_LDO_EN
S8U_ONLY
VREG_S11B_0P95
VREG_S10B_1P8
DISP_TE
GRIP3_LDO_EN
BTP_RST_N
EDP_AUX_N
EDP_AUX_P
USB3.0_SSTX1_M_AP
USB3.0_SSTX1_P_AP
WLAN_COEX_SDXUART_RX
WLAN_COEX_SDXUART_TX
CC_DIR
USB3.0_SSTX0_M_AP
USB3.0_SSTX0_P_AP
SECURE_NVM_SDA_1P8
PEN_SDA_1P8
PEN_SCL_1P8
USB3.0_SSRX0_M_AP
USB3.0_SSRX0_P_AP
LCD_LDO_EN
SECURE_NVM_SCL_1P8
PEN_IRQ
PEN_FWEL
USB3.0_SSRX1_M_AP
USB3.0_SSRX1_P_AP
L220
L221
USB_AP_D_N
USB_AP_D_P
DC_IRQ_N
N_POGO_CHG_EN
S8U_ONLY
PEN_PDCT
CHG_DET
@IFC_SENSE_CC

TITLE TabS7P

SIZE A1 DWG NO PDM NUMBER REV Rev

DRAWN BY User Name SHEET 22 of 15 Date Drawing Date

TAB_S8_S_AP PCB_REV0.5A (Ultra, S8+, S8)

- COREPCB TYPE -

2021.11.24

sheet01 :

sheet02 :

sheet03 :

sheet04 :

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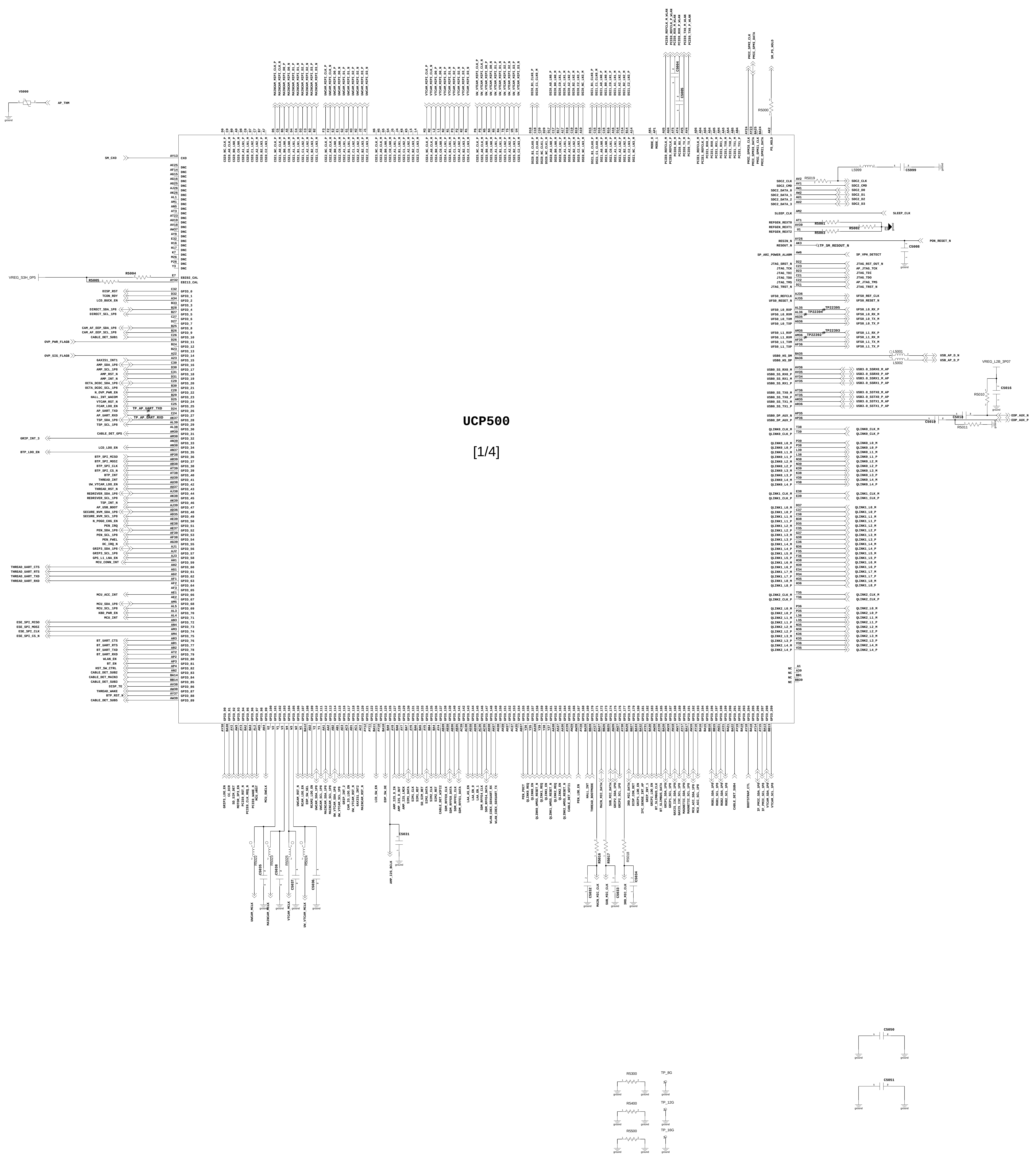
DRAWN BY User Name	SHEET 0 of 5	Date Drawing Date	


UCP500
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V5000
AP_THM
ground
SM_CXO
AY13
CXO
DNC
VREG_S3H_0P5
R5004
R5005
EBI02_CAL
EBI13_CAL
DISP_RST
TCON_RDY
LCD_BUCK_EN
DIRECT_SDA_1P8
DIRECT_SCL_1P8
CAM_AF_EEP_SDA_1P8
CAM_AF_EEP_SCL_1P8
CABLE_DET_SUB1
OVP_PWR_FLAGB
OVP_SIG_FLAGB
6AXIS1_INT1
AMP_SDA_1P8
AMP_SCL_1P8
AMP_RST_N
AMP_INT_N
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
N_OVP_PWR_EN
HALL_INT_WACOM
VTCAM_RST_N
FCAM_LDO_EN
AP_UART_TXD
AP_UART_RXD
TP_AP_UART_TXD
TP_AP_UART_RXD
TSP_SDA_1P8
TSP_SCL_1P8
CABLE_DET_GPS
GRIP_INT_3
LCD_LDO_EN
BTP_LDO_EN
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_SPI_CLK
BTP_SPI_CS_N
BTP_INT
THREAD_INT
UW_VTCAM_LDO_EN
THREAD_RST_N
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
TSP_INT_N
AP_USB_BOOT
SECURE_NVM_SDA_1P8
SECURE_NVM_SCL_1P8
N_POGO_CHG_EN
PEN_IRQ
PEN_SDA_1P8
PEN_SCL_1P8
PEN_FWEL
DC_IRQ_N
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GPS_L1_LNA_EN
MCU_CONN_INT
THREAD_UART_CTS
THREAD_UART_RTS
THREAD_UART_TXD
THREAD_UART_RXD
MCU_ACC_INT
MCU_SDA_1P8
MCU_SCL_1P8
KBD_PWR_EN
MCU_INT
ESE_SPI_MISO
ESE_SPI_MOSI
ESE_SPI_CLK
ESE_SPI_CS_N
BT_UART_CTS
BT_UART_RTS
BT_UART_TXD
BT_UART_RXD
WLAN_EN
BT_EN
HST_SW_CTRL
CABLE_DET_SUB2
CABLE_DET_MAIN3
CABLE_DET_SUB3
DISP_TE
THREAD_WAKE
BTP_RST_N
CABLE_DET_SUB5
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_CLK_N
UWCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_CLK_N
DSI0_B1_CLK0_P
DSI0_C1_CLK0_M
DSI1_B1_CLK0_P
DSI1_C1_CLK0_M
MODE_0
MODE_1
PCIE0_REFCLK_M_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_RX0_P_WLAN
PCIE0_TX0_M_WLAN
PCIE0_TX0_P_WLAN
C5004
C5005
PMIC_SPMI_CLK
PMIC_SPMI_DATA
SM_PS_HOLD
R5000
PS_HOLD
L5999
C5999
R5019
SDC2_CLK
SDC2_CMD
SDC2_DATA_0
SDC2_DATA_1
SDC2_DATA_2
SDC2_DATA_3
SDC2_D0
SDC2_D1
SDC2_D2
SDC2_D3
SLEEP_CLK
REFGEN_REXT0
REFGEN_REXT1
REFGEN_REXT2
R5001
R5002
R5003
RESIN_N
RESOUT_N
TP_SM_RESOUT_N
PON_RESET_N
C5008
SP_ARI_POWER_ALARM
SP_VPH_DETECT
JTAG_SRST_N
JTAG_TCK
JTAG_TDI
JTAG_TDO
JTAG_TMS
JTAG_TRST_N
JTAG_RST_OUT_N
AP_JTAG_TCK
AP_JTAG_TMS
UFS0_REFCLK
UFS0_RESET_N
UFS0_REF_CLK
UFS0_L0_RXP
UFS0_L0_RXM
UFS0_L0_TXM
UFS0_L0_TXP
TP22395
TP22394
TP22393
TP22392
UFS0_L1_RXP
UFS0_L1_RXM
UFS0_L1_TXM
UFS0_L1_TXP
USB0_HS_DM
USB0_HS_DP
L5001
L5002
USB_AP_D_N
USB_AP_D_P
USB0_SS_RX0_M
USB0_SS_RX0_P
USB0_SS_RX1_M
USB0_SS_RX1_P
USB3.0_SSRX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX1_M_AP
USB3.0_SSRX1_P_AP
USB0_SS_TX0_M
USB0_SS_TX0_P
USB0_SS_TX1_M
USB0_SS_TX1_P
USB3.0_SSTX0_M_AP
USB3.0_SSTX0_P_AP
USB3.0_SSTX1_M_AP
USB3.0_SSTX1_P_AP
VREG_L2B_3P07
R5010
C5016
USB0_DP_AUX_N
USB0_DP_AUX_P
C5018
C5019
R5011
EDP_AUX_N
EDP_AUX_P
QLINK0_CLK_M
QLINK0_CLK_P
QLINK1_CLK_M
QLINK1_CLK_P
QLINK2_CLK_M
QLINK2_CLK_P
NC
GRIP3_LDO_EN
CC_DIR
SD_SIM_DET
MOTOR_3P3_EN
PCIE0_RST_N
PCIE0_CLK_REQ_N
PCIE0_WAKE_N
MCU_NRST
MCU_SWCLK
UWCAM_RST_N
RCAM_LDO_EN
EDP_SW_SEL
UWCAM_SDA_1P8
UWCAM_SCL_1P8
MAINCAM_SDA_1P8
MAINCAM_SCL_1P8
UW_VTCAM_SDA_1P8
UW_VTCAM_SCL_1P8
GRIP_INT_2
CAM_IO_LDO_EN
UW_VTCAM_RST_N
6AXIS1_INT2
MAINCAM_RST_N
LCD_SW_EN
EDP_SW_OE
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_LRCK
AMP_I2S_BCLK
C5031
SIM1_DATA
SIM1_CLK
SIM1_RST
SD_SIM_DET
SIM2_DATA
SIM2_CLK
SIM2_RST
CABLE_DET_WIFI2
SDM_RFFE0_CLK
SDM_RFFE0_DATA
SDM_RFFE1_CLK
SDM_RFFE1_DATA
LAA_AS_EN
LAA_EN_0
LAA_EN_1
SDM_RFFE4_CLK
SDM_RFFE4_DATA
WLAN_COEX_SDXUART_RX
WLAN_COEX_SDXUART_TX
UWCAM_MCLK
MAINCAM_MCLK
VTCAM_MCLK
UW_VTCAM_MCLK
R5022
R5023
R5025
R5024
C5035
C5038
C5037
C5036
PEN_PDCT
QLINK0_REQ
QLINK0_EN
QLINK0_WMSS_RESET_N
QLINK1_REQ
QLINK1_EN
QLINK1_WMSS_RESET_N
QLINK2_REQ
QLINK2_EN
QLINK2_WMSS_RESET_N
CABLE_DET_WIFI1
PEN_LDO_EN
HALL_INT
THREAD_BOOTMODE
MAIN_MIC_DATA
SUB_MIC_DATA
GRIP2_SDA_1P8
GRIP2_SCL_1P8
3RD_MIC_DATA
DISP_CON_DET
GRIP1_LDO_EN
IFC_SENSE_INT_AP
GRIP_INT_1
GRIP2_LDO_EN
BT_SLIMBUS_CLK
BT_SLIMBUS_DATA
GRIP1_SDA_1P8
GRIP1_SCL_1P8
6AXIS_I3C_SDA_1P8
6AXIS_I3C_SCL_1P8
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
RGB1_SDA_1P8
RGB1_SCL_1P8
RGB2_SDA_1P8
RGB2_SCL_1P8
CABLE_DET_SUB4
BOOTSTRAP_CTL
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
VTCAM_SDA_1P8
VTCAM_SCL_1P8
R5016
R5017
R5018
C5032
C5033
C5034
MAIN_MIC_CLK
SUB_MIC_CLK
3RD_MIC_CLK
C5050
C5051
R5300
R5400
R5500
TP_8G
TP_12G
TP_16G
K
J
I
H
G
F
E
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UCP500
[2/4]
UCP500
[3/4]
UCP500
[4/4]
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VREG_L4C_1P8_3P0
VREG_S10B_1P8
VREG_L6C_1P8
VREG_S2C_3C_0P5
VREG_S6H_0P82
VREG_S4H_5H_0P75
VREG_S1H_0P75
VREG_S1B_2B_3B_4B_0P75
VREG_S5B_6B_7B_8B_9B_0P67
VREG_L8B_0P75
VREG_L1H_0P75
VREG_L5B_0P88
VREG_L1C_1P8
VREG_L2B_3P07
VREG_L1B_0P91
VREG_L6B_1P2
VREG_S2H_0P95
VREG_L2H_0P91
VREG_L3H_0P91
VREG_S4C_5C_0P75
VREG_S6C_7C_8C_9C_0P82
VREG_S3H_0P5
VREG_S10C_1P05
VREG_L3B_0P9
VDD_MM
VDD_MX_A
VDD_MX_C
VDD_PX3
VDD_APC0
VDD_APC1
VDD_GFX
VDD_LPI_CX
VDD_LPI_MX
VDD_LPI_MX_NAV
VDD_QFPROM
VDD_QFPROM_SP
VDD_A_USB_HS_0P9
VDD_A_USB_HS_1P8
VDD_A_USB_HS_3P1
VDD_A_USB_SS_DP_0P9
VDD_A_USB_SS_DP_1P2
VDD_A_UFS_0_0P9
VDD_A_UFS_0_1P2
VDD_CX
VDD_D_EBI_0
VDD_D_EBI_1
VDD_D_EBI_2
VDD_D_EBI_3
VDD_IO_EBI_0
VDD_IO_EBI_1
VDD_IO_EBI_2
VDD_IO_EBI_3
LPDDR5_VDD1
LPDDR5_VDDQ
LPDDR5_VDD2H
LPDDR5_VDD2L
LPDDR5_ZQ_A_C
LPDDR5_ZQ_B_D
VDD_MODEM
GND
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C6001
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R6002

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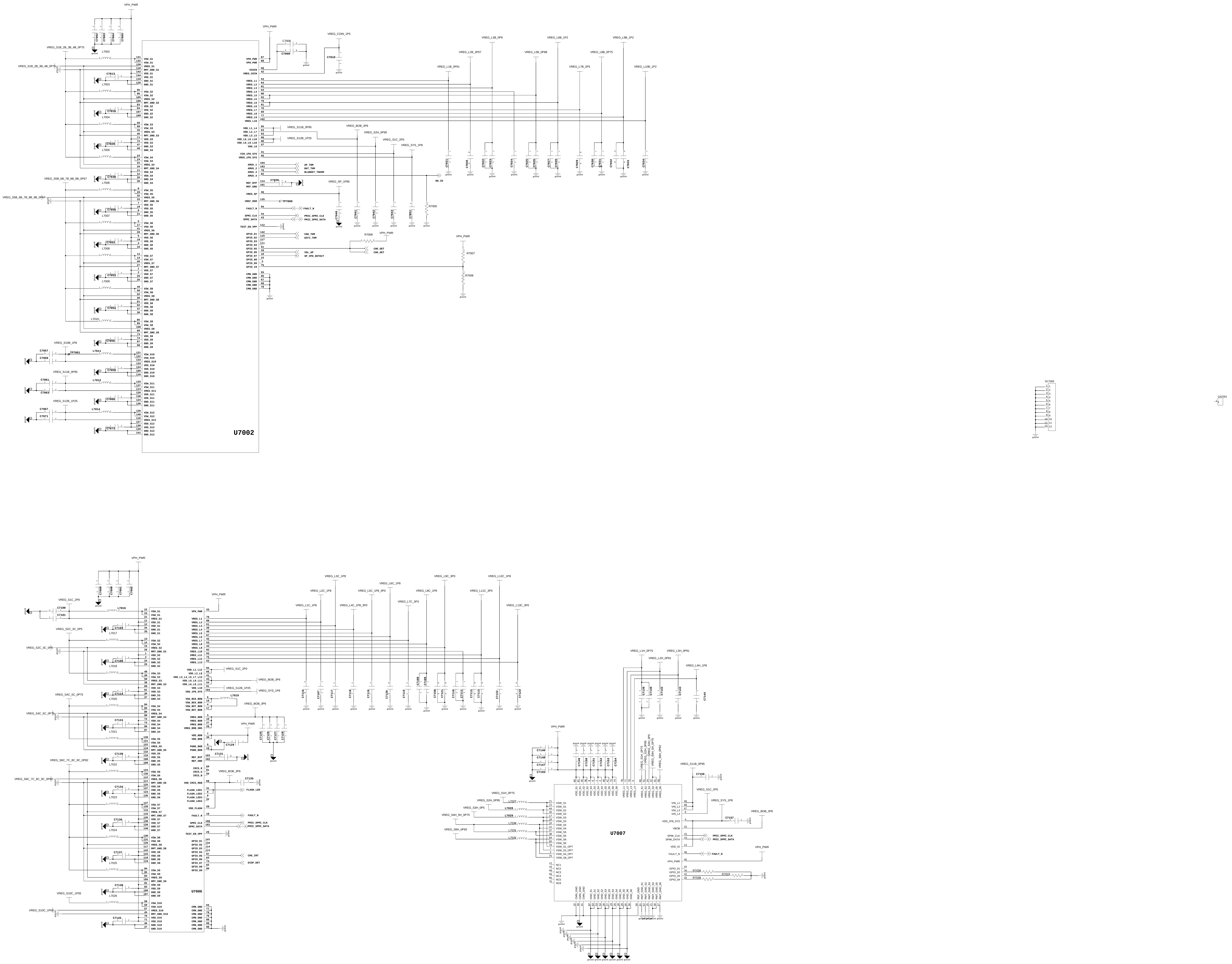
Date Drawing Date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VREG_S12B_1P25
VREG_COIN_1P5
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VREG_L2B_3P07
VREG_L3B_0P9
VREG_L5B_0P88
VREG_L6B_1P2
VREG_L7B_2P5
VREG_L8B_0P75
VREG_L9B_1P2
VREG_L10B_1P2
VREG_BOB_3P6
VREG_S2H_0P95
VREG_S1C_2P0
VREG_SYS_1P8
VREG_SP_1P85
AP_THM
BAT_THM
BLANKET_THERM
HW_ID
FAULT_N
PMIC_SPMI_CLK
PMIC_SPMI_DATA
CHG_THM
WIFI_THM
CHG_DET
VOL_UP
SP_VPH_DETECT
TP7000
TP7001
R7005
R7006
R7007
R7008
U7006
VREG_S1C_2P0
VREG_S2C_3C_0P5
VREG_S4C_5C_0P75
VREG_S6C_7C_8C_9C_0P82
VREG_S10C_1P05
VREG_L1C_1P8
VREG_L2C_1P8
VREG_L3C_1P8
VREG_L4C_1P8_3P0
VREG_L5C_1P8_3P0
VREG_L6C_1P8
VREG_L7C_3P3
VREG_L8C_1P8
VREG_L9C_3P0
VREG_L11C_3P0
VREG_L12C_1P8
VREG_L13C_3P0
VREG_S12B_1P25
FLASH_LED
CHG_INT
DISP_DET
U7007
VREG_S1H_0P75
VREG_S2H_0P95
VREG_S3H_0P5
VREG_S4H_5H_0P75
VREG_S6H_0P82
VREG_L1H_0P75
VREG_L2H_0P91
VREG_L3H_0P91
VREG_L4H_1P8
VREG_S11B_0P95
R7036
R7037
R7038
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SIZE A1 DWG NO REV

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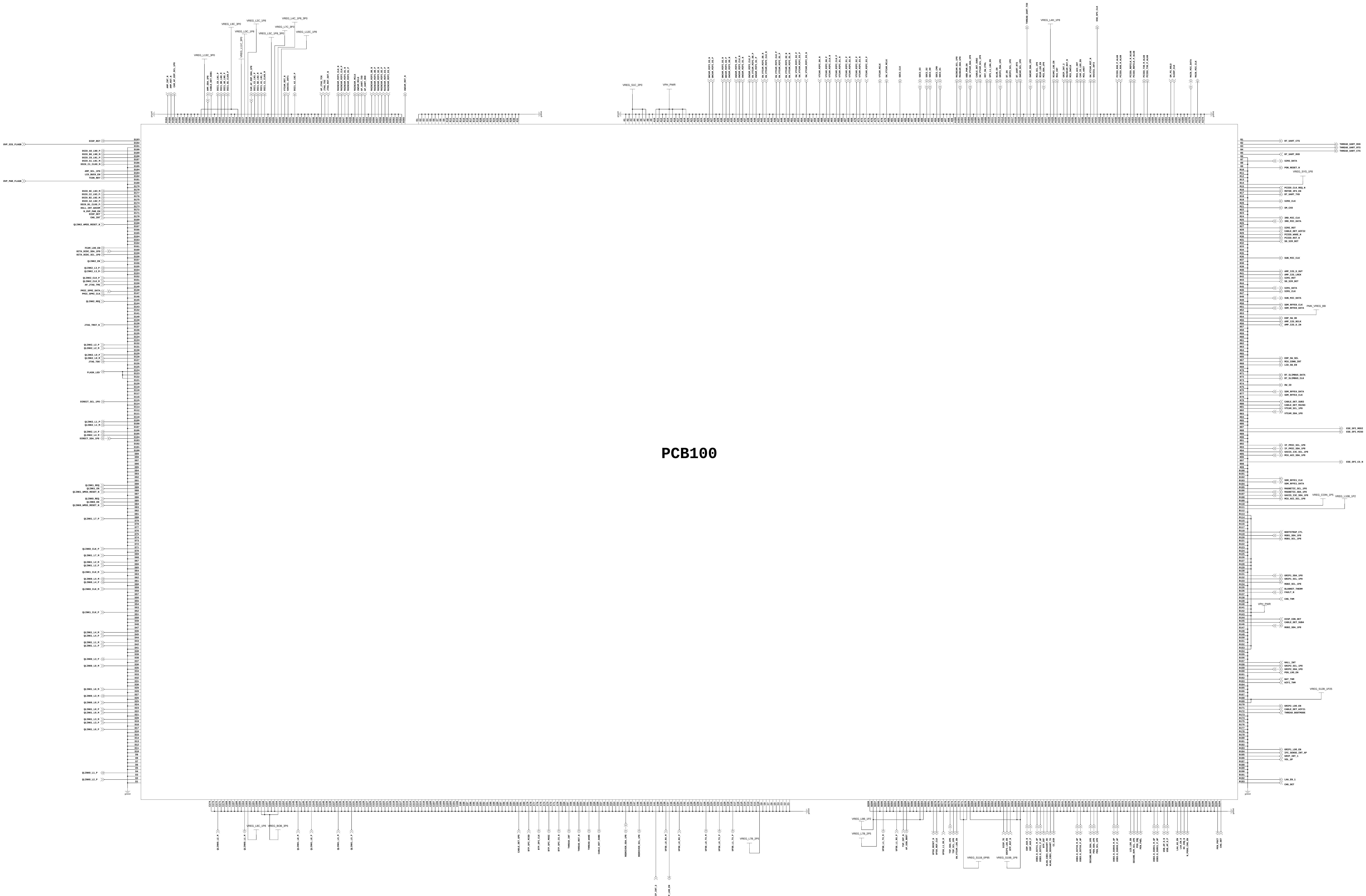
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AMP_SDA_1P8
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DSI1_C0_LN1_P
DSI1_B0_LN0_M
DSI1_A0_LN0_P
DSI1_B1_CLK0_P
VREG_L9C_3P0
VREG_L11C_3P0
VREG_L3C_1P8
VREG_L2C_1P8
CAM_AF_EEP_SDA_1P8
DSI1_C2_LN3_P
DSI1_B2_LN2_M
DSI1_A1_LN1_M
DSI1_C1_CLK0_M
VREG_L5C_1P8_3P0
VREG_L7C_3P3
VREG_L4C_1P8_3P0
VREG_L12C_1P8
VTCAM_RST_N
6AXIS1_INT1
DSI1_A2_LN2_P
AP_JTAG_TCK
JTAG_TDI
JTAG_RST_OUT_N
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MCLK
UWCAM_MCLK
AP_UART_TXD
AP_UART_RXD
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D3_N
UWCAM_RST_N
ground
VREG_S1C_2P0
VPH_PWR
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D0_P
UWCAM_MIPI_D1_P
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
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VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D3_N
VTCAM_MIPI_D3_P
VTCAM_MCLK
UW_VTCAM_MCLK
SDC2_CLK
SDC2_D3
SDC2_D2
SDC2_D0
SDC2_CMD
SDC2_D1
MAINCAM_SCL_1P8
MAINCAM_SDA_1P8
KBD_PWR_EN
UW_VTCAM_SDA_1P8
CABLE_DET_SUB2
UW_VTCAM_SCL_1P8
HST_SW_CTRL
GPS_L1_LNA_EN
WLAN_EN
GRIP3_SDA_1P8
BT_EN
GRIP3_SCL_1P8
BT_UART_RTS
UWCAM_SCL_1P8
THREAD_UART_TXD
UWCAM_SDA_1P8
MCU_SCL_1P8
RCAM_LDO_EN
MCU_SDA_1P8
VREG_L4H_1P8
RCAM2_LDO_EN
MCU_INT
GRIP_INT_2
MAINCAM_RST_N
MCU_SWCLK
MCU_ACC_INT
CAM_IO_LDO_EN
MCU_nRST
UW_VTCAM_RST_N
6AXIS1_INT2
ESE_SPI_CLK
PCIE0_RX0_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_TX0_M_WLAN
PCIE0_TX0_P_WLAN
SM_PS_HOLD
SLEEP_CLK
MAIN_MIC_DATA
MAIN_MIC_CLK
DISP_RST
OVP_SIG_FLAGB
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_C1_CLK0_M
AMP_SCL_1P8
LCD_BUCK_EN
TCON_RDY
OVP_PWR_FLAGB
DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_B2_LN2_M
DSI0_A2_LN2_P
DSI0_B1_CLK0_P
HALL_INT_WACOM
N_OVP_PWR_EN
DISP_DET
CHG_INT
QLINK2_WMSS_RESET_N
FCAM_LDO_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
QLINK2_EN
QLINK2_L3_P
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QLINK2_CLK_M
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PMIC_SPMI_DATA
PMIC_SPMI_CLK
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QLINK2_L0_P
QLINK2_L0_M
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FLASH_LED
DIRECT_SCL_1P8
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QLINK2_L1_M
QLINK2_L4_P
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DIRECT_SDA_1P8
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QLINK1_L1_P
QLINK0_L3_P
QLINK0_L0_M
QLINK1_L6_M
QLINK0_L3_M
QLINK0_L0_P
QLINK1_L0_P
QLINK1_L0_M
QLINK1_L3_M
QLINK1_L3_P
QLINK1_L6_P
QLINK0_L1_P
QLINK0_L2_P
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BT_UART_CTS
THREAD_UART_RXD
THREAD_UART_RTS
THREAD_UART_CTS
BT_UART_RXD
SIM2_DATA
PON_RESET_N
VREG_SYS_1P8
PCIE0_CLK_REQ_N
MOTOR_3P3_EN
BT_UART_TXD
SIM2_CLK
SM_CXO
3RD_MIC_CLK
3RD_MIC_DATA
SIM2_RST
CABLE_DET_WIFI2
PCIE0_WAKE_N
PCIE0_RST_N
SD_SIM_DET
SUB_MIC_CLK
AMP_I2S_D_OUT
AMP_I2S_LRCK
SIM1_RST
SD_SIM_DET
SIM1_DATA
SIM1_CLK
SUB_MIC_DATA
SDM_RFFE0_CLK
SDM_RFFE0_DATA
PMK_VREG_BB
EDP_SW_OE
AMP_I2S_BCLK
AMP_I2S_D_IN
EDP_SW_SEL
MCU_CONN_INT
LCD_SW_EN
BT_SLIMBUS_DATA
BT_SLIMBUS_CLK
HW_ID
SDM_RFFE4_DATA
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CABLE_DET_SUB3
CABLE_DET_MAIN3
VTCAM_SCL_1P8
VTCAM_SDA_1P8
ESE_SPI_MOSI
ESE_SPI_MISO
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
6AXIS_I3C_SCL_1P8
MCU_ACC_SDA_1P8
ESE_SPI_CS_N
SDM_RFFE1_CLK
SDM_RFFE1_DATA
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
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MCU_ACC_SCL_1P8
VREG_COIN_1P5
VREG_L10B_1P2
BOOTSTRAP_CTL
RGB1_SDA_1P8
RGB1_SCL_1P8
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GRIP1_SCL_1P8
RGB2_SCL_1P8
BLANKET_THERM
FAULT_N
CHG_THM
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DISP_CON_DET
CABLE_DET_SUB4
RGB2_SDA_1P8
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GRIP2_SDA_1P8
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QLINK0_L2_M
QLINK0_L1_M
VREG_L8C_1P8
VREG_BOB_3P6
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QLINK1_L8_P
QLINK1_L5_M
QLINK1_L5_P
CABLE_DET_GPS
BTP_SPI_MISO
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_CS_N
THREAD_INT
THREAD_RST_N
THREAD_WAKE
CABLE_DET_SUB5
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
GRIP_INT_3
BTP_LDO_EN
UFS0_L0_RX_M
UFS0_L0_RX_P
UFS0_L0_TX_M
UFS0_L0_TX_P
UFS0_L1_TX_P
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VREG_L9B_1P2
VREG_L7B_2P5
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UFS0_L1_RX_P
TSP_INT_N
AP_USB_BOOT
UFS0_RESET_N
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UFS0_L1_RX_M
TSP_SDA_1P8
TSP_SCL_1P8
UW_VTCAM_LDO_EN
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GRIP3_LDO_EN
BTP_RST_N
VREG_S10B_1P8
EDP_AUX_N
EDP_AUX_P
USB3.0_SSTX1_M_AP
USB3.0_SSTX1_P_AP
BTP_INT
WLAN_COEX_SDXUART_RX
WLAN_COEX_SDXUART_TX
CC_DIR
USB3.0_SSTX0_M_AP
USB3.0_SSTX0_P_AP
SECURE_NVM_SDA_1P8
PEN_SDA_1P8
PEN_SCL_1P8
USB3.0_SSRX0_M_AP
USB3.0_SSRX0_P_AP
LCD_LDO_EN
SECURE_NVM_SCL_1P8
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USB_AP_D_N
USB_AP_D_P
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DC_IRQ_N
N_POGO_CHG_EN
PEN_PDCT
CHG_DET
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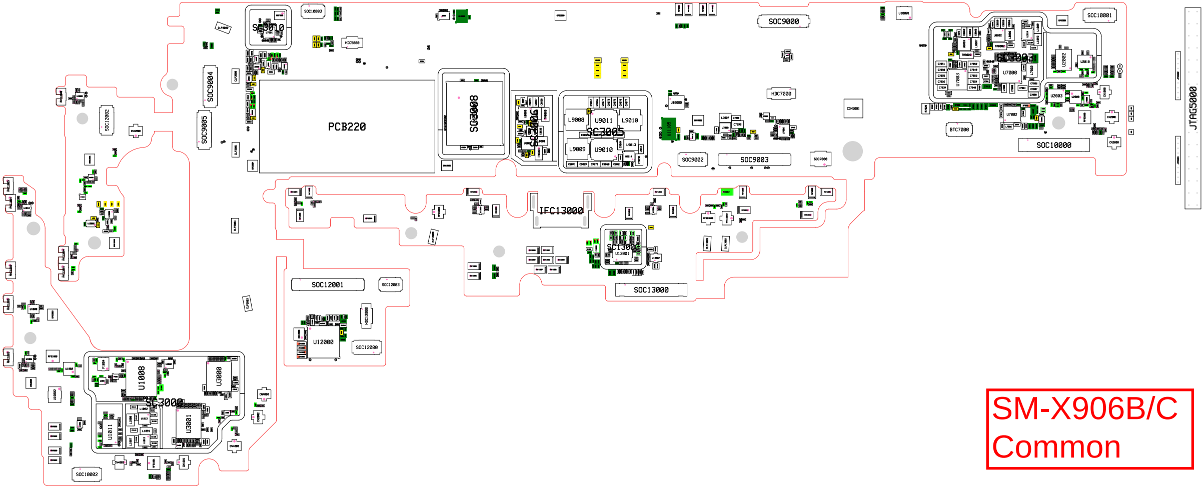
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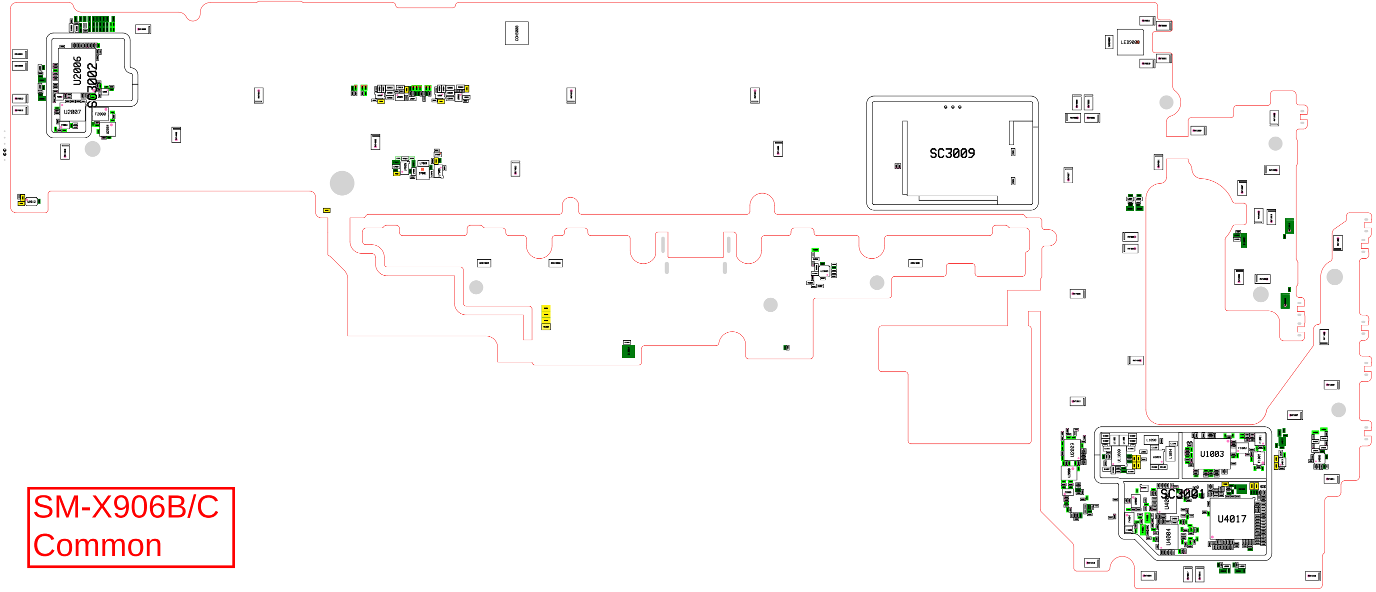
U1013
U1012
U1002
U10002
RFS1000
CN4000
CN4002
CN4003
CN1001
CN2002
CN2000
CN2001
SOC10002

SM-X906B/C
Common


U2006
SC3002
U2007
F2008
U2004
U9013
CON5000
L7009
D7001
SC3009
LED9000
U2009
U2008
F2005
L1090
U11000
U1015
L1094
U1003
F1003
F1001
F1005
SC3001
U4004
U4017

SM-X906B/C
Common

SM-X900_M_REV0.4A

SM-X900 EUR/CHN
Common

2021.11.04

sheet01 : RF (1/3)

sheet02 : RF (2/3)

sheet03 : RF (3/3)

sheet04 : BT / WIFI

sheet05 : AP_SM8450(Secondary PCB), JTAG, UFS

sheet06 : AP_SM8450(Secondary PCB)

sheet07 : PMIC(Secondary PCB), IF PMIC(CC IC), D/C

sheet08 : AUDIO

sheet09 : CAMERA / DISPLAY

sheet10 : SENSOR / SD_SIM / KEY / IF

sheet11 : CP PMIC

sheet12 : WACOM / ANT SUB PCB

sheet13 : USB PCB

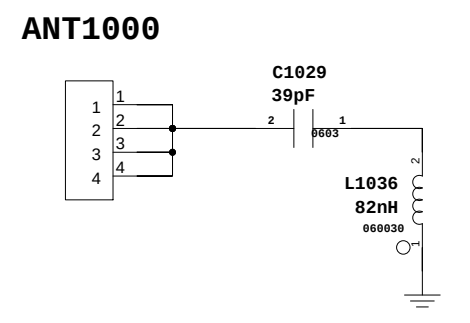
sheet14 : Secondary PCB PAD

TITLE TabS7P
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 0 of 14
Date Drawing Date

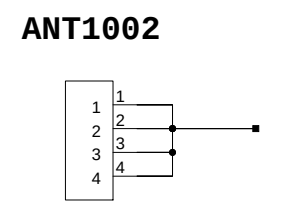
MAIN RF_1

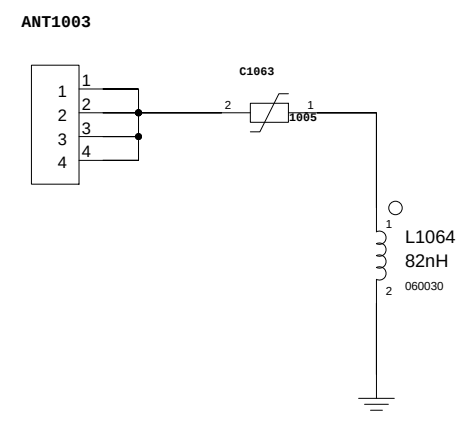
MAIN1 ANT
(TRX_LB/MB/HB / TRX_N41)

WIFI : NC



ANT1000
C1029
39pF
0603
L1036
82nH
060030





ANT1003
C1063
1005
L1064
82nH
060030

MAIN1 ANT XGND1

ENDC GRIP LDO

LB LPAMID

Global B5/8/12/13/17/20/26/28/29, G850/900
NA B5/8/12/13/14/20/26/28/29/71, G850/900

P17 P33

Need to check

B13/B14

Global : B8682(B13 only)
NA : SAYRH757MBA0C3K

REF	NA	EUR/CHN
C1021 C1024	100pF	OPEN

B28

Global only

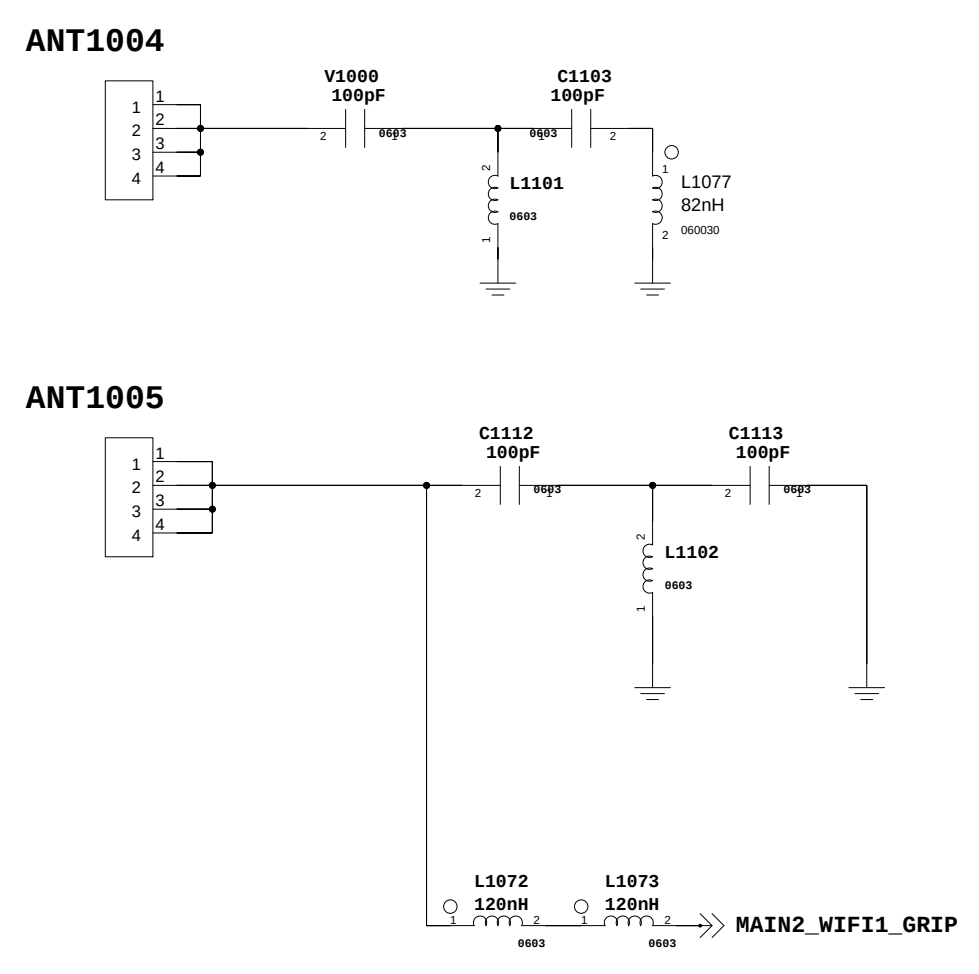
REF	NA	EUR/CHN
C1049 C1059	OPEN	100pF

WIFI : NC

MAIN2 ANT
(TRX_N77/N78)

WIFI : NC

close to PA



ANT1004
V1000
100pF
0603
C1103
100pF
0603
L1101
0603
L1077
82nH
060030
ANT1005
C1112
100pF
0603
C1113
100pF
0603
L1102
0603
L1072
120nH
0603
L1073
120nH
0603
MAIN2_WIFI1_GRIP

UHB PAMID
Global N77/78
NA B48/N77

OMH LPAMID

Global 1/2/3/4/7/25/34/38/39/40/41/66, G1800/1900, N41
NA 1/2/3/4/7/25/30/38/39/40/41/66, G1800/1900, N41

Global only

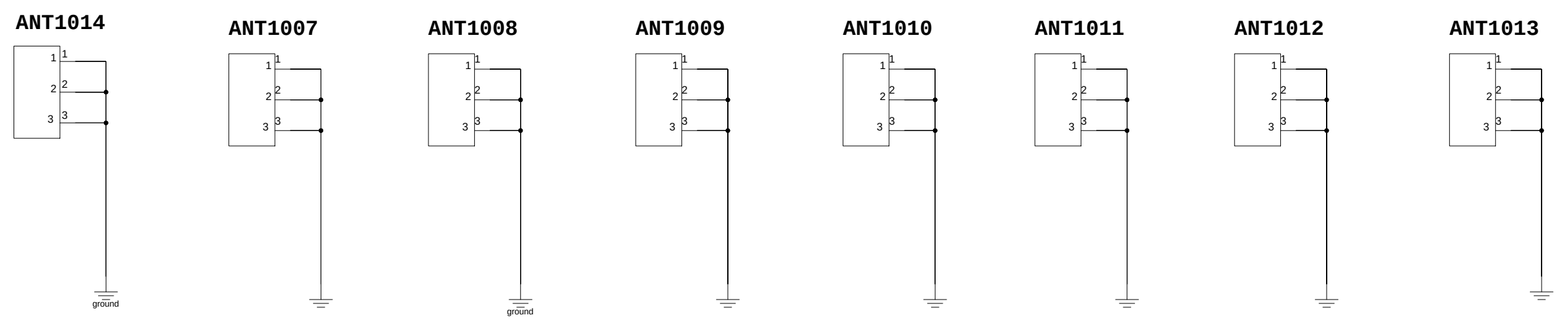
THERMISTERS

WIFI : NC

SUB3 ANT
(DRX MIMO_n77/78)

WIFI : NC

REAR CONTACT



ANT1014
ANT1007
ANT1008
ANT1009
ANT1010
ANT1011
ANT1012
ANT1013
ground

APT Modulator

LB APT Modulator
(QET6105#0)

MHB APT Modulator
(QET6105#1)

ENDC APT Modulator
(QET6105#2)

UHB APT Modulator
(QET6105#3)

IND pin#1 to switching pad

IND pin#1 to switching pad

IND pin#1 to switching pad

IND pin#1 to switching pad

IND pin#1 to switching pad

IND pin#1 to switching pad

IND pin#1 to switching pad

IND pin#1 to switching pad

WIFI : NC

TITLE TabS7P		
SIZE A1	DWG NO PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 1 of 12	Date Drawing Date

MAIN RF_2

WIFI : NC

SUB1 ANT
(DRX_LB/MB/HB / DRX_n41 / ENDC_LB/MB TX)

Close to QDM5877

to QDM5872

SUB2 ANT
(DRX_MIMO_MB/HB, DRX_n77/n78 / DRX_MIMO_n41)

Place neer QDM

SUB4 ANT

(PRX_MIMO_MB/HB / PRX_MIMO_n77/78 / PRX_MIMO_n41)

DRX_LB

(B5/8/12/13/14/17/18/19/20/26/28/29/71_BC0/10_G850/900)

DRX_MHB_2

(B2/4/7/25/30/41/66)

to SUB2 ANT

Global : SFMD6E0K001
NA : SFML8A0K001

Global : SFML6S0V001
NA : QDM3572

ENDC module

(B1/2/3/4/7/25/30/39/41/66_BC1_DCS/PCS)

B32 / n75
Global Only

Global Only

THERMISTERS

UHB

(n77/78)

to SUB4 ANT

Place holder for ESD protection

Global : SFML6S0V001
NA : QDM3572

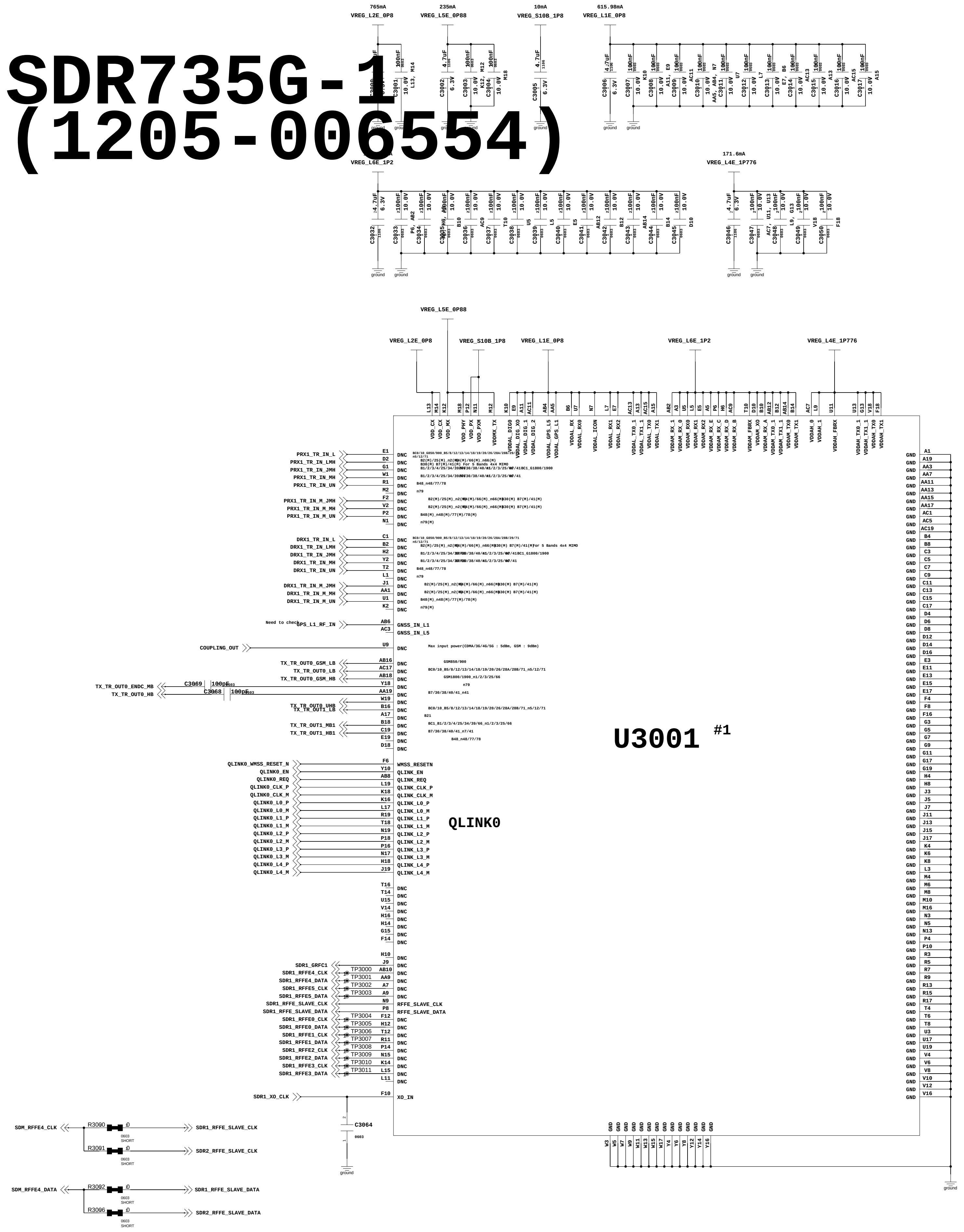
MHB MIMO_LNA OUT

TITLE	TabS7P	
SIZE A1	DWG NO PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 2 of 3	Date Drawing Date

MAIN RF_3

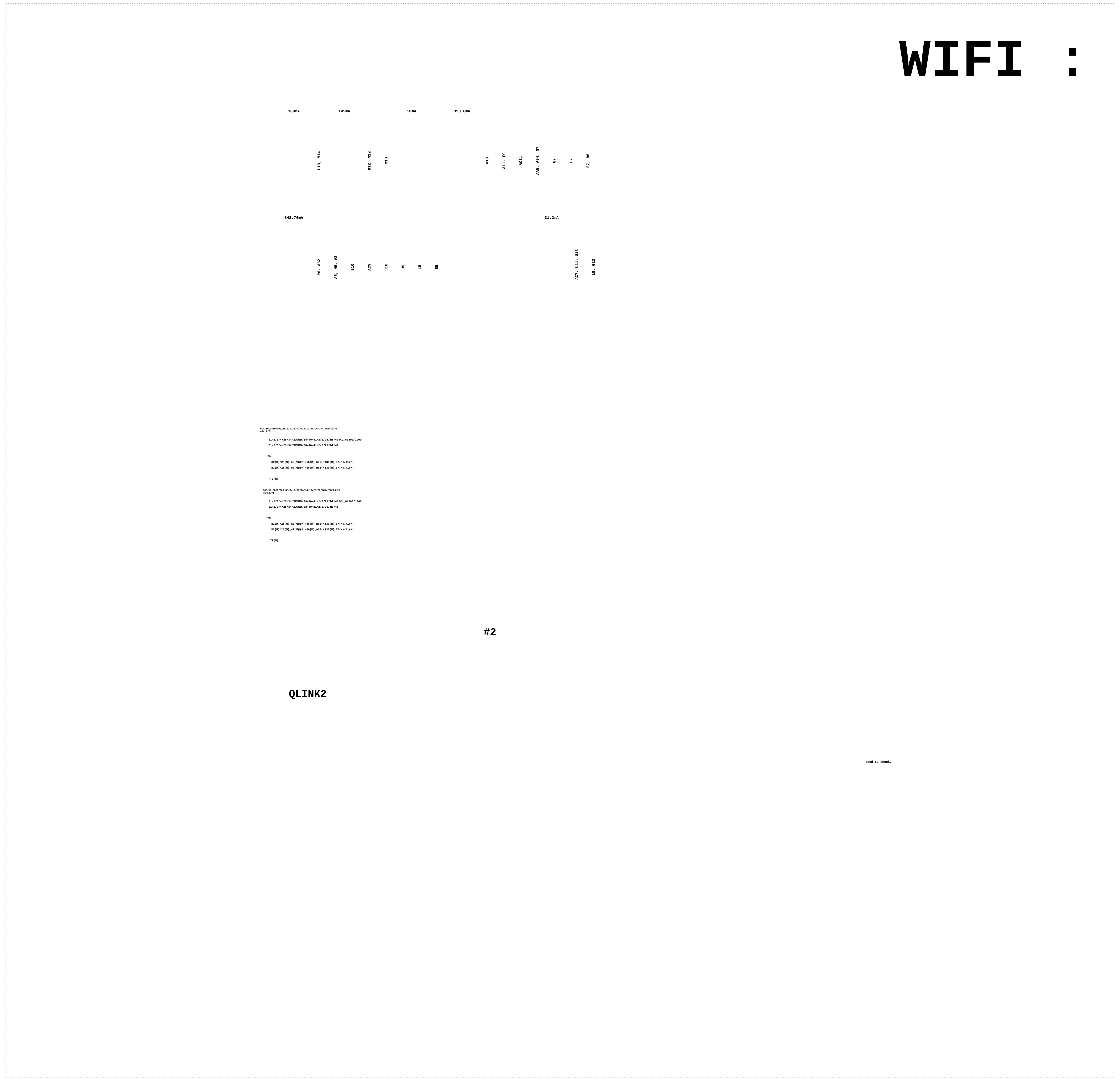
TRANSCEIVER(Dual SDR735)

WIFI : SDR735G-1
(1205-006554)

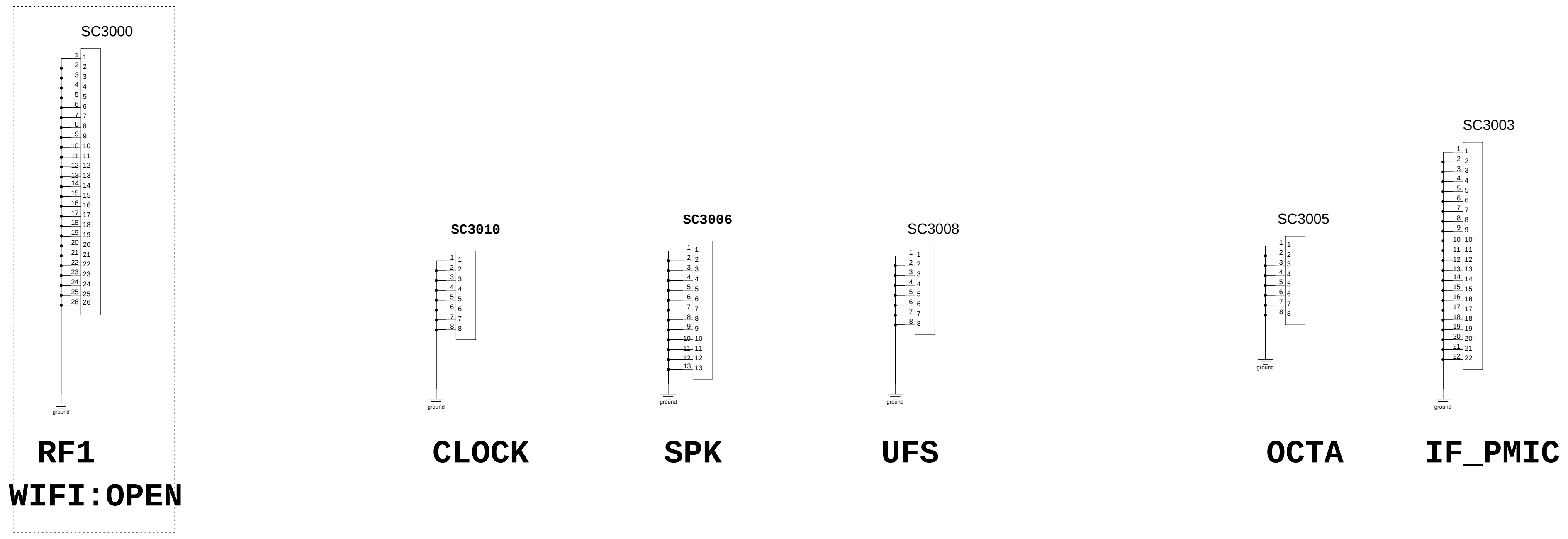


765mA
VREG_L2E_0P8
235mA
VREG_L5E_0P88
10mA
VREG_S10B_1P8
615.98mA
VREG_L1E_0P8
VREG_L6E_1P2
171.6mA
VREG_L4E_1P776
C3000
C3001
C3002
C3003
C3004
C3005
C3006
C3007
C3008
C3009
C3010
C3011
C3012
C3013
C3014
C3015
C3016
C3017
C3032
C3033
C3034
C3035
C3036
C3037
C3038
C3039
C3040
C3041
C3042
C3043
C3044
C3045
C3046
C3047
C3048
C3049
C3050
4.7uF
100nF
6.3V
10.0V
ground
VDD_CX
VDD_MX
VDD_PHY
VDD_PX
VDD_PXM
VDDMX_TX
VDDAL_DIG0
VDDAL_DIG_XO
VDDAL_DIG_1
VDDAL_DIG_2
VDDAL_GPS_L5
VDDAL_GPS_L1
VDDAL_RX
VDDAL_RX0
VDDAL_ICON
VDDAL_RX1
VDDAL_RX2
VDDAL_TX0_1
VDDAL_TX1_1
VDDAL_TX0
VDDAL_TX1
VDDAM_RX_1
VDDAM_RX_0
VDDAM_RX0
VDDAM_RX1
VDDAM_RX2
VDDAM_RX_E
VDDAM_RX_C
VDDAM_RX_D
VDDAM_RX_B
VDDAM_FBRX
VDDAM_XO
VDDAM_RX_A
VDDAM_TX0_1
VDDAM_TX1_1
VDDAM_TX0
VDDAM_TX1
VDDAH_0
VDDAH_1
VDDAH_FBRX
VDDAH_TX0_1
VDDAH_TX1_1
VDDAH_TX0
VDDAH_TX1
PRX1_TR_IN_L
PRX1_TR_IN_LMH
PRX1_TR_IN_JMH
PRX1_TR_IN_MH
PRX1_TR_IN_UN
PRX1_TR_IN_M_JMH
PRX1_TR_IN_M_MH
PRX1_TR_IN_M_UN
DRX1_TR_IN_L
DRX1_TR_IN_LMH
DRX1_TR_IN_JMH
DRX1_TR_IN_MH
DRX1_TR_IN_UN
DRX1_TR_IN_M_JMH
DRX1_TR_IN_M_MH
DRX1_TR_IN_M_UN
Need to check
GPS_L1_RF_IN
GNSS_IN_L1
GNSS_IN_L5
COUPLING_OUT
Max input power(CDMA/3G/4G/5G : 5dBm, GSM : 9dBm)
TX_TR_OUT0_GSM_LB
TX_TR_OUT0_LB
TX_TR_OUT0_GSM_HB
TX_TR_OUT0_ENDC_MB
TX_TR_OUT0_HB
C3069
C3068
100pF
TX_TR_OUT0_UHB
TX_TR_OUT1_LB
TX_TR_OUT1_MB1
TX_TR_OUT1_HB1
QLINK0_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_CLK_P
QLINK0_CLK_M
QLINK0_L0_P
QLINK0_L0_M
QLINK0_L1_P
QLINK0_L1_M
QLINK0_L2_P
QLINK0_L2_M
QLINK0_L3_P
QLINK0_L3_M
QLINK0_L4_P
QLINK0_L4_M
WMSS_RESETN
QLINK_EN
QLINK_REQ
QLINK_CLK_P
QLINK_CLK_M
QLINK_L0_P
QLINK_L0_M
QLINK_L1_P
QLINK_L1_M
QLINK_L2_P
QLINK_L2_M
QLINK_L3_P
QLINK_L3_M
QLINK_L4_P
QLINK_L4_M
QLINK0
SDR1_GRFC1
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
SDR1_RFFE_SLAVE_CLK
SDR1_RFFE_SLAVE_DATA
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
SDR1_RFFE1_CLK
SDR1_RFFE1_DATA
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
SDR1_RFFE3_CLK
SDR1_RFFE3_DATA
TP3000
TP3001
TP3002
TP3003
TP3004
TP3005
TP3006
TP3007
TP3008
TP3009
TP3010
TP3011
RFFE_SLAVE_CLK
RFFE_SLAVE_DATA
SDR1_XO_CLK
XO_IN
C3064
0603
SDM_RFFE4_CLK
R3000
R3001
0
0603
SHORT
SDR1_RFFE_SLAVE_CLK
SDR2_RFFE_SLAVE_CLK
SDM_RFFE4_DATA
R3002
R3006
SDR1_RFFE_SLAVE_DATA
SDR2_RFFE_SLAVE_DATA
DNC
GND
U3001
#1

WIFI : NC



360mA
145mA
10mA
393.6mA
L13, M14
K12, M12
M18
K10
A11, E9
AC11
AA5, AB4, N7
U7
L7
E7, B6
842.76mA
31.2mA
P6, AB2
A5, H6, A3
B10
AC9
D10
U5
L5
E5
AC7, U11, U13
L9, G13
#2
QLINK2
Need to check.



SC3000
SC3010
SC3006
SC3008
SC3005
SC3003
ground

RF1

WIFI:OPEN

CLOCK

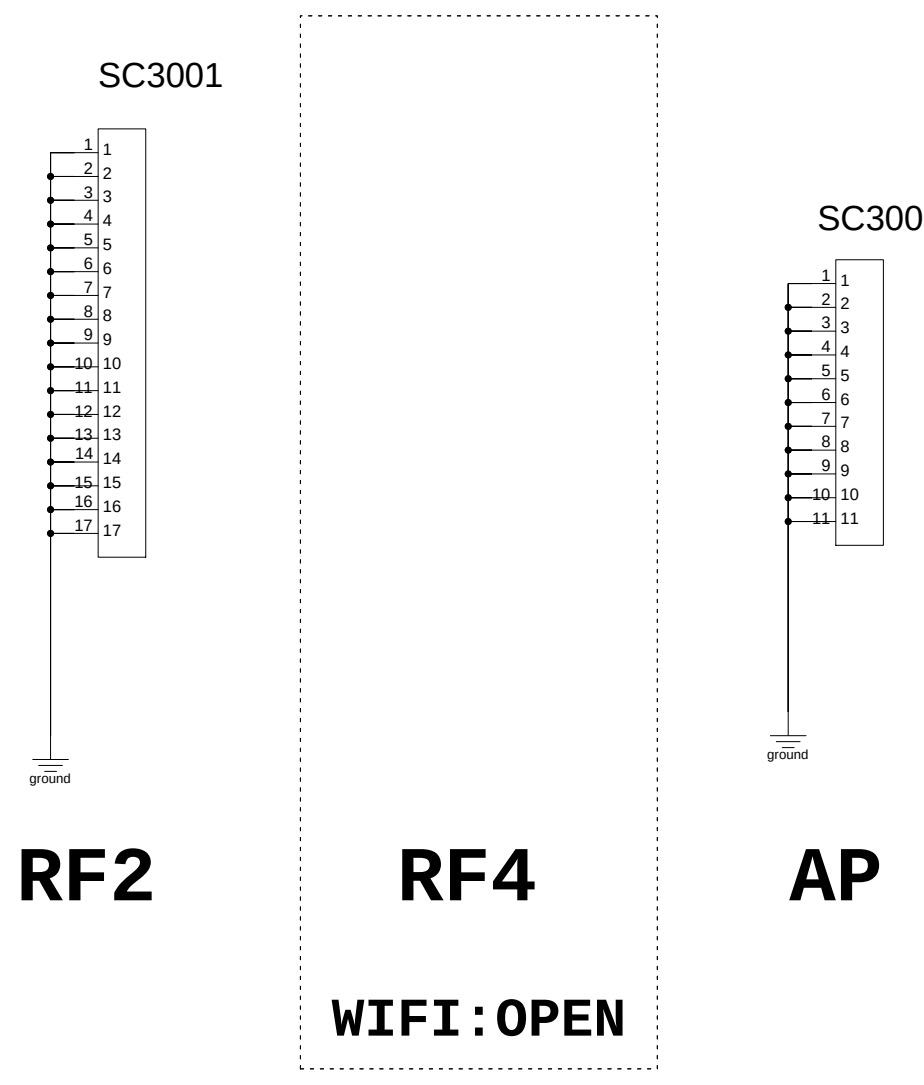
SPK

UFS

OCTA

IF_PMIC

TOP S/C



SC3001
SC3009
ground

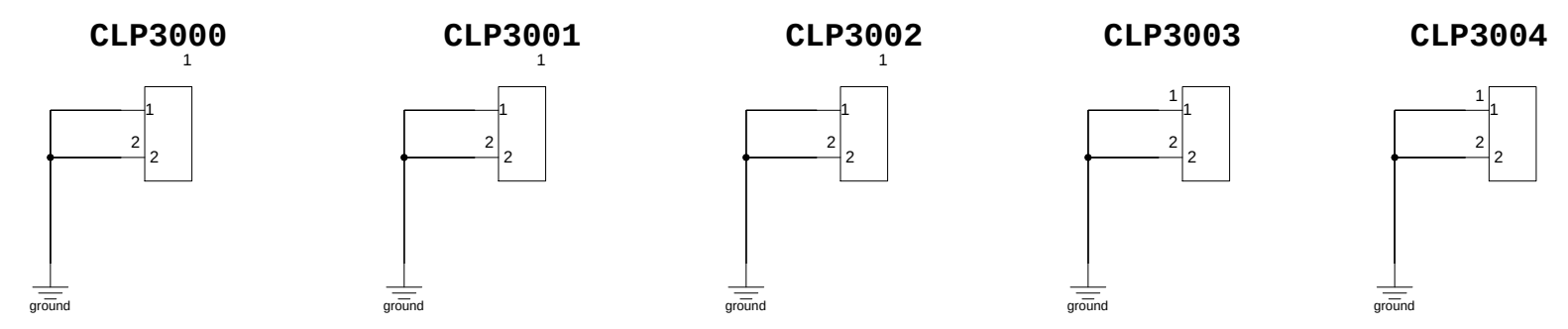
RF2

RF4

WIFI:OPEN

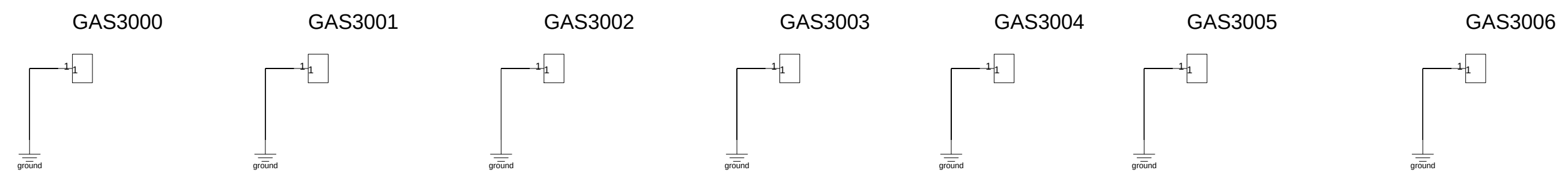
AP

BOTTOM S/C



CLP3000
CLP3001
CLP3002
CLP3003
CLP3004
1
2
ground

Cable CLIP



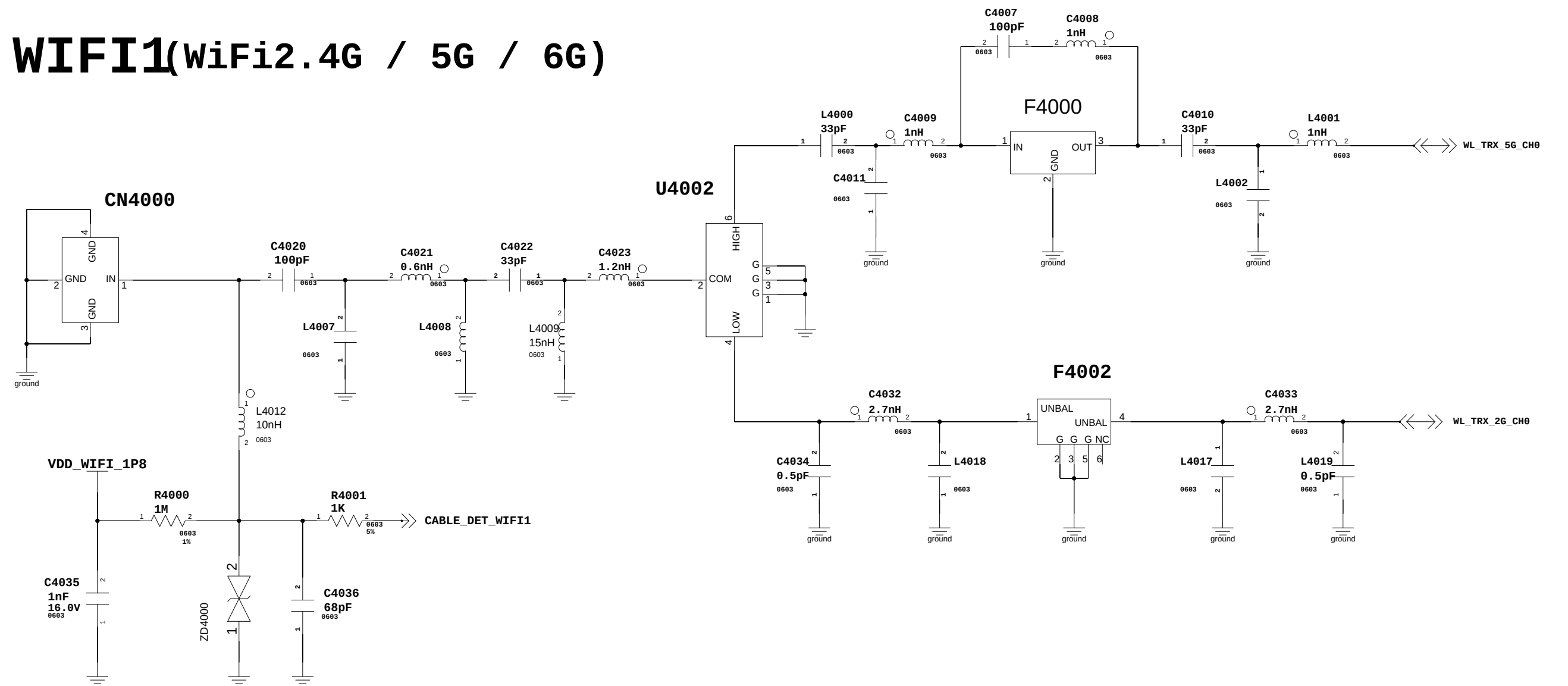
GAS3000
GAS3001
GAS3002
GAS3003
GAS3004
GAS3005
GAS3006
1
ground

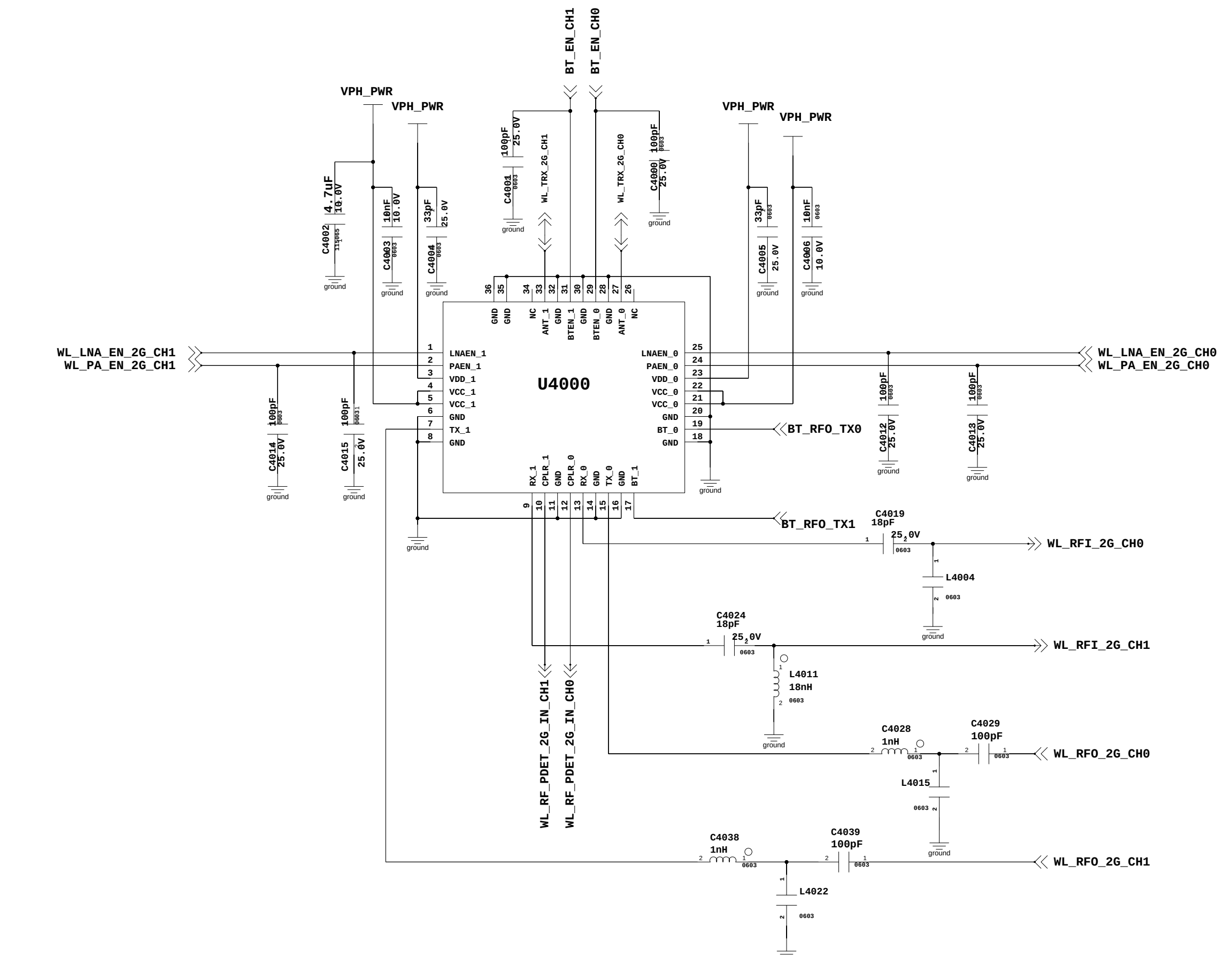
Front CONTACT

TITLE TabS7P
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name SHEET 3 of 16 Date Drawing Date

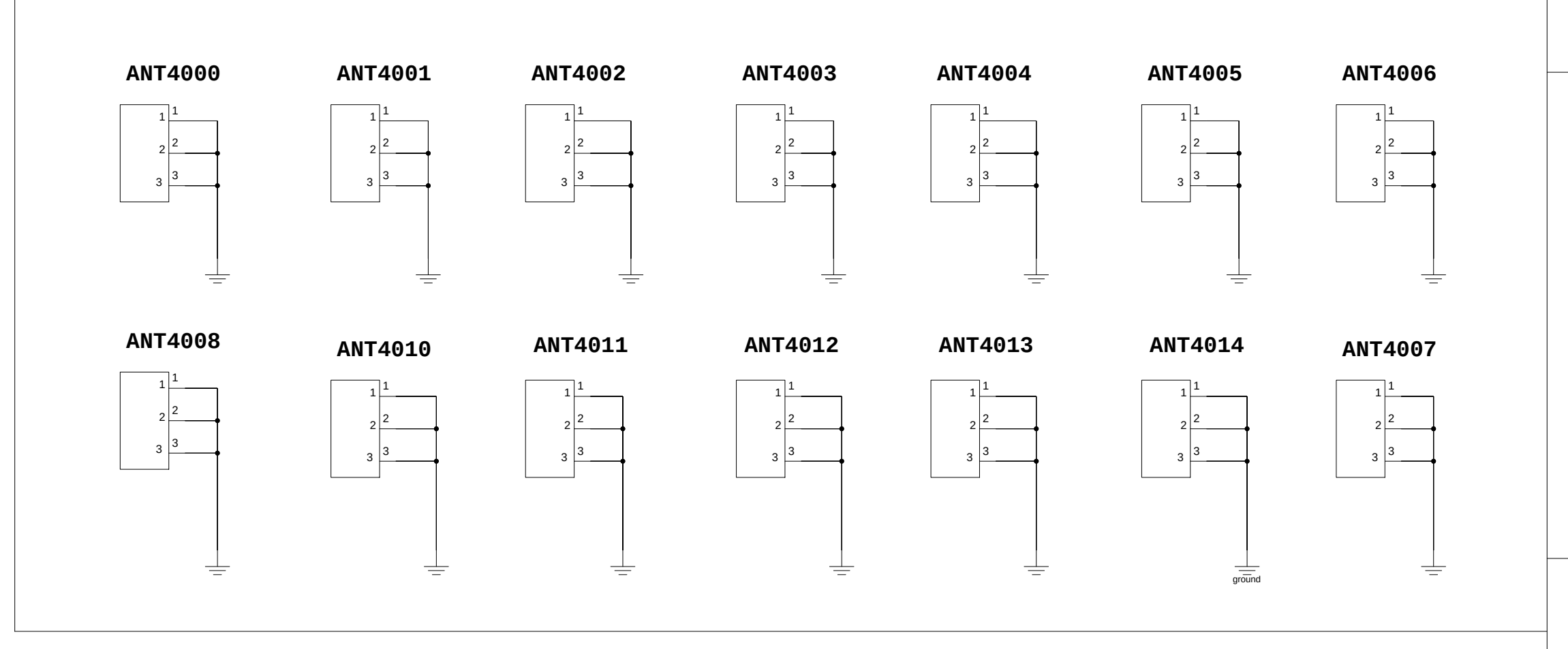
Connectivity

WIFI1(WiFi2.4G / 5G / 6G)

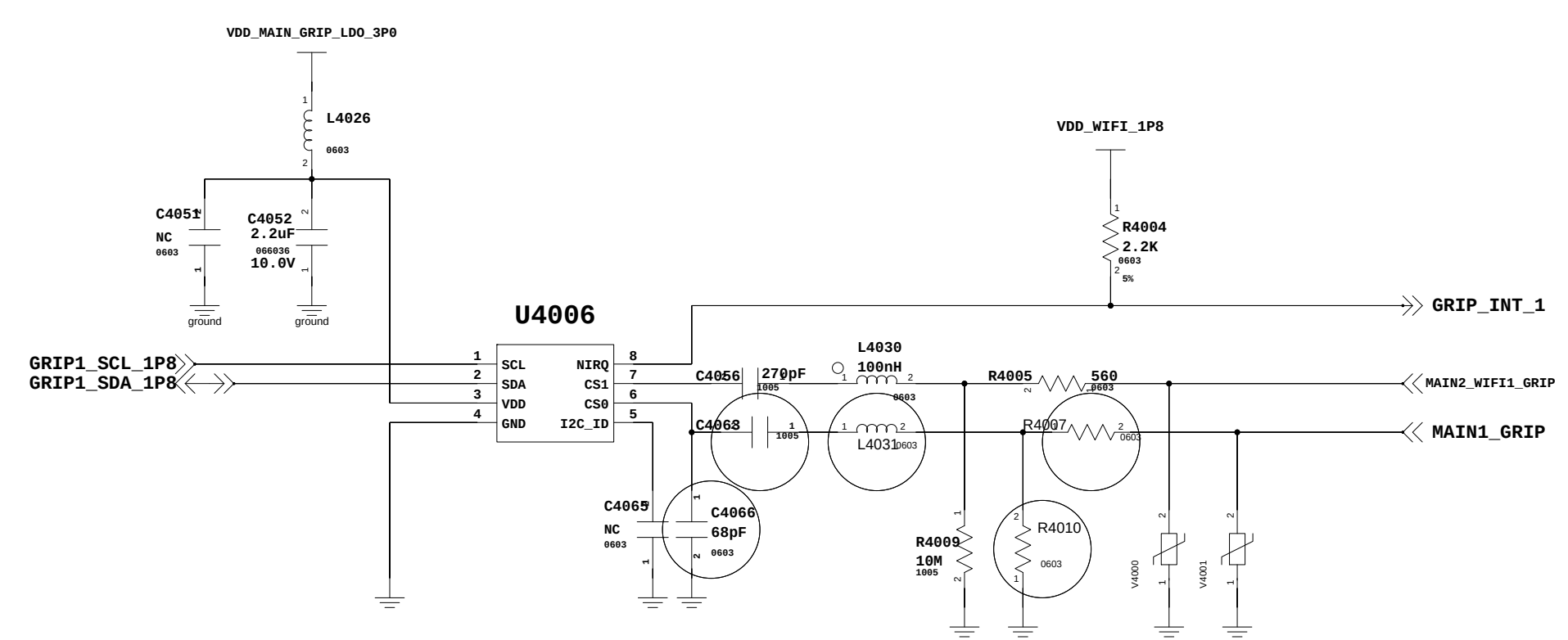

CN4000
C4020 100pF
C4021 0.6nH
C4022 33pF
C4023 1.2nH
L4007
L4008
L4009 15nH
L4012 10nH
U4002
HIGH
COM
LOW
C4007 100pF
C4008 1nH
L4000 33pF
C4009 1nH
F4000
IN
OUT
GND
C4010 33pF
L4001 1nH
C4011
L4002
WL_TRX_5G_CH0
F4002
UNBAL
C4032 2.7nH
C4033 2.7nH
C4034 0.5pF
L4018
L4017
L4019 0.5pF
WL_TRX_2G_CH0
VDD_WIFI_1P8
R4000 1M
R4001 1K
CABLE_DET_WIFI1
C4035 1nF 16.0V
ZD4000
C4036 68pF
ground


BT_EN_CH1
BT_EN_CH0
VPH_PWR
C4002 4.7uF 10.0V
C4003 10nF 10.0V
C4004 33pF 25.0V
C4001 100pF 25.0V
WL_TRX_2G_CH1
WL_TRX_2G_CH0
C4000 100pF 25.0V
C4005 33pF 25.0V
C4006 10nF 10.0V
U4000
WL_LNA_EN_2G_CH1
WL_PA_EN_2G_CH1
LNAEN_1
PAEN_1
VDD_1
VCC_1
TX_1
GND
LNAEN_0
PAEN_0
VDD_0
VCC_0
BT_0
WL_LNA_EN_2G_CH0
WL_PA_EN_2G_CH0
C4014 100pF 25.0V
C4015 100pF 25.0V
C4012 100pF 25.0V
C4013 100pF 25.0V
BT_RFO_TX0
BT_RFO_TX1
C4019 18pF 25.0V
L4004
WL_RFI_2G_CH0
C4024 18pF 25.0V
L4011 18nH
WL_RFI_2G_CH1
C4028 1nH
C4029 100pF
L4015
WL_RFO_2G_CH0
C4038 1nH
C4039 100pF
L4022
WL_RFO_2G_CH1
WL_RF_PDET_2G_IN_CH1
WL_RF_PDET_2G_IN_CH0

REAR CONTACT


ANT4000
ANT4001
ANT4002
ANT4003
ANT4004
ANT4005
ANT4006
ANT4008
ANT4010
ANT4011
ANT4012
ANT4013
ANT4014
ANT4007
ground

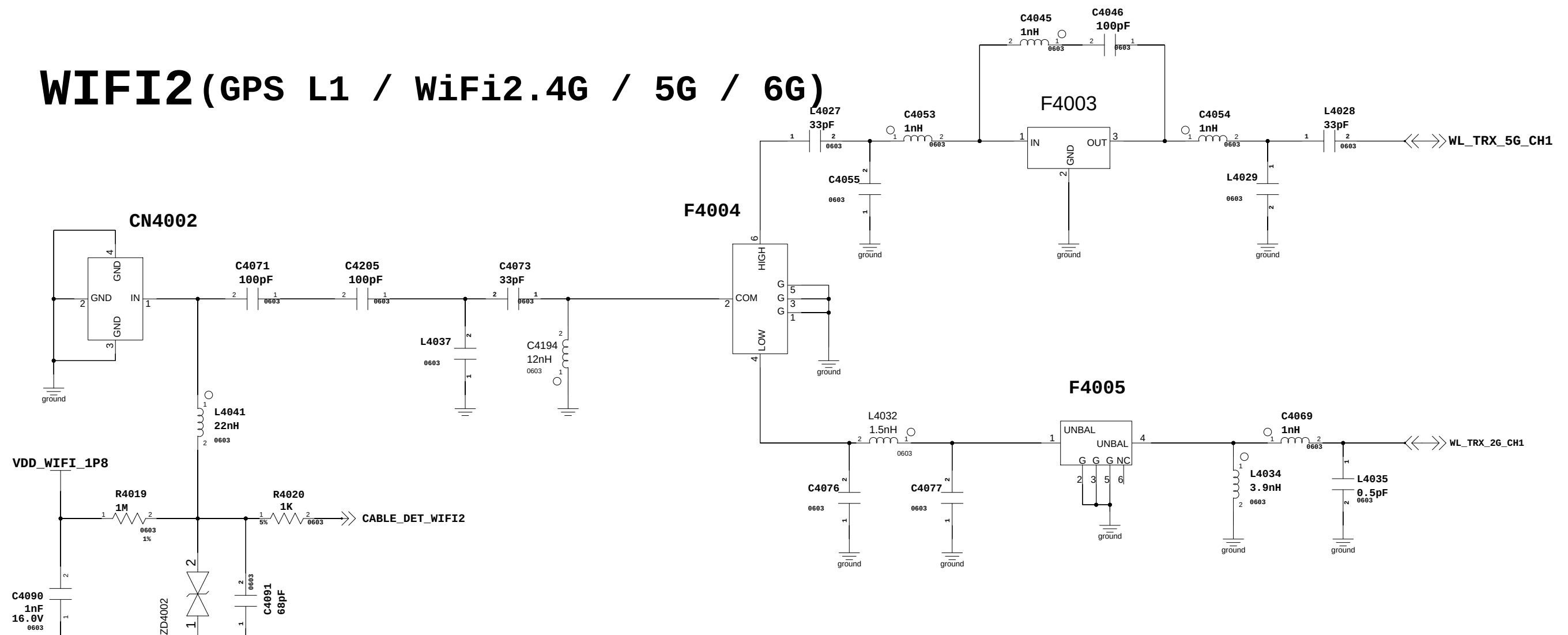
MAIN_GRIP_SENSOR(GRIP #2_#3)

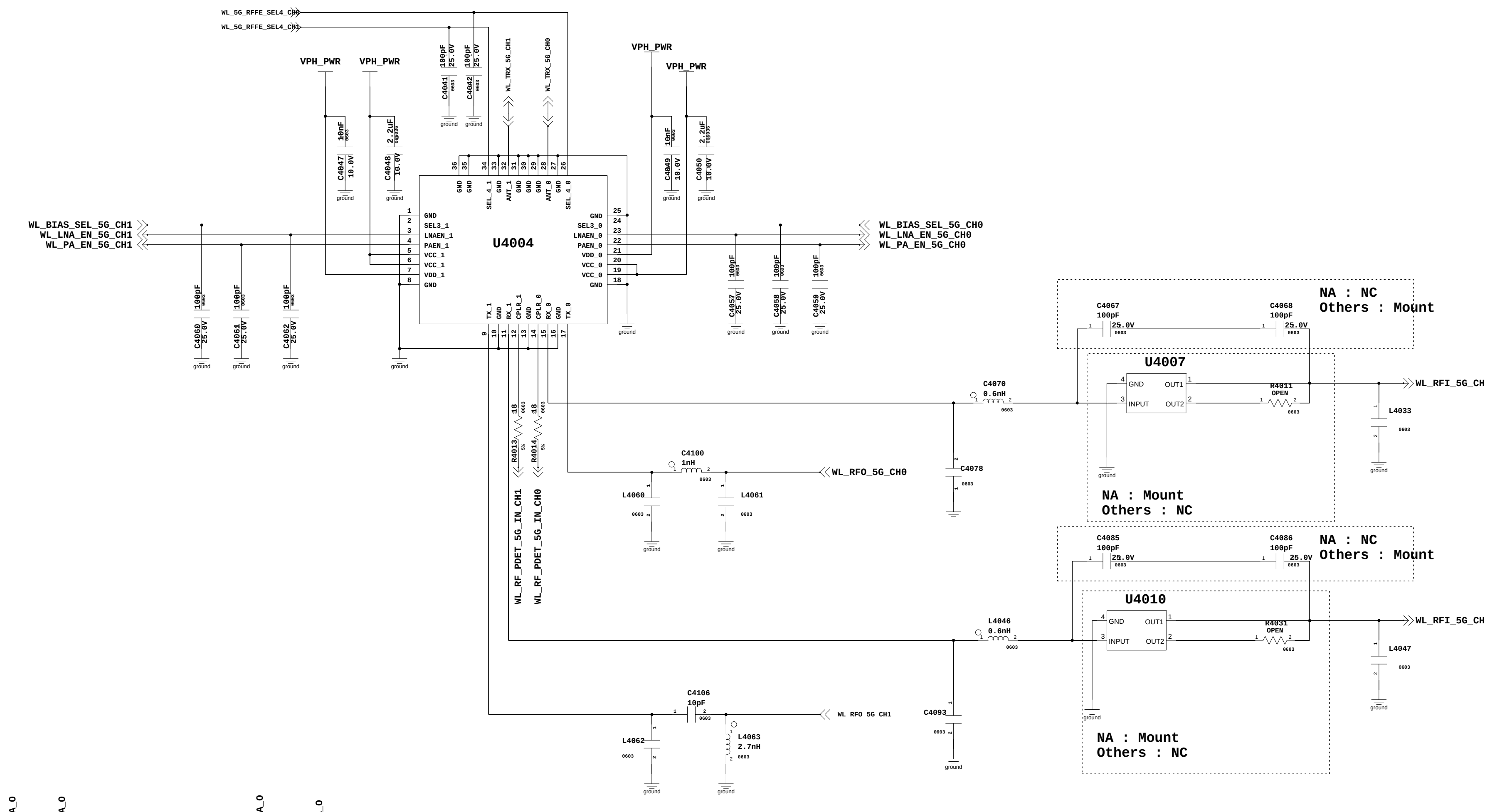

VDD_MAIN_GRIP_LDO_3P0
L4026
C4051 NC
C4052 2.2uF 10.0V
U4006
GRIP1_SCL_1P8
GRIP1_SDA_1P8
SCL
SDA
VDD
GND
NIRQ
CS1
CS0
I2C_ID
VDD_WIFI_1P8
R4004 2.2K
GRIP_INT_1
C4056 270pF
L4030 100nH
L4031
R4005 560
R4007
MAIN2_WIFI1_GRIP
C4068
MAIN1_GRIP
C4065 NC
C4066 68pF
R4009 10M
R4010
V4000
V4001

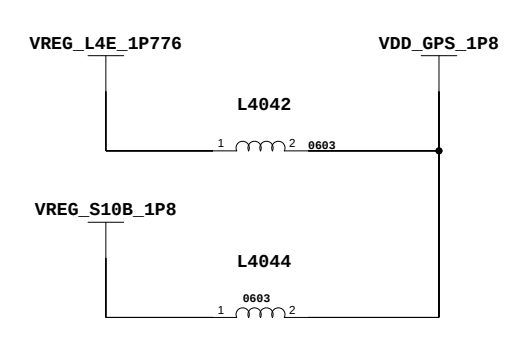
	Sub6	WIFI
C4063	O	X
L4031	O	X
R4010	O	X
R4007	O	X
C4066	X	68pF

WIFI : NC

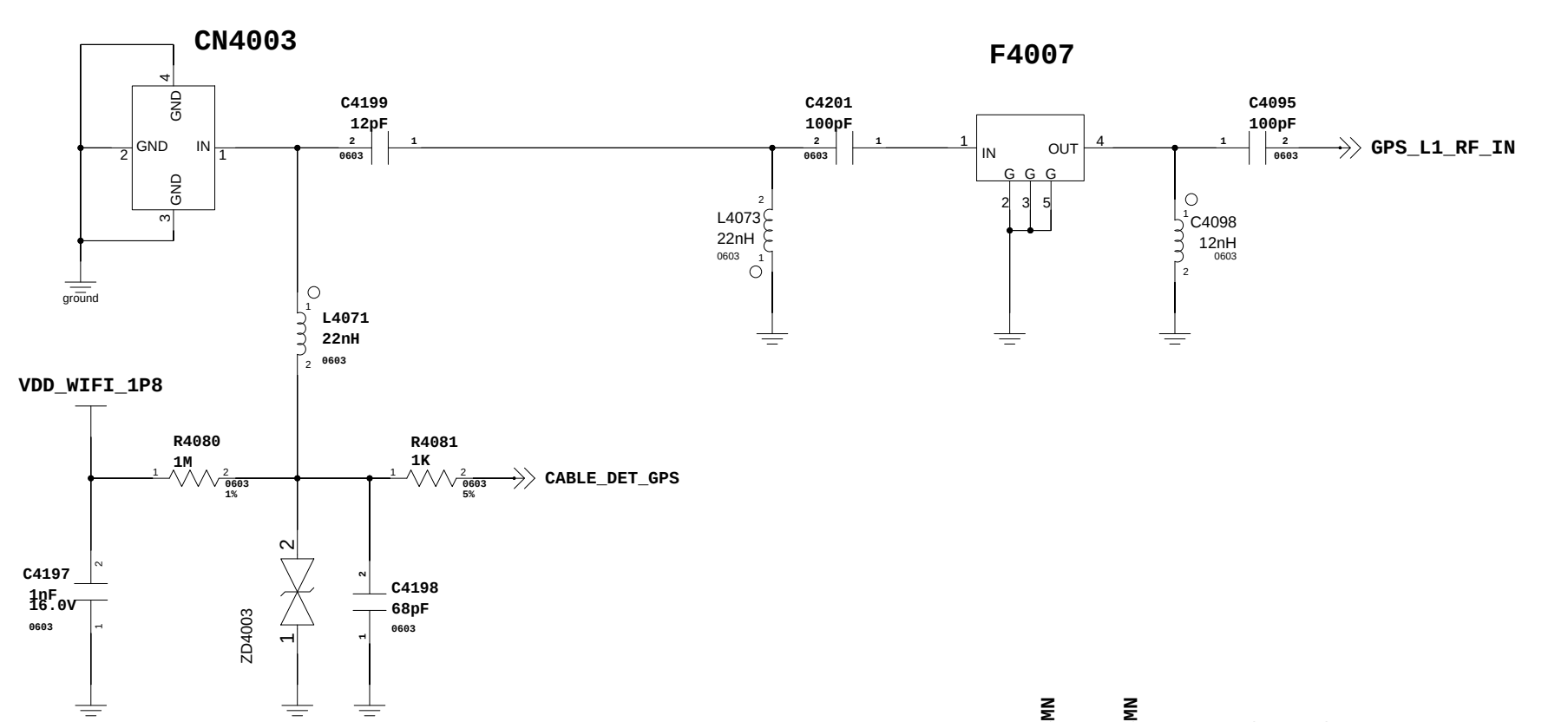
WIFI2(GPS L1 / WiFi2.4G / 5G / 6G)


CN4002
C4071 100pF
C4205 100pF
C4073 33pF
L4037
C4194 12nH
L4041 22nH
F4004
HIGH
COM
LOW
C4045 1nH
C4046 100pF
L4027 33pF
C4053 1nH
F4003
C4054 1nH
L4028 33pF
C4055
L4029
WL_TRX_5G_CH1
F4005
L4032 1.5nH
C4069 1nH
C4076
C4077
L4034 3.9nH
L4035 0.5pF
WL_TRX_2G_CH1
VDD_WIFI_1P8
R4019 1M
R4020 1K
CABLE_DET_WIFI2
C4090 1nF 16.0V
ZD4002
C4091 68pF

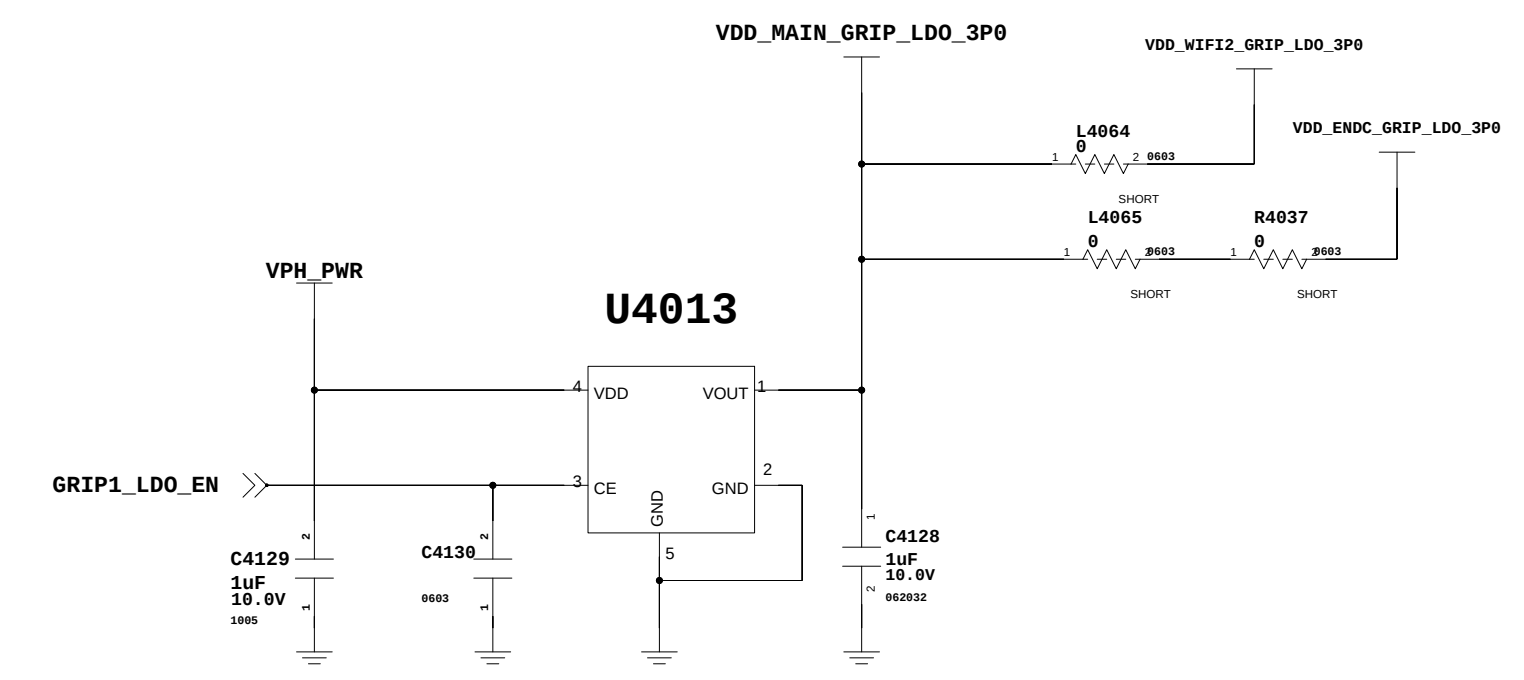

WL_5G_RFFE_SEL4_CH0
WL_5G_RFFE_SEL4_CH1
VPH_PWR
C4041 100pF 25.0V
C4042 100pF 25.0V
WL_TRX_5G_CH1
WL_TRX_5G_CH0
C4047 10nF 10.0V
C4048 2.2uF 10.0V
C4049 10nF
C4050 2.2uF 10.0V
U4004
WL_BIAS_SEL_5G_CH1
WL_LNA_EN_5G_CH1
WL_PA_EN_5G_CH1
WL_BIAS_SEL_5G_CH0
WL_LNA_EN_5G_CH0
WL_PA_EN_5G_CH0
C4060 100pF 25.0V
C4061 100pF 25.0V
C4062 100pF 25.0V
C4057 100pF 25.0V
C4058 100pF 25.0V
C4059 100pF 25.0V
R4013 18
R4014 18
WL_RF_PDET_5G_IN_CH1
WL_RF_PDET_5G_IN_CH0
C4070 0.6nH
C4078
C4067 100pF 25.0V
C4068 100pF 25.0V
NA : NC
Others : Mount
U4007
GND
INPUT
OUT1
OUT2
R4011 OPEN
WL_RFI_5G_CH0
L4033
NA : Mount
Others : NC
C4100 1nH
L4060
L4061
WL_RFO_5G_CH0
C4085 100pF 25.0V
C4086 100pF 25.0V
NA : NC
Others : Mount
U4010
L4046 0.6nH
R4031 OPEN
WL_RFI_5G_CH1
L4047
C4093
NA : Mount
Others : NC
C4106 10pF
L4062
L4063 2.7nH
WL_RFO_5G_CH1


VREG_L4E_1P776
VDD_GPS_1P8
L4042
VREG_S10B_1P8
L4044

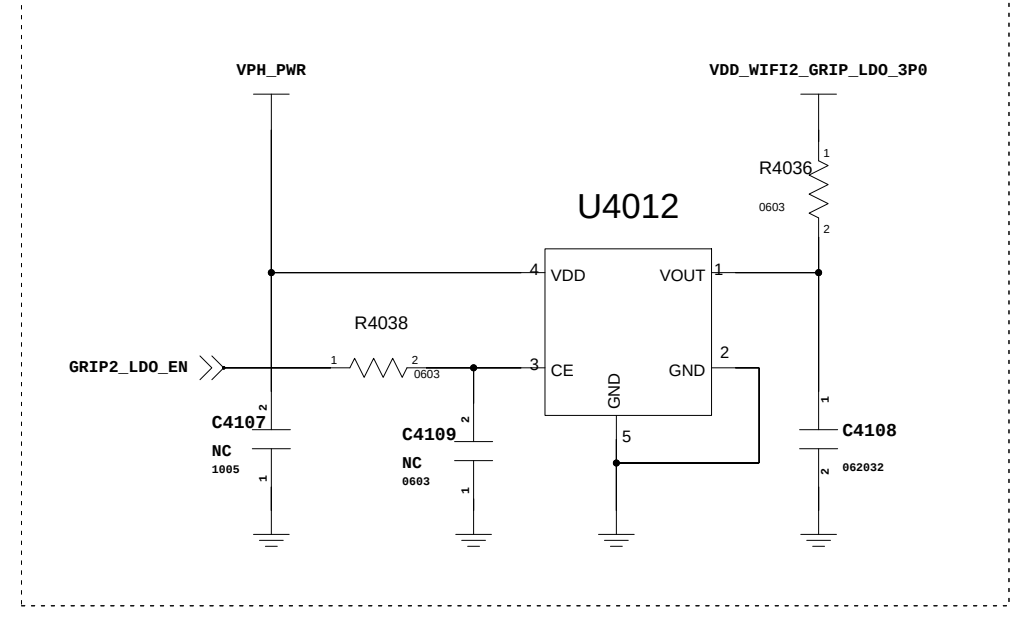
GPS

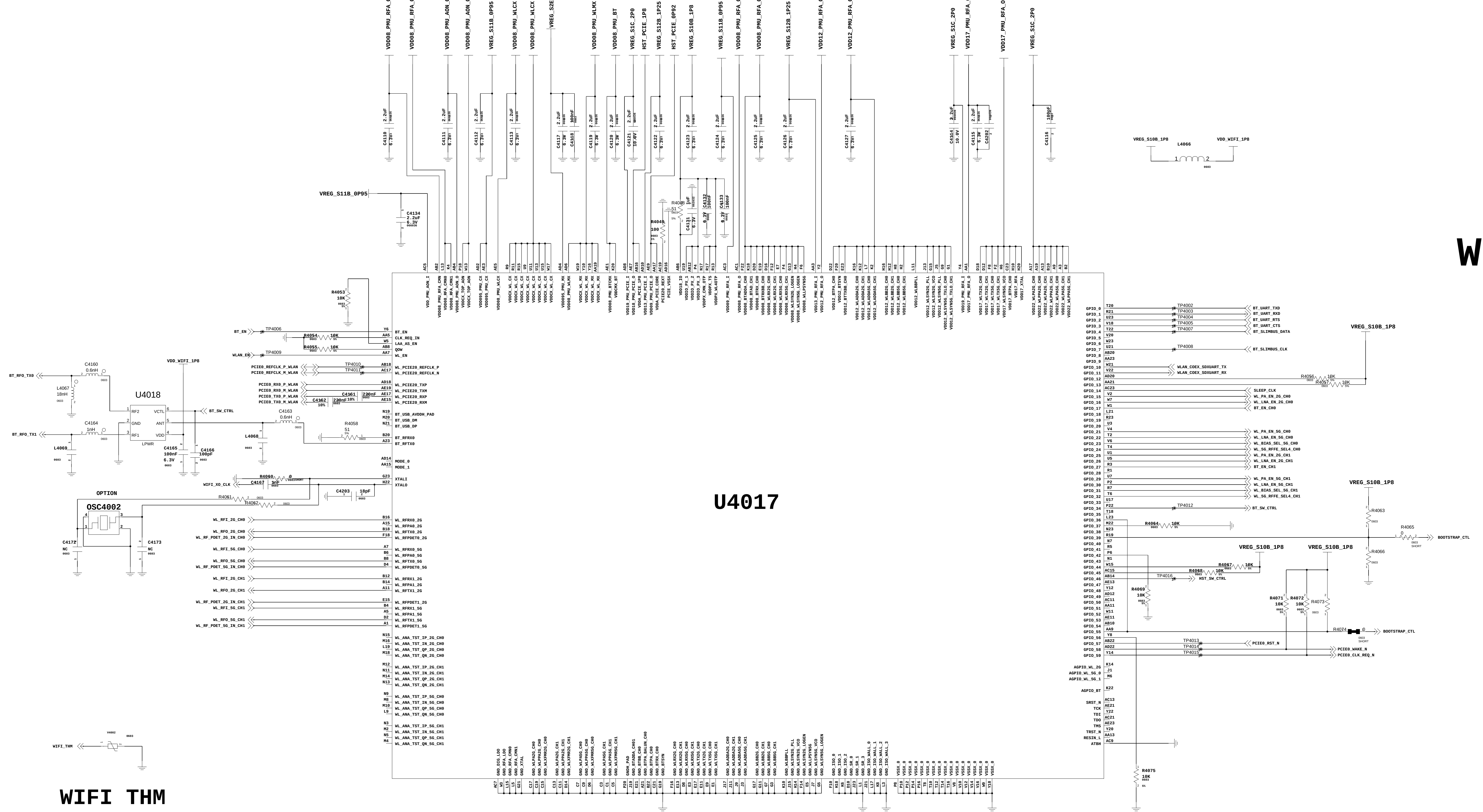

CN4003
C4199 12pF
C4201 100pF
F4007
IN
OUT
C4095 100pF
GPS_L1_RF_IN
L4073 22nH
C4098 12nH
L4071 22nH
VDD_WIFI_1P8
R4080 1M
R4081 1K
CABLE_DET_GPS
C4197 1nF 16.0V
ZD4003
C4198 68pF

MAIN GRIP LDO


VDD_MAIN_GRIP_LDO_3P0
VDD_WIFI2_GRIP_LDO_3P0
VDD_ENDC_GRIP_LDO_3P0
L4064 0
L4065 0
R4037 0
SHORT
VPH_PWR
U4013
VDD
VOUT
CE
GND
GRIP1_LDO_EN
C4129 1uF 10.0V
C4130
C4128 1uF 10.0V

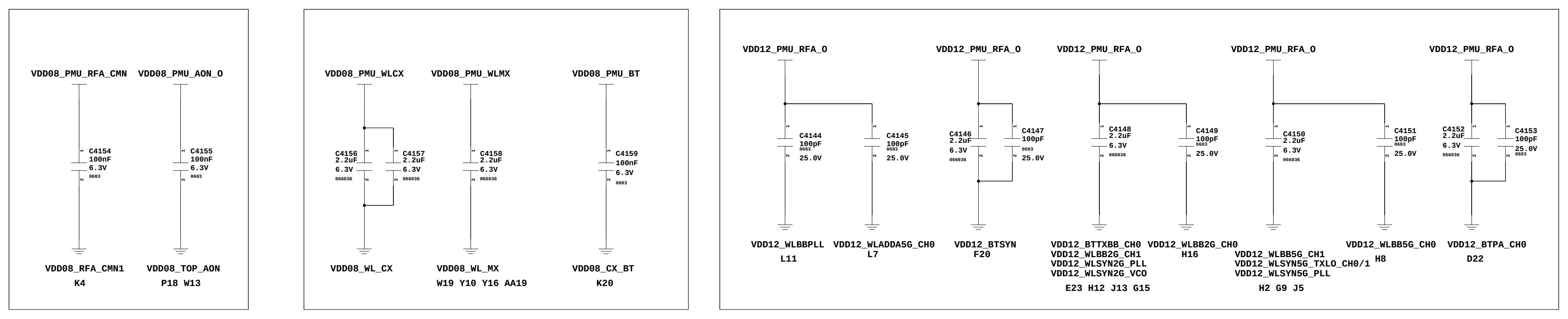
WIFI GRIP LDO

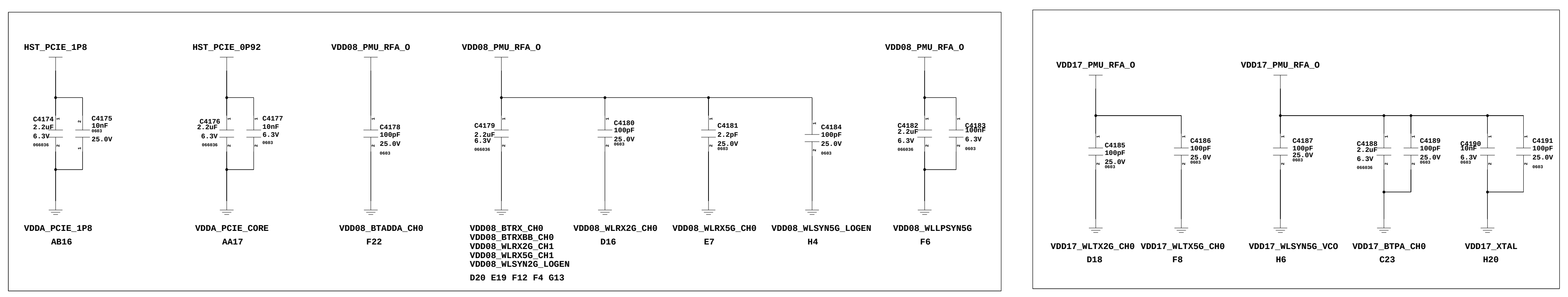

VPH_PWR
VDD_WIFI2_GRIP_LDO_3P0
R4036
U4012
VDD
VOUT
CE
GND
R4038
GRIP2_LDO_EN
C4107 NC
C4109 NC
C4108


VDD08_PMU_RFA_CMN
VDD08_PMU_AON_O
VREG_S11B_0P95
VDD08_PMU_WLCX
VREG_S2E_0P85
VDD08_PMU_WLMX
VDD08_PMU_BT
VREG_S1C_2P0
HST_PCIE_1P8
VREG_S12B_1P25
HST_PCIE_0P92
VREG_S10B_1P8
VDD08_PMU_RFA_O
VDD12_PMU_RFA_O
VDD17_PMU_RFA_O
C4110 2.2uF 6.3V
C4111 2.2uF 6.3V
C4112 2.2uF 6.3V
C4113 2.2uF 6.3V
C4117 2.2uF 6.3V
C4118 100nF
C4119 2.2uF 6.3V
C4120 2.2uF 6.3V
C4121 2.2uF 10.0V
C4122 2.2uF 6.3V
C4123 2.2uF 6.3V
C4124 2.2uF 6.3V
C4125 2.2uF 6.3V
C4126 2.2uF 6.3V
C4127 2.2uF 6.3V
C4114 2.2uF 10.0V
C4115 2.2uF 6.3V
C4202
C4116 100pF
C4134 2.2uF 6.3V
R4048 51
R4049 100
C4131 1uF 6.3V
C4132 100nF
C4133 100nF
VREG_S10B_1P8
L4066
VDD_WIFI_1P8
U4017
R4053 10K
BT_EN
TP4006
R4054 10K
R4055 10K
WLAN_EN
TP4009
PCIE0_REFCLK_P_WLAN
PCIE0_REFCLK_M_WLAN
TP4010
TP4011
PCIE0_RX0_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_TX0_P_WLAN
PCIE0_TX0_M_WLAN
C4161 220nF
C4162 220nF
BT_RFO_TX0
C4160 0.6nH
L4067 18nH
U4018
RF2
GND
RF1
VCTL
ANT
VDD
LPWR
VDD_WIFI_1P8
BT_SW_CTRL
C4164 1nH
BT_RFO_TX1
L4069
C4165 100nF 6.3V
C4166 100pF
C4163 0.6nH
L4068
R4058 51
WIFI_XO_CLK
R4060 0
C4167 1nF
C4203 18pF
R4061
R4062
OPTION
OSC4002
C4172 NC
C4173 NC
WL_RFI_2G_CH0
WL_RFO_2G_CH0
WL_RF_PDET_2G_IN_CH0
WL_RFI_5G_CH0
WL_RFO_5G_CH0
WL_RF_PDET_5G_IN_CH0
WL_RFI_2G_CH1
WL_RFO_2G_CH1
WL_RF_PDET_2G_IN_CH1
WL_RFI_5G_CH1
WL_RFO_5G_CH1
WL_RF_PDET_5G_IN_CH1
GPIO_0 … GPIO_59
TP4002
TP4003
TP4004
TP4005
TP4007
BT_UART_TXD
BT_UART_RXD
BT_UART_RTS
BT_UART_CTS
BT_SLIMBUS_DATA
TP4008
BT_SLIMBUS_CLK
WLAN_COEX_SDXUART_TX
WLAN_COEX_SDXUART_RX
R4056 10K
R4057 10K
VREG_S10B_1P8
SLEEP_CLK
WL_PA_EN_2G_CH0
WL_LNA_EN_2G_CH0
BT_EN_CH0
WL_PA_EN_5G_CH0
WL_LNA_EN_5G_CH0
WL_BIAS_SEL_5G_CH0
WL_5G_RFFE_SEL4_CH0
WL_PA_EN_2G_CH1
WL_LNA_EN_2G_CH1
BT_EN_CH1
WL_PA_EN_5G_CH1
WL_LNA_EN_5G_CH1
WL_BIAS_SEL_5G_CH1
WL_5G_RFFE_SEL4_CH1
TP4012
BT_SW_CTRL
R4064 10K
R4063
R4065 0
R4066
BOOTSTRAP_CTL
R4067 10K
R4068 10K
TP4016
HST_SW_CTRL
R4069 10K
R4071 10K
R4072 10K
R4073
R4074 0
TP4013
TP4014
TP4015
PCIE0_RST_N
PCIE0_WAKE_N
PCIE0_CLK_REQ_N
R4075 10K
V4002
WIFI_THM

WIFI THM

WCN6856 Decap


VDD08_PMU_RFA_CMN
VDD08_PMU_AON_O
C4154 100nF 6.3V
C4155 100nF 6.3V
VDD08_RFA_CMN1
K4
VDD08_TOP_AON
P18 W13
VDD08_PMU_WLCX
VDD08_PMU_WLMX
VDD08_PMU_BT
C4156 2.2uF 6.3V
C4157 2.2uF 6.3V
C4158 2.2uF 6.3V
C4159 100nF 6.3V
VDD08_WL_CX
VDD08_WL_MX
W19 Y10 Y16 AA19
VDD08_CX_BT
K20
VDD12_PMU_RFA_O
C4144 100pF 25.0V
C4145 100pF 25.0V
C4146 2.2uF 6.3V
C4147 100pF 25.0V
C4148 2.2uF 6.3V
C4149 100pF 25.0V
C4150 2.2uF 6.3V
C4151 100pF 25.0V
C4152 2.2uF 6.3V
C4153 100pF 25.0V
VDD12_WLBBPLL
L11
VDD12_WLADDA5G_CH0
L7
VDD12_BTSYN
F20
VDD12_BTTXBB_CH0
VDD12_WLBB2G_CH1
VDD12_WLSYN2G_PLL
VDD12_WLSYN2G_VCO
E23 H12 J13 G15
VDD12_WLBB2G_CH0
H16
VDD12_WLBB5G_CH1
VDD12_WLSYN5G_TXLO_CH0/1
VDD12_WLSYN5G_PLL
H2 G9 J5
VDD12_WLBB5G_CH0
H8
VDD12_BTPA_CH0
D22


HST_PCIE_1P8
C4174 2.2uF 6.3V
C4175 10nF 25.0V
VDDA_PCIE_1P8
AB16
HST_PCIE_0P92
C4176 2.2uF 6.3V
C4177 10nF 6.3V
VDDA_PCIE_CORE
AA17
VDD08_PMU_RFA_O
C4178 100pF 25.0V
VDD08_BTADDA_CH0
F22
C4179 2.2uF 6.3V
C4180 100pF 25.0V
C4181 2.2pF 25.0V
C4184 100pF 25.0V
VDD08_BTRX_CH0
VDD08_BTRXBB_CH0
VDD08_WLRX2G_CH1
VDD08_WLRX5G_CH1
VDD08_WLSYN2G_LOGEN
D20 E19 F12 F4 G13
VDD08_WLRX2G_CH0
D16
VDD08_WLRX5G_CH0
E7
VDD08_WLSYN5G_LOGEN
H4
C4182 2.2uF 6.3V
C4183 100nF 6.3V
VDD08_WLLPSYN5G
F6
VDD17_PMU_RFA_O
C4185 100pF 25.0V
C4186 100pF 25.0V
C4187 100pF 25.0V
C4188 2.2uF 6.3V
C4189 100pF 25.0V
C4190 10nF 6.3V
C4191 100pF 25.0V
VDD17_WLTX2G_CH0
D18
VDD17_WLTX5G_CH0
F8
VDD17_WLSYN5G_VCO
H6
VDD17_BTPA_CH0
C23
VDD17_XTAL
H20

TITLE TabS7P
SIZE A1 PDM NUMBER REV Rev
DRAWN BY User Name SHEET 4 of 14 Date Drawing Date

SM8450

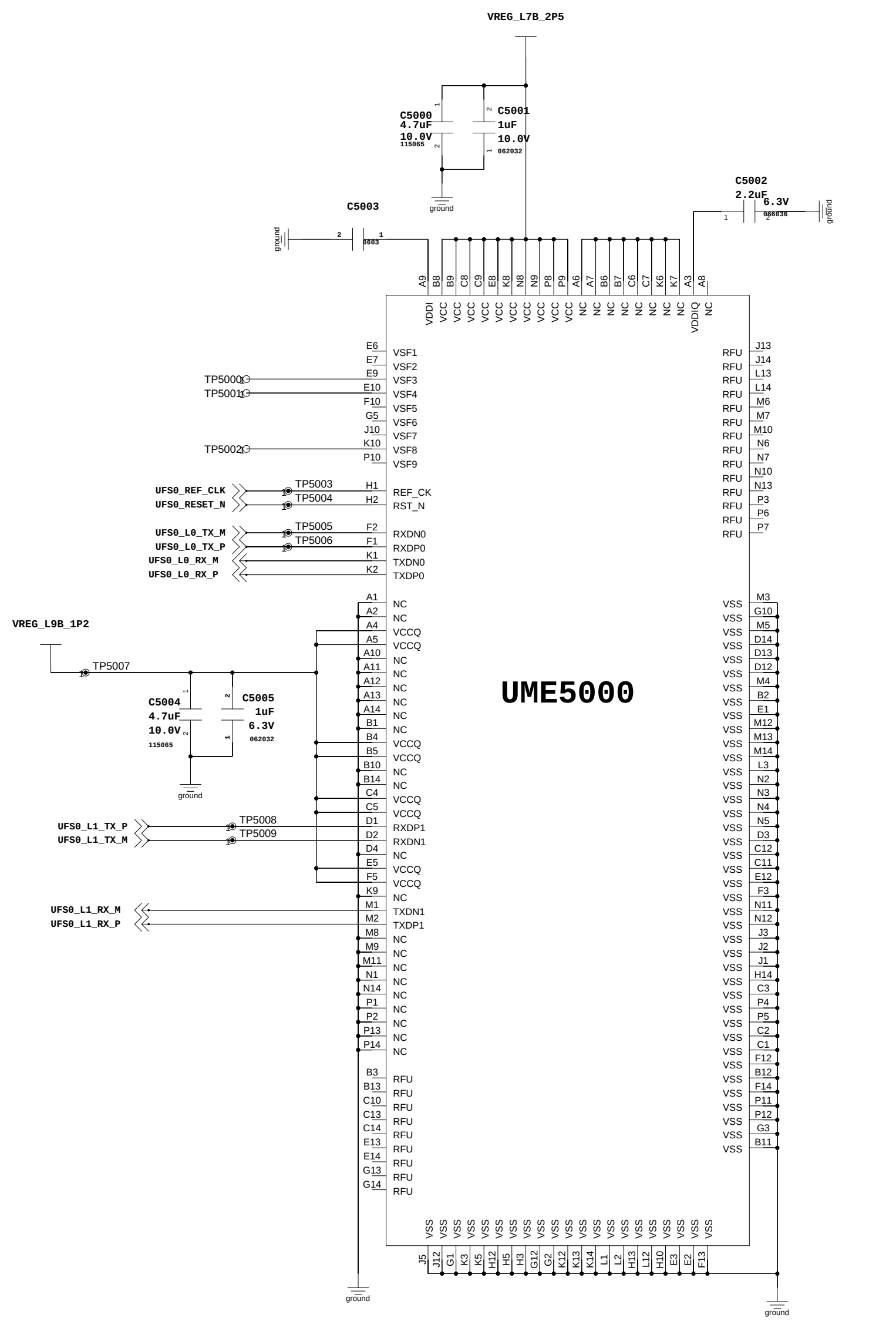
(INTERFACE/GPIO)

System

System

CORE

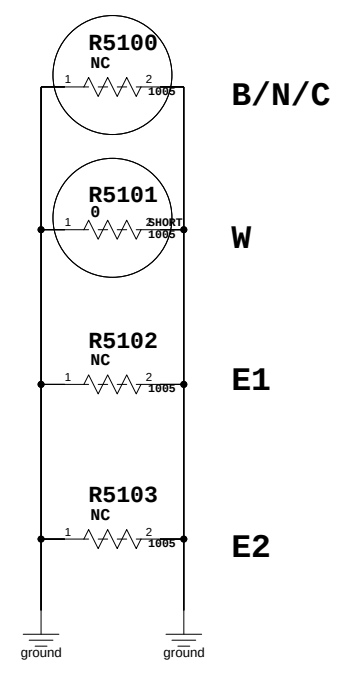
on SECONDARY PCB


VREG_L7B_2P5
C5000
4.7uF
10.0V
C5001
1uF
10.0V
C5002
2.2uF
6.3V
C5003
UME5000
TP5000
TP5001
TP5002
UFS0_REF_CLK
UFS0_RESET_N
TP5003
TP5004
UFS0_L0_TX_M
UFS0_L0_TX_P
UFS0_L0_RX_M
UFS0_L0_RX_P
TP5005
TP5006
VREG_L9B_1P2
TP5007
C5004
4.7uF
10.0V
C5005
1uF
6.3V
UFS0_L1_TX_P
UFS0_L1_TX_M
TP5008
TP5009
UFS0_L1_RX_M
UFS0_L1_RX_P
VDDI
VCC
VDDIQ
NC
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
VSF7
VSF8
VSF9
REF_CK
RST_N
RXDN0
RXDP0
TXDN0
TXDP0
VCCQ
RXDP1
RXDN1
TXDN1
TXDP1
RFU
VSS
ground

UFS : 128G / 256G / 512G / 1TB

REGION

ANT PCB	SHORT
X906(B/N/C)	R5100
X900(W)	R5101
ETC1 (E1)	R5102
ETC2 (E2)	R5103

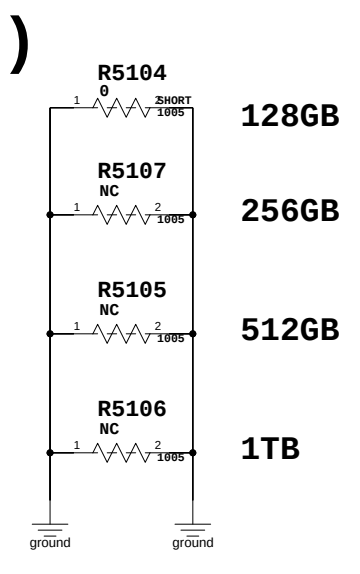

R5100
NC
B/N/C
R5101
0
W
R5102
NC
E1
R5103
NC
E2
ground

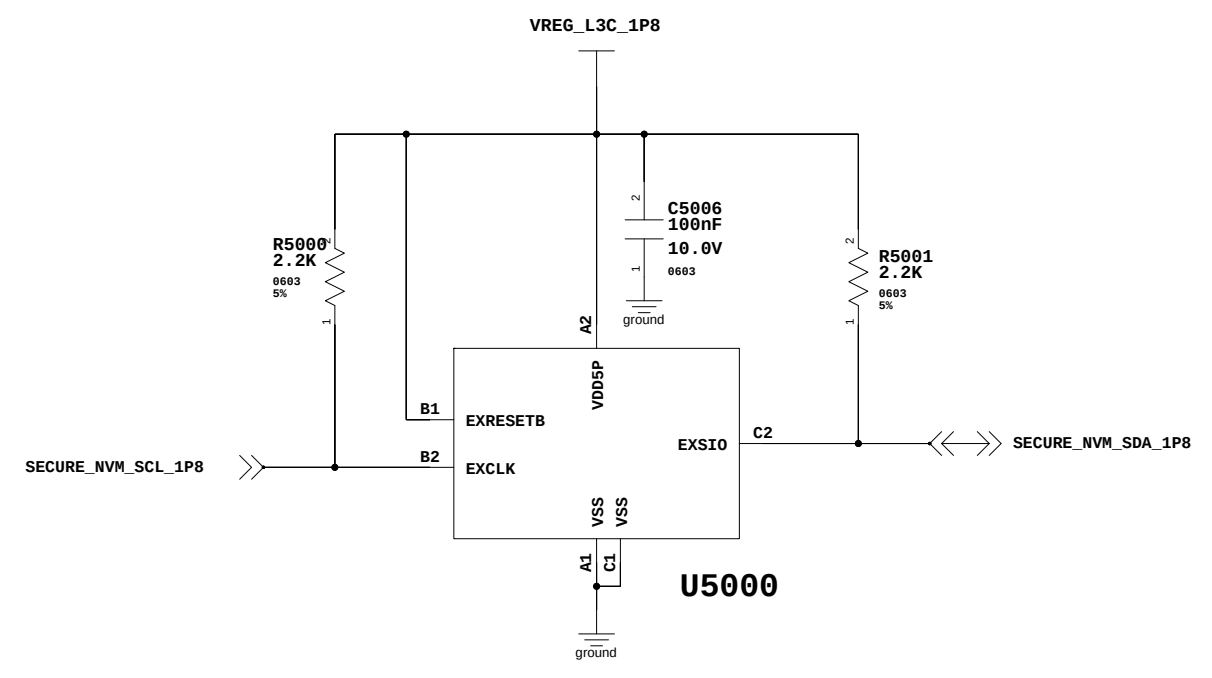
DRAM_POP (S_PCB)

8GB	1105-003137
12GB	1105-003133
16GB	1105-003144

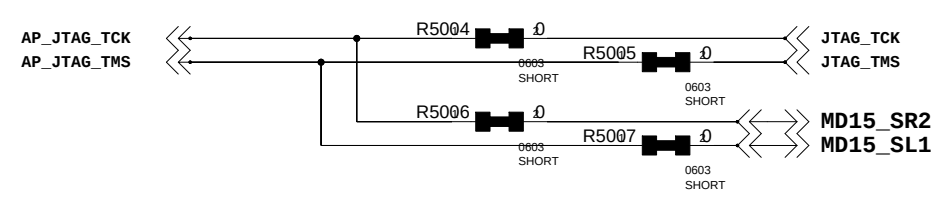
UFS Memory (M_PCB)

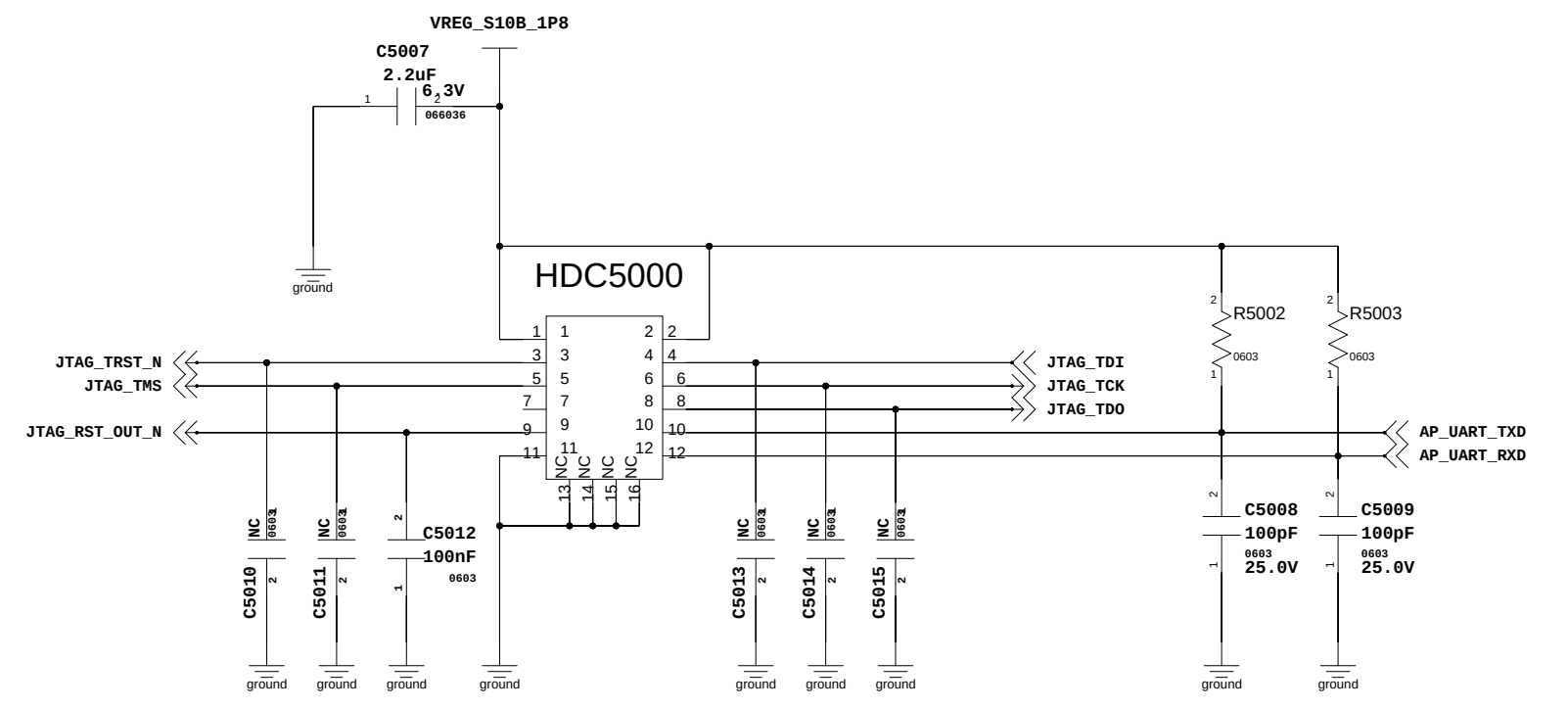
128GB	1107-002751
256GB	1107-002753
512GB	1107-002855
1TB	


R5104
0
128GB
R5107
NC
256GB
R5105
NC
512GB
R5106
NC
1TB
ground


VREG_L3C_1P8
R5000
2.2K
C5006
100nF
10.0V
R5001
2.2K
SECURE_NVM_SCL_1P8
SECURE_NVM_SDA_1P8
EXRESETB
EXCLK
EXSIO
VDDSP
VSS
U5000
ground

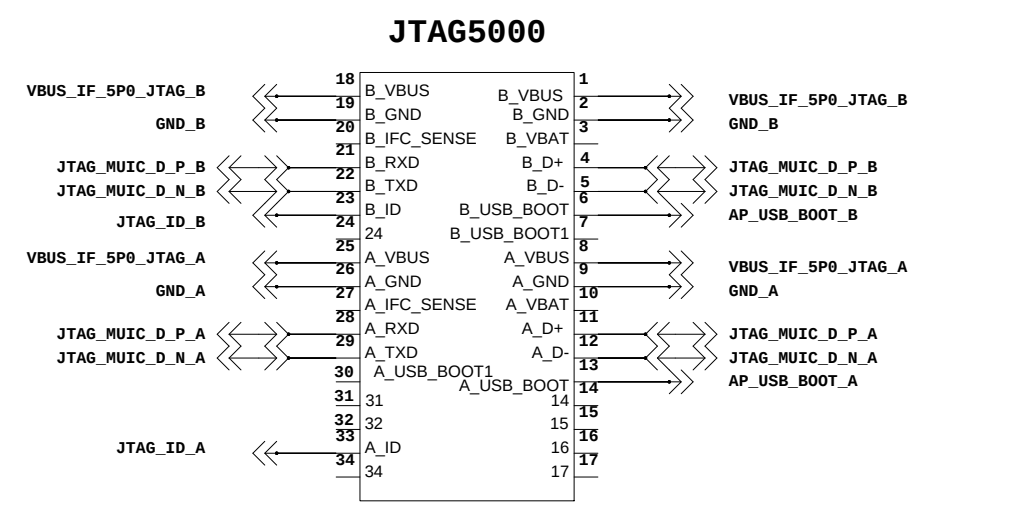
SECURE NVM


AP_JTAG_TCK
AP_JTAG_TMS
R5094
R5095
R5096
R5097
0
0603
SHORT
JTAG_TCK
JTAG_TMS
MD15_SR2
MD15_SL1

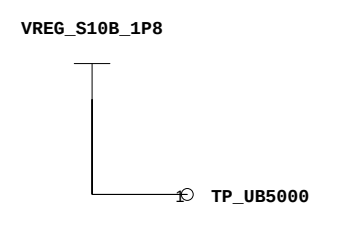

VREG_S10B_1P8
C5007
2.2uF
6.3V
HDC5000
JTAG_TRST_N
JTAG_TMS
JTAG_RST_OUT_N
JTAG_TDI
JTAG_TCK
JTAG_TDO
R5002
R5003
AP_UART_TXD
AP_UART_RXD
C5010
C5011
C5012
100nF
C5013
C5014
C5015
NC
C5008
100pF
25.0V
C5009
100pF
25.0V
ground

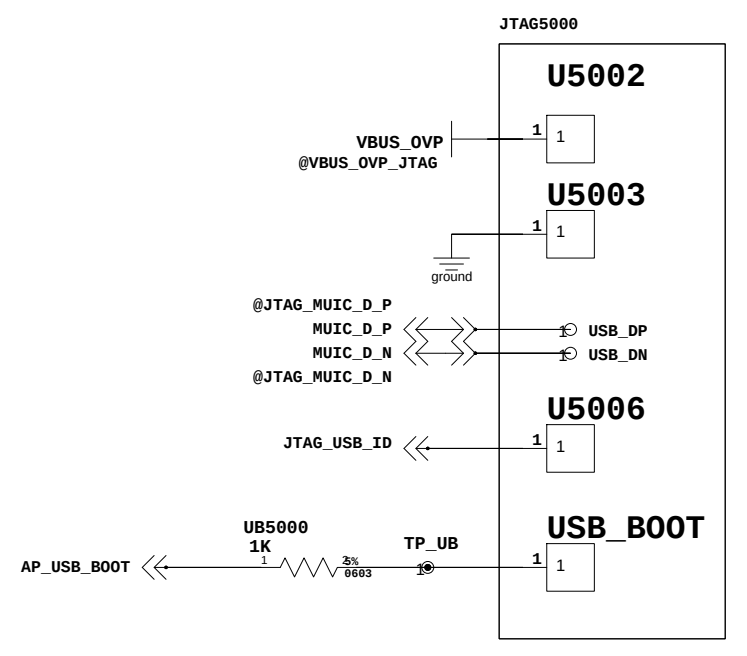
AP JTAG CONNECTOR

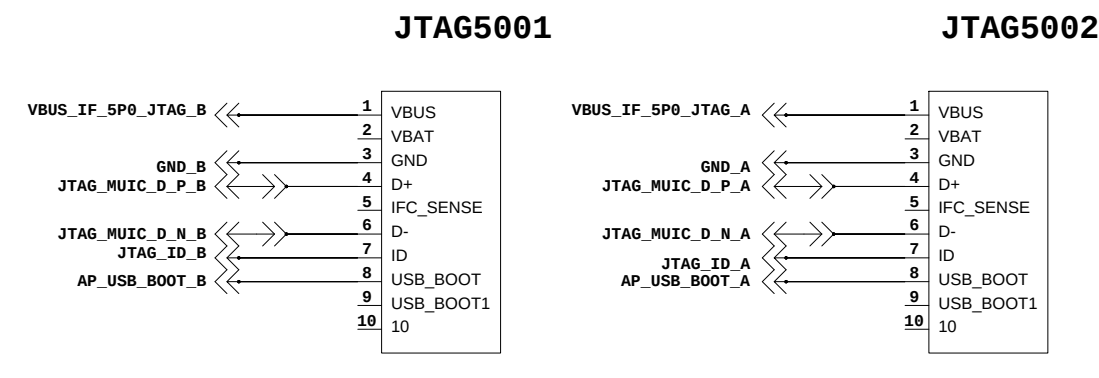
(3711-007173 / AXE612124)


JTAG5000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
JTAG_ID_A
AP_USB_BOOT_B
AP_USB_BOOT_A
B_VBUS
B_GND
B_IFC_SENSE
B_RXD
B_TXD
B_ID
B_VBAT
B_D+
B_D-
B_USB_BOOT
B_USB_BOOT1
A_VBUS
A_GND
A_IFC_SENSE
A_RXD
A_TXD
A_USB_BOOT1
A_ID
A_VBAT
A_D+
A_D-
A_USB_BOOT

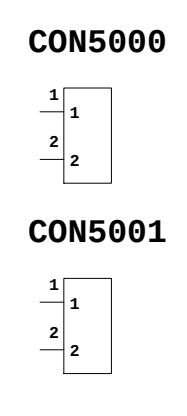
CAUTION: PLACEMENT
Near to TP AP_USB_BOOT


VREG_S10B_1P8
TP_UB5000


JTAG5000
U5002
VBUS_OVP
@VBUS_OVP_JTAG
U5003
ground
@JTAG_MUIC_D_P
MUIC_D_P
MUIC_D_N
@JTAG_MUIC_D_N
USB_DP
USB_DN
U5006
JTAG_USB_ID
USB_BOOT
UB5000
1K
TP_UB
AP_USB_BOOT


JTAG5001
JTAG5002
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
AP_USB_BOOT_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
JTAG_ID_A
AP_USB_BOOT_A
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT
USB_BOOT1

PCB array JTAG

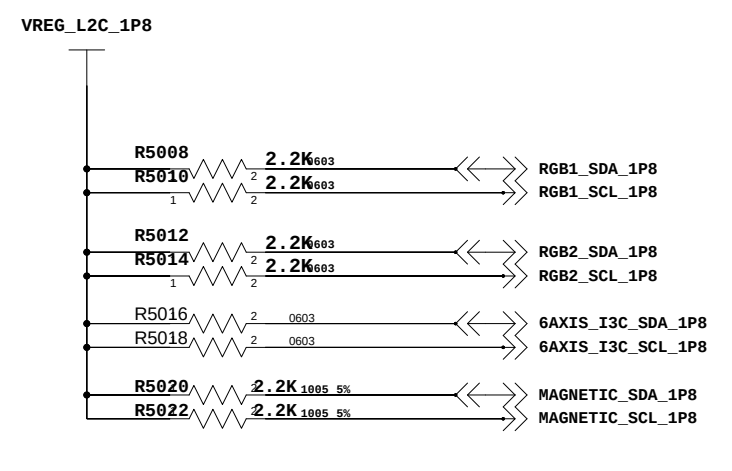

CON5000
CON5001

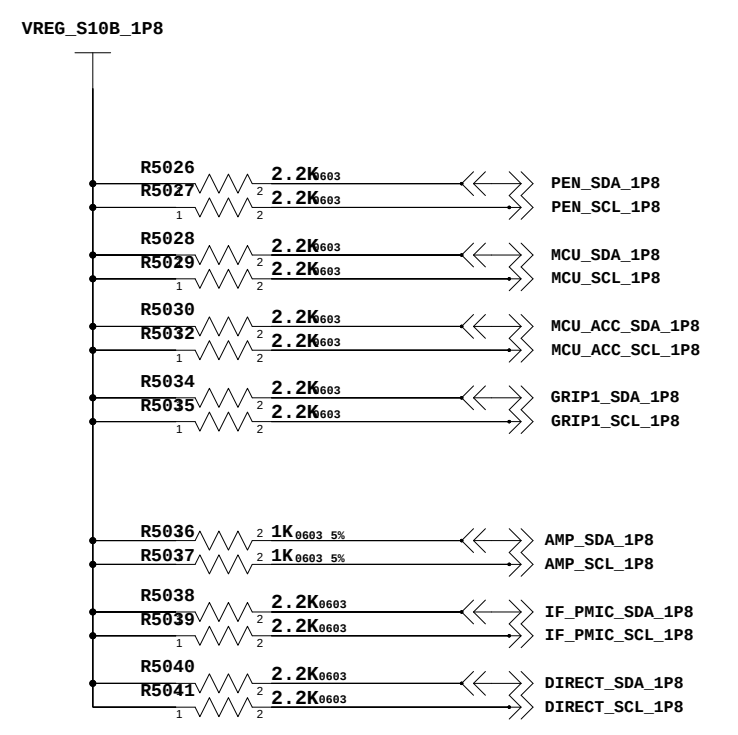
2D BARCODE

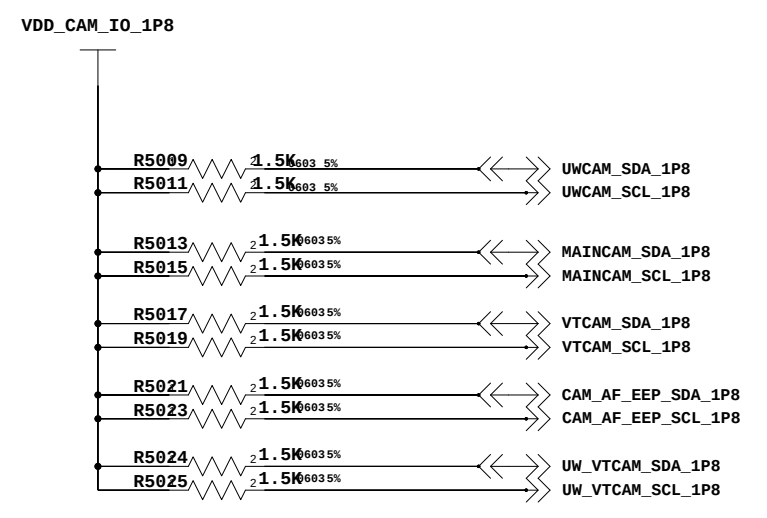
SMD NC

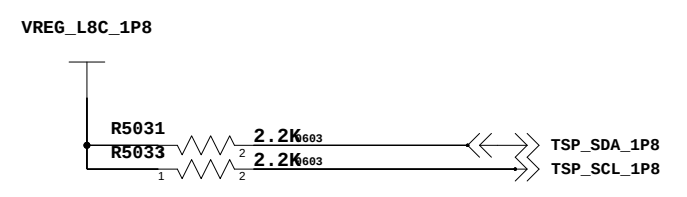
DUMMY CAP

2.2uF
C5053
2.2uF
C5054
1005
ground


VREG_L2C_1P8
R5008
R5010
2.2K
RGB1_SDA_1P8
RGB1_SCL_1P8
R5012
R5014
2.2K
RGB2_SDA_1P8
RGB2_SCL_1P8
R5016
R5018
6AXIS_I3C_SDA_1P8
6AXIS_I3C_SCL_1P8
R5020
R5022
2.2K
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8


VREG_S10B_1P8
R5026
R5027
2.2K
PEN_SDA_1P8
PEN_SCL_1P8
R5028
R5029
2.2K
MCU_SDA_1P8
MCU_SCL_1P8
R5030
R5032
2.2K
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
R5034
R5035
2.2K
GRIP1_SDA_1P8
GRIP1_SCL_1P8
R5036
R5037
1K
AMP_SDA_1P8
AMP_SCL_1P8
R5038
R5039
2.2K
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
R5040
R5041
2.2K
DIRECT_SDA_1P8
DIRECT_SCL_1P8


VDD_CAM_IO_1P8
R5009
R5011
1.5K
UWCAM_SDA_1P8
UWCAM_SCL_1P8
R5013
R5015
1.5K
MAINCAM_SDA_1P8
MAINCAM_SCL_1P8
R5017
R5019
1.5K
VTCAM_SDA_1P8
VTCAM_SCL_1P8
R5021
R5023
1.5K
CAM_AF_EEP_SDA_1P8
CAM_AF_EEP_SCL_1P8
R5024
R5025
1.5K
UW_VTCAM_SDA_1P8
UW_VTCAM_SCL_1P8


VREG_L8C_1P8
R5031
R5033
2.2K
TSP_SDA_1P8
TSP_SCL_1P8

TITLE	TabS7P		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 5 of 15	Date Drawing Date	

SM8450 Power

(POWER, GND)

CORE

on SECONDARY PCB

TITLE Tab S7+
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name SHEET 6 of 14 Date Drawing Date

P PMIC set & CLK Generator

PMIC (PM8450/8350/8350C)

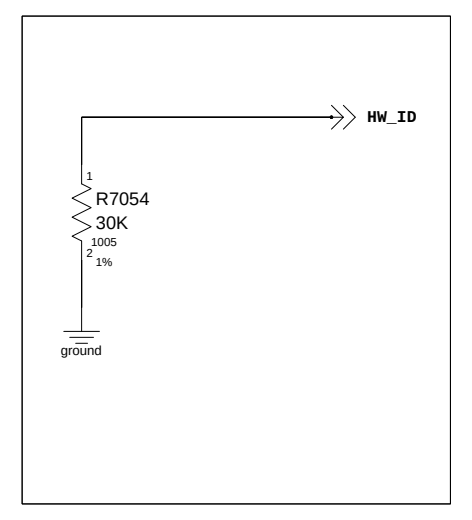
!! CAUTION !!
Place Cin 10uF Bulk Caps at corner of PM8350 and as close to PM8350 as possible to achieve this requirement.

CORE

on SECONDARY PCB

Power System

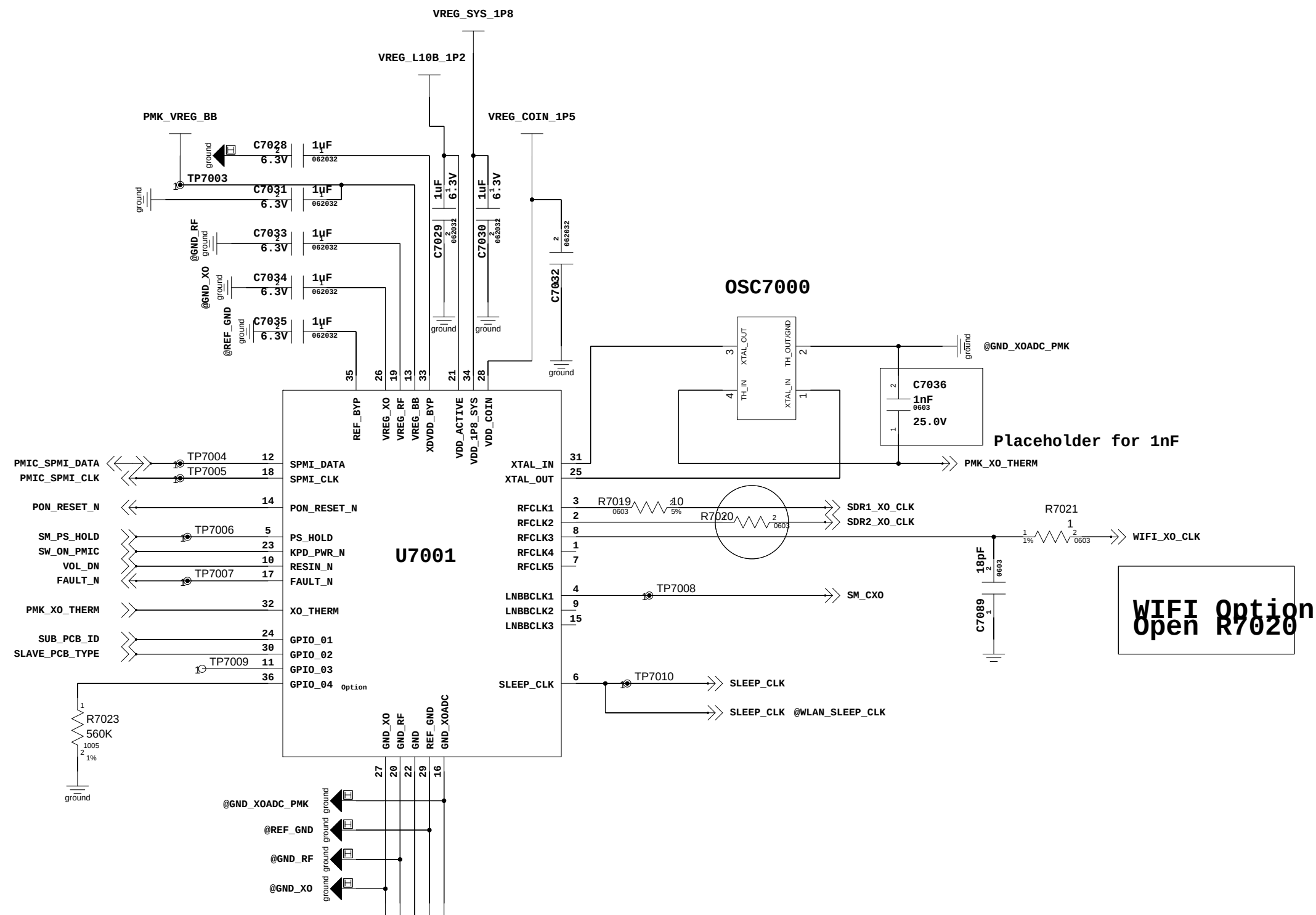
H/W ID



HW_ID
R7054
30K
1005
1%
ground

H/W ID table(PM8350 AMUX_2)

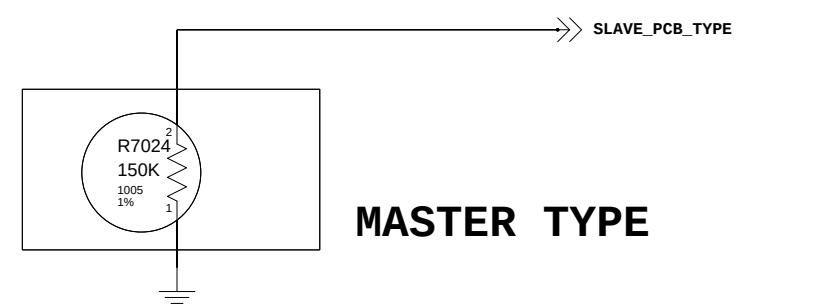
HW ID	BOARD REVISION	Value	Mech
00	TAB S8 ULTRA 14 REV0.0(SW bring-up)	8.04K	
01	TAB S8 ULTRA 14 REV0.1	9.1K	
02	TAB S8 ULTRA 14 REV0.2	12.4K	
03	TAB S8 ULTRA 14 REV0.3	18K	
04	TAB S8 ULTRA 14 REV0.3_CP	22K	
05	TAB S8 ULTRA 14 REV0.4 / REV0.4A	30K	
06		39K	
07		47K	
08		56K	
09		68K	
10		82K	
11		100K	
12		120K	
13		150K	
14		180K	
15		200K	



VREG_SYS_1P8
VREG_L10B_1P2
VREG_COIN_1P5
PMK_VREG_BB
TP7003
C7028 1uF 6.3V
C7031 1uF 6.3V
C7033 1uF 6.3V
C7034 1uF 6.3V
C7035 1uF 6.3V
@GND_RF
@GND_XO
@REF_GND
C7029 1uF 6.3V
C7030 1uF 6.3V
C7032
OSC7000
XTAL_OUT
TH_OUT/GND
TH_IN
XTAL_IN
@GND_XOADC_PMK
C7036 1nF 25.0V
Placeholder for 1nF
PMK_XO_THERM
REF_BYP
VREG_XO
VREG_RF
VREG_BB
XOVDD_BYP
VDD_ACTIVE
VDD_1P8_SYS
VDD_COIN
PMIC_SPMI_DATA
PMIC_SPMI_CLK
TP7004
TP7005
SPMI_DATA
SPMI_CLK
PON_RESET_N
SM_PS_HOLD
SW_ON_PMIC
VOL_DN
FAULT_N
TP7006
TP7007
PS_HOLD
KPD_PWR_N
RESIN_N
XO_THERM
SUB_PCB_ID
SLAVE_PCB_TYPE
TP7009
GPIO_01
GPIO_02
GPIO_03
GPIO_04
Option
U7001
XTAL_IN
XTAL_OUT
RFCLK1
RFCLK2
RFCLK3
RFCLK4
RFCLK5
R7019 10
R7020
SDR1_XO_CLK
SDR2_XO_CLK
R7021 1
WIFI_XO_CLK
C7089 18pF
WIFI Option
Open R7020
LNBBCLK1
LNBBCLK2
LNBBCLK3
TP7008
SM_CXO
SLEEP_CLK
TP7010
SLEEP_CLK @WLAN_SLEEP_CLK
R7023 560K 1005 1%
GND_XO
GND_RF
GND
REF_GND
GND_XOADC
@GND_XOADC_PMK
@REF_GND
@GND_RF
@GND_XO
@GND_XOADC_PMK
ground

Option pin state (PMK8450; 76.8MHz XO)

OPTION	LPDDR5	SD card LDO	GPIO05B Active	R value
8	O	enable	HIGH	120K
9	O	enable	LOW	150K
10 (Reserved)				
11 (Reserved)				
12	O	skip	HIGH	360K
13	O	skip	LOW	560K
14 (Reserved)				
15 (Reserved)				



SLAVE_PCB_TYPE
R7024
150K
1005
1%
MASTER TYPE
ground

MASTER_TYPE

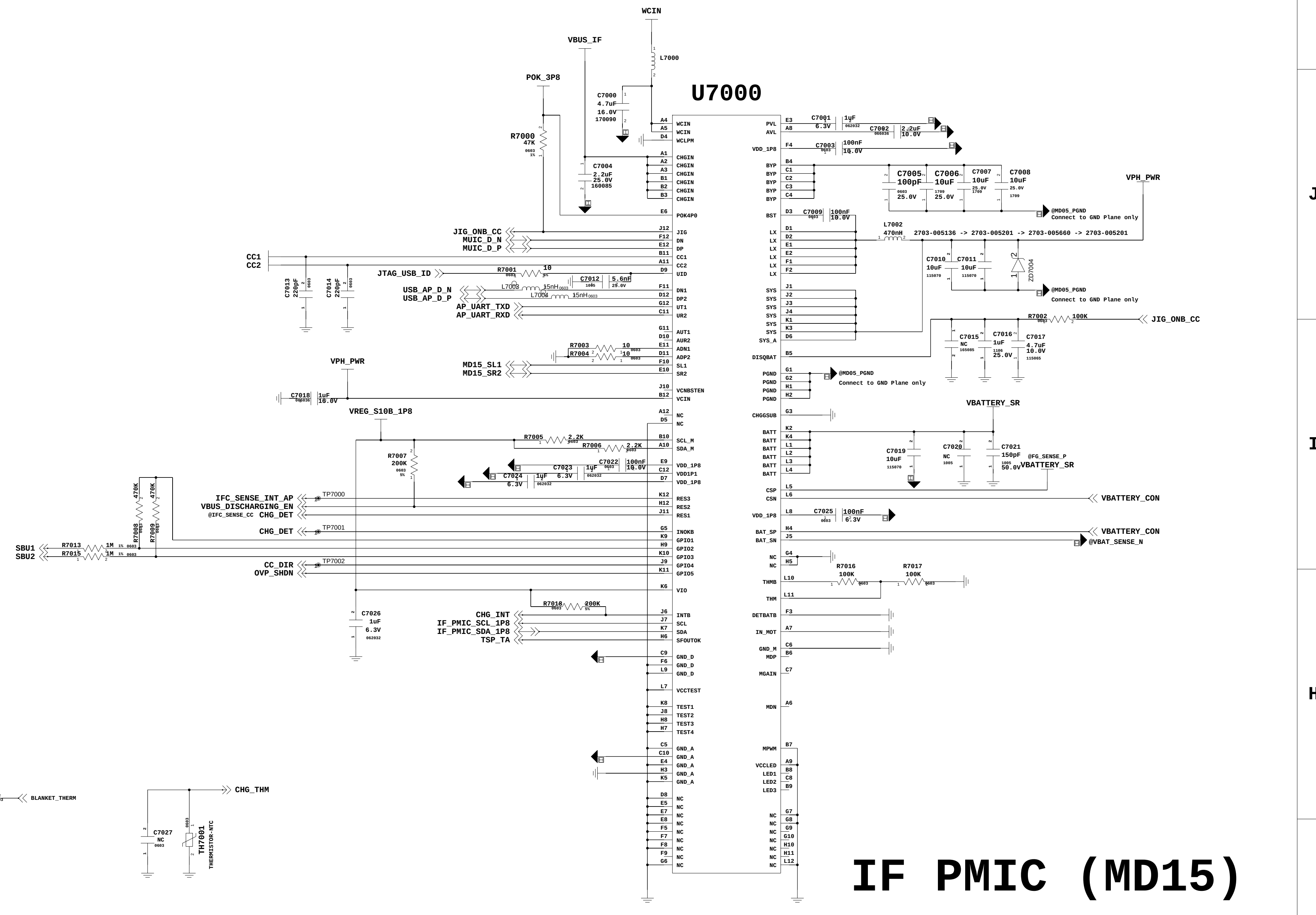
PMK8450 GPIO_02	ADC
EUR(Sub_6)	68K
USA(Sub_6)	100K
WIFI	150K
mmWave	200K
EUR with eSE	560K
For JIG BINARY	OPEN

!! CAUTION !!
Place Cin 10uF Bulk Caps at corner of PM8350C and as close to PM8350C as possible to achieve this requirement.

CORE

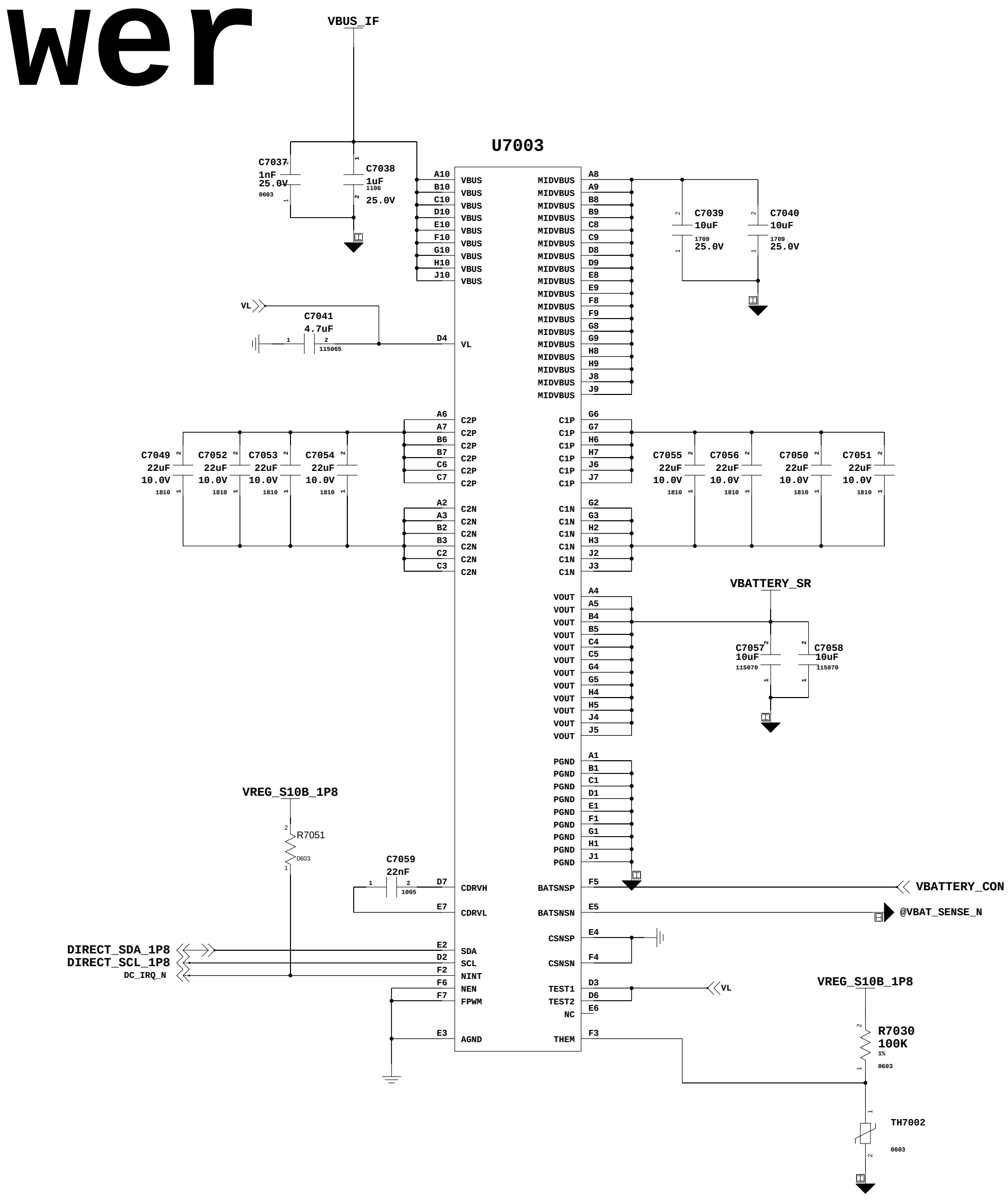
on SECONDARY PCB

Power



WCIN
L7000
VBUS_IF
POK_3P8
C7000 4.7uF 16.0V
R7000 47K
C7004 2.2uF 25.0V
U7000
WCIN
WCLPM
CHGIN
POK4P0
JIG_ONB_CC
MUIC_D_N
MUIC_D_P
JIG
DN
DP
CC1
CC2
C7013 220pF
C7014 220pF
JTAG_USB_ID
R7001 10
C7012 5.6nF
UID
USB_AP_D_N
USB_AP_D_P
L7003 15nH
L7004 15nH
AP_UART_TXD
AP_UART_RXD
DN1
DP2
UT1
UR2
AUT1
AUR2
ADN1
ADP2
R7003 10
R7004 10
MD15_SL1
MD15_SR2
SL1
SR2
VPH_PWR
C7018 1uF 16.0V
VCNBSTEN
VCIN
VREG_S10B_1P8
R7005 2.2K
R7006 2.2K
SCL_M
SDA_M
R7007 200K
C7022 100nF 10.0V
C7023 1uF 6.3V
C7024 1uF 6.3V
VDD_1P8
VDD1P1
IFC_SENSE_INT_AP
VBUS_DISCHARGING_EN
@IFC_SENSE_CC
CHG_DET
TP7000
TP7001
TP7002
RES3
RES2
RES1
INOKB
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
R7008 470K
R7009 470K
SBU1
SBU2
R7013 1M
R7015 1M
CC_DIR
OVP_SHDN
VIO
C7026 1uF 6.3V
R7018 200K
CHG_INT
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
TSP_TA
INTB
SCL
SDA
SFOUTOK
GND_D
VCCTEST
TEST1
TEST2
TEST3
TEST4
GND_A
NC
TH7000
BLANKET_THERM
CHG_THM
C7027 NC
TH7001
THERMISTOR-NTC
PVL
AVL
VDD_1P8
C7001 1uF 6.3V
C7002 2.2uF 10.0V
C7003 100nF 10.0V
BYP
C7005 100pF 25.0V
C7006 10uF 25.0V
C7007 10uF 25.0V
C7008 10uF 25.0V
VPH_PWR
BST
C7009 100nF 10.0V
@MD05_PGND
Connect to GND Plane only
LX
L7002 470nH
2703-005136 -> 2703-005201 -> 2703-005660 -> 2703-005201
C7010 10uF
C7011 10uF
ZD7004
SYS
SYS_A
DISQBAT
R7002 100K
JIG_ONB_CC
C7015 NC
C7016 1uF 25.0V
C7017 4.7uF 10.0V
PGND
@MD05_PGND
Connect to GND Plane only
CHGGSUB
VBATTERY_SR
BATT
C7019 10uF
C7020 NC
C7021 150pF 50.0V
@FG_SENSE_P
VBATTERY_SR
CSP
CSN
VBATTERY_CON
VDD_1P8
C7025 100nF 6.3V
BAT_SP
BAT_SN
VBATTERY_CON
@VBAT_SENSE_N
THMB
THM
R7016 100K
R7017 100K
DETBATB
IN_MOT
GND_M
MDP
MGAIN
MDN
MPWM
VCCLED
LED1
LED2
LED3
IF PMIC (MD15)

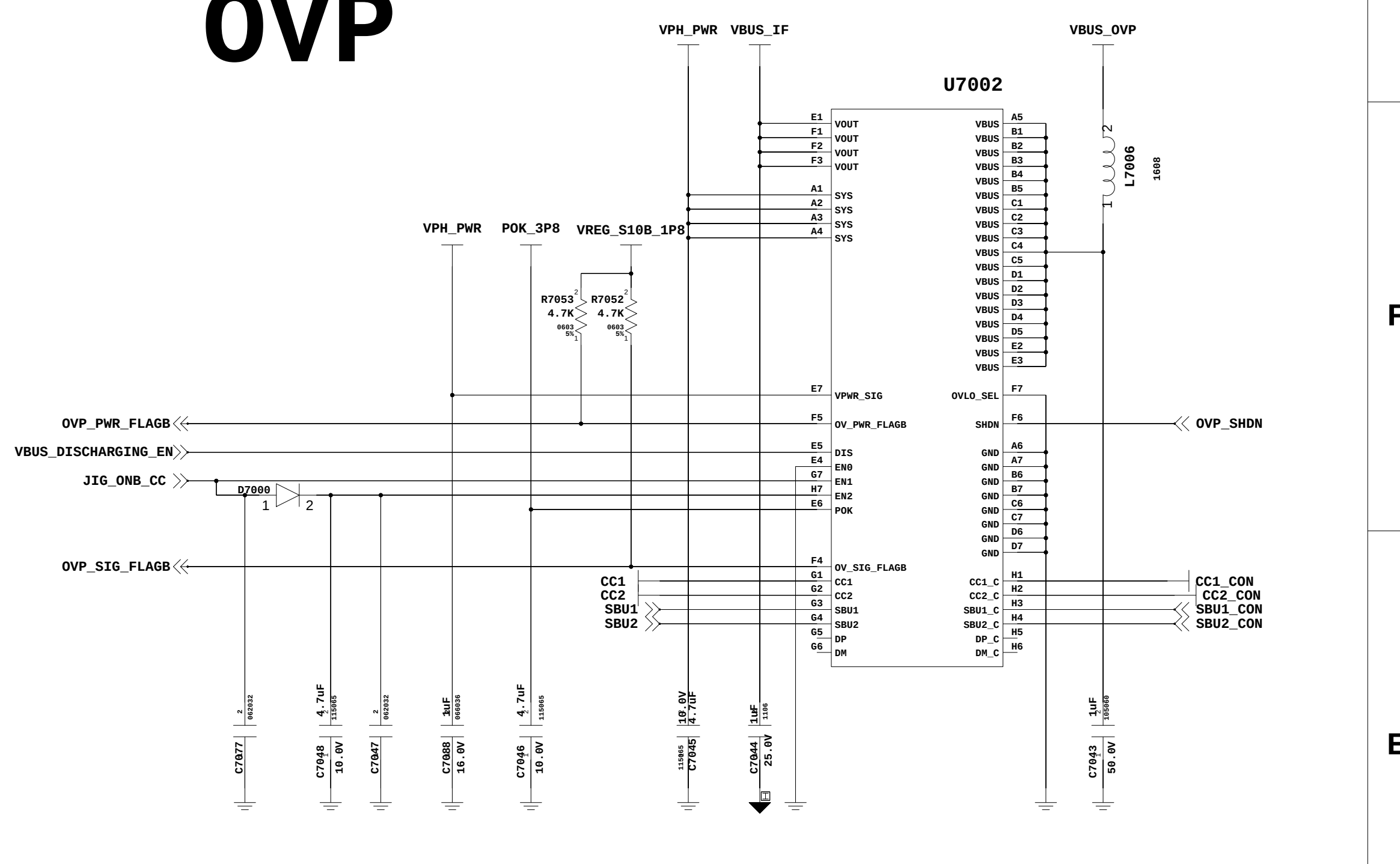
Power



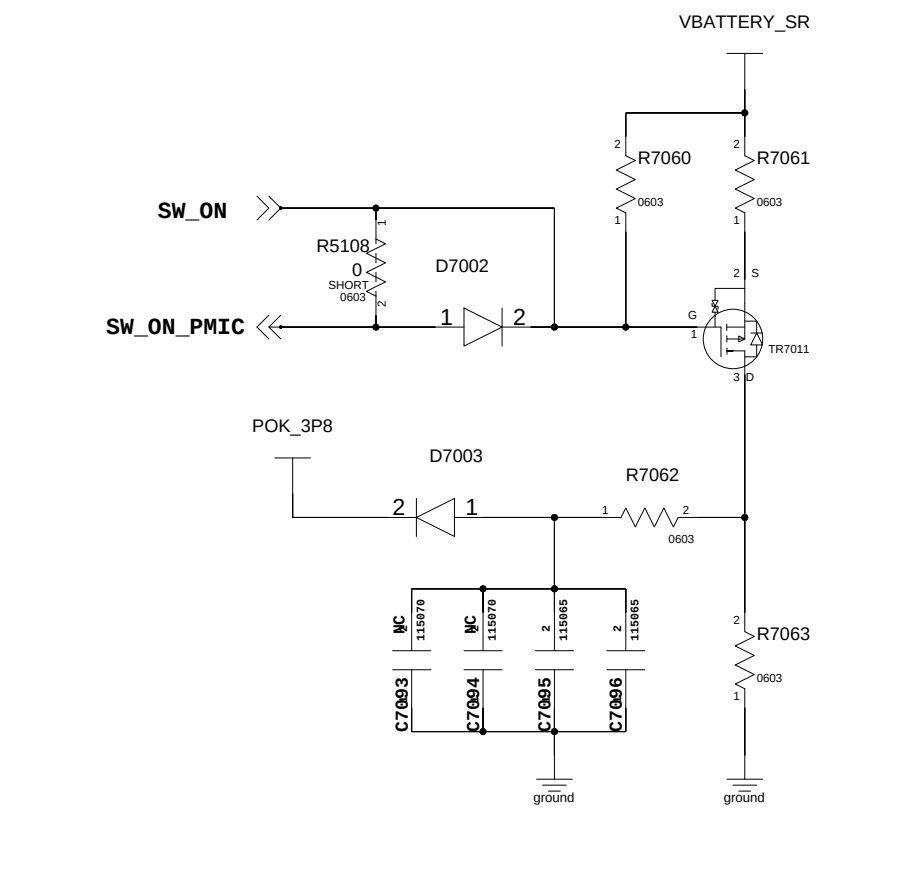
VBUS_IF
C7037 1nF 25.0V
C7038 1uF 25.0V
U7003
VBUS
MIDVBUS
C7039 10uF 25.0V
C7040 10uF 25.0V
VL
C7041 4.7uF
VL
C2P
C1P
C7049 22uF 10.0V
C7052 22uF 10.0V
C7053 22uF 10.0V
C7054 22uF 10.0V
C7055 22uF 10.0V
C7056 22uF 10.0V
C7050 22uF 10.0V
C7051 22uF 10.0V
C2N
C1N
VOUT
VBATTERY_SR
C7057 10uF
C7058 10uF
PGND
VREG_S10B_1P8
R7051
C7059 22nF
CDRVH
CDRVL
BATSNSP
BATSNSN
VBATTERY_CON
@VBAT_SENSE_N
CSNSP
CSNSN
DIRECT_SDA_1P8
DIRECT_SCL_1P8
DC_IRQ_N
SDA
SCL
NINT
NEN
FPWM
TEST1
TEST2
NC
VL
VREG_S10B_1P8
R7030 100K
TH7002
AGND
THEM

Direct Charging

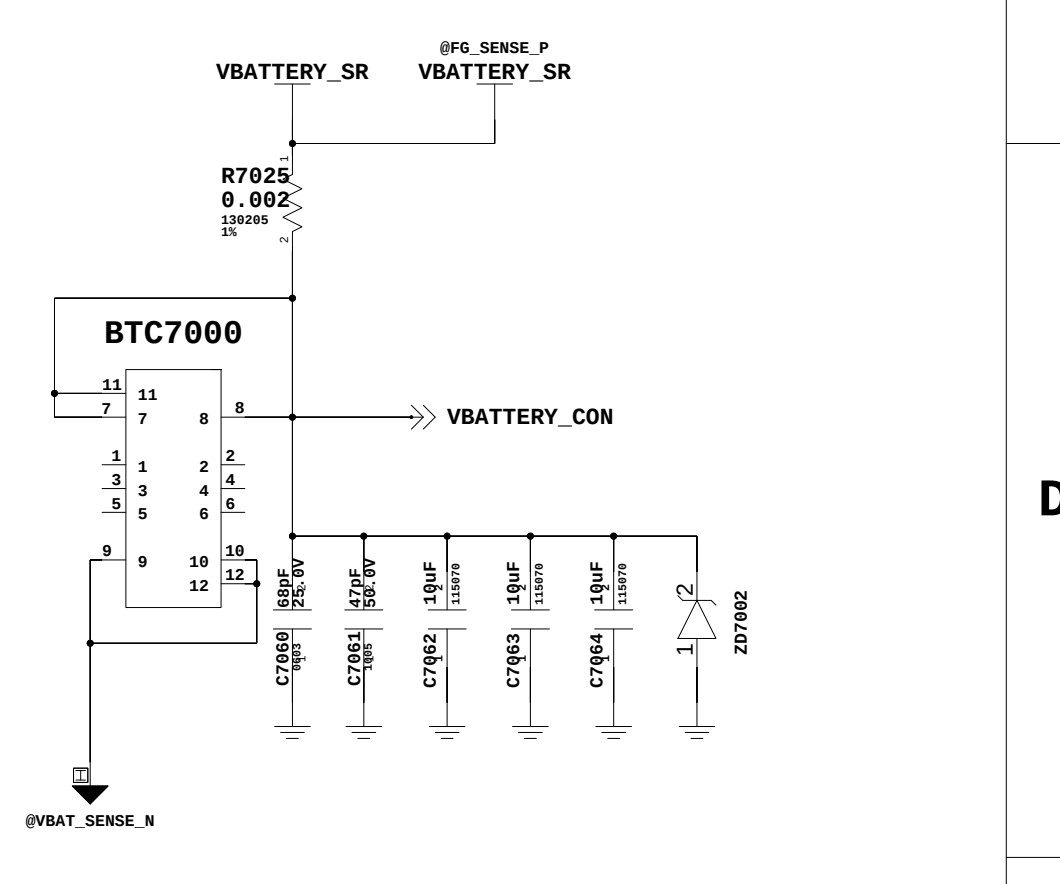
OVP



VPH_PWR
VBUS_IF
VBUS_OVP
U7002
VOUT
SYS
VBUS
L7006
VPH_PWR
POK_3P8
VREG_S10B_1P8
R7053 4.7K
R7052 4.7K
VPWR_SIG
OVLO_SEL
OVP_PWR_FLAGB
OV_PWR_FLAGB
SHDN
OVP_SHDN
VBUS_DISCHARGING_EN
DIS
EN0
EN1
EN2
POK
JIG_ONB_CC
D7000
GND
OVP_SIG_FLAGB
OV_SIG_FLAGB
CC1
CC2
SBU1
SBU2
DP
DM
CC1_C
CC2_C
SBU1_C
SBU2_C
DP_C
DM_C
CC1_CON
CC2_CON
SBU1_CON
SBU2_CON
C7077
C7048 4.7uF 10.0V
C7047
C7088 1uF 16.0V
C7046 4.7uF 10.0V
C7045 4.7uF 10.0V
C7044 1uF 25.0V
C7043 1uF 50.0V



VBATTERY_SR
SW_ON
R5108 0 SHORT
D7002
SW_ON_PMIC
R7060
R7061
TR7011
POK_3P8
D7003
R7062
R7063
C7003 NC
C7004 NC
C7005
C7006
ground

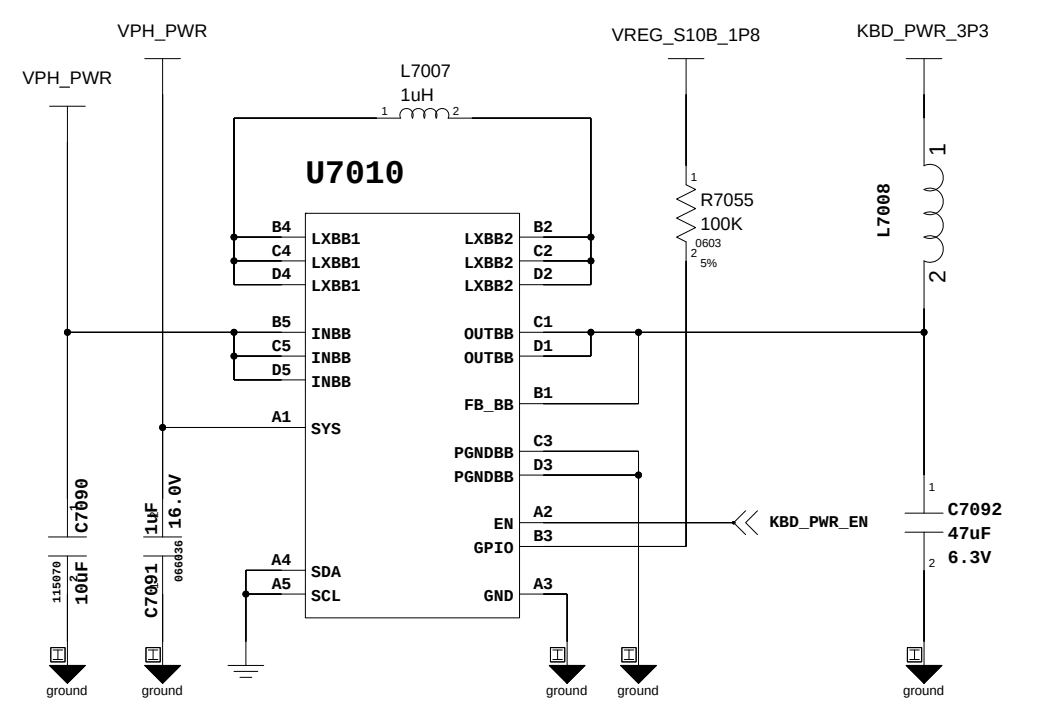


@FG_SENSE_P
VBATTERY_SR
VBATTERY_SR
R7025 0.002 1%
BTC7000
VBATTERY_CON
C7060 68pF 25.0V
C7061 47pF 50.0V
C7062 10uF
C7063 10uF
C7064 10uF
ZD7002
@VBAT_SENSE_N

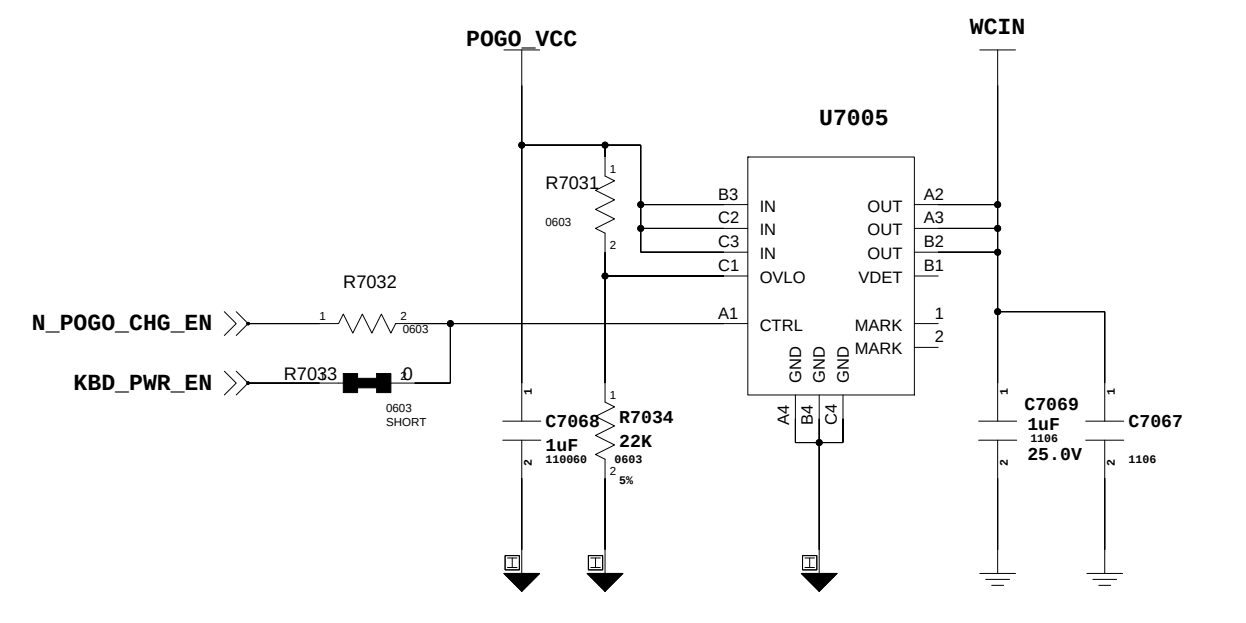
Ship Mode Off

Battery Con

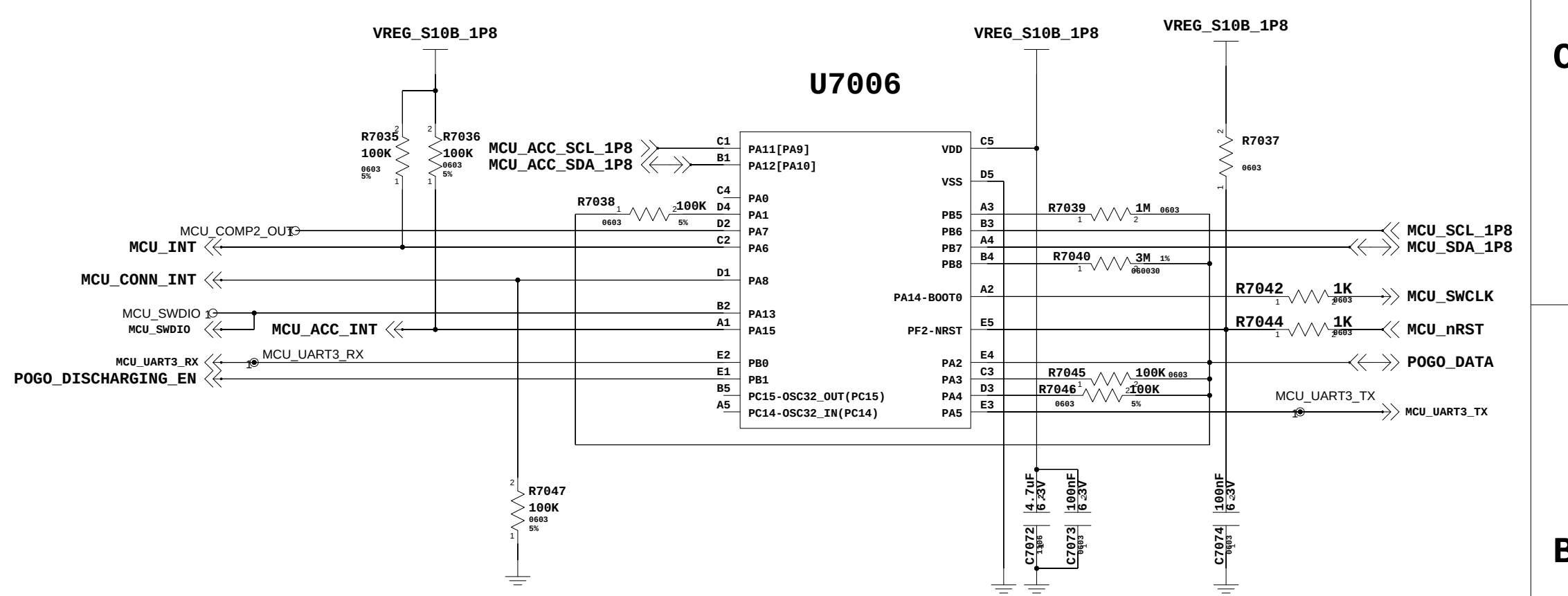
POGO



VPH_PWR
VPH_PWR
L7007 1uH
VREG_S10B_1P8
KBD_PWR_3P3
U7010
LXBB1
LXBB2
INBB
OUTBB
SYS
FB_BB
PGNDBB
EN
GPIO
SDA
SCL
GND
R7055 100K
L7008
KBD_PWR_EN
C7092 47uF 6.3V
C7090 10uF
C7091 1uF 16.0V
ground

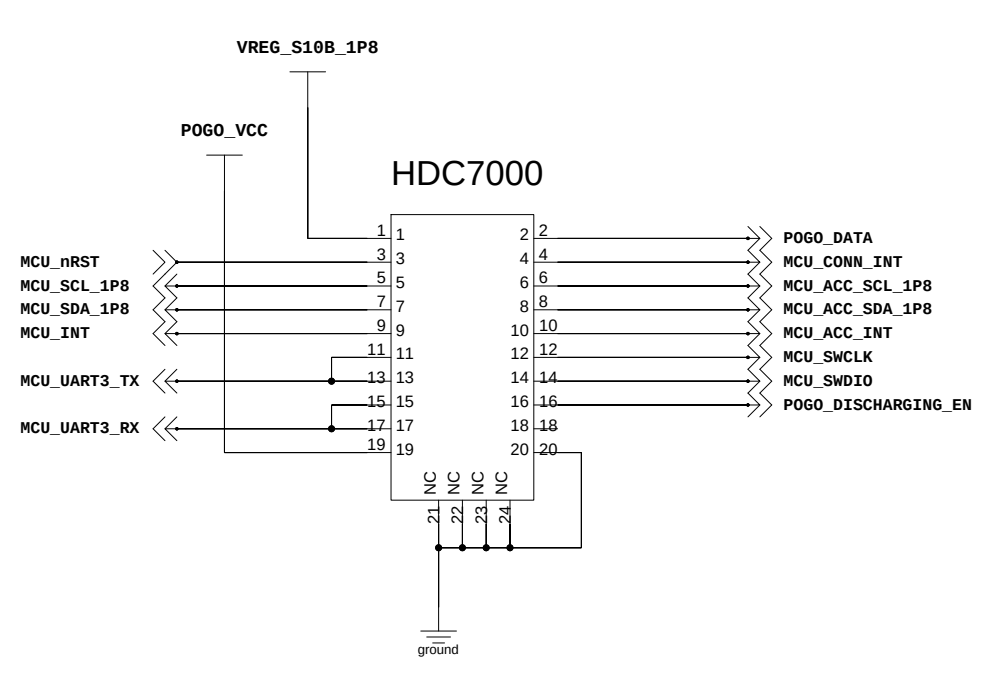


POGO_VCC
WCIN
U7005
R7031
IN
OVLO
OUT
VDET
N_POGO_CHG_EN
R7032
CTRL
MARK
KBD_PWR_EN
R7033 0 SHORT
GND
C7068 1uF
R7034 22K
C7069 1uF 25.0V
C7067



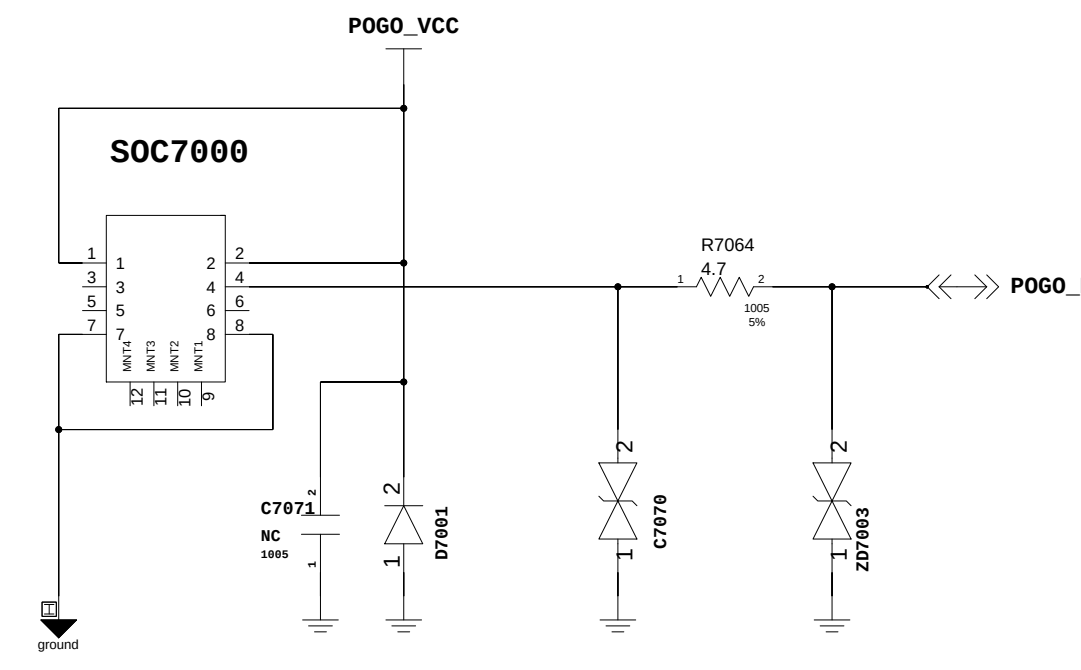
VREG_S10B_1P8
VREG_S10B_1P8
VREG_S10B_1P8
U7006
R7035 100K
R7036 100K
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
PA11[PA9]
PA12[PA10]
VDD
VSS
R7037
R7038 100K
MCU_COMP2_OUT
MCU_INT
PA0
PA1
PA7
PA6
R7039 1M
R7040 3M
PB5
PB6
PB7
PB8
MCU_SCL_1P8
MCU_SDA_1P8
MCU_CONN_INT
PA8
PA14-BOOT0
R7042 1K
MCU_SWCLK
MCU_SWDIO
MCU_ACC_INT
PA13
PA15
PF2-NRST
R7044 1K
MCU_nRST
MCU_UART3_RX
POGO_DISCHARGING_EN
PB0
PB1
PC15-OSC32_OUT(PC15)
PC14-OSC32_IN(PC14)
PA2
PA3
PA4
PA5
R7045 100K
R7046 100K
POGO_DATA
MCU_UART3_TX
R7047 100K
C7072 4.7uF 6.3V
C7073 100nF 6.3V
C7074 100nF 6.3V

POGO MCU



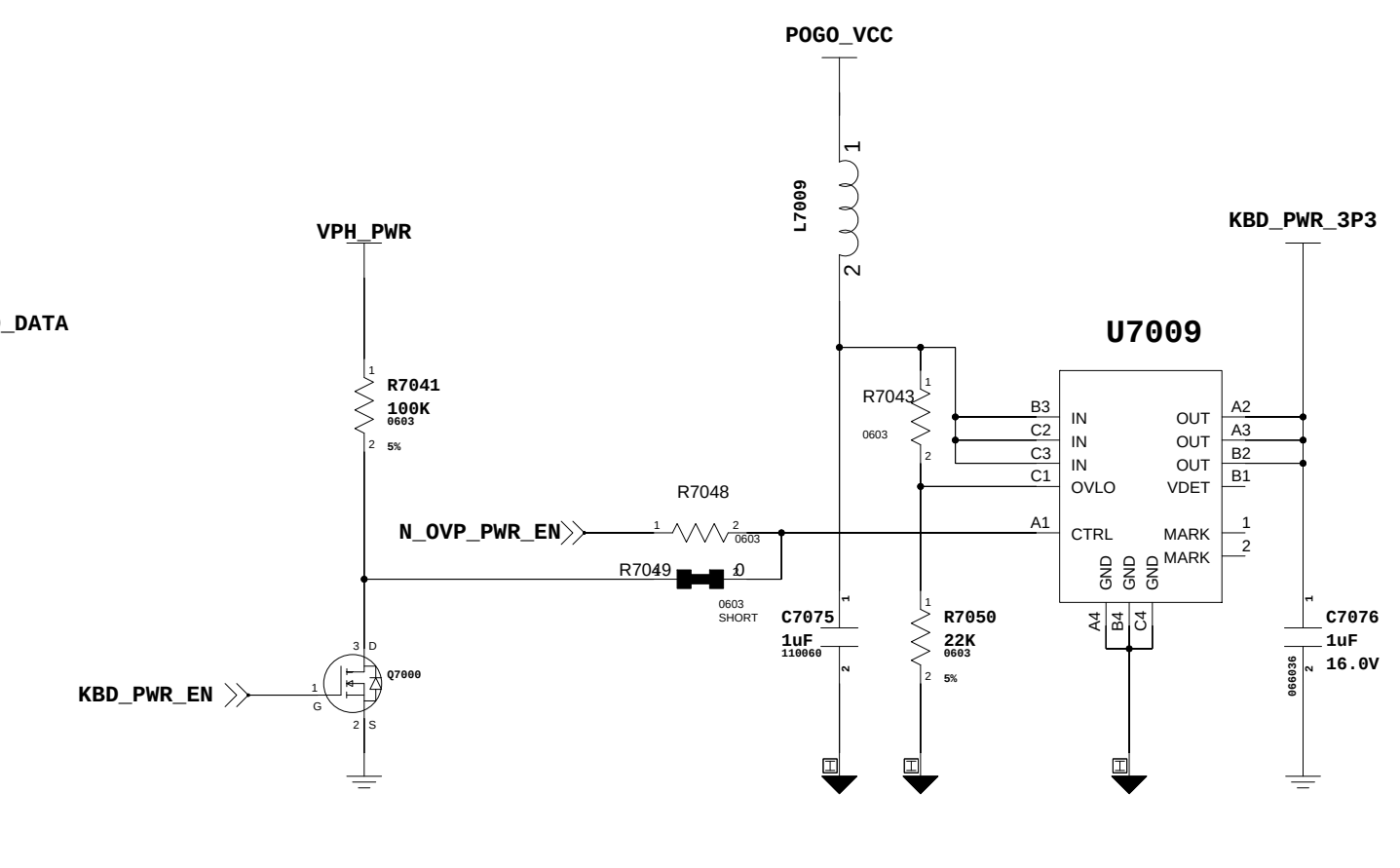
VREG_S10B_1P8
POGO_VCC
HDC7000
MCU_nRST
MCU_SCL_1P8
MCU_SDA_1P8
MCU_INT
MCU_UART3_TX
MCU_UART3_RX
POGO_DATA
MCU_CONN_INT
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_ACC_INT
MCU_SWCLK
MCU_SWDIO
POGO_DISCHARGING_EN
NC
ground

MCU DEBUG CON

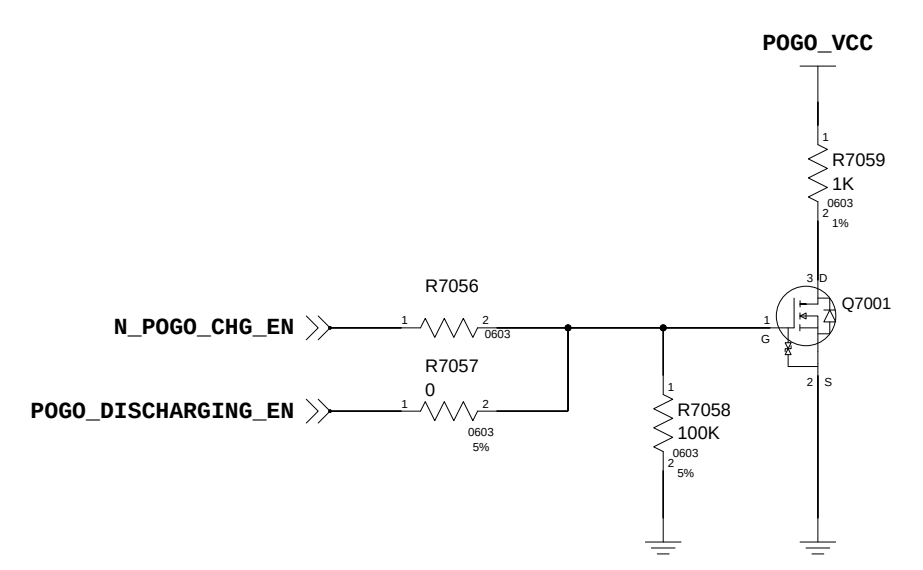


POGO_VCC
SOC7000
R7064 4.7
POGO_DATA
C7071 NC
D7001
C7070
ZD7003
ground

POGO BtoB Con.



POGO_VCC
L7009
KBD_PWR_3P3
U7009
VPH_PWR
R7041 100K
R7043
IN
OVLO
OUT
VDET
N_OVP_PWR_EN
R7048
CTRL
MARK
R7049 0 SHORT
GND
C7075 1uF
R7050 22K
C7076 1uF 16.0V
KBD_PWR_EN
Q7000



POGO_VCC
R7059 1K
Q7001
N_POGO_CHG_EN
R7056
POGO_DISCHARGING_EN
R7057 0
R7058 100K

TITLE	MODEL NAME		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 1 of 2	Date Drawing Date	

AUDIO

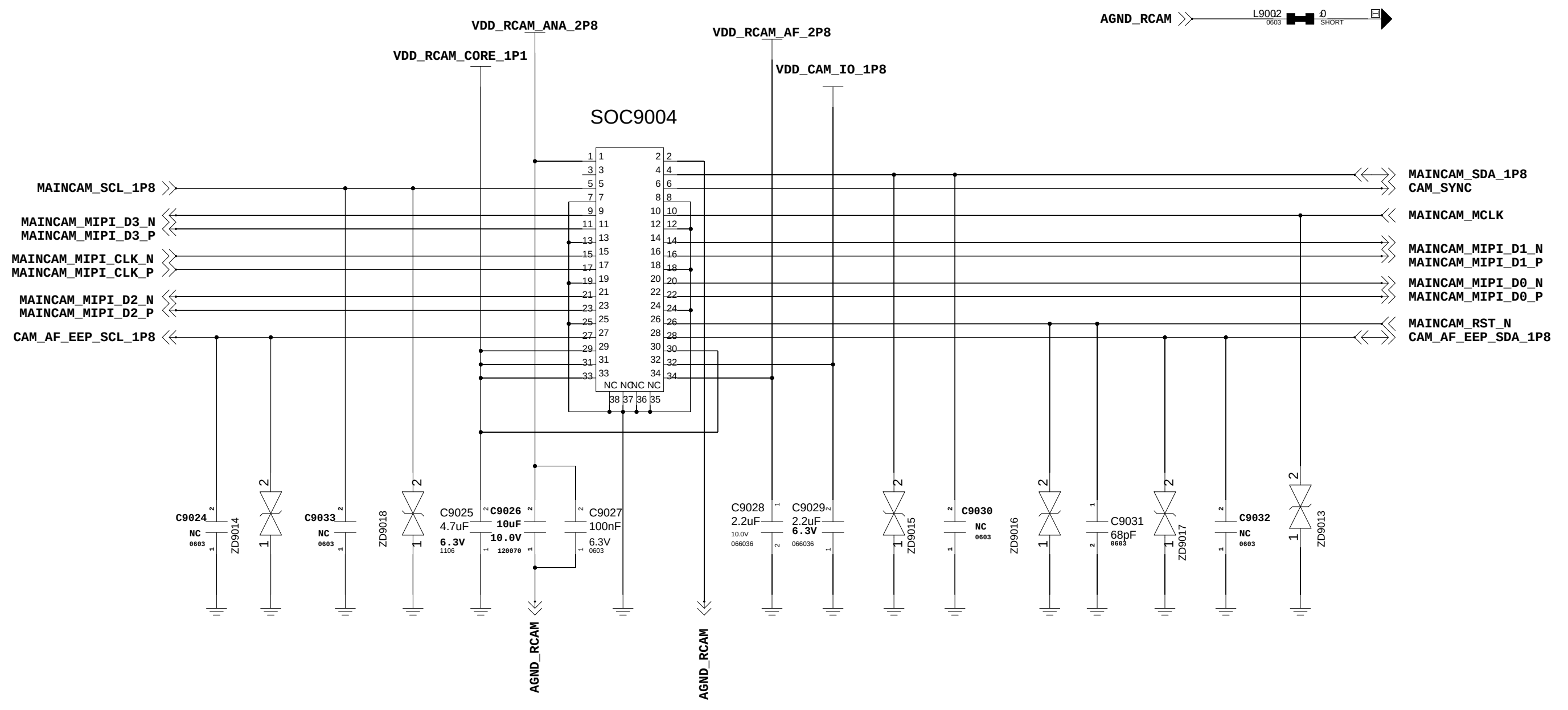
MAIN MIC
VREG_L12C_1P8
ANT8004
ANT8005
ANT8006
ANT8007
C8001
NC
0603
R8002
30
0603
C8003
NC
0603
ZD8001
ZD8003
MAIN_MIC_DATA
MAIN_MIC_CLK
SUB MIC
VREG_L12C_1P8
ANT8002
ANT8003
ANT8008
ANT8009
C8004
NC
0603
R8001
30
0603
C8002
56pF
0603
ZD8000
ZD8004
SUB_MIC_DATA
SUB_MIC_CLK
3rd MIC
VREG_L12C_1P8
ANT8000
ANT8001
ANT8010
ANT8011
C8000
NC
0603
R8000
30
0603
C8005
56pF
0603
ZD8002
ZD8005
3RD_MIC_DATA
3RD_MIC_CLK
VBATTERY_SR
VPH_PWR
R8003
0
0603
SHORT
Place On array dummy
(SPK AMP SMD F/T)
SPK TOP LEFT AMP
U8001
VREG_S10B_1P8
R8006
100K
0603
5%
C8008
100nF
0603
C8009
10uF
C8011
1uF
6.3V
R8005
0
SHORT
0603
C8013
1uF
6.3V
C8015
470nF
6.3V
L8001
1uH
210170
C8018
100nF
25.0V
C8019
10uF
TA8001
47uF
8.0V
C8032
10uF
C8033
100nF
25.0V
C8034
10uF
C8035
10uF
C8031
NC
R8007
R8008
R8010
1
1%
R8011
0
SHORT
R8013
0
SHORT
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
RESET
I2C_SCL/SPI_MISO
I2C_SDA/SPI_MOSI
SPI_SS
SPI_SCK
ASP_DOUT
ASP_DIN
ASP_BCLK
ASP_FSYNC
INT/GPIO2
SYNC/GPIO1
VDD_LV
VREF_FILT
VDD_BST
SWIRE_CLK
SWIRE_DATA
ADDR
VDD_A
LDO_FILT
VDD_BATT
SW
VDD_SPK
OUTP_SPK
OUTN_SPK
VSNSP_SPK
VSNSN_SPK
GPIO3
GND_A
GND_B
GND_P
VBATTERY_SR
SPK_TL_OUT_P
SPK_TL_OUT_N
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_SENSE_N
SPK TOP RIGHT AMP
U8000
C8006
100nF
C8007
10uF
C8010
1uF
6.3V
R8004
0
SHORT
C8012
1uF
6.3V
C8014
470nF
6.3V
L8000
1uH
C8016
100nF
C8017
10uF
TA8000
47uF
C8027
10uF
C8028
100nF
25.0V
C8029
10uF
C8030
10uF
C8020
NC
R8009
1
R8012
0
SHORT
SPK_TR_OUT_P
SPK_TR_OUT_N
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_SENSE_N
SPK BOTTOM LEFT AMP
U8003
C8044
100nF
C8045
10uF
C8047
1uF
6.3V
R8015
5.1K
1005
1%
C8049
1uF
6.3V
C8051
470nF
6.3V
L8007
1uH
C8054
100nF
C8055
10uF
TA8003
47uF
C8066
10uF
C8067
100nF
25.0V
C8068
10uF
C8069
10uF
C8071
NC
C8072
NC
C8075
NC
C8073
NC
C8074
NC
L8010
L8011
1005
TP8000
TP8001
TP8002
R8017
1
R8019
0
SHORT
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
SPK_BL_OUT_P
SPK_BL_OUT_N
SPK BOTTOM RIGHT AMP
U8002
C8042
100nF
C8043
10uF
C8046
1uF
6.3V
R8014
5.1K
1005
1%
C8048
1uF
6.3V
C8050
470nF
6.3V
L8006
1uH
C8052
100nF
C8053
10uF
TA8002
47uF
C8056
10uF
C8057
100nF
25.0V
C8058
10uF
C8059
10uF
C8070
NC
R8016
1
R8018
0
SHORT
SPK_BR_OUT_P
SPK_BR_OUT_N
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_SENSE_N
SPK BR CON
ANT8012
ANT8013
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_P
L8002
82nH
1005
C8021
NC
C8022
56pF
C8023
NC
ZD8006
VOLT
ZD8007
VOLT
C8024
NC
C8025
56pF
L8003
82nH
C8026
NC
SPK_BR_OUT_SENSE_N
SPK_BR_OUT_N
SPK TR CON
ANT8014 ANT8015
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_P
L8004
82nH
C8036
NC
C8037
56pF
C8038
NC
ZD8008
ZD8009
C8039
NC
C8040
56pF
L8005
82nH
C8041
NC
SPK_TR_OUT_SENSE_N
SPK_TR_OUT_N
SPK TL CON
ANT8016 ANT8017
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_P
L8008
82nH
C8060
NC
C8061
56pF
C8062
NC
ZD8010
ZD8011
C8063
NC
C8064
56pF
L8009
82nH
C8065
NC
SPK_TL_OUT_SENSE_N
SPK_TL_OUT_N

TITLE TabS7P

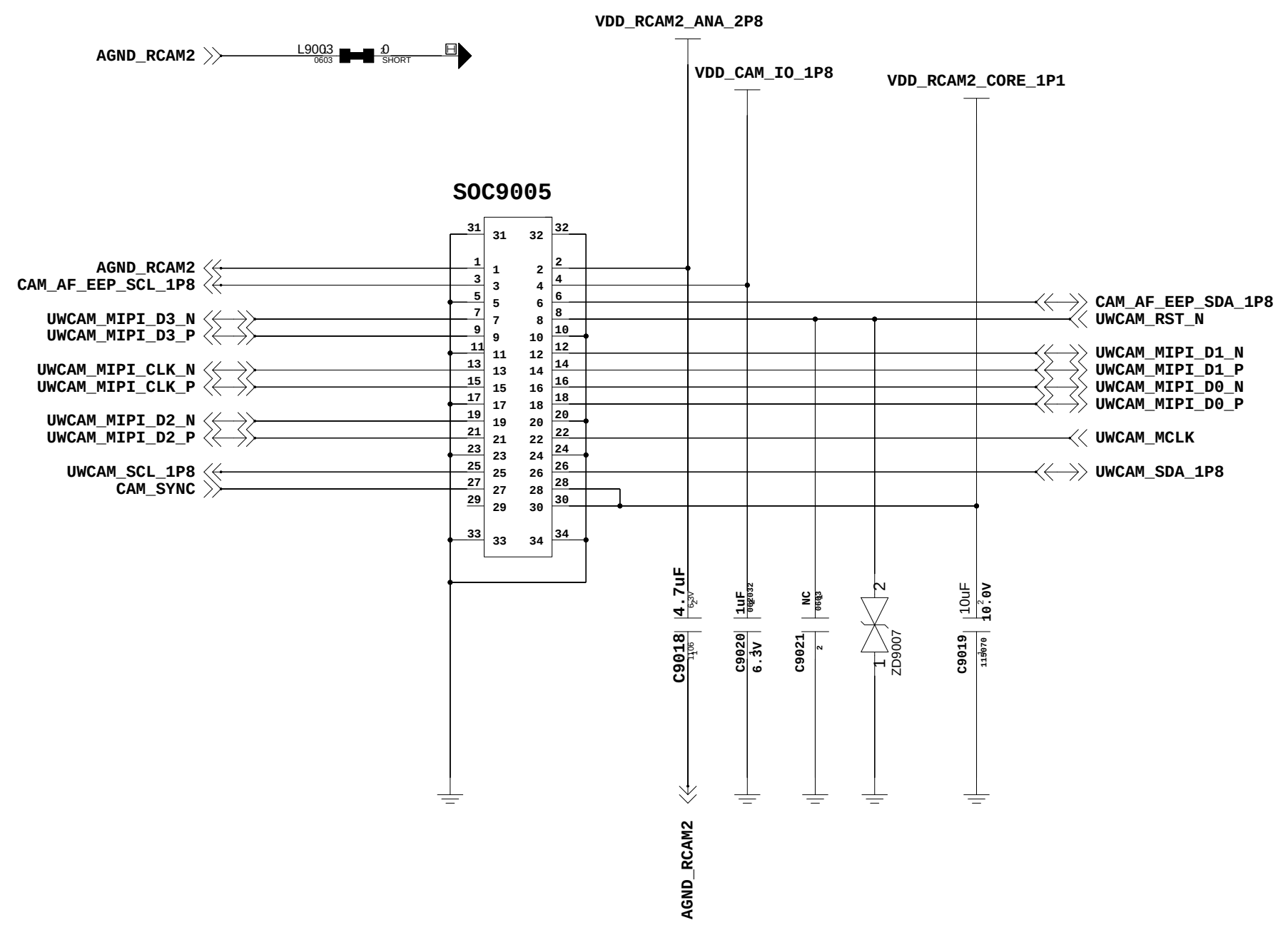
SIZE A1 DWG NO PDM NUMBER REV Rev

DRAWN BY User Name SHEET 8 of 16 Date Drawing Date

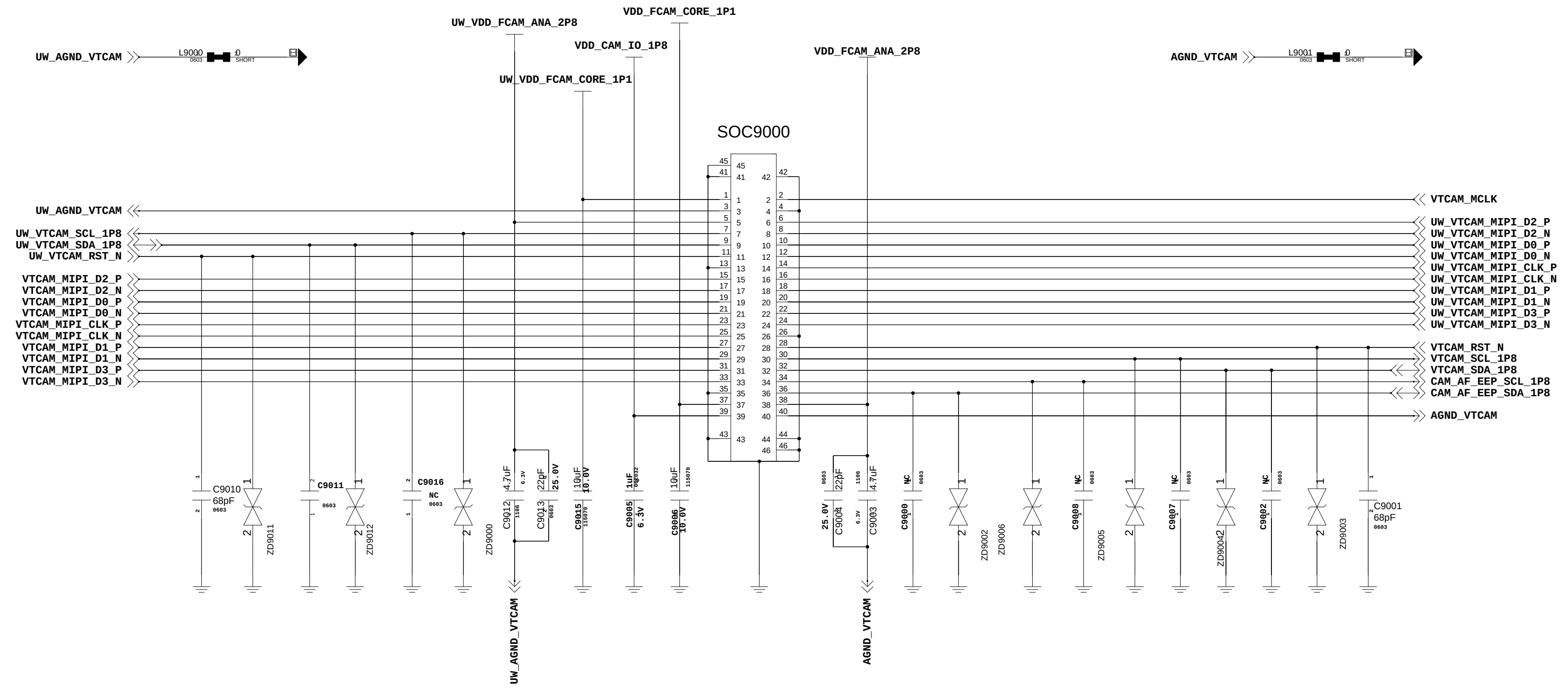
CAMERA


SOC9004
VDD_RCAM_ANA_2P8
VDD_RCAM_CORE_1P1
VDD_RCAM_AF_2P8
VDD_CAM_IO_1P8
AGND_RCAM
L9002
0
0603
SHORT
MAINCAM_SCL_1P8
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
CAM_AF_EEP_SCL_1P8
MAINCAM_SDA_1P8
CAM_SYNC
MAINCAM_MCLK
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_RST_N
CAM_AF_EEP_SDA_1P8
C9024 NC 0603
ZD9014
C9033 NC 0603
ZD9018
C9025 4.7uF 6.3V 1106
C9026 10uF 10.0V 120070
C9027 100nF 6.3V 0603
C9028 2.2uF 10.0V 066036
C9029 2.2uF 6.3V 066036
ZD9015
C9030 NC 0603
ZD9016
C9031 68pF 0603
ZD9017
C9032 NC 0603
ZD9013
AGND_RCAM
AGND_RCAM

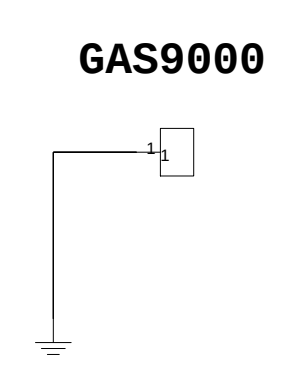
13M_CAM (RCAM)

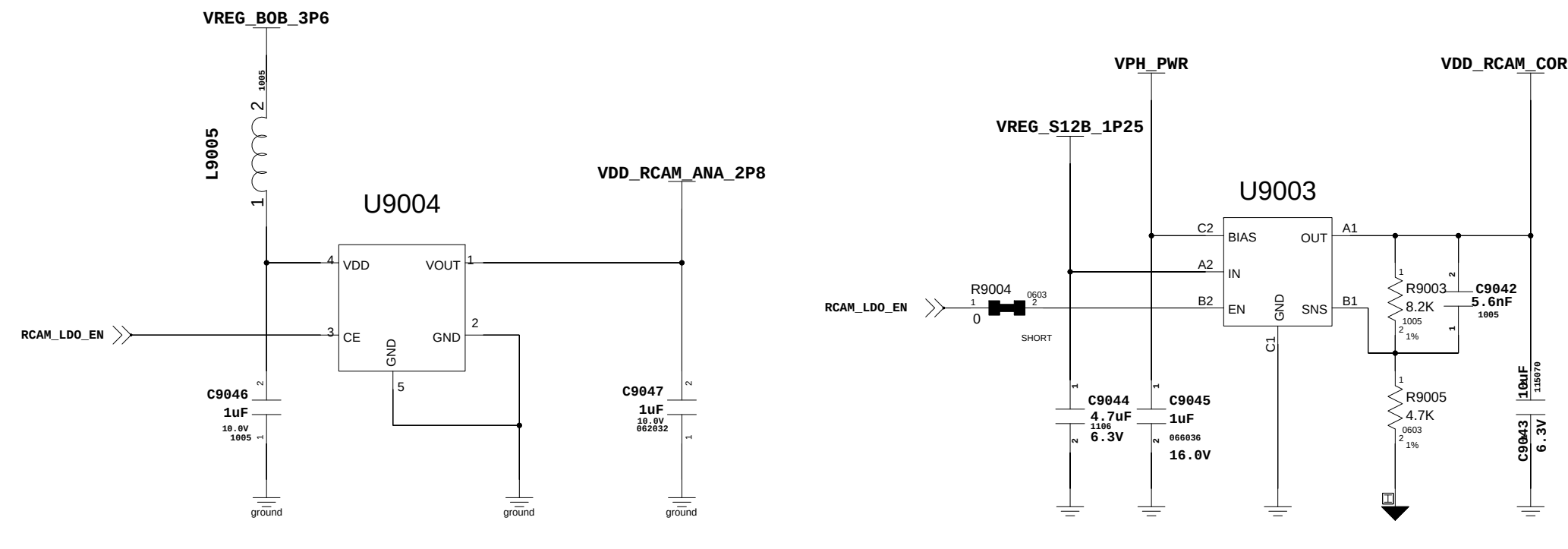

SOC9005
AGND_RCAM2
L9003
0
0603
SHORT
VDD_RCAM2_ANA_2P8
VDD_CAM_IO_1P8
VDD_RCAM2_CORE_1P1
AGND_RCAM2
CAM_AF_EEP_SCL_1P8
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_CLK_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_SCL_1P8
CAM_SYNC
CAM_AF_EEP_SDA_1P8
UWCAM_RST_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_D1_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
UWCAM_MCLK
UWCAM_SDA_1P8
C9018 4.7uF 1106
C9020 1uF 6.3V
C9021 NC 0603
ZD9007
C9019 10uF 10.0V 115070
AGND_RCAM2

6M_UW CAM (RCAM2)

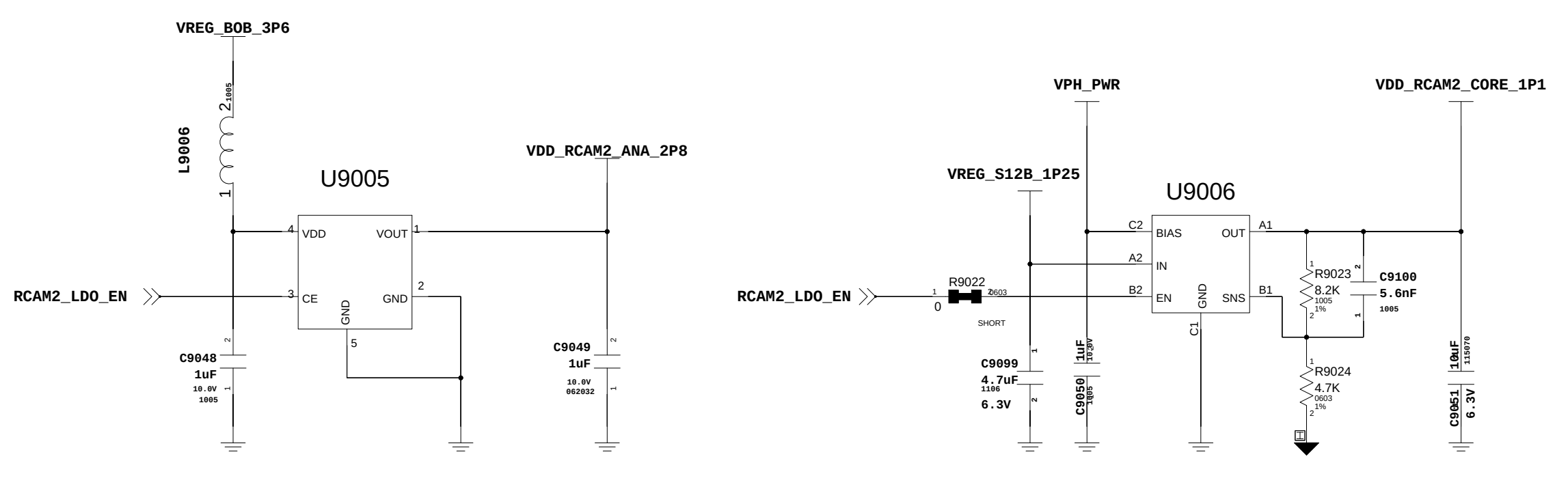

SOC9000
UW_AGND_VTCAM
L9000
0
0603
SHORT
UW_VDD_FCAM_ANA_2P8
VDD_FCAM_CORE_1P1
VDD_CAM_IO_1P8
UW_VDD_FCAM_CORE_1P1
VDD_FCAM_ANA_2P8
AGND_VTCAM
L9001
0
0603
SHORT
UW_AGND_VTCAM
UW_VTCAM_SCL_1P8
UW_VTCAM_SDA_1P8
UW_VTCAM_RST_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_D0_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D3_P
VTCAM_MIPI_D3_N
VTCAM_MCLK
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D1_P
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_RST_N
VTCAM_SCL_1P8
VTCAM_SDA_1P8
CAM_AF_EEP_SCL_1P8
CAM_AF_EEP_SDA_1P8
AGND_VTCAM
C9010 68pF 0603
ZD9011
C9011 0603
ZD9012
C9016 NC 0603
ZD9000
C9012 4.7uF 6.3V 1106
C9013 22pF 25.0V 0603
C9015 10uF 16.0V 115070
C9005 1uF 6.3V
C9006 10uF 10.0V 115070
C9004 22pF 25.0V 0603
C9003 4.7uF 6.3V 1106
C9000 NC 0603
ZD9002
ZD9006
C9008 NC 0603
ZD9005
C9007 NC 0603
ZD9004
C9002 NC 0603
ZD9003
C9001 68pF 0603
UW_AGND_VTCAM
AGND_VTCAM

12M Dual FCAM

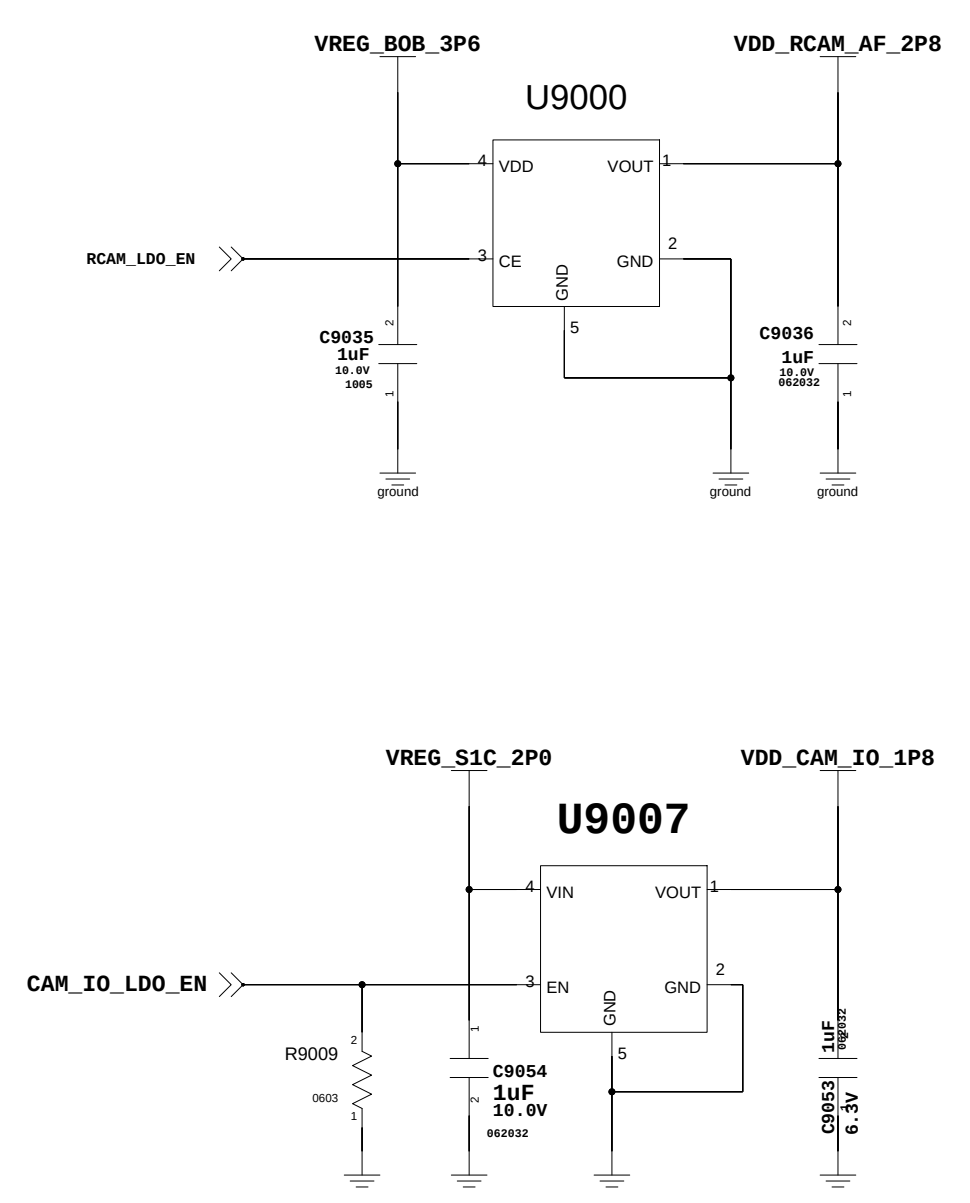


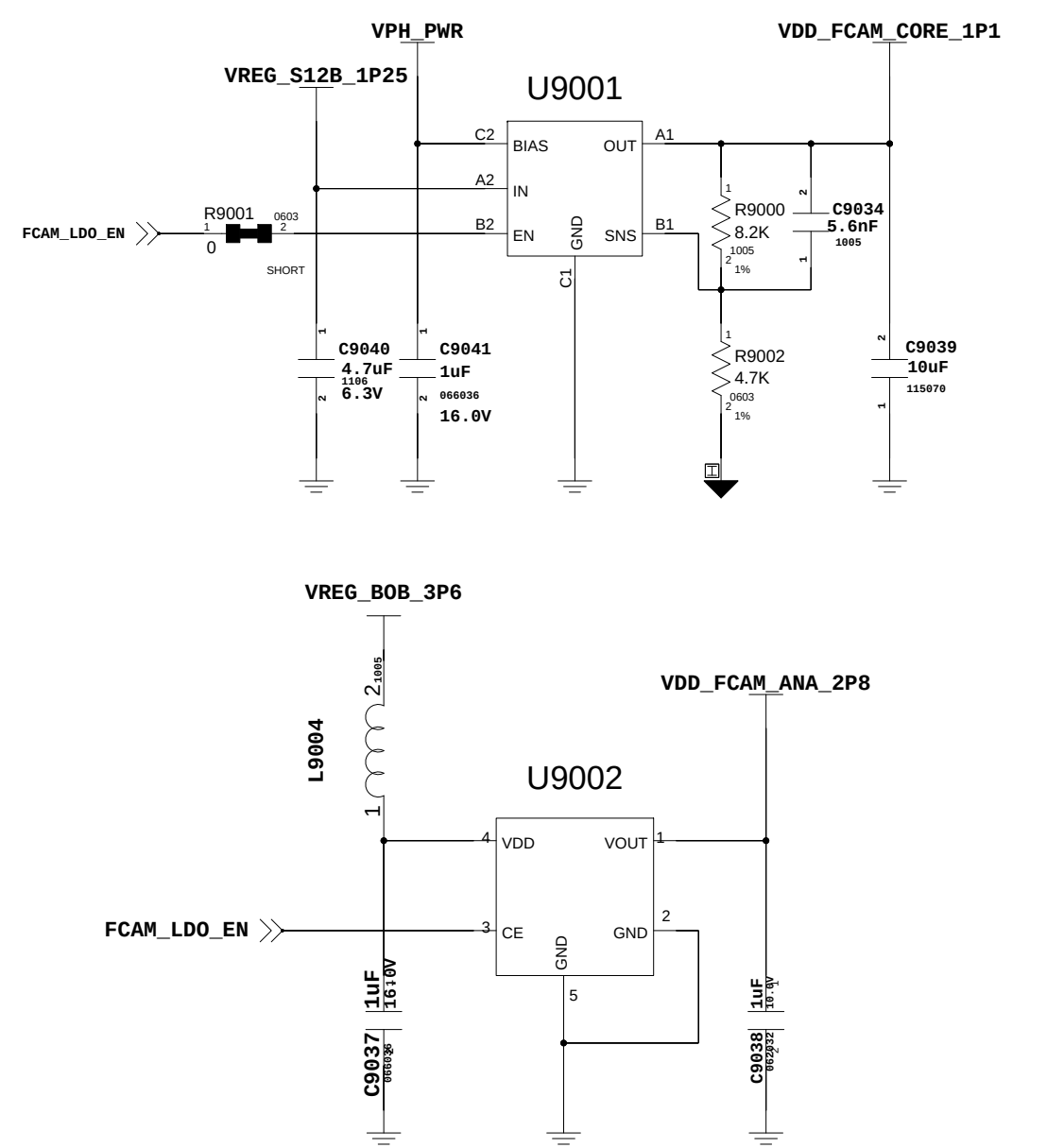

VREG_BOB_3P6
L9005
1005
U9004
VDD
VOUT
CE
GND
VDD_RCAM_ANA_2P8
RCAM_LDO_EN
C9046 1uF 10.0V 1005
C9047 1uF 10.0V 062032
ground
VPH_PWR
VREG_S12B_1P25
U9003
BIAS
IN
EN
OUT
SNS
GND
VDD_RCAM_CORE_1P1
RCAM_LDO_EN
R9004 0 0603 SHORT
R9003 8.2K 1005 1%
C9042 5.6nF 1005
R9005 4.7K 0603 1%
C9044 4.7uF 1106 6.3V
C9045 1uF 066036 16.0V
C9043 10uF 115070 6.3V

13M_CAM (RCAM)_LDO

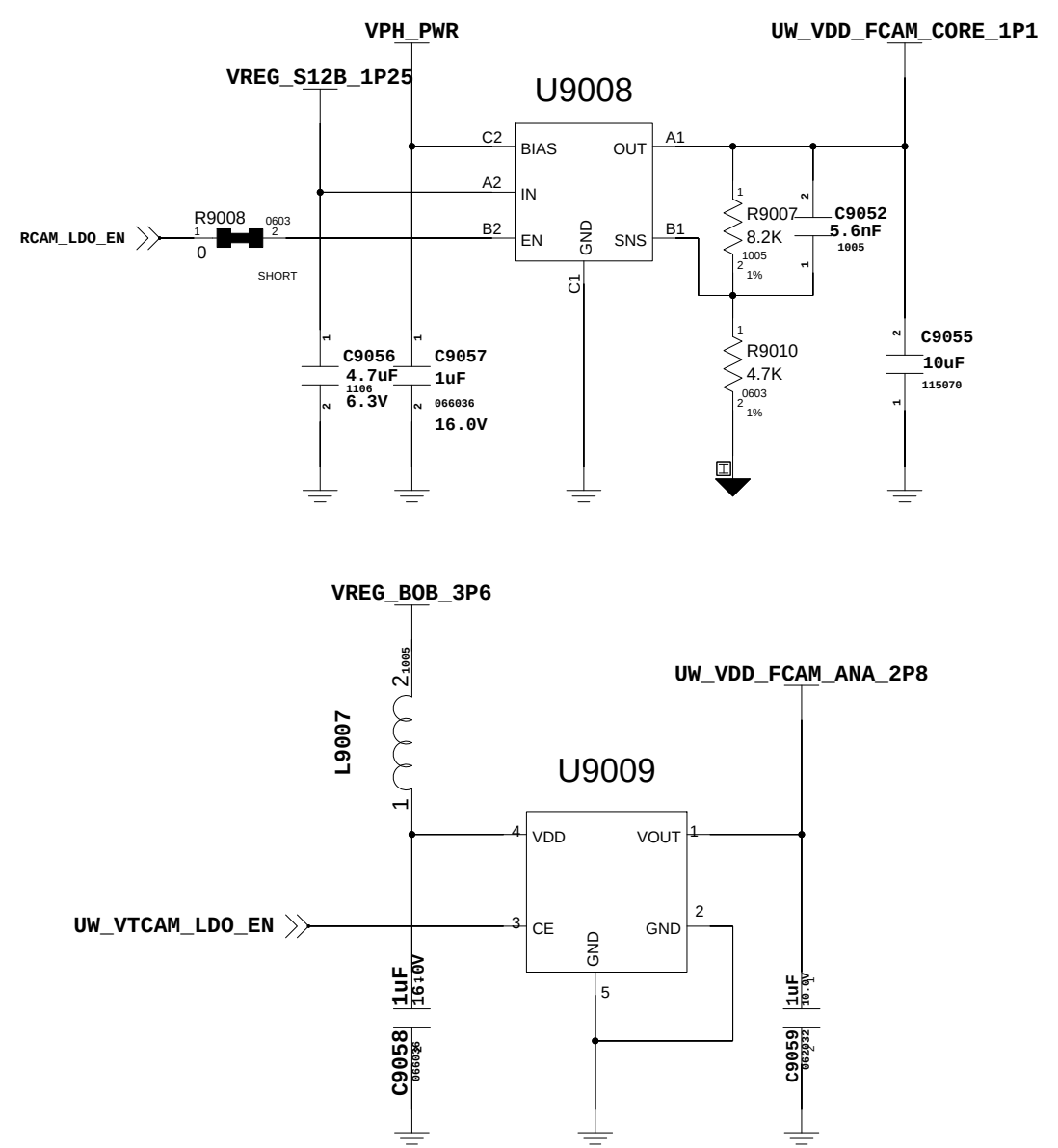

VREG_BOB_3P6
L9006
U9005
VDD
VOUT
CE
GND
VDD_RCAM2_ANA_2P8
RCAM2_LDO_EN
C9048 1uF 10.0V 1005
C9049 1uF 10.0V 062032
VPH_PWR
VREG_S12B_1P25
U9006
BIAS
IN
EN
OUT
SNS
GND
VDD_RCAM2_CORE_1P1
RCAM2_LDO_EN
R9022 0 0603 SHORT
R9023 8.2K 1005 1%
C9100 5.6nF 1005
R9024 4.7K 0603 1%
C9099 4.7uF 1106 6.3V
C9050 1uF 16.0V
C9051 10uF 115070 6.3V

6M_UW CAM (RCAM2)_LDO

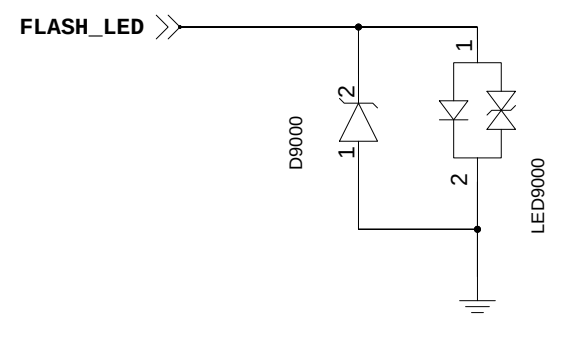

VREG_BOB_3P6
U9000
VDD
VOUT
CE
GND
VDD_RCAM_AF_2P8
RCAM_LDO_EN
C9035 1uF 10.0V 1005
C9036 1uF 10.0V 062032
ground
VREG_S1C_2P0
VDD_CAM_IO_1P8
U9007
VIN
VOUT
EN
GND
CAM_IO_LDO_EN
R9009 0603
C9054 1uF 10.0V 062032
C9053 1uF 6.3V


VPH_PWR
VREG_S12B_1P25
U9001
BIAS
IN
EN
OUT
SNS
GND
VDD_FCAM_CORE_1P1
FCAM_LDO_EN
R9001 0 0603 SHORT
R9000 8.2K 1005 1%
C9034 5.6nF 1005
R9002 4.7K 0603 1%
C9039 10uF 115070
C9040 4.7uF 1106 6.3V
C9041 1uF 066036 16.0V
VREG_BOB_3P6
L9004
U9002
VDD
VOUT
CE
GND
VDD_FCAM_ANA_2P8
FCAM_LDO_EN
C9037 1uF 16.0V
C9038 1uF

VTCAM_LDO

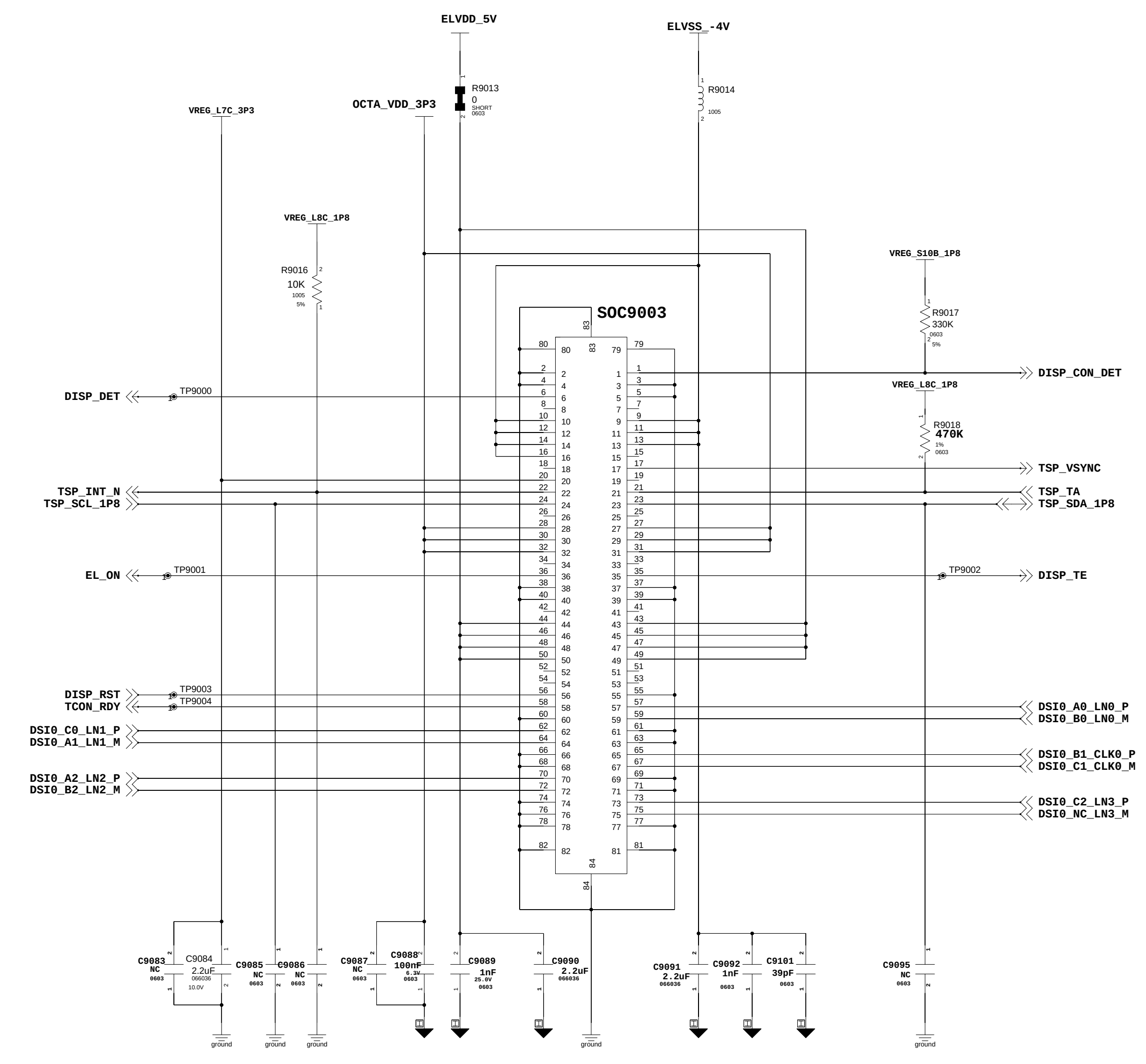

VPH_PWR
VREG_S12B_1P25
U9008
BIAS
IN
EN
OUT
SNS
GND
UW_VDD_FCAM_CORE_1P1
RCAM_LDO_EN
R9008 0 0603 SHORT
R9007 8.2K 1005 1%
C9052 5.6nF 1005
R9010 4.7K 0603 1%
C9055 10uF 115070
C9056 4.7uF 1106 6.3V
C9057 1uF 066036 16.0V
VREG_BOB_3P6
L9007
U9009
VDD
VOUT
CE
GND
UW_VDD_FCAM_ANA_2P8
UW_VTCAM_LDO_EN
C9058 1uF 16.0V
C9059 1uF

UW VT CAM_LDO

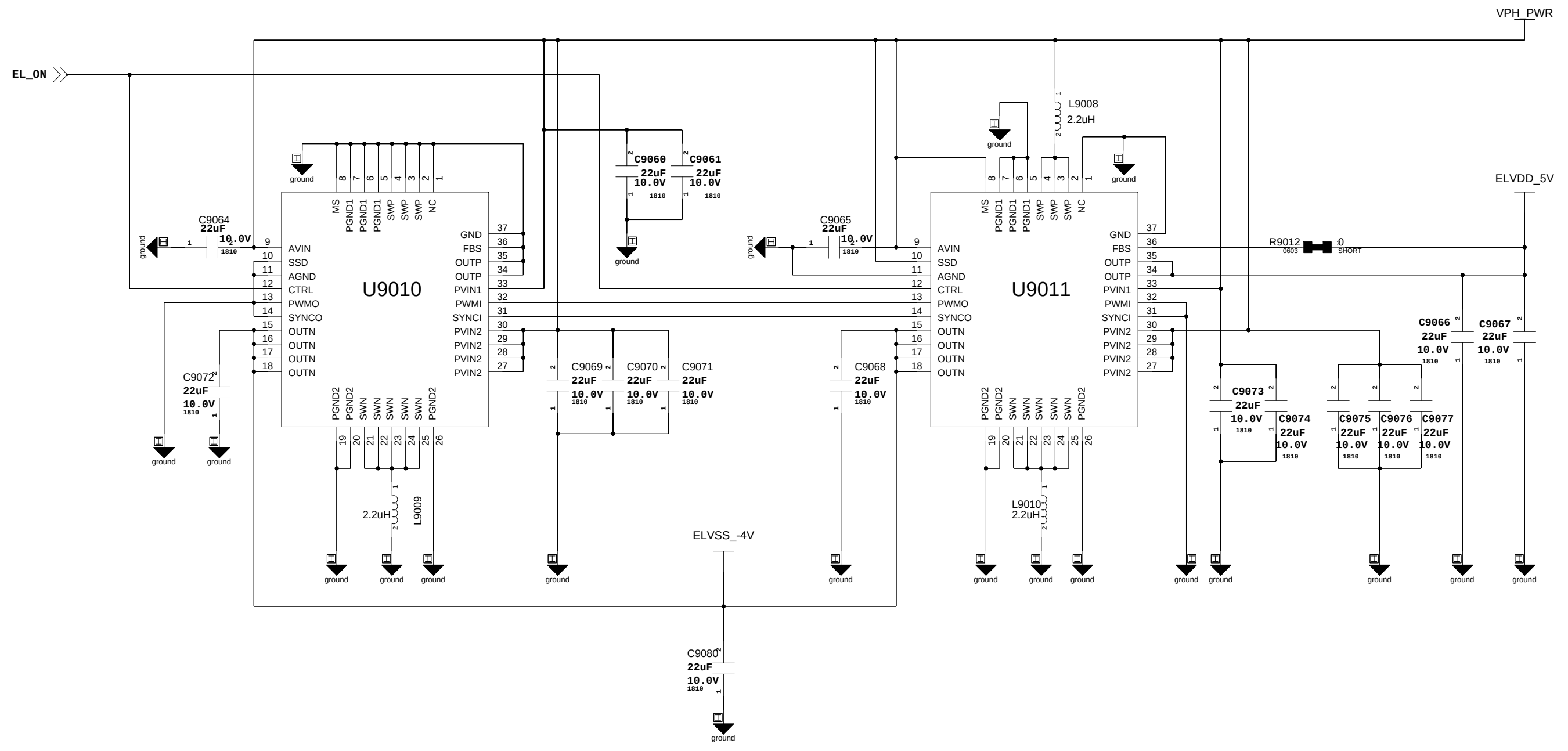

FLASH_LED
D9000
LED9000

FLASH LED

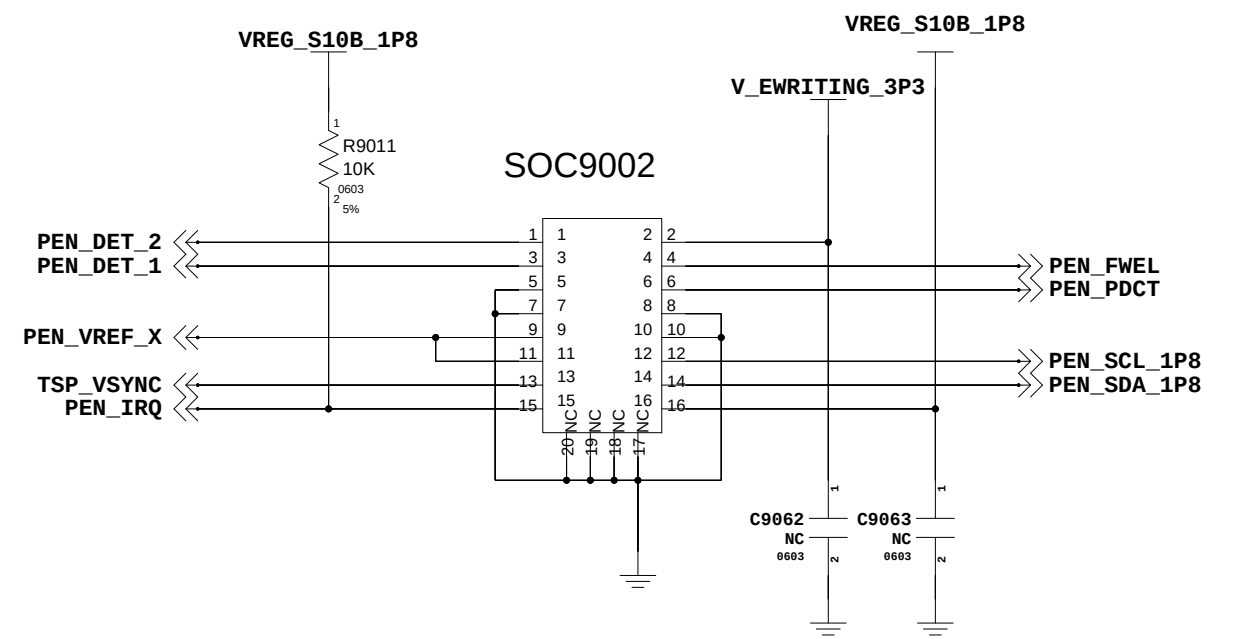
DISPLAY


SOC9003
ELVDD_5V
ELVSS_-4V
R9013 0 SHORT 0603
R9014 1005
OCTA_VDD_3P3
VREG_L7C_3P3
VREG_L8C_1P8
R9016 10K 1005 5%
VREG_S10B_1P8
R9017 330K 0603 5%
VREG_L8C_1P8
R9018 470K 1% 0603
DISP_DET
TP9000
TSP_INT_N
TSP_SCL_1P8
EL_ON
TP9001
DISP_RST
TP9003
TCON_RDY
TP9004
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_A2_LN2_P
DSI0_B2_LN2_M
DISP_CON_DET
TSP_VSYNC
TSP_TA
TSP_SDA_1P8
DISP_TE
TP9002
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_B1_CLK0_P
DSI0_C1_CLK0_M
DSI0_C2_LN3_P
DSI0_NC_LN3_M
C9083 NC 0603
C9084 2.2uF 066036 10.0V
C9085 NC 0603
C9086 NC 0603
C9087 NC 0603
C9088 100nF 6.3V 0603
C9089 1nF 25.0V 0603
C9090 2.2uF 066036
C9091 2.2uF 066036
C9092 1nF 0603
C9101 39pF 0603
C9095 NC 0603
ground

OCTA Con.

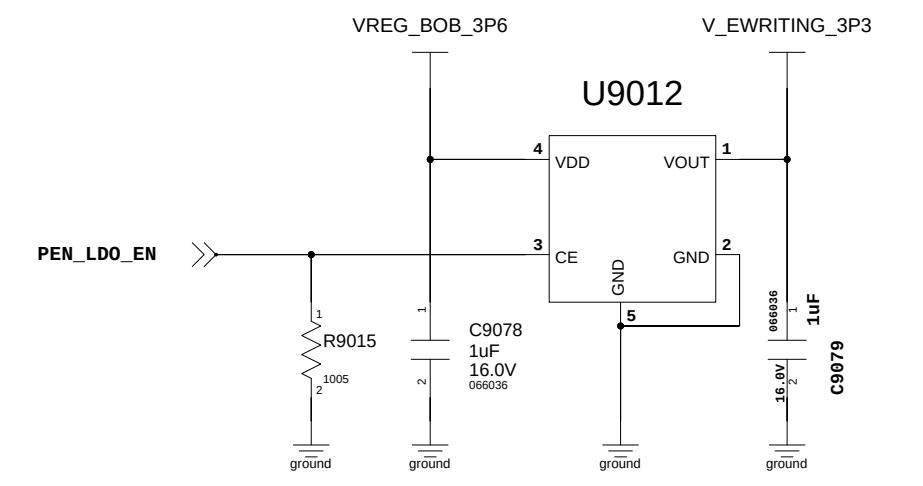

EL_ON
VPH_PWR
U9010
U9011
AVIN
SSD
AGND
CTRL
PWMO
SYNCO
OUTN
GND
FBS
OUTP
PVIN1
PWMI
SYNCI
PVIN2
MS
PGND1
SWP
NC
PGND2
SWN
C9064 22uF 10.0V 1810
C9065 22uF 10.0V 1810
C9072 22uF 10.0V 1810
C9060 22uF 10.0V 1810
C9061 22uF 10.0V 1810
C9069 22uF 10.0V 1810
C9070 22uF 10.0V 1810
C9071 22uF 10.0V 1810
C9068 22uF 10.0V 1810
L9009 2.2uH
L9008 2.2uH
L9010 2.2uH
R9012 0 0603 SHORT
ELVDD_5V
C9066 22uF 10.0V 1810
C9067 22uF 10.0V 1810
C9073 22uF 10.0V 1810
C9074 22uF 10.0V 1810
C9075 22uF 10.0V 1810
C9076 22uF 10.0V 1810
C9077 22uF 10.0V 1810
ELVSS_-4V
C9080 22uF 10.0V 1810
ground

OCTA POWER

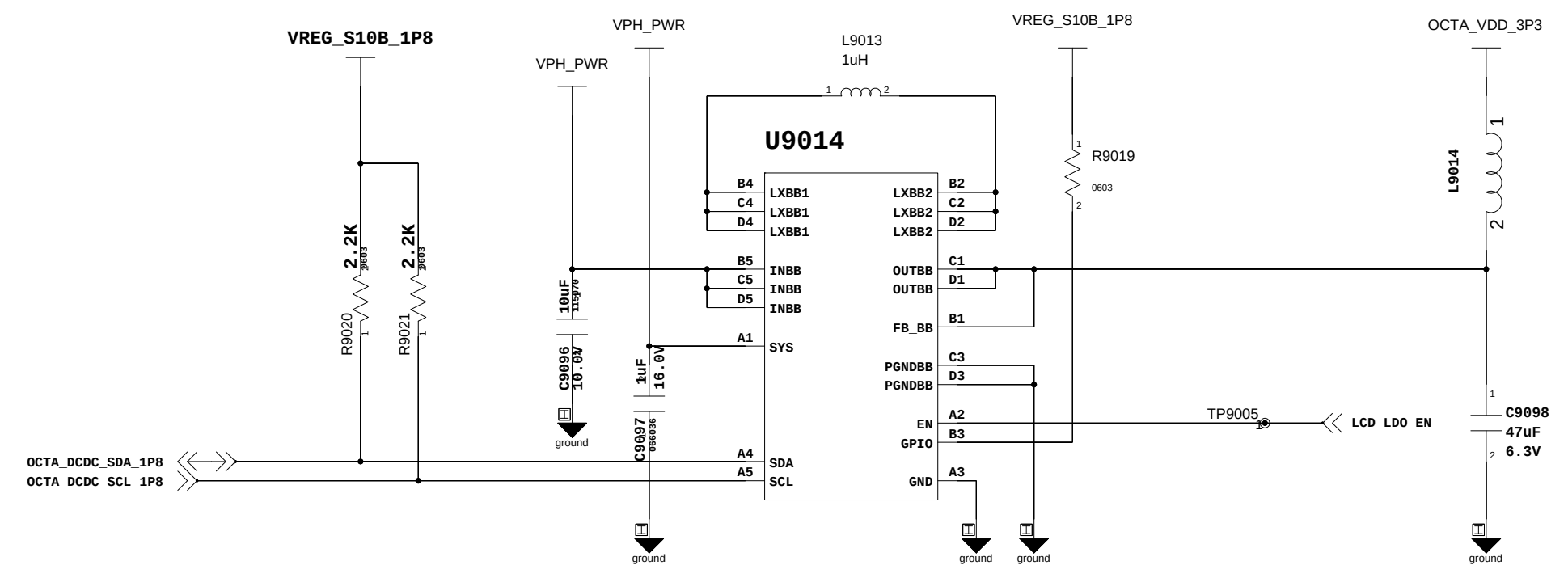

VREG_S10B_1P8
R9011 10K 0603 5%
SOC9002
V_EWRITING_3P3
VREG_S10B_1P8
PEN_DET_2
PEN_DET_1
PEN_VREF_X
TSP_VSYNC
PEN_IRQ
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
NC
C9062 NC 0603
C9063 NC 0603

WACOM CONN

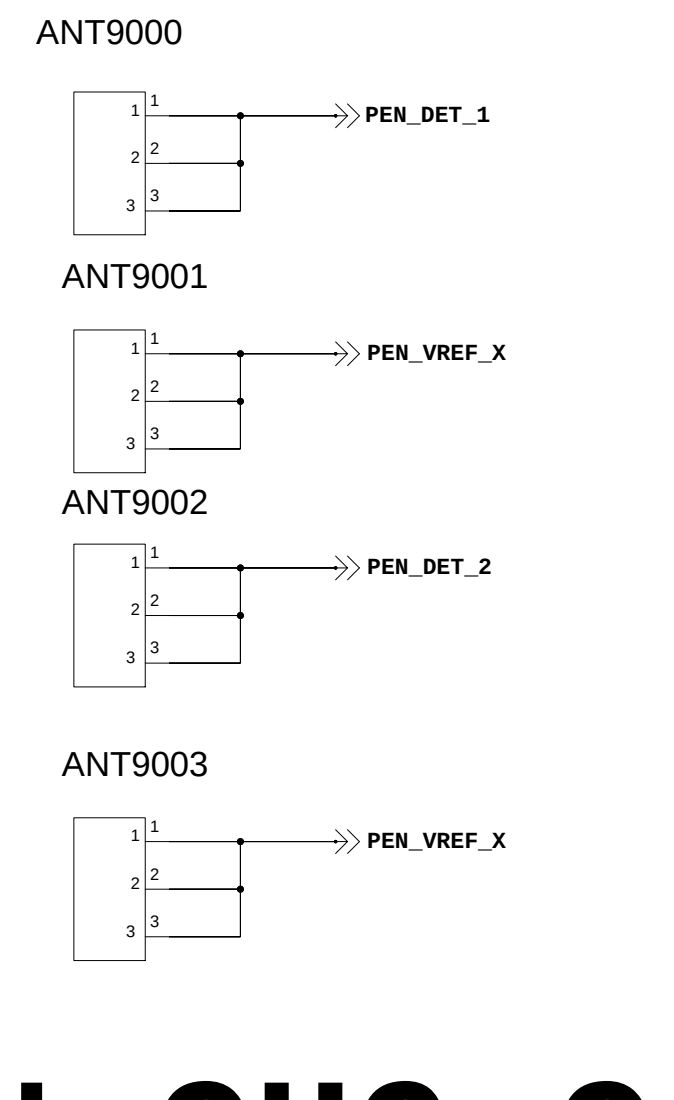
(MAIN <-> OCTA / WACOM)


VREG_BOB_3P6
V_EWRITING_3P3
U9012
VDD
VOUT
CE
GND
PEN_LDO_EN
R9015 1005
C9078 1uF 16.0V 066036
C9079 1uF 16.0V 066036
ground

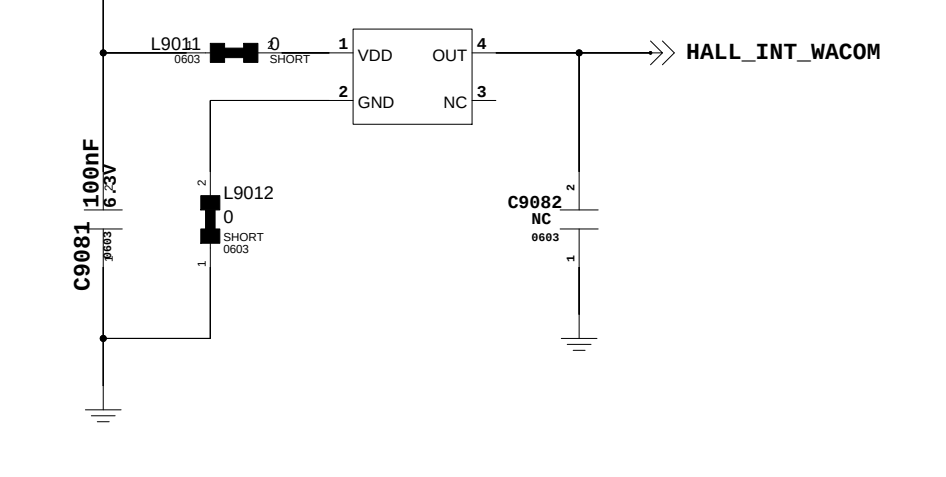
PEN LDO


VREG_S10B_1P8
VPH_PWR
VPH_PWR
L9013 1uH
VREG_S10B_1P8
OCTA_VDD_3P3
U9014
LXBB1
INBB
SYS
SDA
SCL
LXBB2
OUTBB
FB_BB
PGNDBB
EN
GPIO
GND
R9020 2.2K 0603
R9021 2.2K 0603
C9096 10uF 10.0V
C9097 1uF 16.0V
R9019 0603
L9014
TP9005
LCD_LDO_EN
C9098 47uF 6.3V
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
ground

TCON BUCK BOOSTER


ANT9000
PEN_DET_1
ANT9001
PEN_VREF_X
ANT9002
PEN_DET_2
ANT9003
PEN_VREF_X

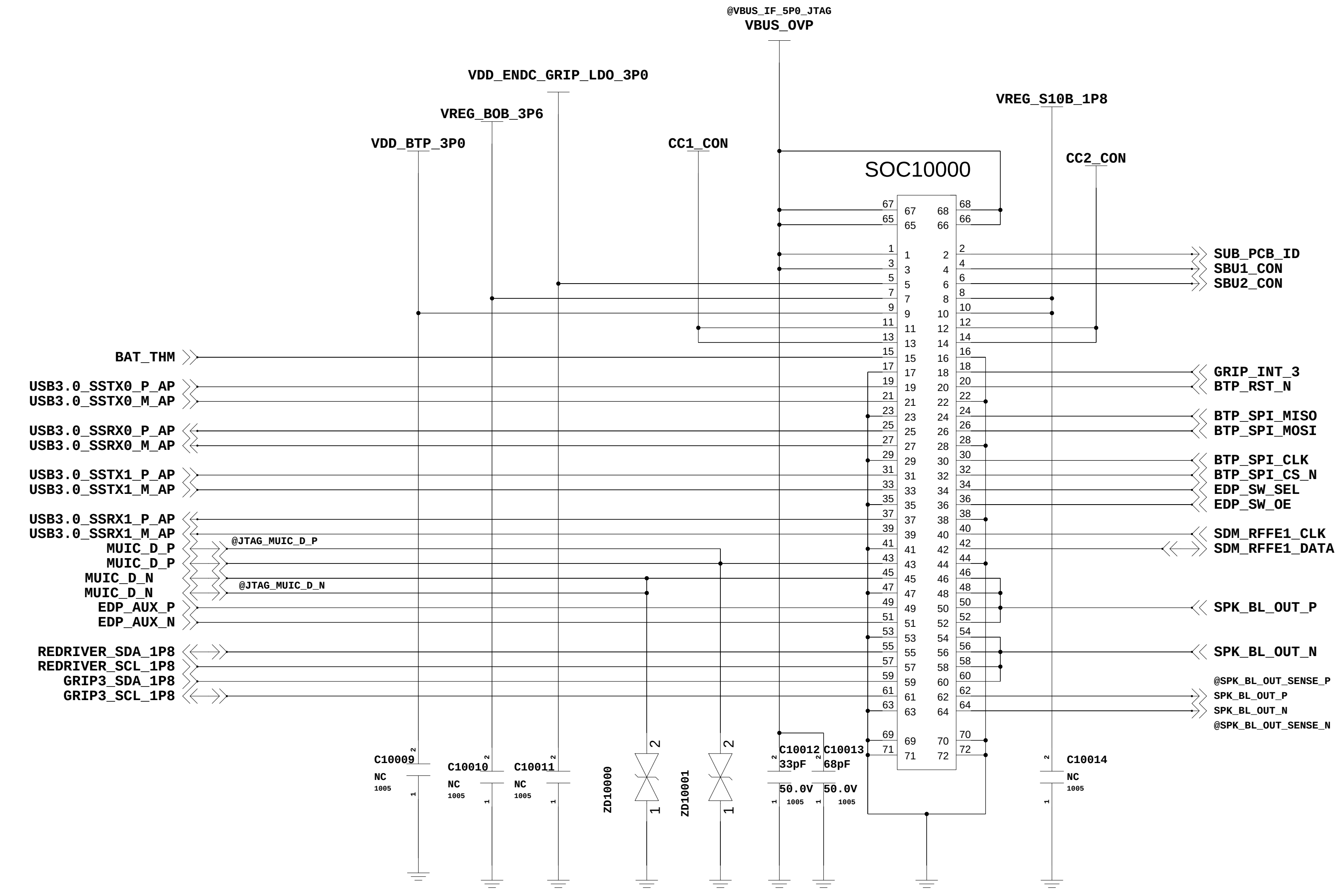
PEN CHG C/T


VREG_S10B_1P8
U9013
VDD
OUT
GND
NC
L9011 0 0603 SHORT
L9012 0 SHORT 0603
HALL_INT_WACOM
C9081 100nF 6.3V 0603
C9082 NC 0603

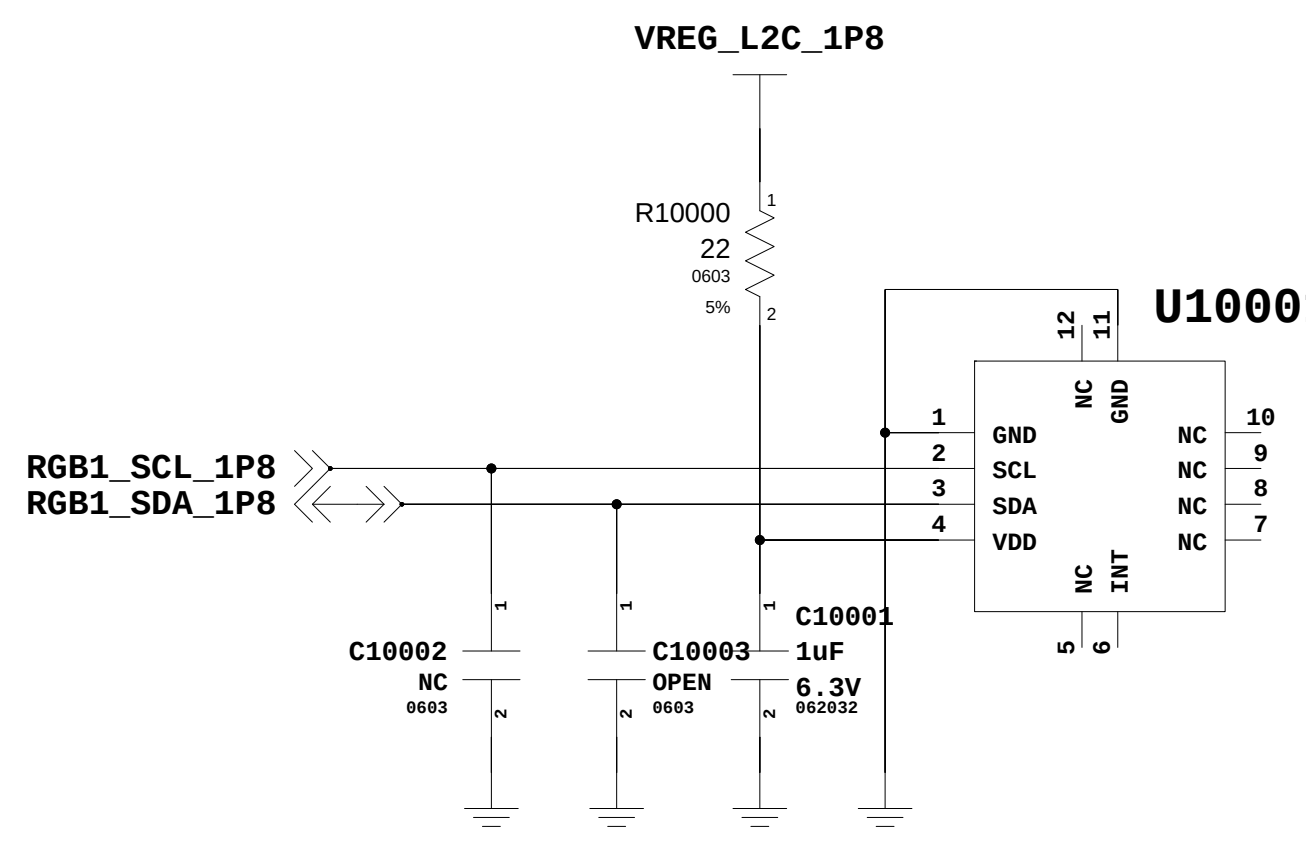
COVER_DET HALL_IC

TITLE TabS7P
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name SHEET 9 of 14 Date Drawing Date

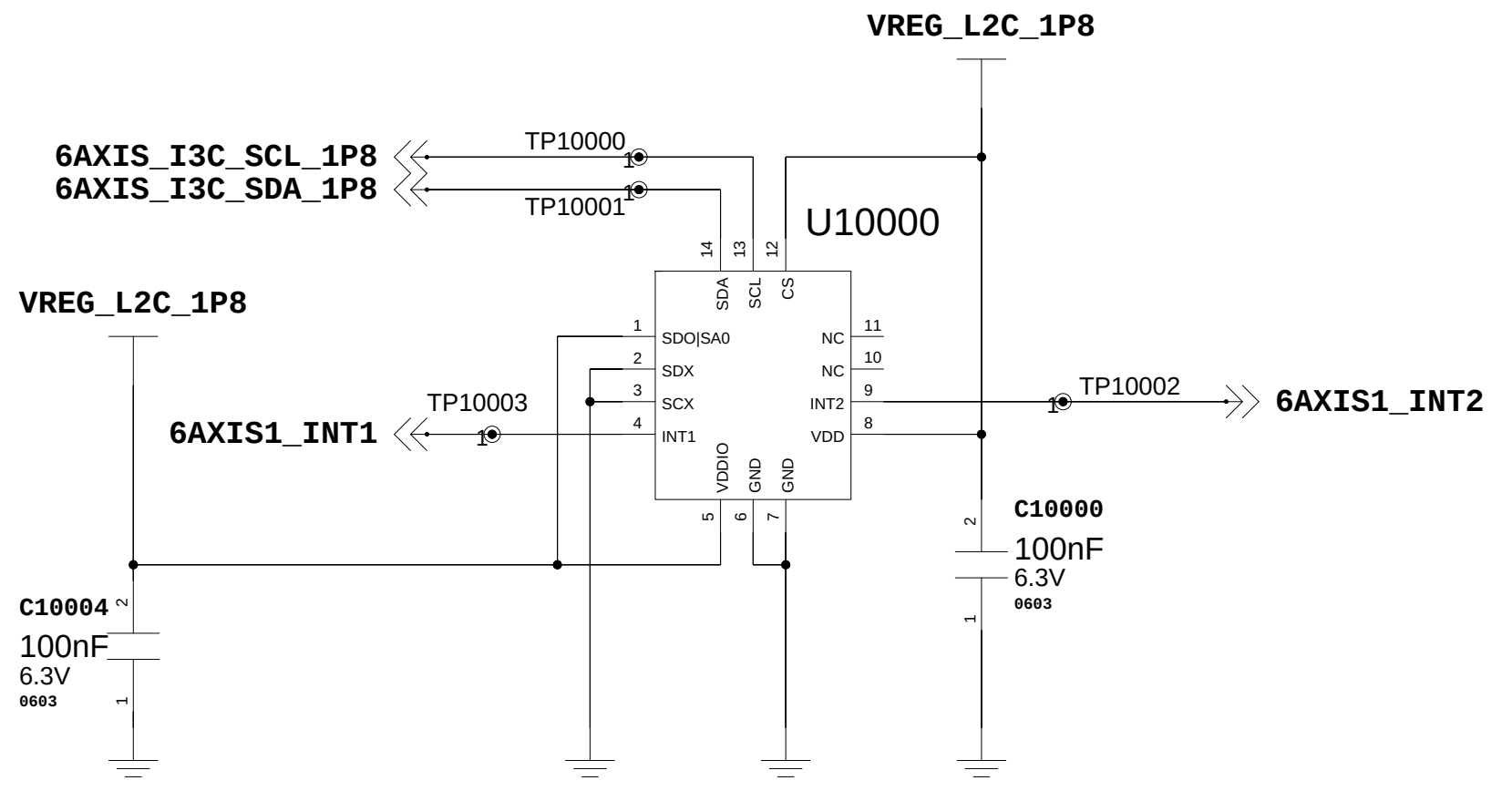
SENSOR / SD/SIM / KEY / I/F


@VBUS_IF_5P0_JTAG
VBUS_OVP
VDD_ENDC_GRIP_LDO_3P0
VREG_BOB_3P6
VDD_BTP_3P0
CC1_CON
VREG_S10B_1P8
CC2_CON
SOC10000
SUB_PCB_ID
SBU1_CON
SBU2_CON
BAT_THM
USB3.0_SSTX0_P_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX0_M_AP
USB3.0_SSTX1_P_AP
USB3.0_SSTX1_M_AP
USB3.0_SSRX1_P_AP
USB3.0_SSRX1_M_AP
MUIC_D_P
MUIC_D_P
MUIC_D_N
MUIC_D_N
@JTAG_MUIC_D_P
@JTAG_MUIC_D_N
EDP_AUX_P
EDP_AUX_N
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GRIP_INT_3
BTP_RST_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_SPI_CLK
BTP_SPI_CS_N
EDP_SW_SEL
EDP_SW_OE
SDM_RFFE1_CLK
SDM_RFFE1_DATA
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
C10009 NC 1005
C10010 NC 1005
C10011 NC 1005
ZD10000
ZD10001
C10012 33pF 50.0V 1005
C10013 68pF 50.0V 1005
C10014 NC 1005

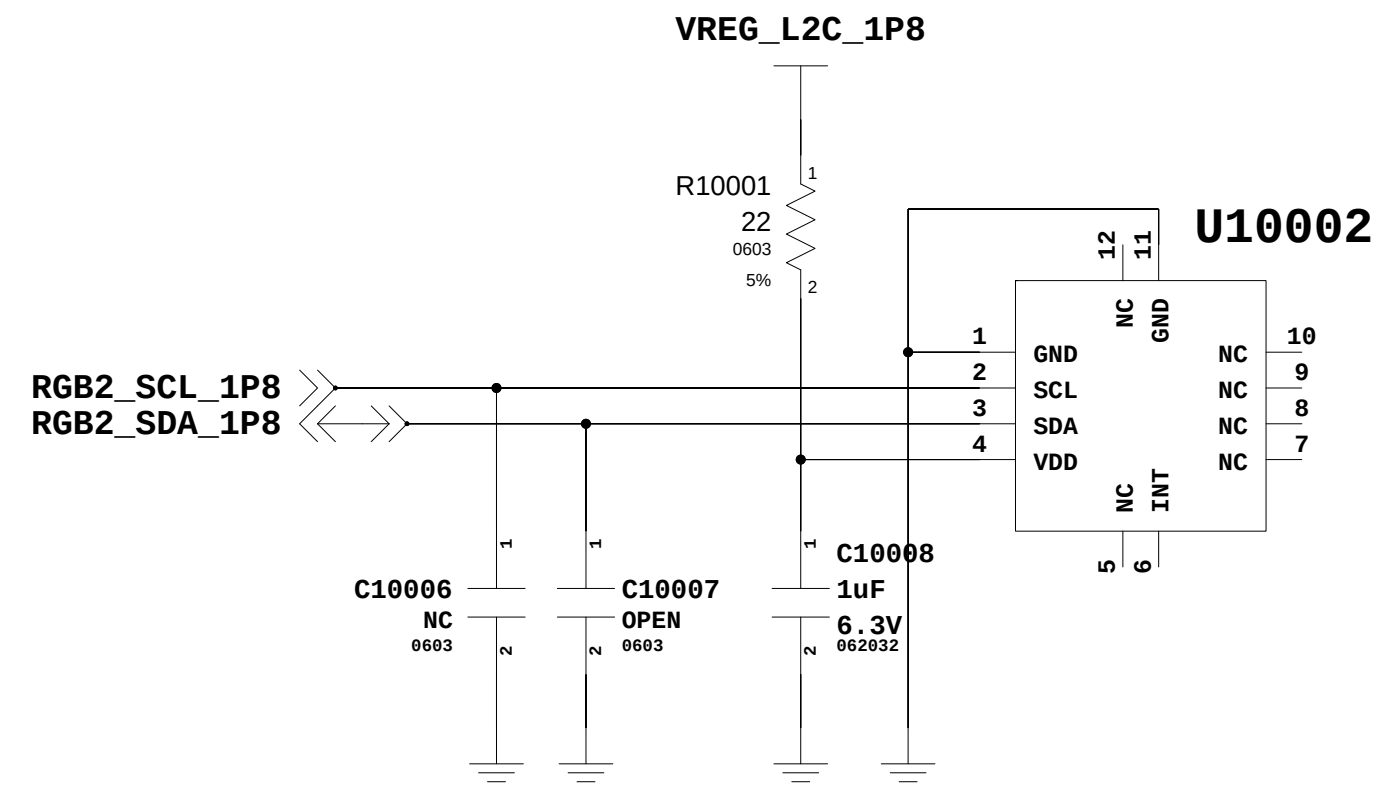
I/F CONN
(MAIN <-> USB)


VREG_L2C_1P8
R10000 22 0603 5%
U10001
RGB1_SCL_1P8
RGB1_SDA_1P8
GND
SCL
SDA
VDD
NC
INT
C10002 NC 0603
C10003 OPEN 0603
C10001 1uF 6.3V 062032

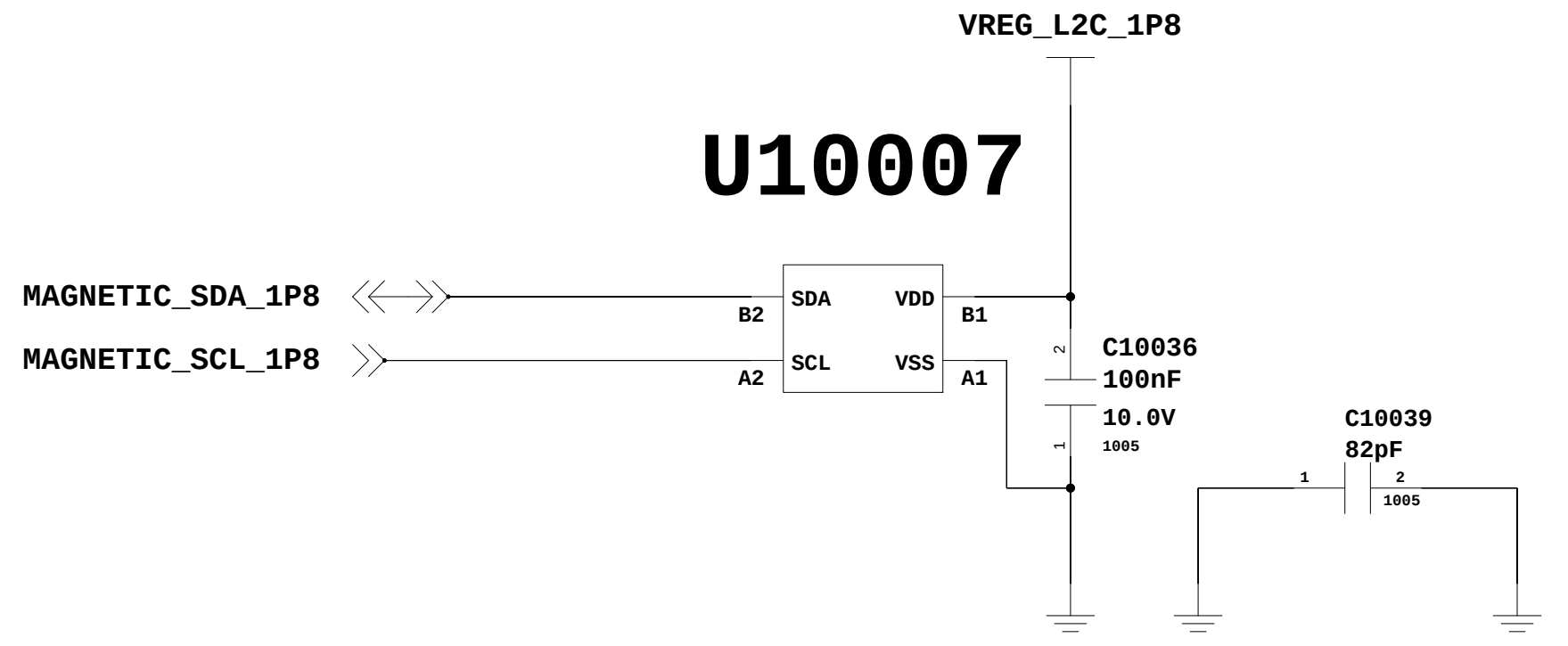
RGB Sensor 1


VREG_L2C_1P8
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
TP10000
TP10001
U10000
VREG_L2C_1P8
6AXIS1_INT1
TP10003
TP10002
6AXIS1_INT2
SDO|SA0
SDX
SCX
INT1
VDDIO
GND
SDA
SCL
CS
NC
INT2
VDD
C10004 100nF 6.3V 0603
C10000 100nF 6.3V 0603

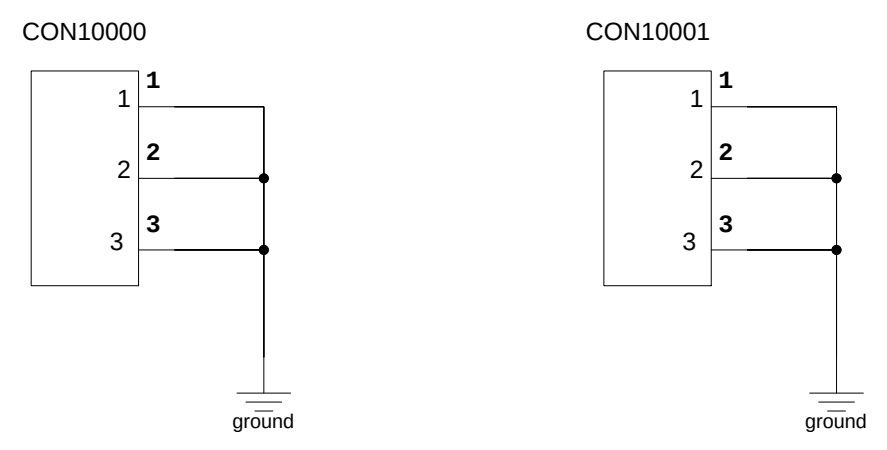
6 Axis (GYRO,ACCEL)


VREG_L2C_1P8
R10001 22 0603 5%
U10002
RGB2_SCL_1P8
RGB2_SDA_1P8
GND
SCL
SDA
VDD
NC
INT
C10006 NC 0603
C10007 OPEN 0603
C10008 1uF 6.3V 062032

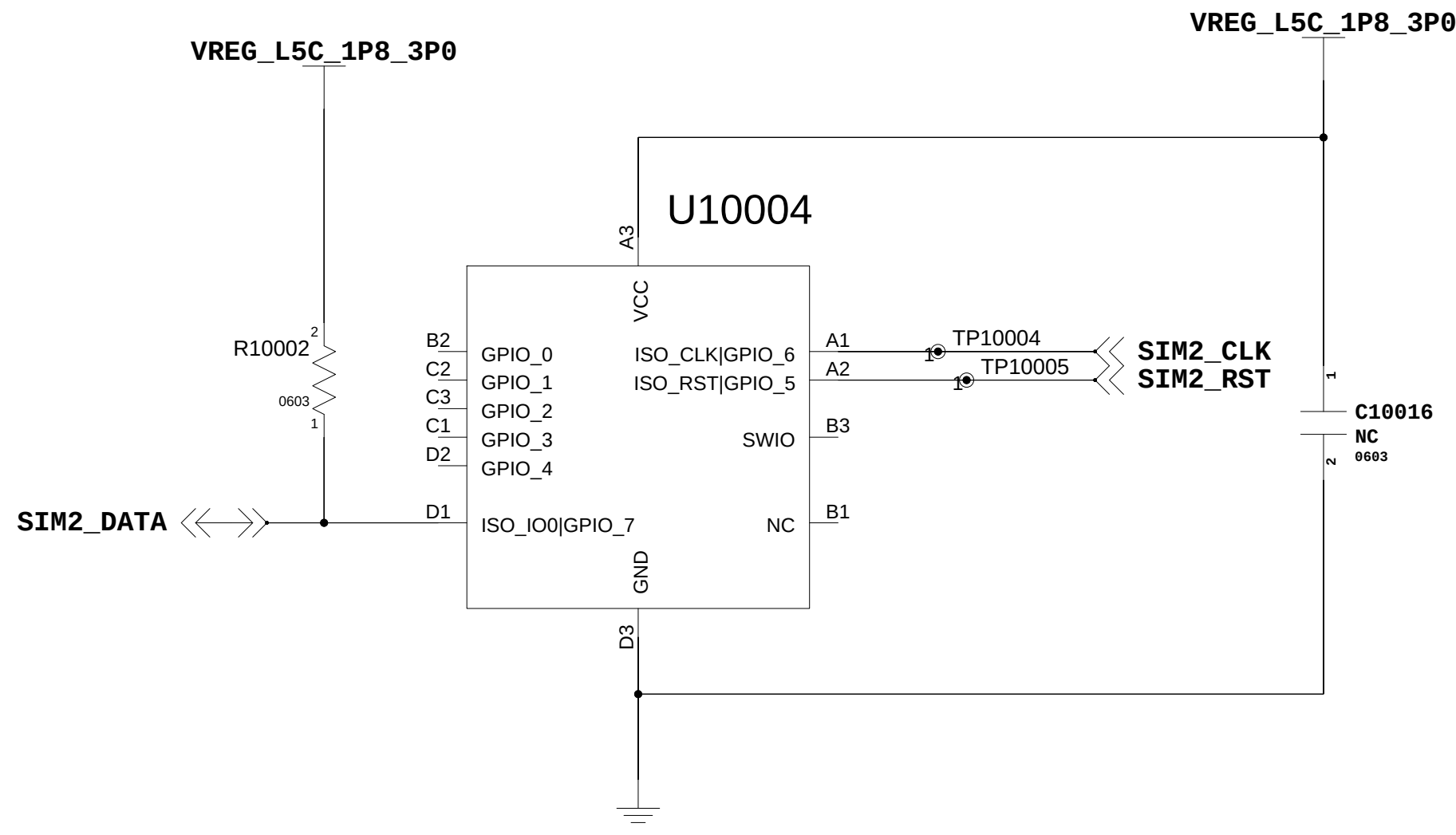
RGB Sensor 2


VREG_L2C_1P8
U10007
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
SDA
SCL
VDD
VSS
C10036 100nF 10.0V 1005
C10039 82pF 1005

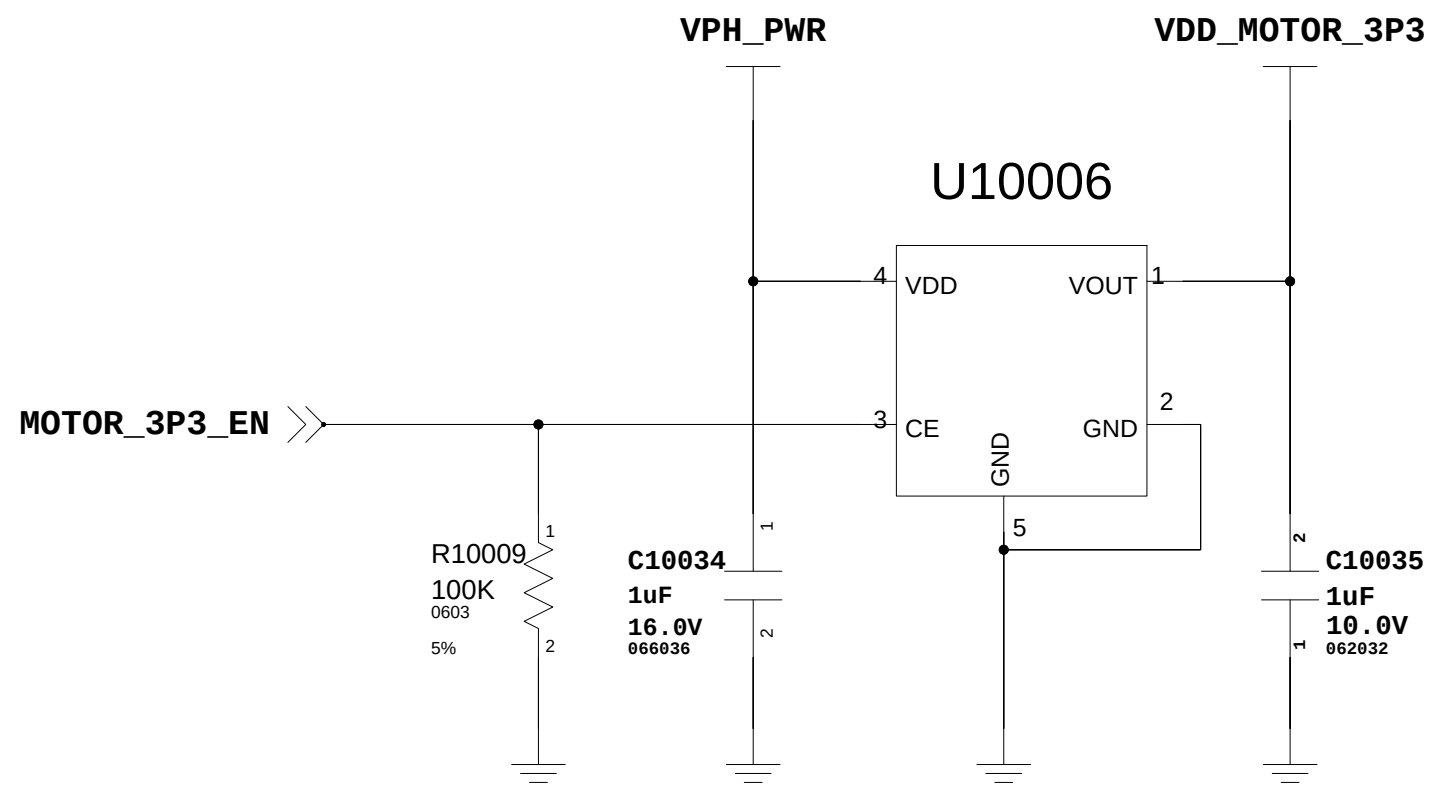
Magnetic SENSOR


CON10000
CON10001
ground
ground

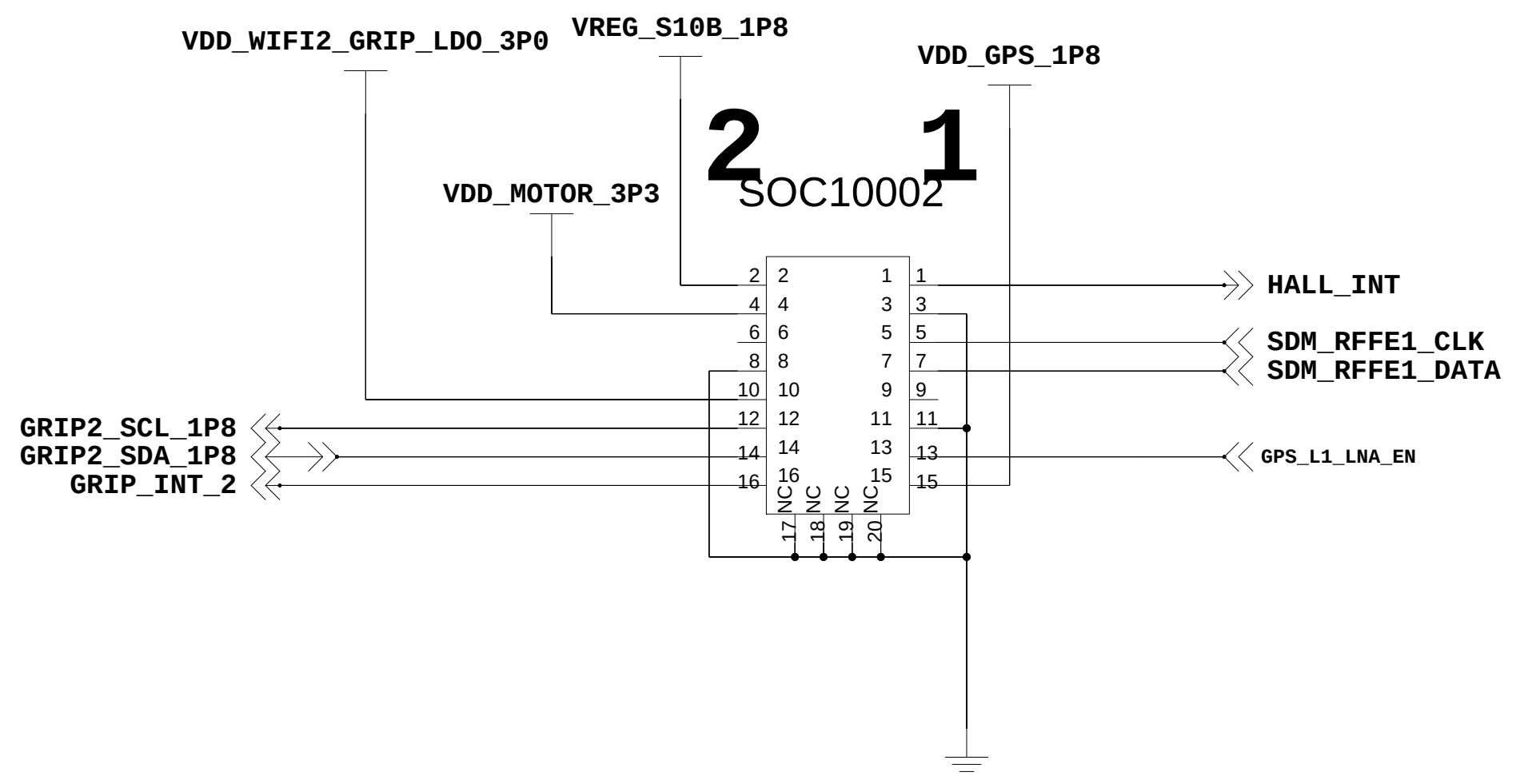
SD/SIM FPCB CONTACT


VREG_L5C_1P8_3P0
VREG_L5C_1P8_3P0
U10004
R10002 0603
SIM2_DATA
GPIO_0
GPIO_1
GPIO_2
GPIO_3
GPIO_4
ISO_IO0|GPIO_7
VCC
ISO_CLK|GPIO_6
ISO_RST|GPIO_5
SWIO
NC
GND
TP10004
TP10005
SIM2_CLK
SIM2_RST
C10016 NC 0603

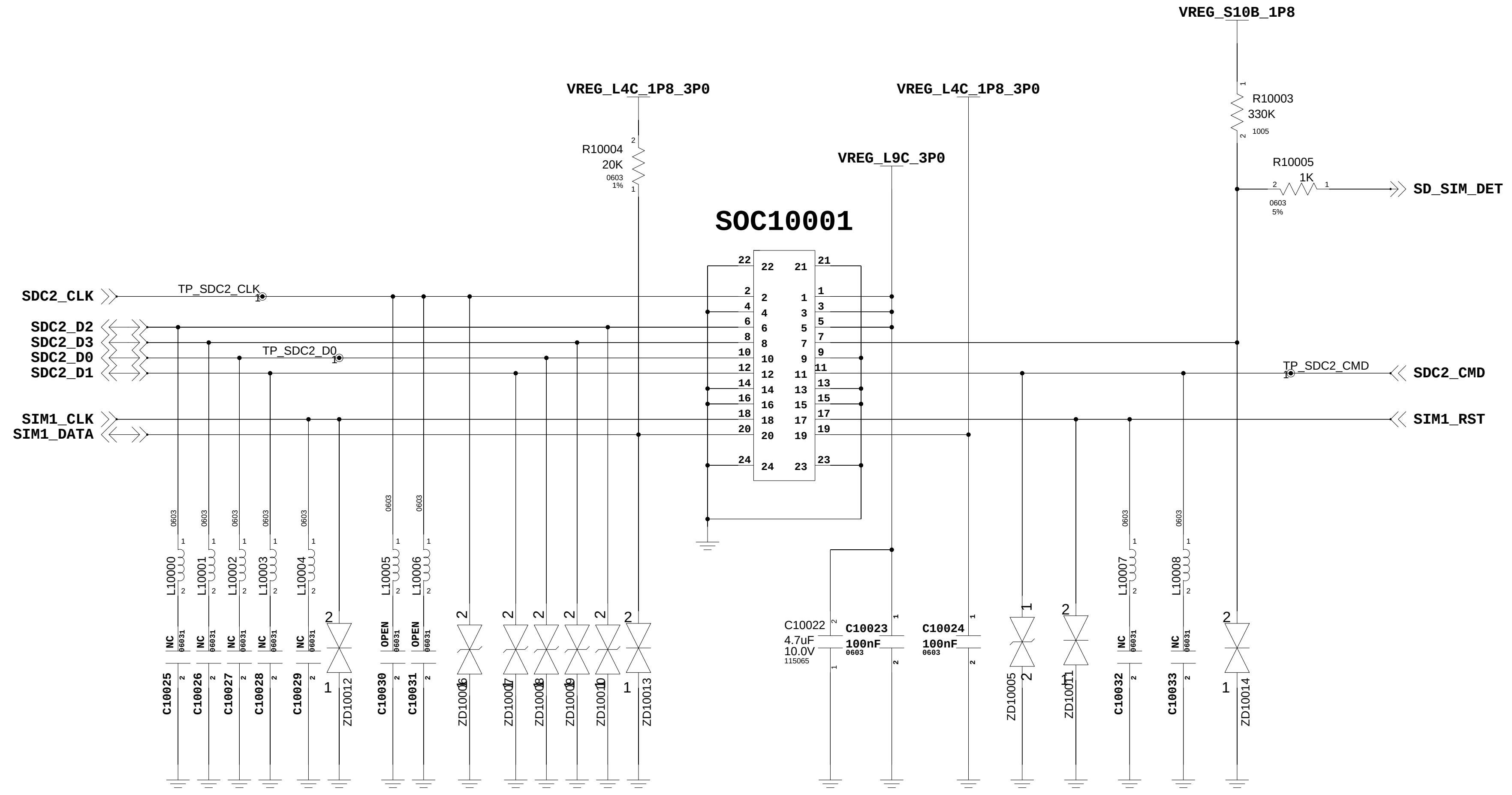
eSIM - NC


VPH_PWR
VDD_MOTOR_3P3
U10006
MOTOR_3P3_EN
VDD
CE
VOUT
GND
GND
R10009 100K 0603 5%
C10034 1uF 16.0V 066036
C10035 1uF 10.0V 062032

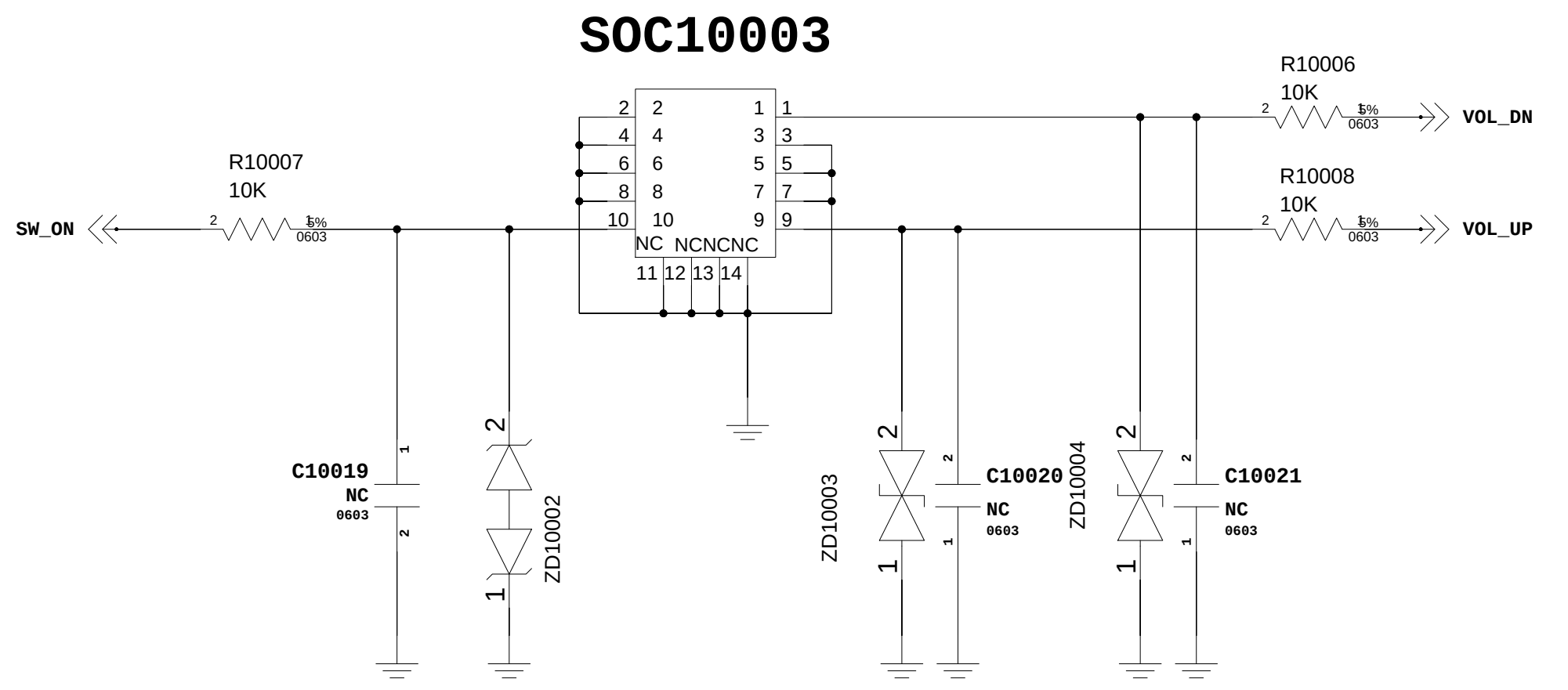
MOTOR LDO


VDD_WIFI2_GRIP_LDO_3P0
VREG_S10B_1P8
VDD_GPS_1P8
VDD_MOTOR_3P3
2
1
SOC10002
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
HALL_INT
SDM_RFFE1_CLK
SDM_RFFE1_DATA
GPS_L1_LNA_EN

SUB CONN
(MAIN <-> ANT SUB)


VREG_S10B_1P8
R10003 330K 1005
R10005 1K 0603 5%
SD_SIM_DET
VREG_L4C_1P8_3P0
R10004 20K 0603 1%
VREG_L4C_1P8_3P0
VREG_L9C_3P0
SOC10001
SDC2_CLK
TP_SDC2_CLK
SDC2_D2
SDC2_D3
SDC2_D0
SDC2_D1
TP_SDC2_D0
SIM1_CLK
SIM1_DATA
TP_SDC2_CMD
SDC2_CMD
SIM1_RST
L10000
L10001
L10002
L10003
L10004
L10005
L10006
L10007
L10008
C10025 NC
C10026 NC
C10027 NC
C10028 NC
C10029 NC
C10030 OPEN
C10031 OPEN
C10032 NC
C10033 NC
ZD10012
ZD10006
ZD10007
ZD10008
ZD10009
ZD10010
ZD10013
ZD10005
ZD10011
ZD10014
C10022 4.7uF 10.0V 115065
C10023 100nF 0603
C10024 100nF 0603

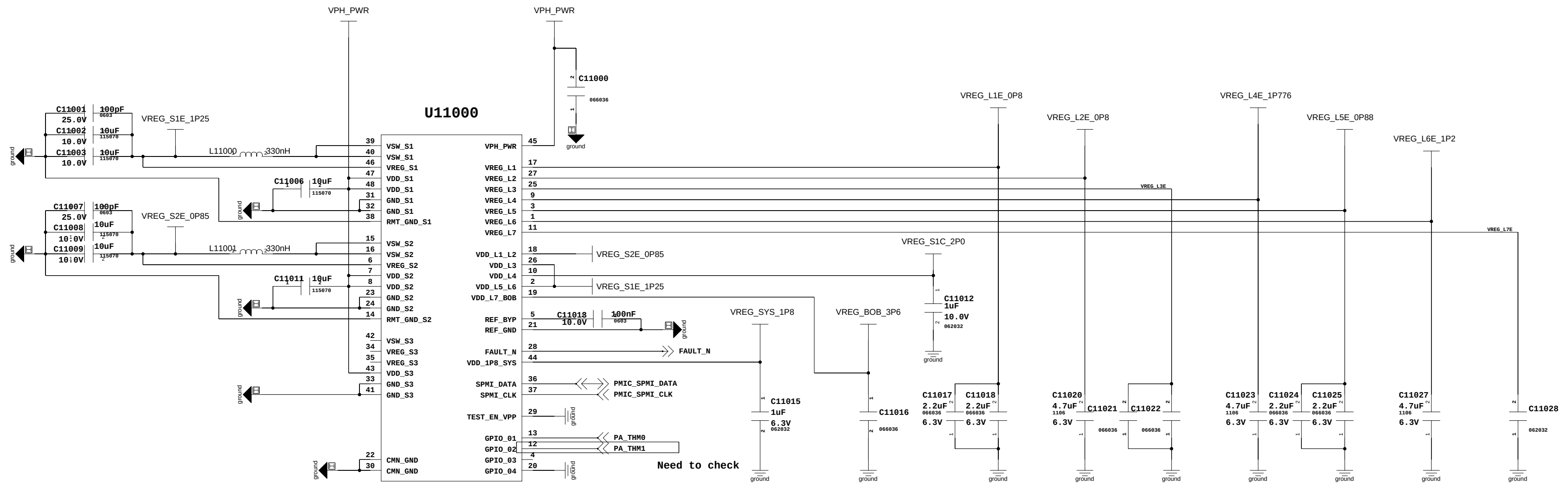
SIM/SD SOCKET Conn.


SOC10003
R10007 10K
SW_ON
R10006 10K
VOL_DN
R10008 10K
VOL_UP
C10019 NC 0603
ZD10002
ZD10003
C10020 NC 0603
ZD10004
C10021 NC 0603

Key CONN

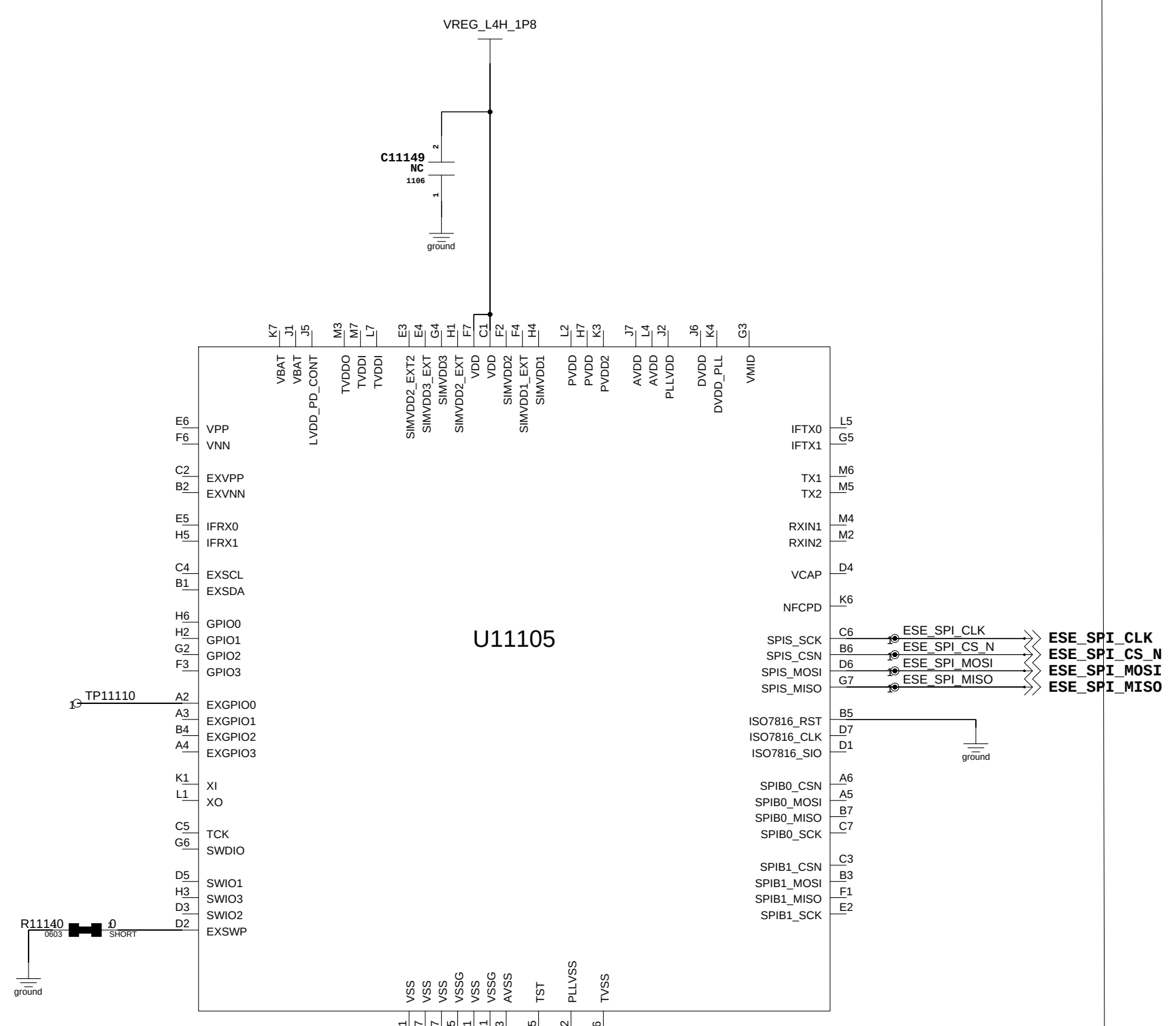
TITLE	TabS7P	
SIZE A1	DWG NO PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 10 of 15	Date Drawing Date

RF PMIC


U11000
VPH_PWR
VPH_PWR
C11000
066036
ground
C11001 100pF 25.0V 0603
C11002 10uF 10.0V 115070
C11003 10uF 10.0V 115070
VREG_S1E_1P25
L11000 330nH
C11006 10uF 115070
C11007 100pF 25.0V 0603
C11008 10uF 10.0V 115070
C11009 10uF 10.0V 115070
VREG_S2E_0P85
L11001 330nH
C11011 10uF 115070
39 VSW_S1
40 VSW_S1
46 VREG_S1
47 VDD_S1
48 VDD_S1
31 GND_S1
32 GND_S1
38 RMT_GND_S1
15 VSW_S2
16 VSW_S2
6 VREG_S2
7 VDD_S2
8 VDD_S2
23 GND_S2
24 GND_S2
14 RMT_GND_S2
42 VSW_S3
34 VREG_S3
35 VREG_S3
43 VDD_S3
33 GND_S3
41 GND_S3
22 CMN_GND
30 CMN_GND
VPH_PWR 45
VREG_L1 17
VREG_L2 27
VREG_L3 25
VREG_L4 9
VREG_L5 3
VREG_L6 1
VREG_L7 11
VDD_L1_L2 18
VDD_L3 26
VDD_L4 10
VDD_L5_L6 2
VDD_L7_BOB 19
REF_BYP 5
REF_GND 21
FAULT_N 28
VDD_1P8_SYS 44
SPMI_DATA 36
SPMI_CLK 37
TEST_EN_VPP 29
GPIO_01 13
GPIO_02 12
GPIO_03 4
GPIO_04 20
VREG_S2E_0P85
VREG_S1E_1P25
C11018 100nF 10.0V 0603
FAULT_N
PMIC_SPMI_DATA
PMIC_SPMI_CLK
PA_THM0
PA_THM1
Need to check
VREG_S1C_2P0
C11012 1uF 10.0V 062032
VREG_SYS_1P8
VREG_BOB_3P6
C11015 1uF 6.3V 062032
C11016 066036
VREG_L1E_0P8
VREG_L2E_0P8
VREG_L3E
VREG_L4E_1P776
VREG_L5E_0P88
VREG_L6E_1P2
VREG_L7E
C11017 2.2uF 066036 6.3V
C11018 2.2uF 066036 6.3V
C11020 4.7uF 1106 6.3V
C11021 066036
C11022 066036
C11023 4.7uF 1106 6.3V
C11024 2.2uF 066036 6.3V
C11025 2.2uF 066036 6.3V
C11027 4.7uF 1106 6.3V
C11028 062032
ground

PMR735A

ESE


VREG_L4H_1P8
C11149 NC 1106
ground
U11105
VBAT
VBAT
LVDD_PD_CONT
TVDDO
TVDDI
TVDDI
SIMVDD2_EXT2
SIMVDD3_EXT
SIMVDD3
SIMVDD2_EXT
VDD
VDD
SIMVDD2
SIMVDD1_EXT
SIMVDD1
PVDD
PVDD
PVDD2
AVDD
AVDD
PLLVDD
DVDD
DVDD_PLL
VMID
E6 VPP
F6 VNN
C2 EXVPP
B2 EXVNN
E5 IFRX0
H5 IFRX1
C4 EXSCL
B1 EXSDA
H6 GPIO0
H2 GPIO1
G2 GPIO2
F3 GPIO3
TP11110
A2 EXGPIO0
A3 EXGPIO1
B4 EXGPIO2
A4 EXGPIO3
K1 XI
L1 XO
C5 TCK
G6 SWDIO
D5 SWIO1
H3 SWIO3
D3 SWIO2
D2 EXSWP
R11140 0 0603 SHORT
IFTX0 L5
IFTX1 G5
TX1 M6
TX2 M5
RXIN1 M4
RXIN2 M2
VCAP D4
NFCPD K6
SPIS_SCK C6
SPIS_CSN B6
SPIS_MOSI D6
SPIS_MISO G7
ISO7816_RST B5
ISO7816_CLK D7
ISO7816_SIO D1
SPIB0_CSN A6
SPIB0_MOSI A5
SPIB0_MISO B7
SPIB0_SCK C7
SPIB1_CSN C3
SPIB1_MOSI B3
SPIB1_MISO F1
SPIB1_SCK E2
ESE_SPI_CLK
ESE_SPI_CS_N
ESE_SPI_MOSI
ESE_SPI_MISO
VSS
VSS
VSS
VSSG
VSS
VSSG
AVSS
TST
PLLVSS
TVSS
ground

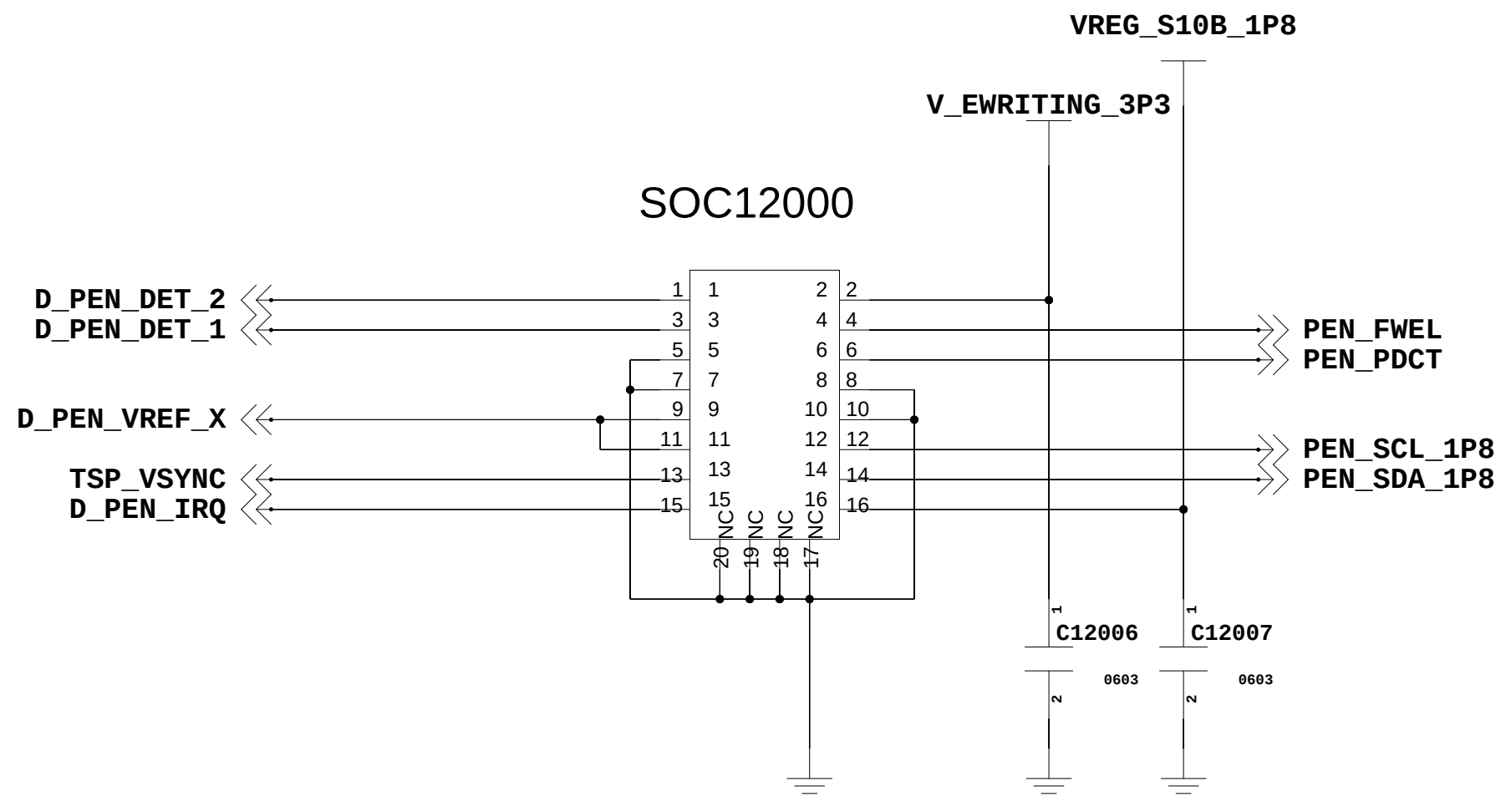
eSE (SUB6_B2B)

TITLE TabS7P

SIZE A1 DWG NO PDM NUMBER REV Rev

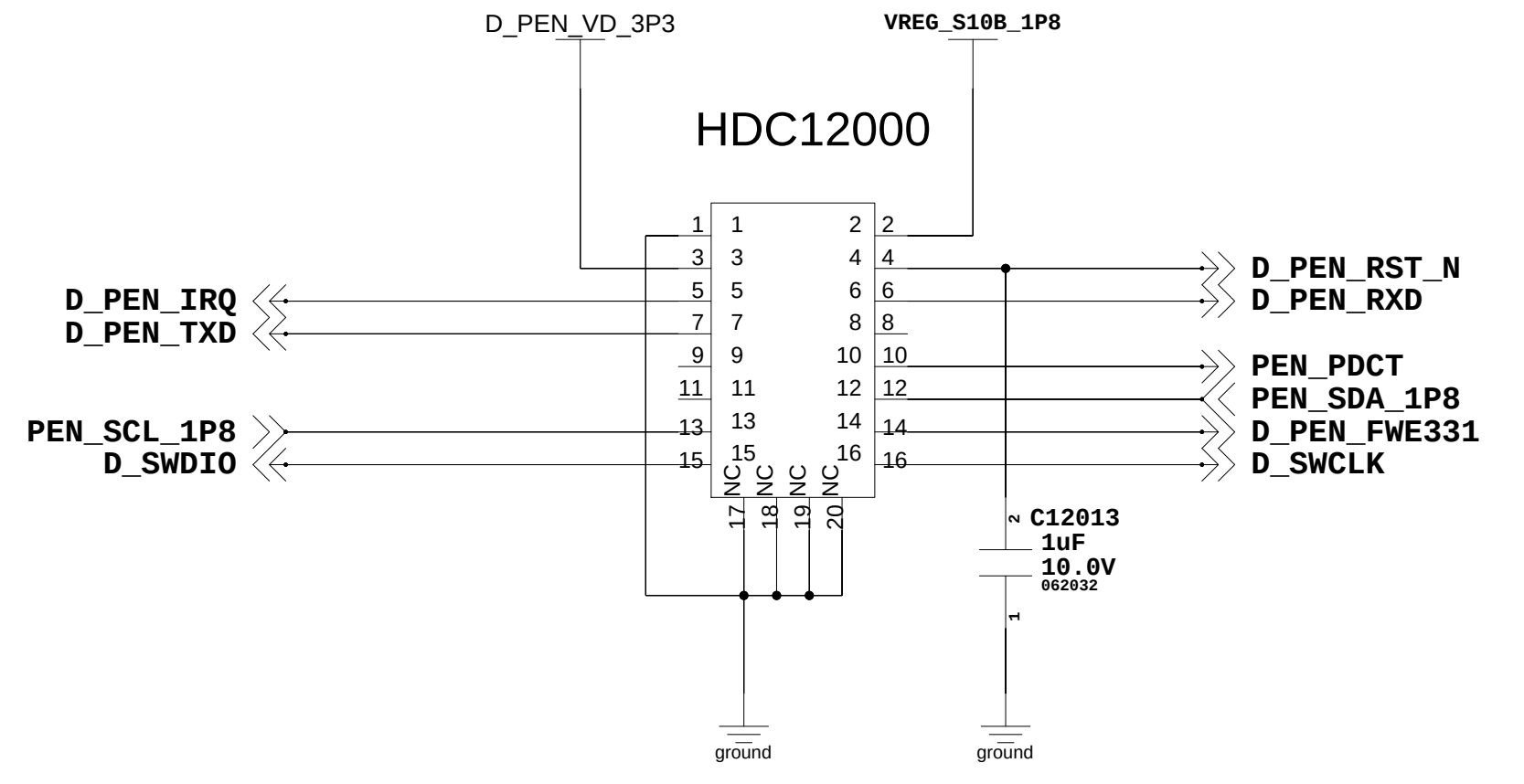
DRAWN BY User Name SHEET 11 of 12 Date Drawing Date

WACOM PCB

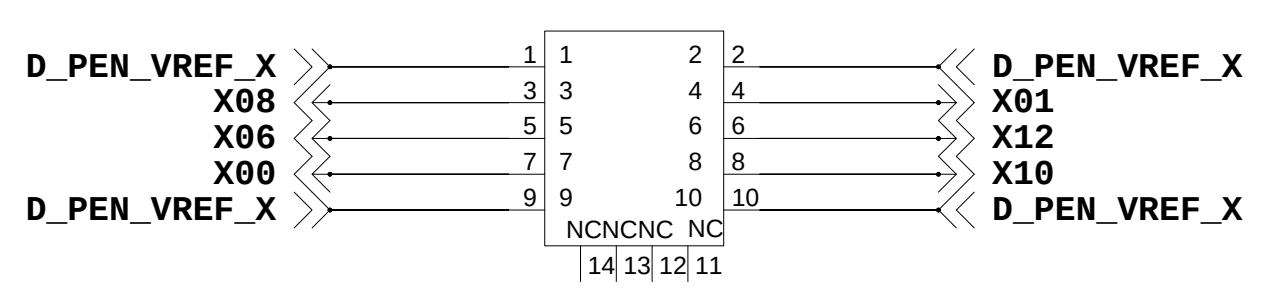

VREG_S10B_1P8
V_EWRITING_3P3
SOC12000
D_PEN_DET_2
D_PEN_DET_1
D_PEN_VREF_X
TSP_VSYNC
D_PEN_IRQ
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
C12006
0603
C12007
0603

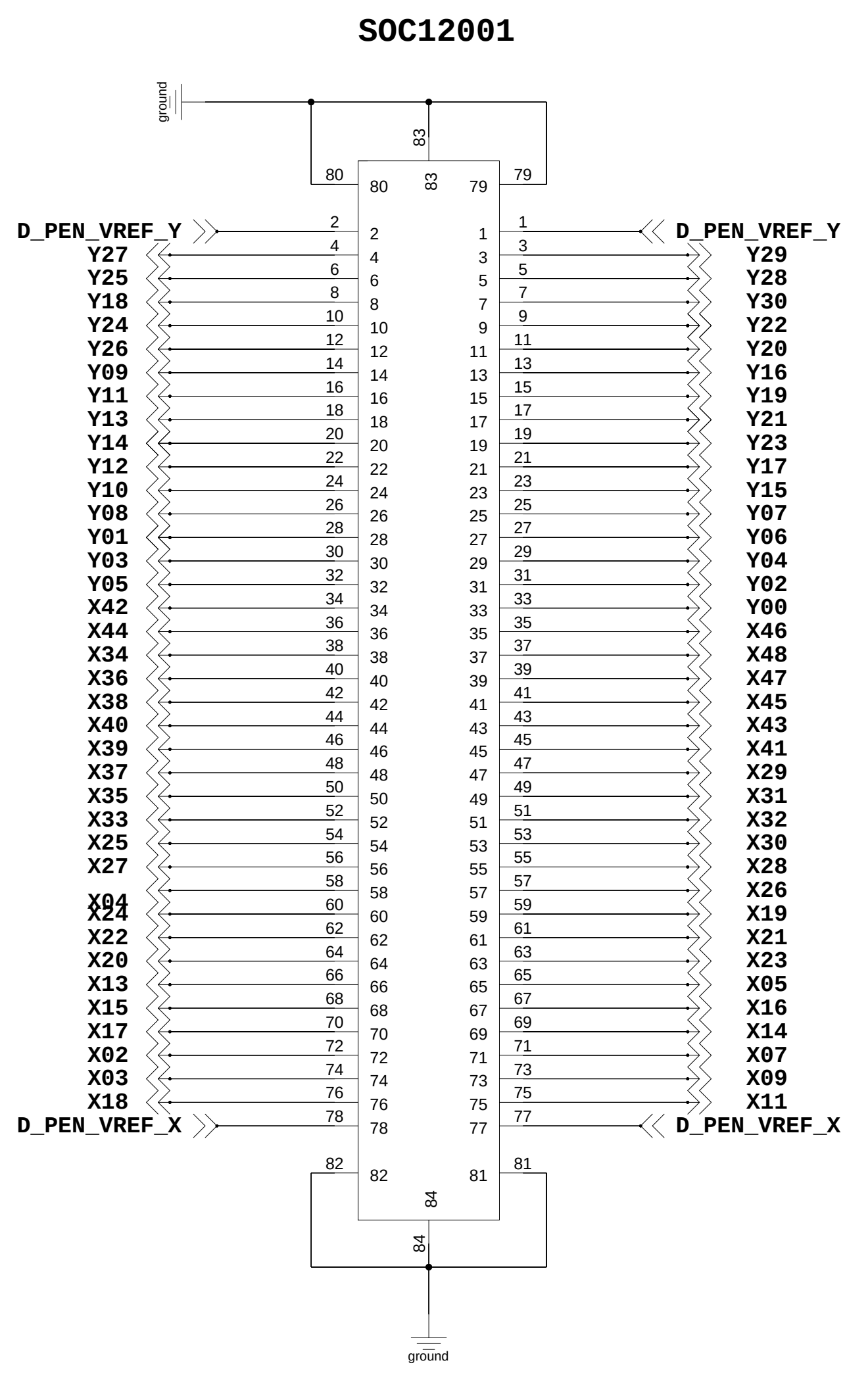
WACOM I/F CON.

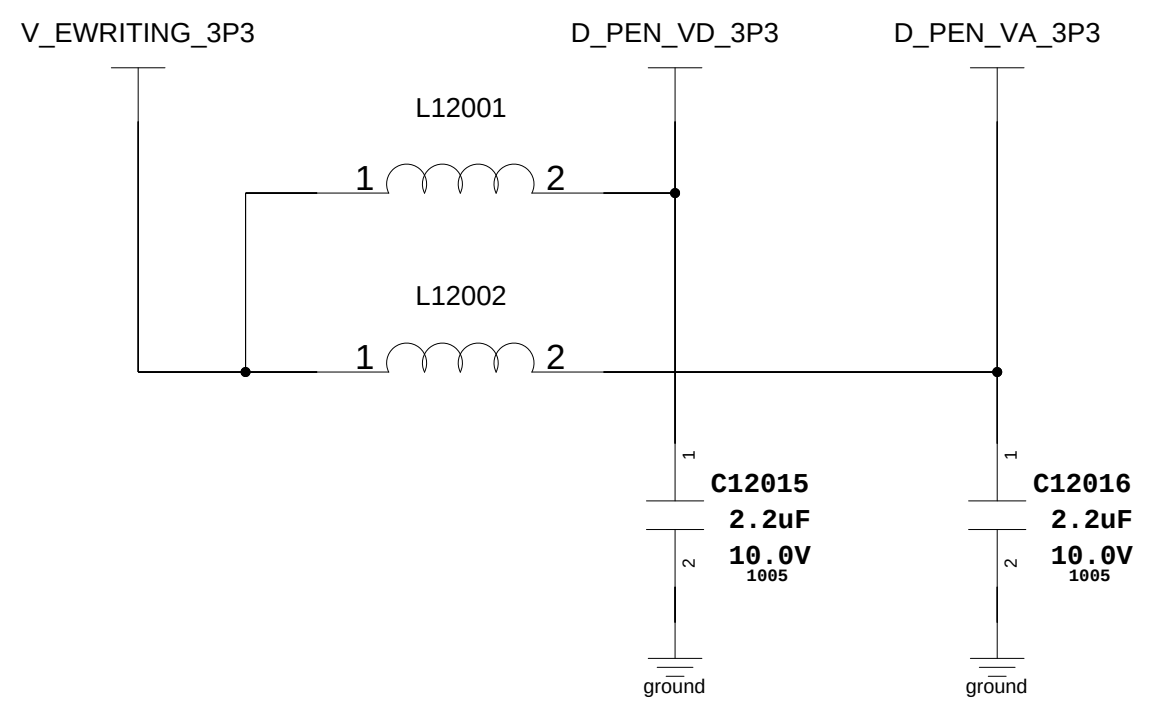
MBOM : NC


D_PEN_VD_3P3
VREG_S10B_1P8
HDC12000
D_PEN_IRQ
D_PEN_TXD
PEN_SCL_1P8
D_SWDIO
D_PEN_RST_N
D_PEN_RXD
PEN_PDCT
PEN_SDA_1P8
D_PEN_FWE331
D_SWCLK
C12013
1uF
10.0V
062032
ground
ground

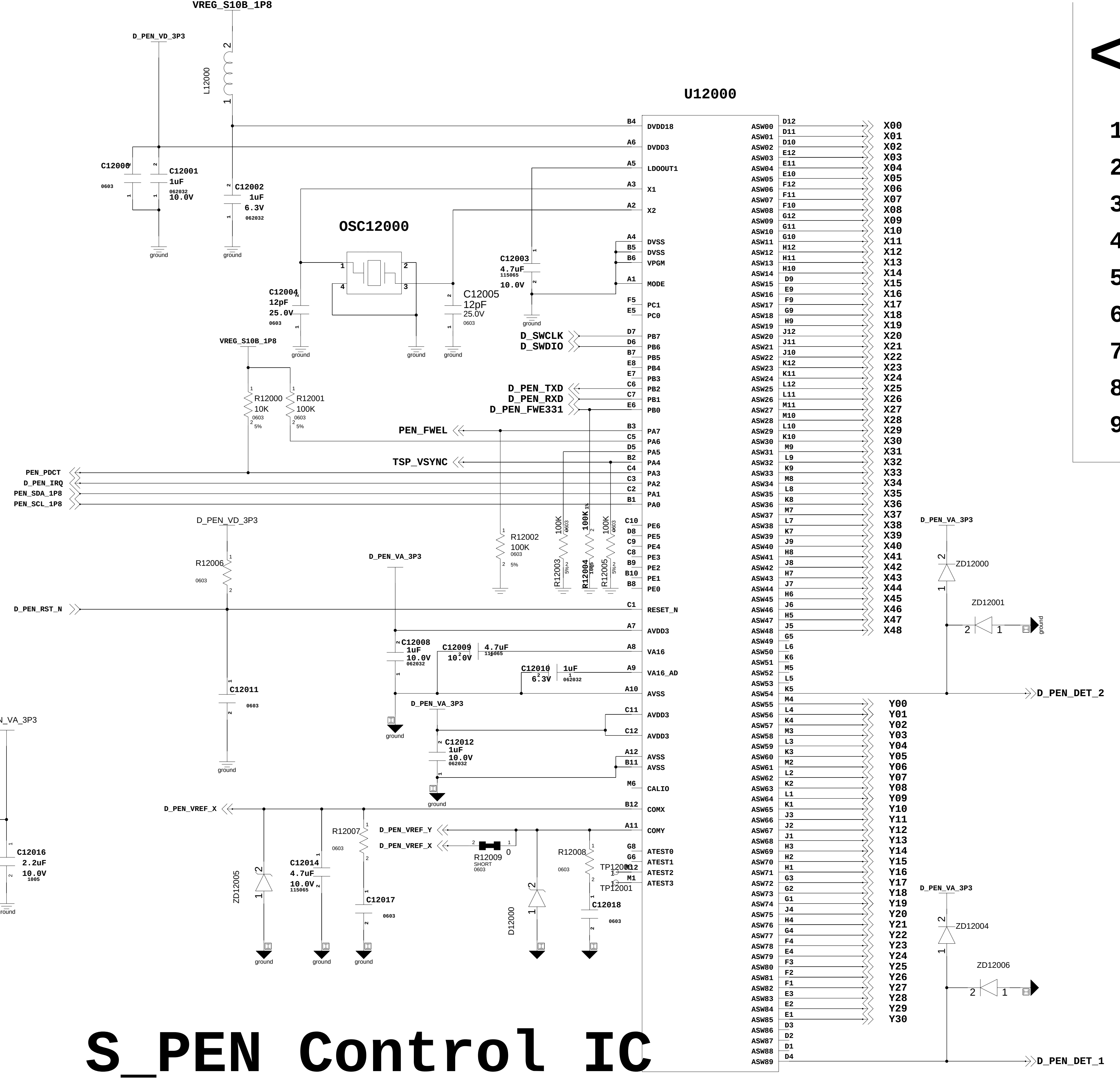
DEBUGER


SOC12003
D_PEN_VREF_X
X08
X06
X00
D_PEN_VREF_X
D_PEN_VREF_X
X01
X12
X10
D_PEN_VREF_X
NCNCNC NC
14 13 12 11


SOC12001
ground
D_PEN_VREF_Y
Y27
Y25
Y18
Y24
Y26
Y09
Y11
Y13
Y14
Y12
Y10
Y08
Y01
Y03
Y05
X42
X44
X34
X36
X38
X40
X39
X37
X35
X33
X25
X27
X04
X24
X22
X20
X13
X15
X17
X02
X03
X18
D_PEN_VREF_X
D_PEN_VREF_Y
Y29
Y28
Y30
Y22
Y20
Y16
Y19
Y21
Y23
Y17
Y15
Y07
Y06
Y04
Y02
Y00
X46
X48
X47
X45
X43
X41
X29
X31
X32
X30
X28
X26
X19
X21
X23
X05
X16
X14
X07
X09
X11
D_PEN_VREF_X
ground


V_EWRITING_3P3
D_PEN_VD_3P3
D_PEN_VA_3P3
L12001
L12002
C12015
2.2uF
10.0V
1005
C12016
2.2uF
10.0V
1005
ground
ground

78+16 pin S-PEN Connector


VREG_S10B_1P8
D_PEN_VD_3P3
L12000
C12000
0603
C12001
1uF
062032
10.0V
C12002
1uF
6.3V
062032
OSC12000
C12004
12pF
25.0V
0603
C12005
12pF
25.0V
0603
C12003
4.7uF
115065
10.0V
U12000
DVDD18
DVDD3
LDOOUT1
X1
X2
DVSS
DVSS
VPGM
MODE
PC1
PC0
PB7
PB6
PB5
PB4
PB3
PB2
PB1
PB0
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
PE6
PE5
PE4
PE3
PE2
PE1
PE0
RESET_N
AVDD3
VA16
VA16_AD
AVSS
AVDD3
AVDD3
AVSS
AVSS
CALIO
COMX
COMY
ATEST0
ATEST1
ATEST2
ATEST3
D_SWCLK
D_SWDIO
D_PEN_TXD
D_PEN_RXD
D_PEN_FWE331
PEN_FWEL
TSP_VSYNC
PEN_PDCT
D_PEN_IRQ
PEN_SDA_1P8
PEN_SCL_1P8
R12000
10K
R12001
100K
R12002
100K
R12003
100K
R12004
100K
R12005
100K
D_PEN_VD_3P3
R12006
D_PEN_RST_N
C12011
D_PEN_VA_3P3
C12008
1uF
10.0V
C12009
4.7uF
C12010
1uF
6.3V
C12012
1uF
10.0V
D_PEN_VREF_X
D_PEN_VREF_Y
ZD12005
C12014
4.7uF
10.0V
R12007
C12017
R12009
SHORT
0603
D12000
R12008
C12018
TP12001
ASW00–ASW89
X00 X01 X02 X03 X04 X05 X06 X07 X08 X09 X10 X11 X12 X13 X14 X15 X16 X17 X18 X19 X20 X21 X22 X23 X24 X25 X26 X27 X28 X29 X30 X31 X32 X33 X34 X35 X36 X37 X38 X39 X40 X41 X42 X43 X44 X45 X46 X47 X48
Y00 Y01 Y02 Y03 Y04 Y05 Y06 Y07 Y08 Y09 Y10 Y11 Y12 Y13 Y14 Y15 Y16 Y17 Y18 Y19 Y20 Y21 Y22 Y23 Y24 Y25 Y26 Y27 Y28 Y29 Y30
D_PEN_VA_3P3
ZD12000
ZD12001
D_PEN_DET_2
ZD12004
ZD12006
D_PEN_DET_1
ground

S_PEN Control IC

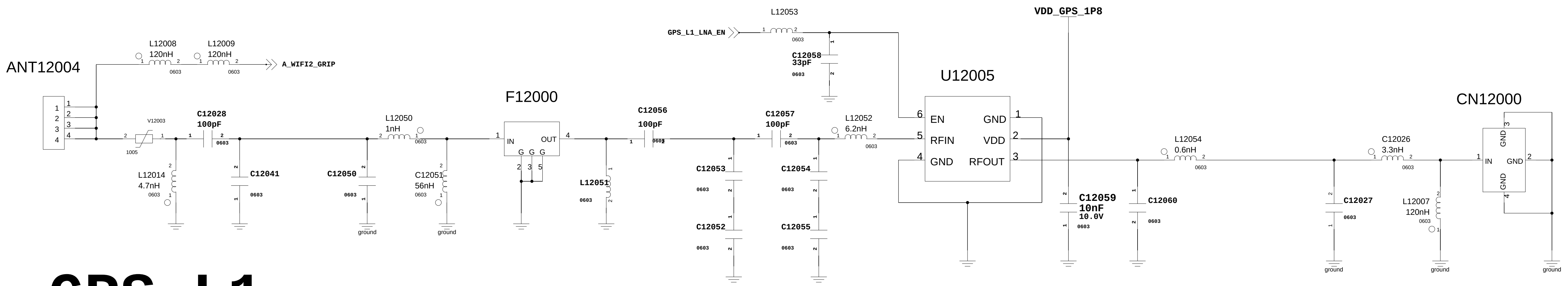
(DC mode)

< SMD F/T >

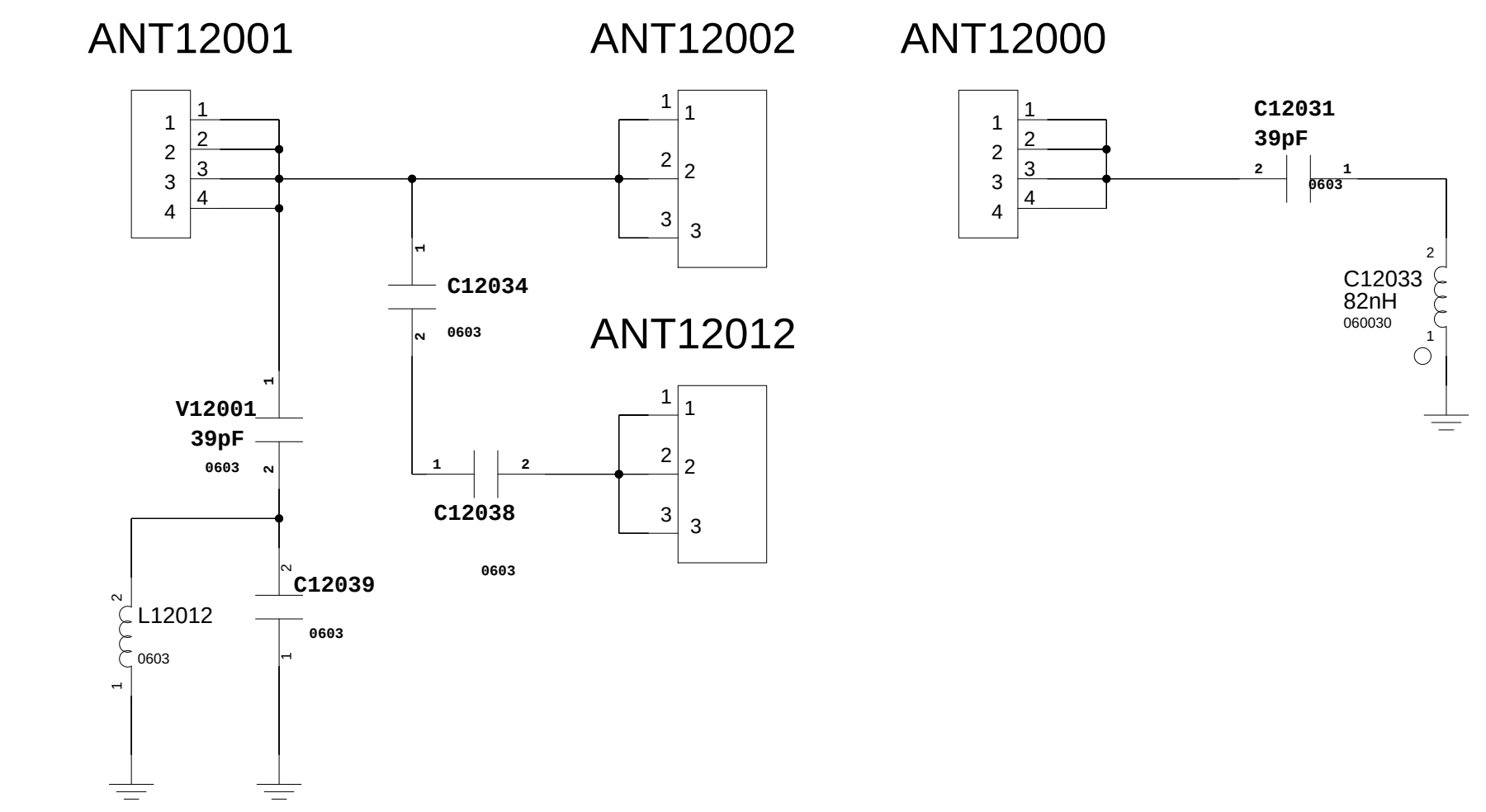
1. V_EWRITING_3P3
2. VREG_S10B_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GPS_L1_LNA_EN
8. VDD_GPS_1P8
9. GND

ANT SUB PCB

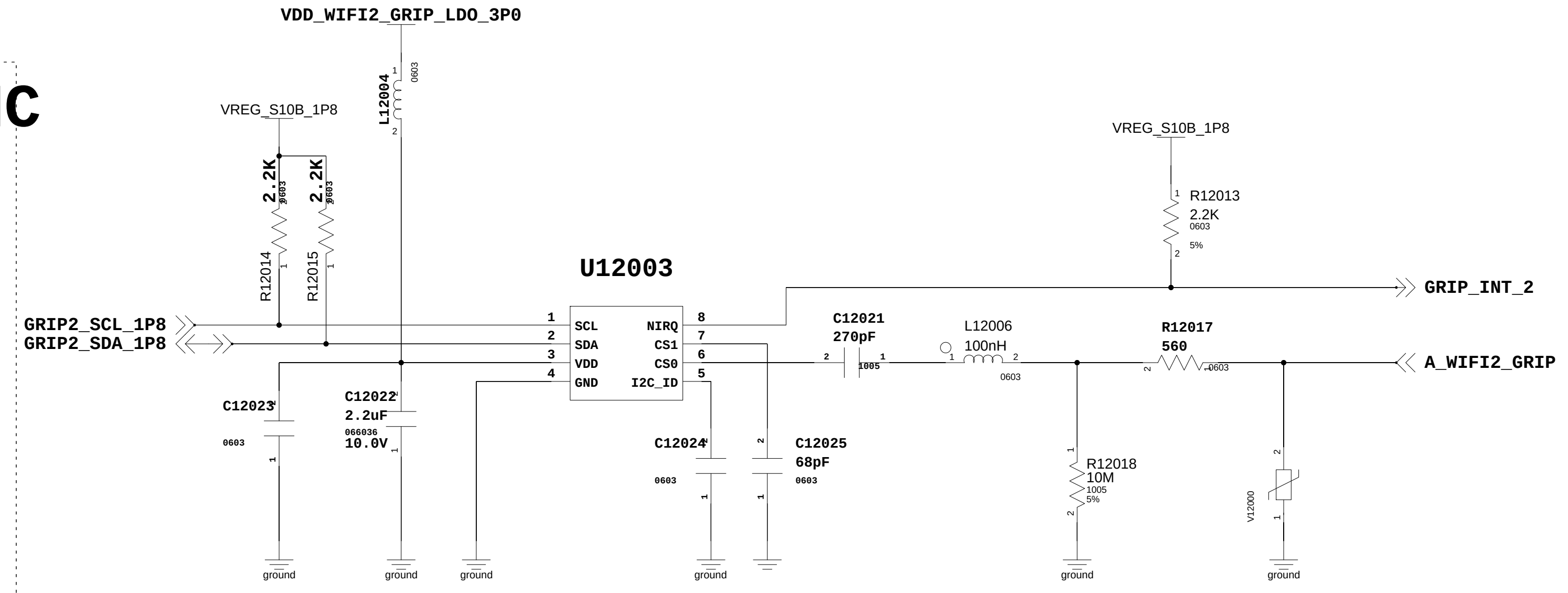
(MAIN1 ANT XGND / WIFI2 ANT GRIP)


ANT12004
L12008
120nH
L12009
120nH
A_WIFI2_GRIP
V12003
C12028
100pF
L12014
4.7nH
C12041
C12050
L12050
1nH
C12051
56nH
F12000
IN
OUT
L12051
C12056
100pF
C12053
C12052
GPS_L1_LNA_EN
L12053
C12058
33pF
C12057
100pF
C12054
C12055
L12052
6.2nH
U12005
EN
RFIN
GND
GND
VDD
RFOUT
VDD_GPS_1P8
C12059
10nF
10.0V
C12060
L12054
0.6nH
C12027
C12026
3.3nH
L12007
120nH
CN12000
IN
GND
ground

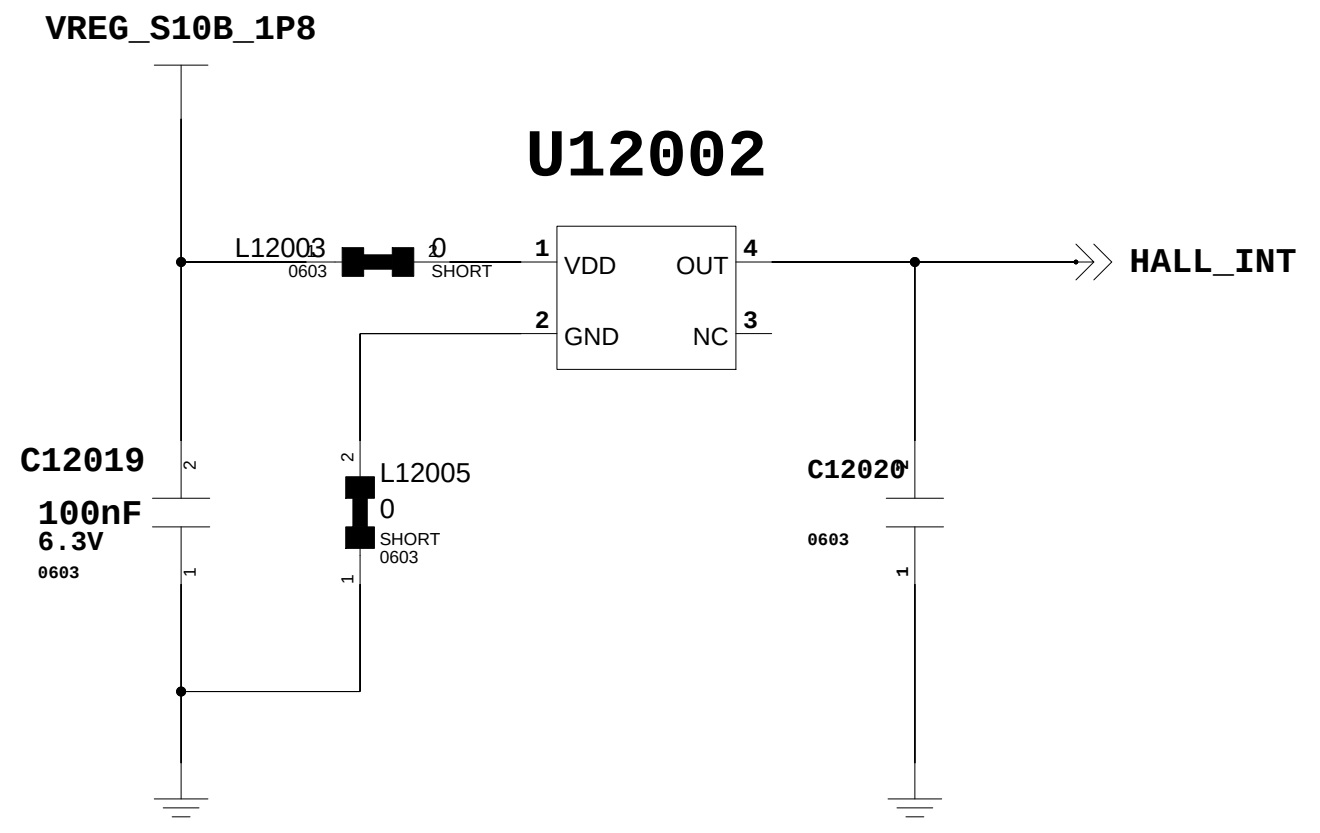
GPS_L1


ANT12001
ANT12002
ANT12012
ANT12000
C12034
V12001
39pF
C12038
C12039
L12012
C12031
39pF
C12033
82nH

WIFI : NC


VDD_WIFI2_GRIP_LDO_3P0
L12004
VREG_S10B_1P8
2.2K
2.2K
R12014
R12015
GRIP2_SCL_1P8
GRIP2_SDA_1P8
U12003
SCL
SDA
VDD
GND
NIRQ
CS1
CS0
I2C_ID
C12023
C12022
2.2uF
10.0V
C12024
C12025
68pF
C12021
270pF
L12006
100nH
VREG_S10B_1P8
R12013
2.2K
GRIP_INT_2
R12017
560
A_WIFI2_GRIP
R12018
10M
V12000
ground

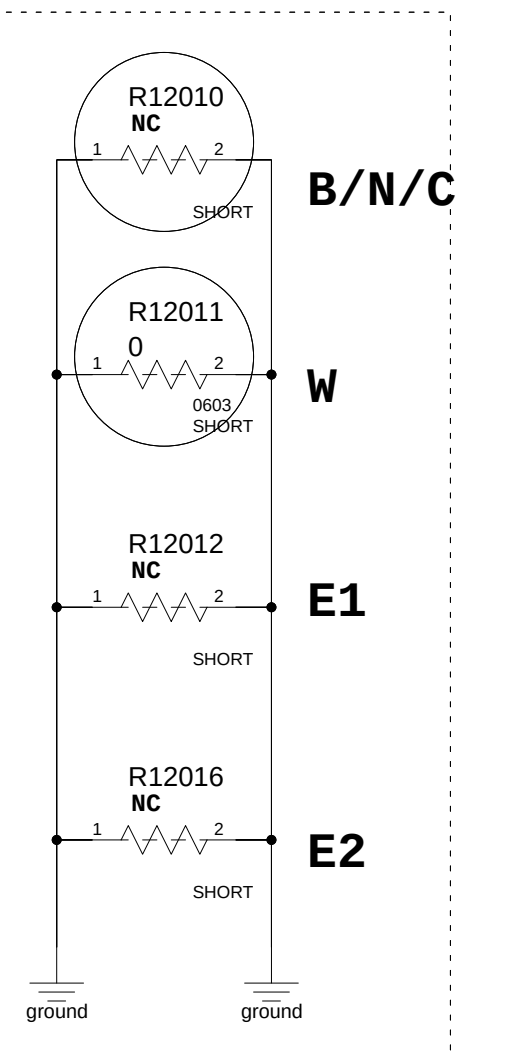
WIFI2_GRIP_SENSOR (GRIP#1)


VREG_S10B_1P8
U12002
L12003
0
SHORT
VDD
GND
OUT
NC
HALL_INT
C12019
100nF
6.3V
0603
L12005
0
SHORT
C12020
0603

COVER_DET HALL_IC

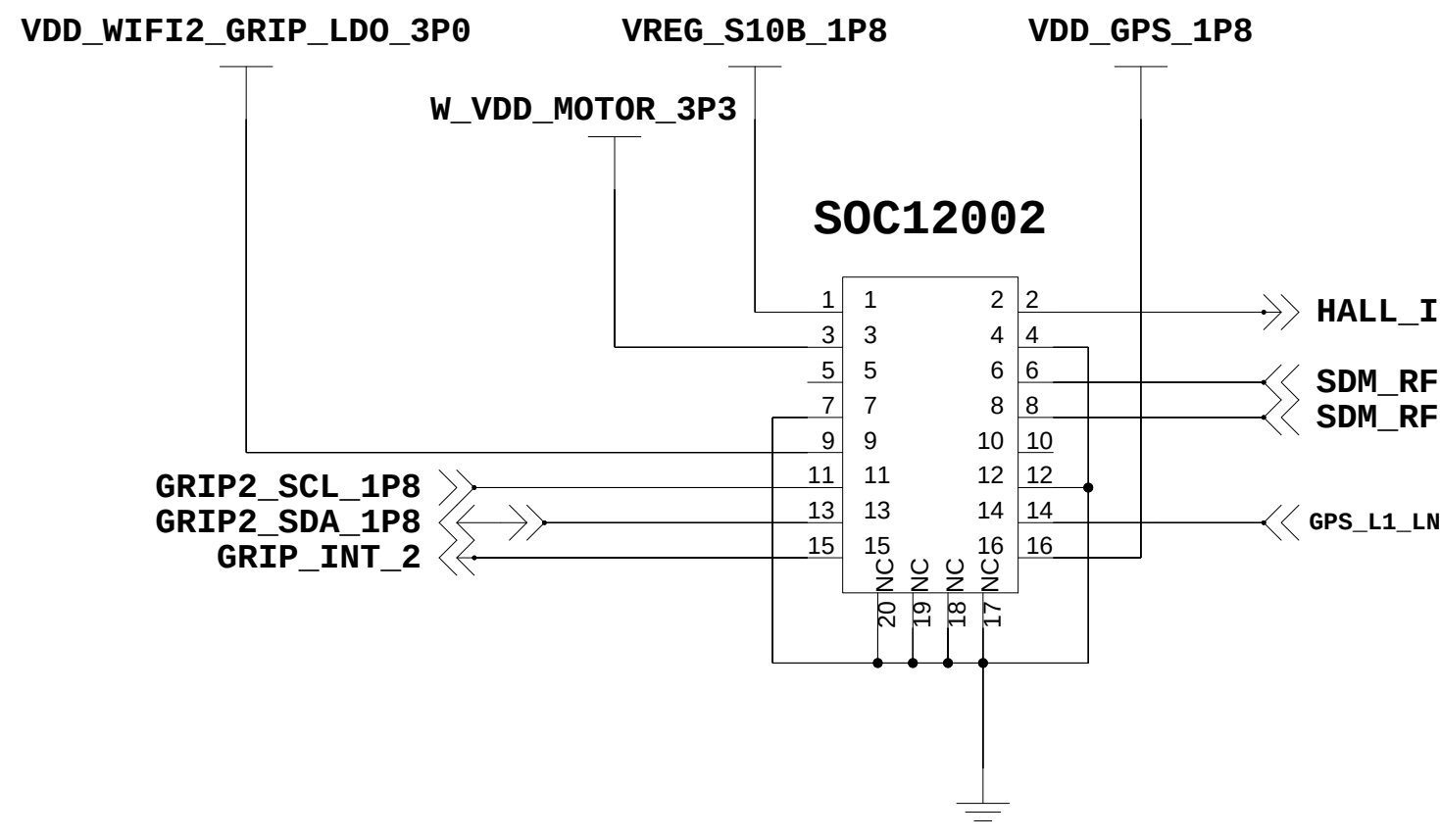
ANT PCB REGION

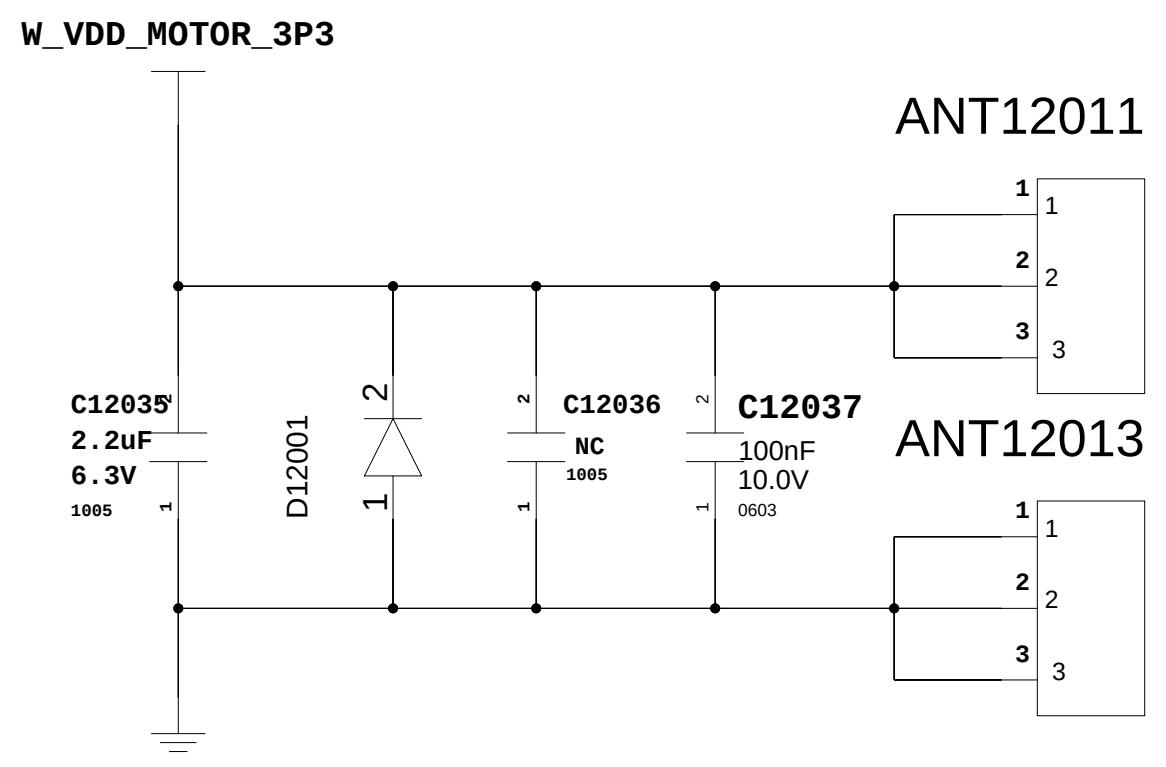
ANT PCB	SHORT
X906B(B/N/C)	R12010
X900(W)	R12011
ETC_1(E1)	R12012
ETC_2(E2)	R12016


R12010
NC
SHORT
B/N/C
R12011
0
SHORT
W
R12012
NC
SHORT
E1
R12016
NC
SHORT
E2
ground
ground

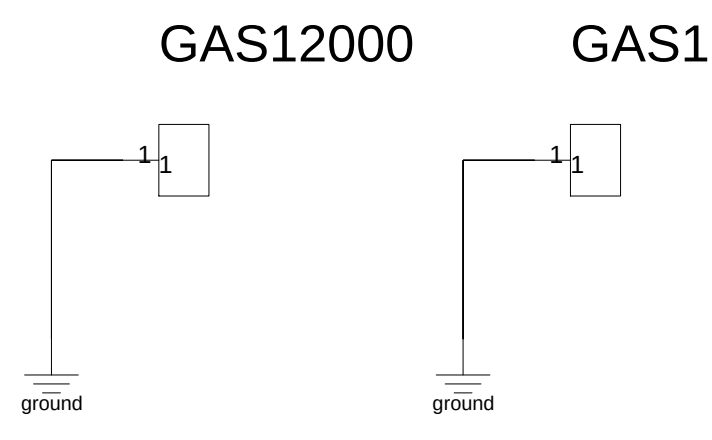
< SMD F/T >

1. VDD_WIFI2_GRIP_LDO_3P0
2. GRIP2_SCL_1P8
3. GRIP2_SDA_1P8
4. GRIP_INT_2
5. SDM_RFFE1_CLK
6. SDM_RFFE1_DATA
7. VREG_S10B_1P8
8. HALL_INT
9. GPS_L1_LNA_EN
10. VDD_GPS_1P8
11. GND

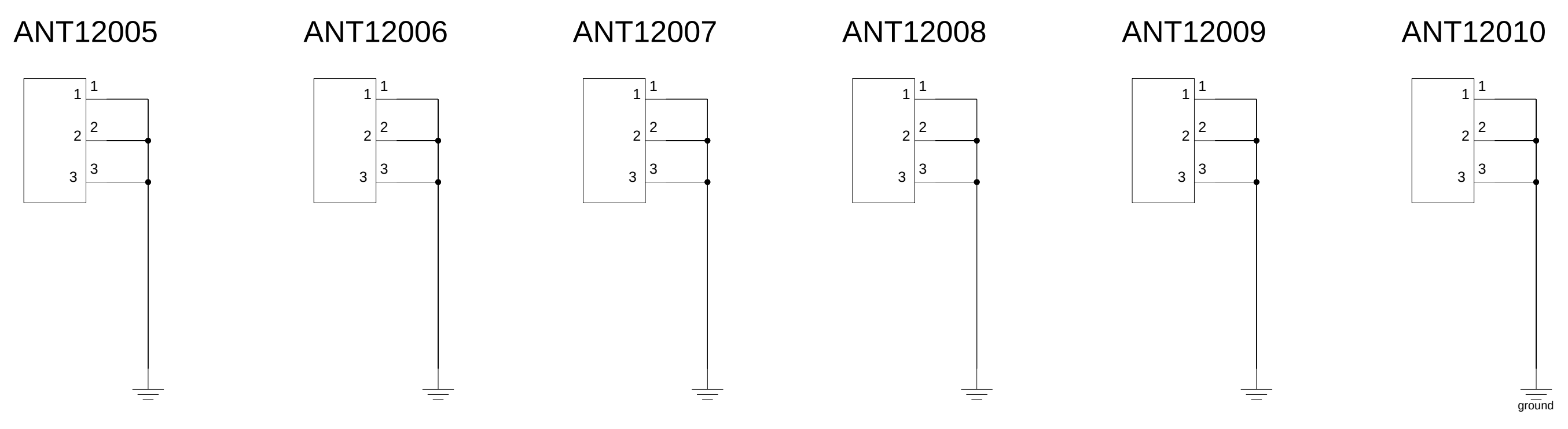

VDD_WIFI2_GRIP_LDO_3P0
VREG_S10B_1P8
VDD_GPS_1P8
W_VDD_MOTOR_3P3
SOC12002
HALL_INT
SDM_RFFE1_CLK
SDM_RFFE1_DATA
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
GPS_L1_LNA_EN


W_VDD_MOTOR_3P3
ANT12011
ANT12013
C12035
2.2uF
6.3V
D12001
C12036
NC
C12037
100nF
10.0V

MOTOR


GAS12000
GAS12001
ground
ground

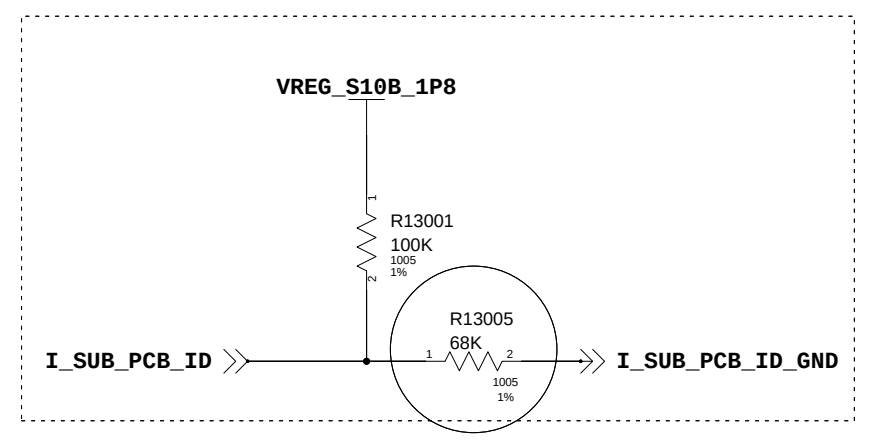
Front CONTACT


ANT12005
ANT12006
ANT12007
ANT12008
ANT12009
ANT12010
ground

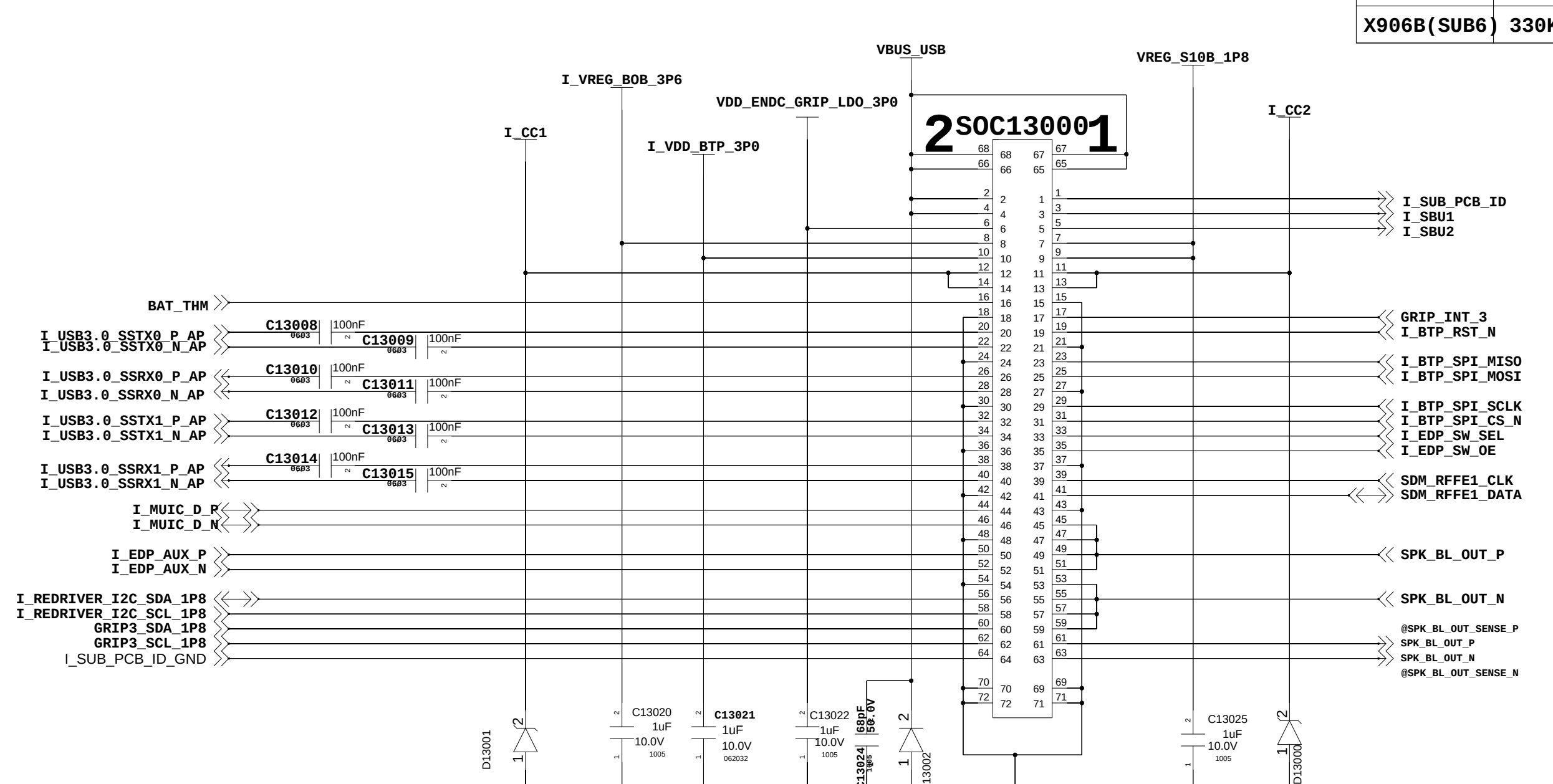
REAR CONTACT

TITLE TabS7P
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 12 of 14
Date Drawing Date

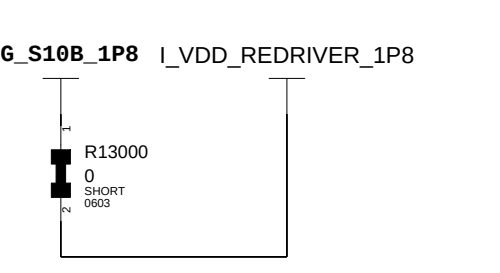
USB PCB

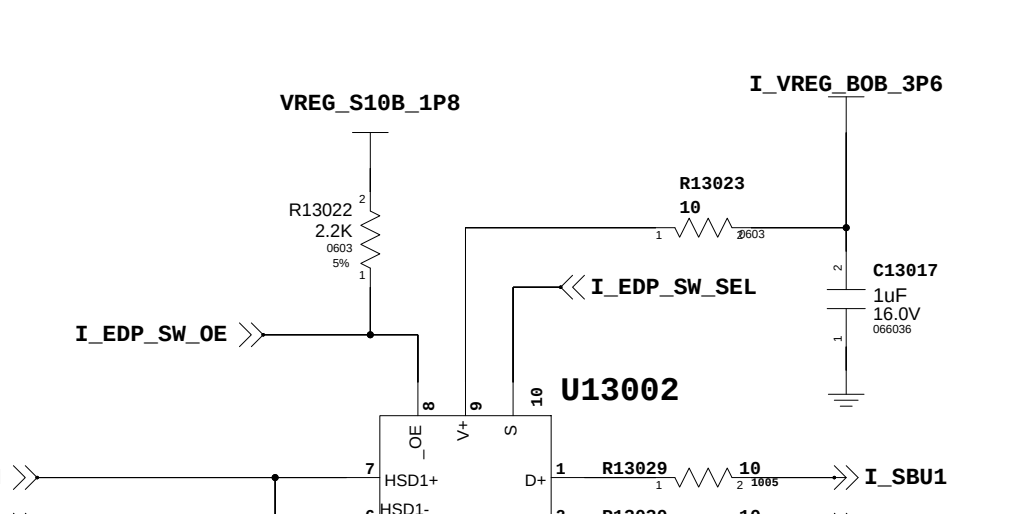

VREG_S10B_1P8
R13001
100K
I_SUB_PCB_ID
R13005
68K
I_SUB_PCB_ID_GND

USB PCB	R13005	ADC	Range	ID
	22K	324	0 - 600	0
X900(WIFI)	68K	728	601 - 900	4
	150K	1080	901 - 1200	5
X906B(SUB6)	330K	1381	1201 - 1500	6

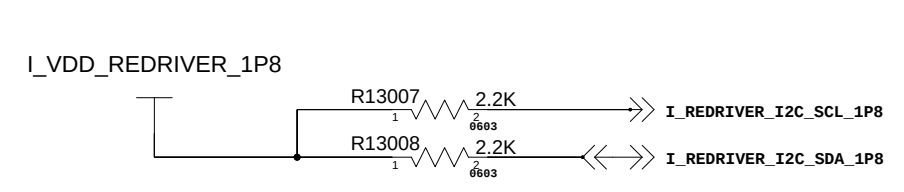

VBUS_USB
I_VREG_BOB_3P6
VDD_ENDC_GRIP_LDO_3P0
VREG_S10B_1P8
I_CC1
I_VDD_BTP_3P0
I_CC2
2 SOC13000 1
BAT_THM
I_USB3.0_SSTX0_P_AP
I_USB3.0_SSTX0_N_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSRX0_N_AP
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_N_AP
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_N_AP
C13008
C13009
C13010
C13011
C13012
C13013
C13014
C13015
100nF
I_MUIC_D_P
I_MUIC_D_N
I_EDP_AUX_P
I_EDP_AUX_N
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
GRIP3_SDA_1P8
GRIP3_SCL_1P8
I_SUB_PCB_ID_GND
I_SUB_PCB_ID
I_SBU1
I_SBU2
GRIP_INT_3
I_BTP_RST_N
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
I_EDP_SW_SEL
I_EDP_SW_OE
SDM_RFFE1_CLK
SDM_RFFE1_DATA
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
D13001
C13020
C13021
C13022
C13024
68pF
50.0V
D13002
C13025
D13000
1uF
10.0V

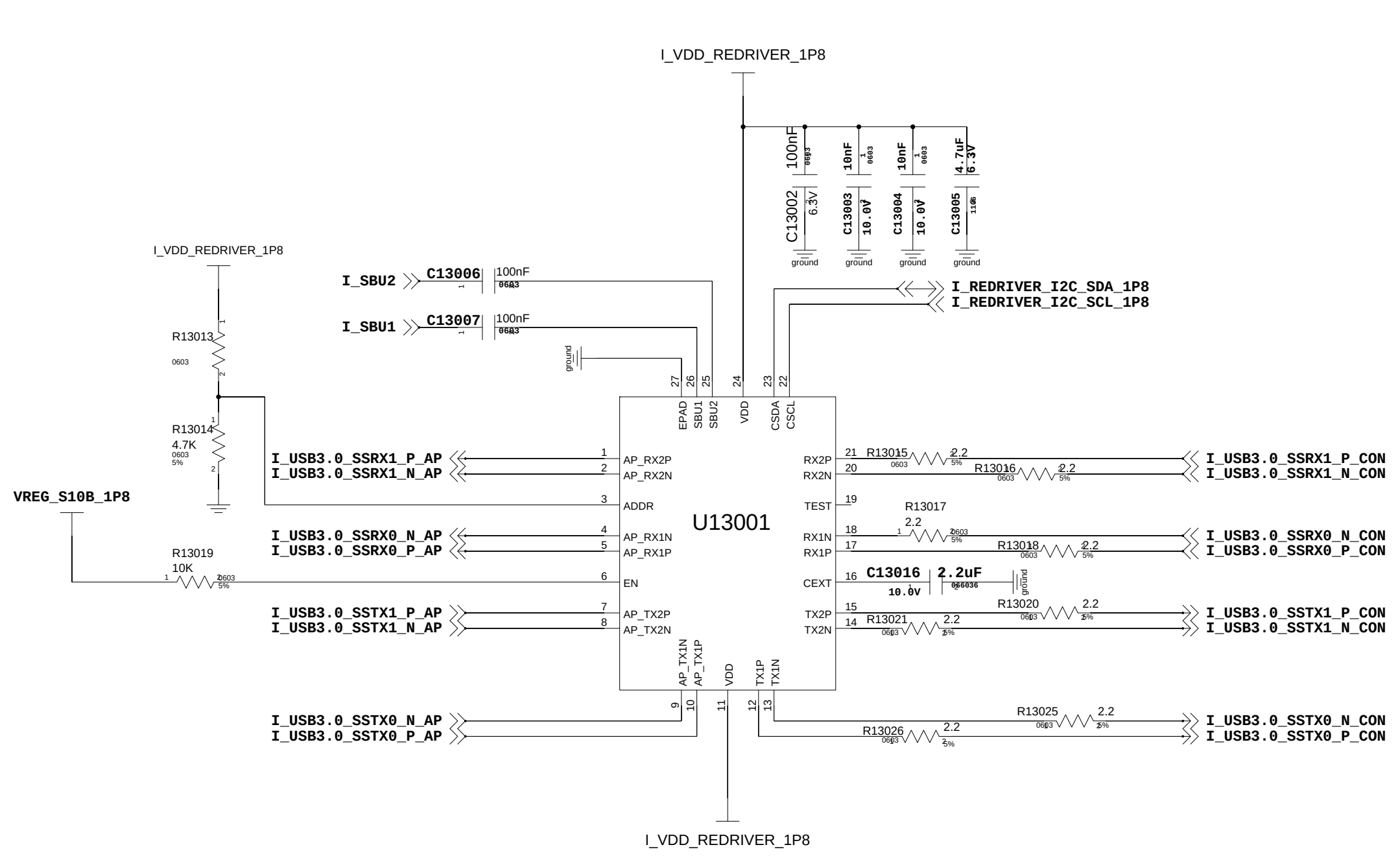
I/F CONN (USB)


VREG_S10B_1P8
I_VDD_REDRIVER_1P8
R13000
0
SHORT

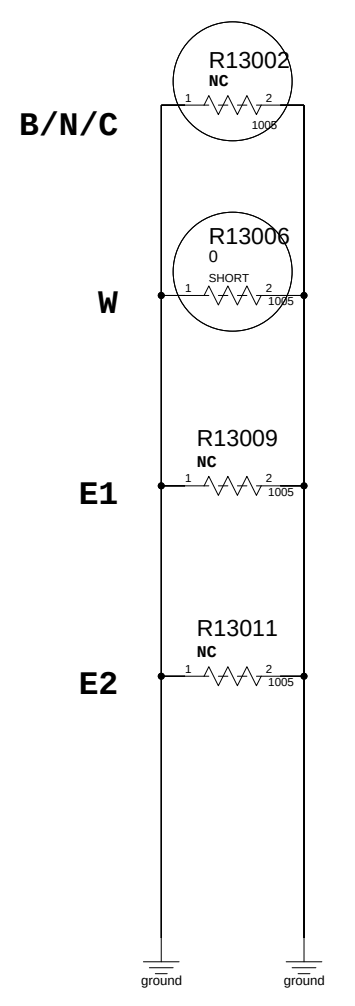

VREG_S10B_1P8
I_VREG_BOB_3P6
R13022
2.2K
R13023
10
C13017
1uF
16.0V
I_EDP_SW_OE
I_EDP_SW_SEL
U13002
I_EDP_AUX_N
I_EDP_AUX_P
HSD1+
HSD1-
HSD2+
HSD2-
OE
V+
S
D+
D-
GND
R13029
R13030
10
I_SBU1
I_SBU2

DP Switch


I_VDD_REDRIVER_1P8
R13007
2.2K
R13008
2.2K
I_REDRIVER_I2C_SCL_1P8
I_REDRIVER_I2C_SDA_1P8


I_VDD_REDRIVER_1P8
C13002
100nF
6.3V
C13003
10nF
10.0V
C13004
10nF
10.0V
C13005
4.7uF
6.3V
I_SBU2
C13006
100nF
I_SBU1
C13007
100nF
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
R13013
R13014
4.7K
VREG_S10B_1P8
R13019
10K
U13001
EPAD
SBU1
SBU2
VDD
CSDA
CSCL
AP_RX2P
AP_RX2N
ADDR
AP_RX1N
AP_RX1P
EN
AP_TX2P
AP_TX2N
AP_TX1N
AP_TX1P
TX1P
TX1N
RX2P
RX2N
TEST
RX1N
RX1P
CEXT
TX2P
TX2N
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_N_AP
I_USB3.0_SSRX0_N_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_N_AP
I_USB3.0_SSTX0_N_AP
I_USB3.0_SSTX0_P_AP
R13015
R13016
R13017
R13018
C13016
2.2uF
10.0V
R13020
R13021
R13025
R13026
2.2
I_USB3.0_SSRX1_P_CON
I_USB3.0_SSRX1_N_CON
I_USB3.0_SSRX0_N_CON
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSTX1_P_CON
I_USB3.0_SSTX1_N_CON
I_USB3.0_SSTX0_N_CON
I_USB3.0_SSTX0_P_CON
I_VDD_REDRIVER_1P8

REDRIVER & DP

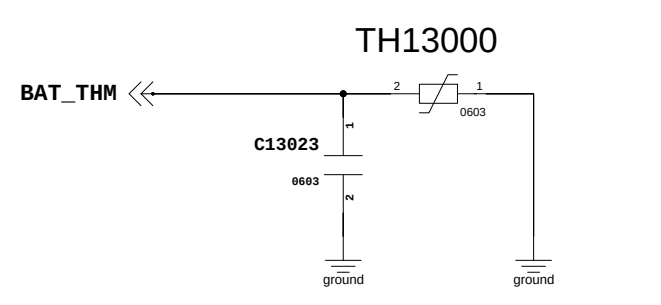

B/N/C
R13002
NC
W
R13006
0
SHORT
E1
R13009
NC
E2
R13011
NC

USB PCB REGION

USB PCB	SHORT
SUB6(B/N/C)	R13002
WIFI(W)	R13006
ETC1(E1)	R13009
ETC2(E2)	R13011

< USB PCB - SMD F/T

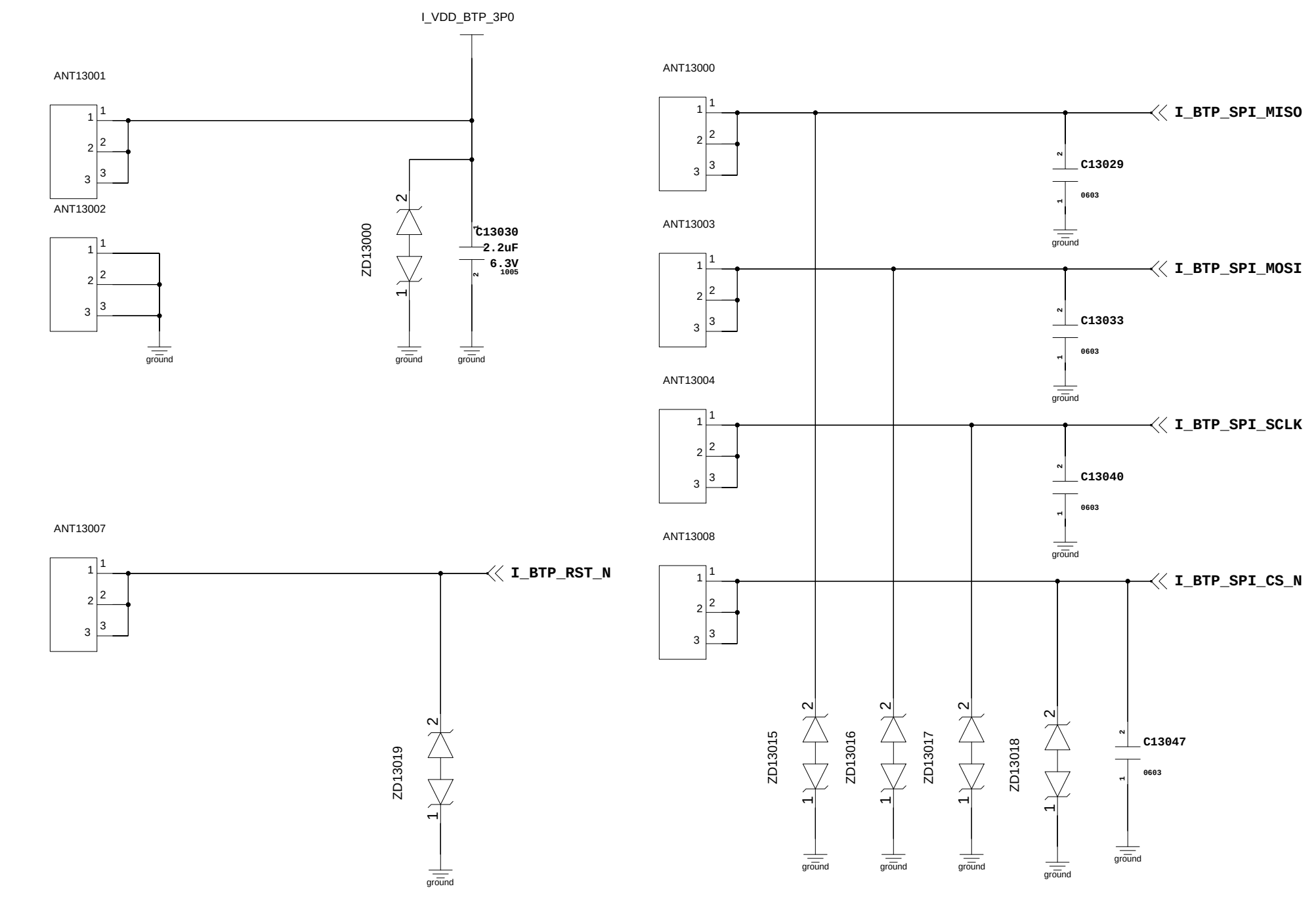
1. VREG_S10B_1P8
2. BAT_THM
3. SDM_RFFE1_CLK
4. SDM_RFFE1_DATA
5. GND
6. SPK_BL_OUT_P
7. SPK_BL_OUT_SENSE_P
8. SPK_BL_OUT_N
9. SPK_BL_OUT_SENSE_N
10. VDD_ENDC_GRIP_LDO_3P0
11. GRIP_INT_3
12. GRIP_SDA_1P8
13. GRIP_SCL_1P8


TH13000
BAT_THM
C13023

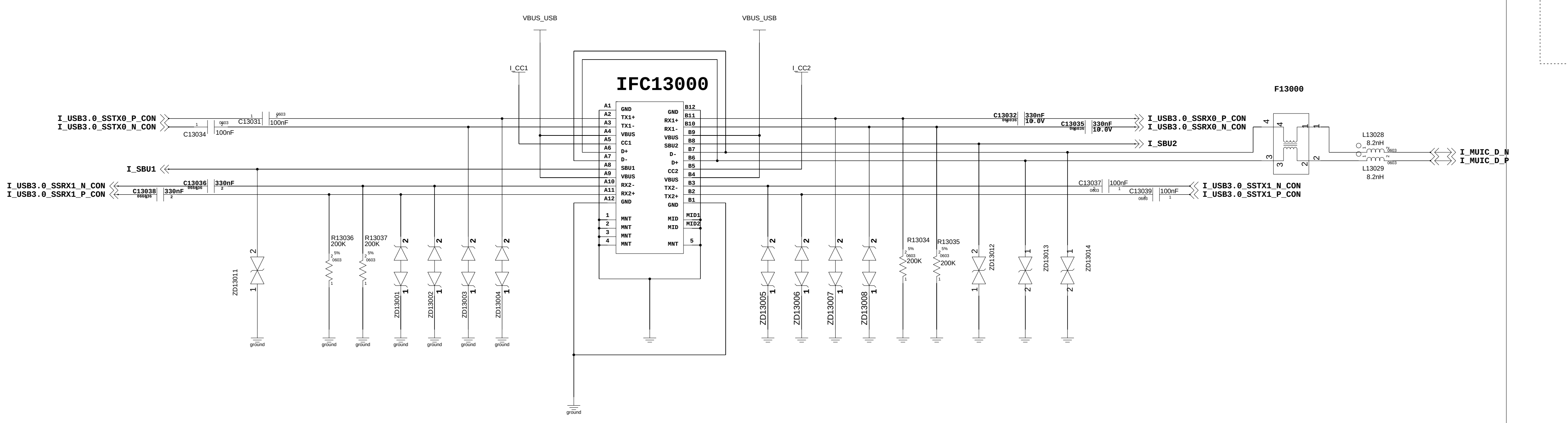
BAT THM

ENDC_GRIP_SENSOR(GRIP #4)

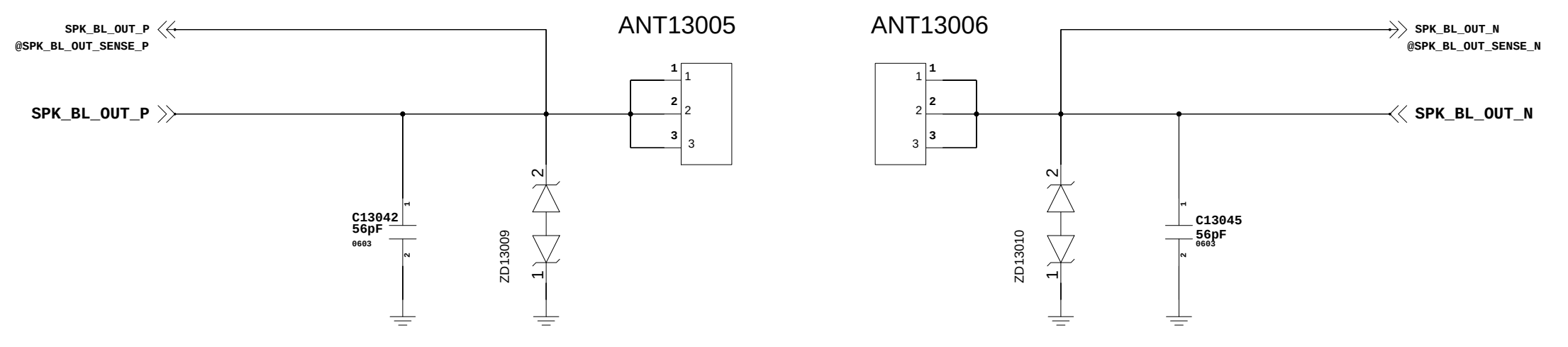
WIFI : NC


I_VDD_BTP_3P0
ANT13001
ANT13002
ZD13000
C13030
2.2uF
6.3V
ANT13000
ANT13003
ANT13004
ANT13008
ANT13007
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
I_BTP_RST_N
C13029
C13033
C13040
C13047
ZD13019
ZD13015
ZD13016
ZD13017
ZD13018

FOD CON.

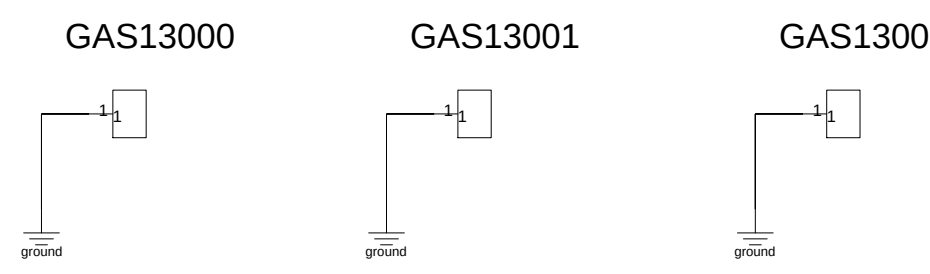

VBUS_USB
VBUS_USB
I_CC1
I_CC2
IFC13000
I_USB3.0_SSTX0_P_CON
I_USB3.0_SSTX0_N_CON
C13031
C13034
100nF
I_SBU1
I_USB3.0_SSRX1_N_CON
I_USB3.0_SSRX1_P_CON
C13036
C13038
330nF
GND
TX1+
TX1-
VBUS
CC1
D+
D-
SBU1
RX2-
RX2+
MNT
RX1+
RX1-
SBU2
CC2
TX2-
TX2+
MID
C13032
C13035
330nF
10.0V
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSRX0_N_CON
I_SBU2
C13037
C13039
100nF
I_USB3.0_SSTX1_N_CON
I_USB3.0_SSTX1_P_CON
F13000
L13028
L13029
8.2nH
I_MUIC_D_N
I_MUIC_D_P
ZD13011
R13036
R13037
200K
ZD13001
ZD13002
ZD13003
ZD13004
ZD13005
ZD13006
ZD13007
ZD13008
R13034
R13035
200K
ZD13012
ZD13013
ZD13014

Type-C Conn.

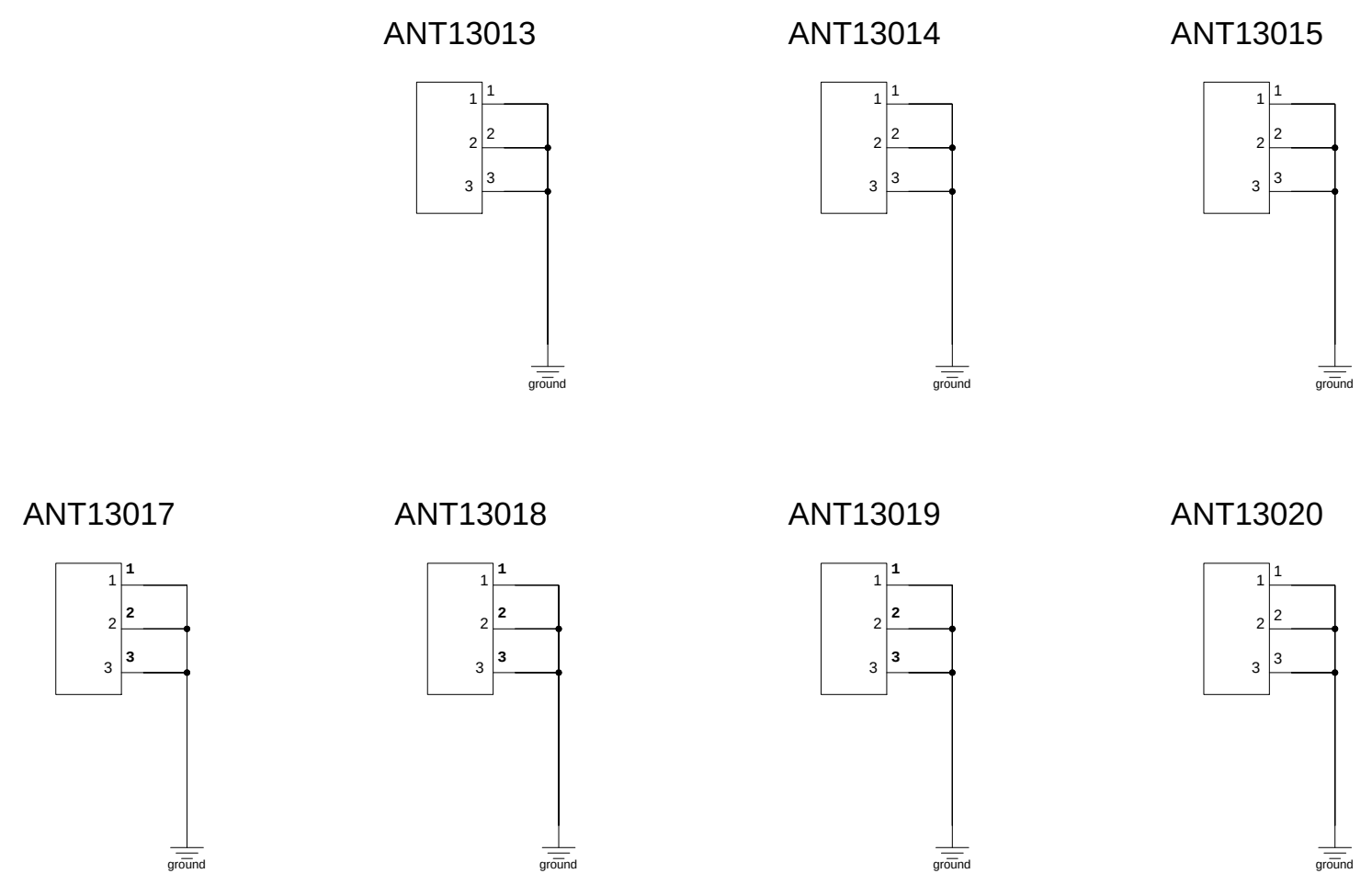

SPK_BL_OUT_P
@SPK_BL_OUT_SENSE_P
SPK_BL_OUT_P
ANT13005
ANT13006
C13042
56pF
ZD13009
ZD13010
C13045
56pF
SPK_BL_OUT_N
@SPK_BL_OUT_SENSE_N
SPK_BL_OUT_N

SPK BL CON.

BOTHHOLE/CONTACT

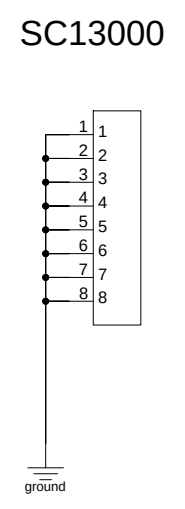

GAS13000
GAS13001
GAS13002

Front CONTACT


ANT13013
ANT13014
ANT13015
ANT13017
ANT13018
ANT13019
ANT13020

REAR CONTACT(0.9t)

SHIELD CAN



Re-Driver

< BOTTOM S/C >

SUB1 ANT

(DRX_LB/MB/HB / DRX_n41 / ENDC_LB/MB TX)

WIFI : NC

SUB2 ANT

(DRX_MIMO_MB/HB / DRX_n77/n78 / DRX_MIMO_n41)

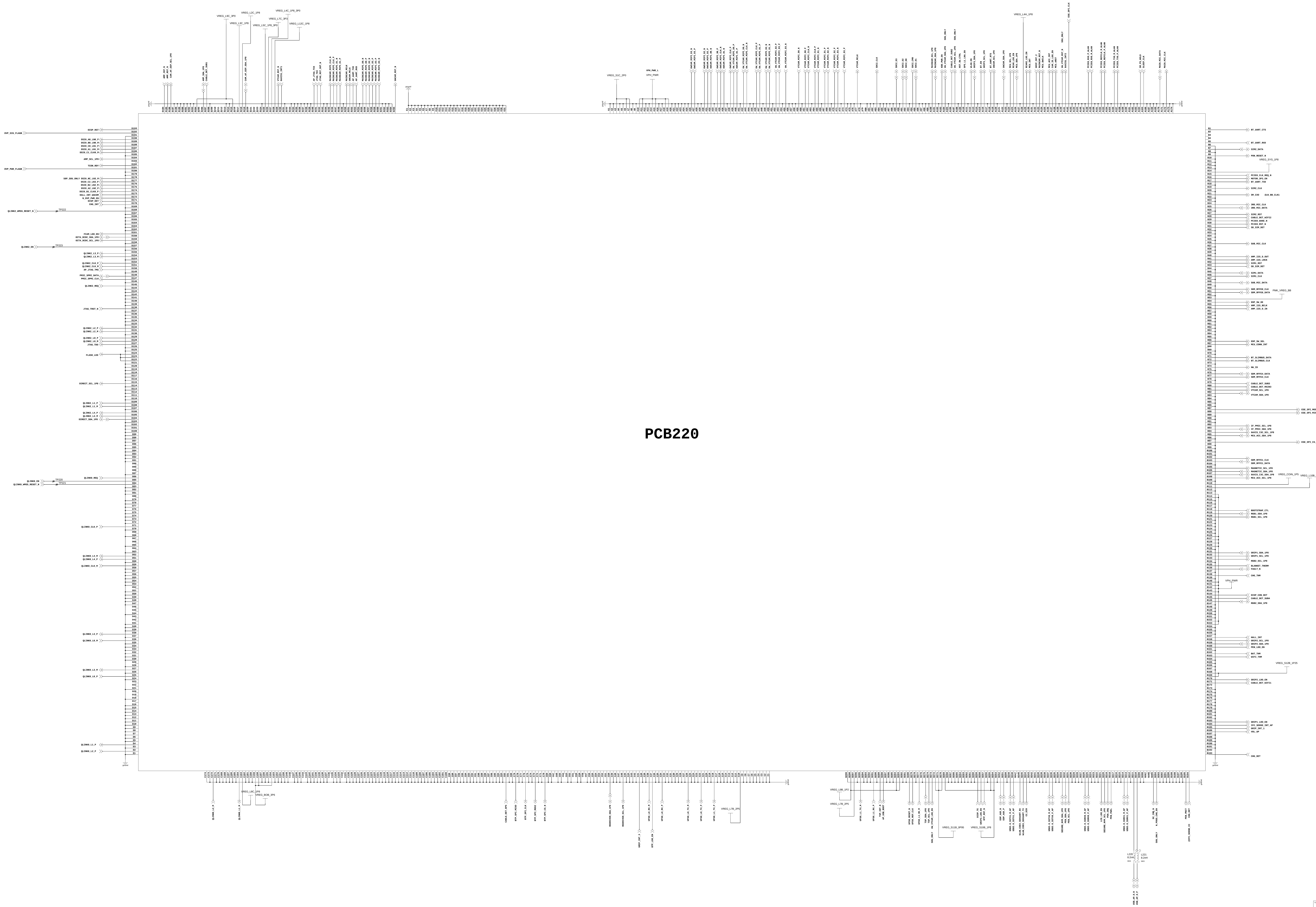
Cable CLIP

SUB4 ANT

(PRX_MIMO_MB/HB / PRX_MIMO_n77/n78 / PRX_MIMO_n41)

TITLE	TabS7P
SIZE	A1
DWG NO	PDM NUMBER
REV	Rev
SHEET	13 of 17
Date	Drawing Date
DRAWN BY	User Name

COREPCB PAD



PCB220
VREG_L9C_3P0
VREG_L2C_1P8
VREG_L3C_1P8
VREG_L4C_1P8_3P0
VREG_L7C_3P3
VREG_L5C_1P8_3P0
VREG_L12C_1P8
AMP_INT_N
AMP_RST_N
CAM_AF_EEP_SCL_1P8
AMP_SDA_1P8
CABLE_DET_SUB1
CAM_AF_EEP_SDA_1P8
VTCAM_RST_N
6AXIS1_INT1
AP_JTAG_TCK
JTAG_TDI
JTAG_RST_OUT_N
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MCLK
UWCAM_MCLK
AP_UART_TXD
AP_UART_RXD
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D3_N
UWCAM_RST_N
ground
VREG_S1C_2P0
VPH_PWR_L
VPH_PWR
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D0_P
UWCAM_MIPI_D1_P
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D1_P
VTCAM_MIPI_D2_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_D3_N
VTCAM_MIPI_D3_P
VTCAM_MCLK
SDC2_CLK
SDC2_D3
SDC2_D2
SDC2_D0
SDC2_CMD
SDC2_D1
MAINCAM_SCL_1P8
MAINCAM_SDA_1P8
S8U_ONLY
KBD_PWR_EN
UW_VTCAM_SDA_1P8
CABLE_DET_SUB2
UW_VTCAM_SCL_1P8
HST_SW_CTRL
GPS_L1_LNA_EN
WLAN_EN
GRIP3_SDA_1P8
BT_EN
GRIP3_SCL_1P8
BT_UART_RTS
UWCAM_SCL_1P8
UWCAM_SDA_1P8
MCU_SCL_1P8
RCAM_LDO_EN
MCU_SDA_1P8
VREG_L4H_1P8
RCAM2_LDO_EN
MCU_INT
GRIP_INT_2
MAINCAM_RST_N
MCU_SWCLK
MCU_ACC_INT
CAM_IO_LDO_EN
MCU_NRST
UW_VTCAM_RST_N
6AXIS1_INT2
ESE_SPI_CLK
PCIE0_RX0_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_TX0_M_WLAN
PCIE0_TX0_P_WLAN
SM_PS_HOLD
SLEEP_CLK
MAIN_MIC_DATA
MAIN_MIC_CLK
DISP_RST
OVP_SIG_FLAGB
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_C1_CLK0_M
AMP_SCL_1P8
TCON_RDY
OVP_PWR_FLAGB
S8P_S8U_ONLY DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_B2_LN2_M
DSI0_A2_LN2_P
DSI0_B1_CLK0_P
HALL_INT_WACOM
N_OVP_PWR_EN
DISP_DET
CHG_INT
QLINK2_WMSS_RESET_N
TP222
FCAM_LDO_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
QLINK2_EN
TP223
QLINK2_L3_P
QLINK2_L3_M
QLINK2_CLK_P
QLINK2_CLK_M
AP_JTAG_TMS
PMIC_SPMI_DATA
PMIC_SPMI_CLK
QLINK2_REQ
JTAG_TRST_N
QLINK2_L2_P
QLINK2_L2_M
QLINK2_L0_P
QLINK2_L0_M
JTAG_TDO
FLASH_LED
DIRECT_SCL_1P8
QLINK2_L1_P
QLINK2_L1_M
QLINK2_L4_P
QLINK2_L4_M
DIRECT_SDA_1P8
QLINK0_REQ
QLINK0_EN
TP220
QLINK0_WMSS_RESET_N
TP221
QLINK0_CLK_P
QLINK0_L4_M
QLINK0_L4_P
QLINK0_CLK_M
QLINK0_L3_P
QLINK0_L0_M
QLINK0_L3_M
QLINK0_L0_P
QLINK0_L1_P
QLINK0_L2_P
BT_UART_CTS
BT_UART_RXD
SIM2_DATA
PON_RESET_N
VREG_SYS_1P8
PCIE0_CLK_REQ_N
MOTOR_3P3_EN
BT_UART_TXD
SIM2_CLK
SM_CXO
@LN_BB_CLK1
3RD_MIC_CLK
3RD_MIC_DATA
SIM2_RST
CABLE_DET_WIFI2
PCIE0_WAKE_N
PCIE0_RST_N
SD_SIM_DET
SUB_MIC_CLK
AMP_I2S_D_OUT
AMP_I2S_LRCK
SIM1_RST
SD_SIM_DET
SIM1_DATA
SIM1_CLK
SUB_MIC_DATA
SDM_RFFE0_CLK
SDM_RFFE0_DATA
PMK_VREG_BB
EDP_SW_OE
AMP_I2S_BCLK
AMP_I2S_D_IN
EDP_SW_SEL
MCU_CONN_INT
BT_SLIMBUS_DATA
BT_SLIMBUS_CLK
HW_ID
SDM_RFFE4_DATA
SDM_RFFE4_CLK
CABLE_DET_SUB3
CABLE_DET_MAIN3
VTCAM_SCL_1P8
VTCAM_SDA_1P8
ESE_SPI_MOSI
ESE_SPI_MISO
IF_PMIC_SCL_1P8
IF_PMIC_SDA_1P8
6AXIS_I3C_SCL_1P8
MCU_ACC_SDA_1P8
ESE_SPI_CS_N
SDM_RFFE1_CLK
SDM_RFFE1_DATA
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
6AXIS_I3C_SDA_1P8
MCU_ACC_SCL_1P8
VREG_COIN_1P5
VREG_L10B_1P2
BOOTSTRAP_CTL
RGB1_SDA_1P8
RGB1_SCL_1P8
GRIP1_SDA_1P8
GRIP1_SCL_1P8
RGB2_SCL_1P8
BLANKET_THERM
FAULT_N
CHG_THM
VPH_PWR
DISP_CON_DET
CABLE_DET_SUB4
RGB2_SDA_1P8
HALL_INT
GRIP2_SCL_1P8
GRIP2_SDA_1P8
PEN_LDO_EN
BAT_THM
WIFI_THM
VREG_S12B_1P25
GRIP2_LDO_EN
CABLE_DET_WIFI1
GRIP1_LDO_EN
IFC_SENSE_INT_AP
GRIP_INT_1
VOL_UP
CHG_DET
QLINK0_L2_M
QLINK0_L1_M
VREG_L8C_1P8
VREG_BOB_3P6
CABLE_DET_GPS
BTP_SPI_MISO
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_CS_N
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
GRIP_INT_3
BTP_LDO_EN
UFS0_L0_RX_M
UFS0_L0_RX_P
UFS0_L0_TX_M
UFS0_L0_TX_P
UFS0_L1_TX_P
VREG_L7B_2P5
VREG_L9B_1P2
VREG_L7B_2P5
UFS0_L1_TX_M
UFS0_L1_RX_P
TSP_INT_N
AP_USB_BOOT
UFS0_RESET_N
UFS0_REF_CLK
UFS0_L1_RX_M
TSP_SDA_1P8
TSP_SCL_1P8
UW_VTCAM_LDO_EN
S8U_ONLY
VREG_S11B_0P95
DISP_TE
GRIP3_LDO_EN
BTP_RST_N
VREG_S10B_1P8
EDP_AUX_N
EDP_AUX_P
USB3.0_SSTX1_M_AP
USB3.0_SSTX1_P_AP
WLAN_COEX_SDXUART_RX
WLAN_COEX_SDXUART_TX
CC_DIR
USB3.0_SSTX0_M_AP
USB3.0_SSTX0_P_AP
SECURE_NVM_SDA_1P8
PEN_SDA_1P8
PEN_SCL_1P8
USB3.0_SSRX0_M_AP
USB3.0_SSRX0_P_AP
LCD_LDO_EN
SECURE_NVM_SCL_1P8
PEN_IRQ
PEN_FWEL
USB3.0_SSRX1_M_AP
USB3.0_SSRX1_P_AP
L220
8.2nH
0603
L221
8.2nH
0603
USB_AP_D_N
USB_AP_D_P
DC_IRQ_N
N_POGO_CHG_EN
S8U_ONLY
PEN_PDCT
CHG_DET
@IFC_SENSE_CC

TITLE
TabS7P
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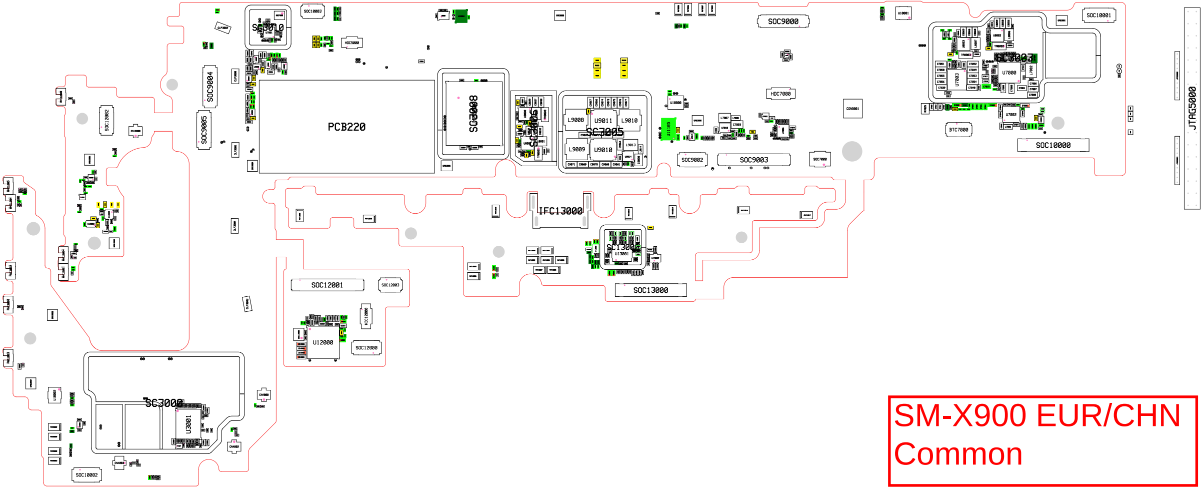
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 22 of 15
Date Drawing Date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SC3006
SC3005
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U9011
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L9013
U11105
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HDC7000
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TA8003
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C7050
C7055
C7056
C7052
C7049
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C7054
C7086
C7087
C7088
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R7025
BTC7000
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JTAG5000
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CN12000
IFC13000
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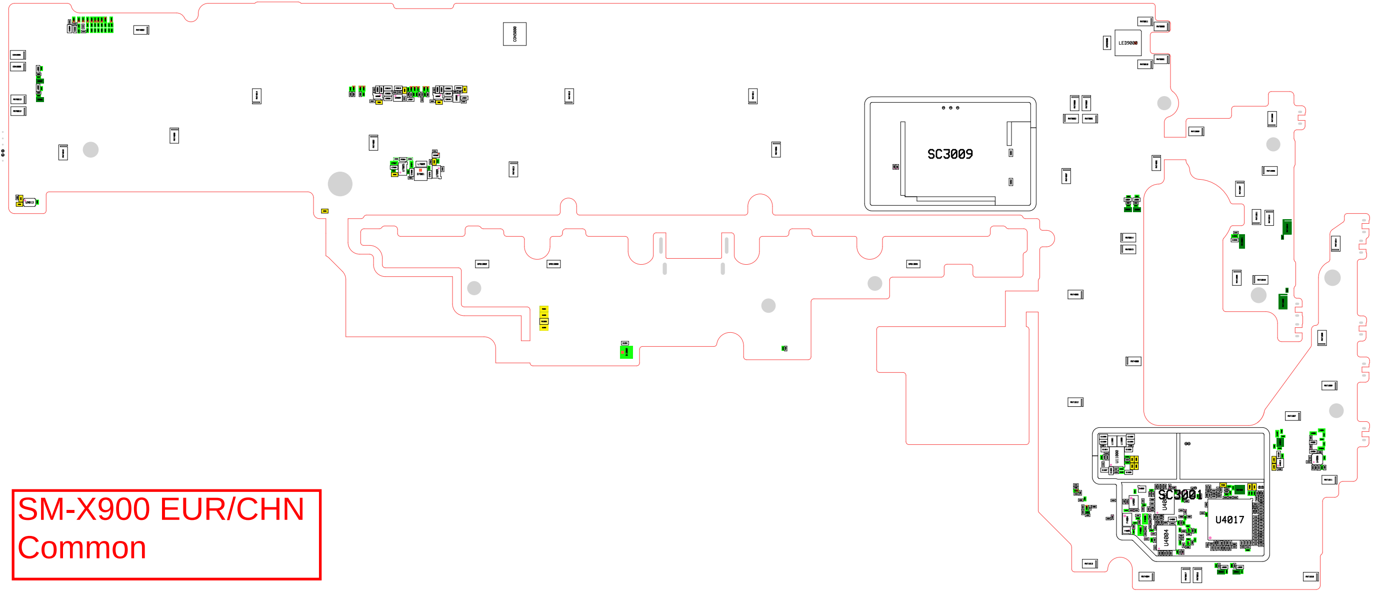
CN4003
SOC10002

SM-X900 EUR/CHN
Common


CON5800
LED9000
SC3009
U9013
L7009
D7001
U7005
U7009
U11000
SC3001
U4017
U4004

SM-X900 EUR/CHN
Common